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I Introduction

1. Historical background

Irish is a Celtic[1] language which is the earliest recorded language in Ireland. It belongs to a group of four languages which are still extant (the other three are Scottish Gaelic, Welsh and Breton) and two more which exist in revived forms (Cornish and Manx which died out in the eighteenth and mid twentieth centuries respectively, George and Broderick 1993). Linguistically, it can be grouped with Scottish Gaelic (Watson 1994) and Manx (Broderick 1984-6, Thomson 1992, N. Williams 1994a) both of which derive from transported forms of Irish taken to Scotland and the Isle of Man during the first millennium CE.

The term *Gaelic* /ge:lɪk/, as a label for the language, is not frequent in Ireland. The designation *Scottish Gaelic* is used for the Celtic language which is still spoken in Scotland, mostly along the west coast, and which is historically derived from northern forms of Irish (see Appendix 2). The Scots themselves refer to their variety of Gaelic by their pronunciation of the word, i.e. /galɪk/, written *Gaidhlig* in Scottish Gaelic[2]. The word for the language is feminine in Irish (and Scottish Gaelic) so that the initial sound changes to a velar fricative on using the article before it: *An Ghaeilge* /ən ɣeːlʲgʲə/[3] 'the Irish language, Irish'.

In linguistic literature, Irish (D. Ó Baoill 2009), Scottish Gaelic (Gillies 2009) and Manx (Broderick 2009) are referred to collectively as *Q-Celtic* (Eska and D. E. Evans 1993). This term derives from the fact that Indo-European $*k^w$/$*k$ (de

Bernardo Stempel 2013; Fife and King 1998) remained unchanged in this section of Insular Celtic (Isaac 1993, 2007), having shifted to *p* in Welsh (Thomas 1992a; T. A. Watkins 1993), Cornish (Thomas 1992b, George 1993) and Breton (Ternes 1992; Humphreys 1993; Stephens 1993), hence the designation *P-Celtic* for the latter three languages (Pedersen 1909-13, Lewis and Pedersen 1962 [1937], Russell 1995), compare Irish *ceann* 'head', *ceathair* 'four', *mac* 'son' with Welsh *pen, pedwar, mab* (← /map/) 'son' or Latin *planta* 'plant' with early Irish *cland*, later *clann* 'family'.

The proto-Celtic language appeared on the European mainland in the last millennium BCE. Continental Celtic (Eska and D. E. Evans 1993: 26-43) is a term which is given to Celtic as spoken chiefly in Gaul and on the Iberian Peninsula (Tovar 1961: 76-90). Between roughly 500 and 300 BCE Celtic speakers moved across to the British Isles. The Celts who came to Ireland were speakers of Q-Celtic and possibly came through different routes and at separate times from the P-Celtic speakers who settled in Britain.

A distinctive feature of Celtic vis à vis most of the remaining Indo-European languages is the loss of initial *p* in all positions except adjacent to a homorganic obstruent (*t, n, s*, Hamp 1951: 230). This can be seen by comparing many Latin or Greek forms with their Irish cognates, e.g. Latin *pisces* 'fish' and Irish *iasc*, Greek *polu-* and Irish *il-* 'poly-, multi-'. The relationship of Celtic to other branches of Indo-European is unclear. Formerly scholars thought that there was an earlier unity between Italic and Celtic on certain morphological grounds, but this has not been established with certainty (see the discussions in Kortlandt 2007).

The location of the Celts in Europe can be determined by considering two archaeologically defined cultures in the first millennium BCE. The first is the Hallstatt culture, named after a location in the Salzkammergut in Austria, and the second is the La Tène culture, after a location just north of Lake Neuchâtel in Switzerland. The Hallstatt culture flourished during the early Iron Age (*c* 800-450 BCE) and the La Tène culture somewhat later (*c* 450-100 BCE).

1.1. Ogam (400-600)

After the coming of the Celts to Britain and Ireland towards the end of the first millennium BCE there are no records of their language for many centuries. In Ireland the first remains are stone inscriptions – referred to collectively as Ogam – which appear in the first centuries CE and which provide a glimpse of how Q-Celtic had evolved after its separation from continental forms of Celtic.

Ogam is a form of writing in which letters are represented by a series of horizontal or slanted notches on stone. Other materials such as wood and bone may have been used for short texts. A number of stone remains of Ogam are extant, chiefly in the South and South-West of Ireland, a few in South-West Wales and one or two remains in North-East Scotland (McManus 1991, Ziegler 1994). The period to which the Ogam stones belong is known as Primitive Old Irish. The majority of inscriptions which document this period stem from the fifth and sixth centuries and consist of personal names in the genitive found in formulae like ‘in memory of X’ or ‘dedicated to X’.

Even at this early stage there was a tendency to weaken consonants in intervocalic position. This was a low-level phonetic phenomenon, as yet without any consequences for the language, given that the inherited inflections remained intact. There were also geminate symbols which point to phonological length for consonants at that period of Irish (Harvey 1987a, 1987b).

1.2. Old Irish (600-900)

The period of early Irish for which remains are available in the Roman alphabet begins after the christianisation of Ireland in the fifth century (McCone 1994). The first documents are glosses and marginalia from the mid-eighth century contained in manuscripts found at Irish monastic sites on mainland Europe, above all in Germany (*Codex Paulinus* in Würzburg), Switzerland (*Codex Sangallensis* in St. Gall which contains a glossed version of Priscian's grammar) and Northern Italy (*Codex Ambrosianus* in Milan). This period lasted until the end of the ninth century. With the Viking invasions, which began in the late eighth century, insular Old Irish society came under considerable pressure. Monastic life, and with it literacy in the Old Irish language, suffered considerable setbacks.

By comparing a random selection of Latin loanwords in Irish one can see that part of the phonological makeup of the language was the lenition which had begun during the Ogam period. Thus one has *lebor* /ljevər/ later *leabhar* /ljaur/ from *liber* 'book', *sacart* /sagart/ from *sacerdos* 'priest'. Only medial geminates are resistant to lenition: *peccad* /pjekað/ (← */pekkad/) later *peaca* /pjakə/ from *peccatum* 'sin'.

The same applies to the Scandinavian loanwords towards the end of the first millennium (Cahill 1936), e.g. *margadh* /margað/, later /margə/, from *markaðr* 'market'. Cluster simplification can be seen in words like *fuinneog* /fɪnjo:g/ from *vindauga* 'window'. These developments continue well into the Middle Ages so that with Anglo-Norman loanwords from the thirteenth and fourteenth centuries one has similar lenition, e.g. *bagún* from *bacun* 'bacon', *buidéal* from *botel* 'bottle' (Risk 1971, 1974).

In Old Irish there was also a process termed vowel colouring or vowel affection. By this is meant a change in vowel height depending on the vowel in the following syllable, a type of umlaut which remained a characteristic of the language for a considerable time. For instance, /o/ shifts to /u/ before a following /i/ and /u/ becomes /o/ before a following /a/ or /o/ (Thurneysen 1946: 50-55). However, if one compares Irish with Germanic, one sees that these vowel changes, while occurring in clear phonological environments, never attained grammatical status as umlaut did in Germanic (except Gothic). There is a reason for this which has to do with the general makeup of the language at the time. Irish was losing vowels due to apocopate (loss of inflections) and syncope (contraction in polysyllabic forms) so that there was a concentration on consonants as the carriers of grammatical information. In Germanic, the status of umlaut in the formation of plurals was also supported by the status of vowel alternations (ablaut) in the group of so-called strong verbs. In present-day Irish there are few instances of grammatical information being carried by vowels alone. Furthermore, these may vary as a consequence of consonantal change. Vowel alternations on their own are not usually the signals for grammatical categories in Modern Irish.

1.3. Middle Irish (900-1200)

This is the period in which the classical standard of Old Irish declined and spoken forms slowly came to pervade written Irish. However, dialect formation is not yet evident in Middle Irish. There is some confusion in morphology with writers less and less sure about what constitutes classical Old Irish. The period draws to a close with the coming of the Normans at the end of the twelfth century, in 1169 in the South-East of the country (Dolley 1972). As with Old Irish, the manuscripts containing attestations of Middle Irish generally date from after this period (L. Breatnach 1994).

The simplification of the inflectional system was continued throughout the Middle Irish period with the nominal and verbal system of Old Irish very much reduced (Strachan 1905, L. Breatnach 1994). By the end of the Middle Irish period there was no longer a distinction between genitive and dative with most nouns and the complex system of verb prefixes had been greatly simplified either by these being dropped or by being absorbed into the stem of a verb and becoming opaque in form.

As part of the general direction of typological change, independent forms of personal pronouns (Ó Sé 1996) develop during this period. The old infixed pronouns are replaced by post-posed independent pronouns and synthetic forms of pronoun and copula are replaced by an invariable form of the copula followed by a personal pronoun.

Changes in the sound system led to certain realignments. The loss of /θ/ and /ð/ – probably in the thirteenth century (see the discussion in O'Rahilly 1926: 183-185) – resulted in different outcomes for lenition as an initial mutation. The lenition of /t/ resulted in /h/ and that of /d/ yielded /ɣ/.

1.4. Classical Modern Irish (1200-1600)

This period begins with the arrival of the Normans in the late twelfth century and finishes with the collapse of Irish aristocratic society in the early seventeenth century as a consequence of English military successes and the increasing anglicisation of Ireland. These developments signalled the end of an independent Gaelic society.

The old form of Irish society in which poets still had a place and a public function came to an end so that there was no continuation of a single written standard. Indeed it is unlikely that such a standard would have survived as it was long since remote from spoken varieties of the language. The seventeenth-century anglicisation of Ireland accelerated a process which had begun long before.

It is in this period that a series of instructions for poets were composed intending to act as guidelines for those wishing to use the classical standard for poetic composition at a time when the latter was no longer spoken. These are collectively known as the *Bardic Syntactical Tracts* (L. McKenna 1979 [1944]) or as *Irish Grammatical Tracts* (Bergin 1915-1925, 1969) and date back to the fifteenth century or earlier (Ó Cuív 1965: 142).

This period is characterised by the language of a professional class of poets, called *filí*, and is known as Classical Modern Irish (McManus 1994). The writers of the period were mostly employees of Irish courts (witness the quantity of praise-poetry produced, Ó Cuív 1965: 143). who were not necessarily conversant with the norms of Middle or indeed Old Irish which they emulated and the result was a form of language hampered by its own artificiality. In this period the dichotomy between the older norm and contemporary usage led to a tension between what was called *ceart na bhfileadh* 'the poets' standard' and *canamhain* 'speech', i.e. the spoken Irish of the time.

Linguistically, the Classical Modern Irish period is when the changes, initiated and partially carried through in the Middle Irish period, were consolidated. The verb prefixes are greatly reduced in number, for example *do-*, *ad-, no-* and *ro-* frequently level to *do-*, which was retained up to recently as a marker of the past (and still is found in Southern Irish as a general pre-verbal preterite marker). Independent personal pronouns became normal with a single form of the copula verb. Indeed this pattern, invariant verb form and independent personal pronoun, spread to other verbs and has become common in Modern Irish, despite some synthetic forms which were retained.

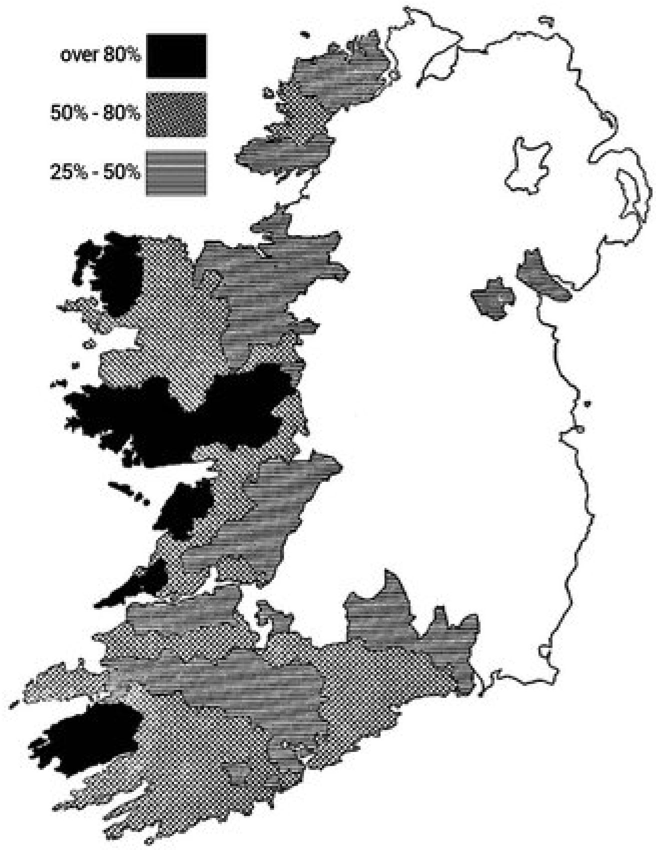
1.5. Early Modern Irish (1600-1900)

When the old Gaelic order came to an end in the early seventeenth century the system of patronage for Irish poets and scholars also declined rapidly (Cahill 1939, 1940). With that the use of classical written Irish declined and in the course of this century traces of dialects appear increasingly in documents (N. Williams 1994b: 447). It is certain that Irish had already become dialectally diverse but because of the nature of the textual record, many features of the dialects did not show up in writing.

The exclusion of Irish from public life resulted from the Penal Laws, a collective term for anti-Catholic, i.e. anti-Irish, legislation which greatly diminished the standing of the language and its speakers in Irish society. With further developments of the seventeenth century, notably the campaigns and expulsions by Oliver Cromwell in the late 1640s and early 1650s, language shift from Irish to English began in earnest. This was accompanied by bilingualism and in the seventeenth, eighteenth and nineteenth centuries there are macaronic texts which testify to this, particularly in the eighteenth century (Mac Mathúna 2007: 183-217), but the dominance of English in the nineteenth century led to a demise in bilingual texts.

The language shift was a process which was never to be reversed. Other major demographic events, especially the Great Famine of the late 1840s and the massive emigration this triggered, led to a serious drop in the numbers of Irish speakers (de Fréine 1966, 1977) so that by the beginning of the twentieth century the Irish-speaking districts were fragmented into three areas, Cork-Kerry, Galway-Mayo and Donegal with a small enclave in Waterford and a later transported one in Co. Meath (from the late 1920s onwards).

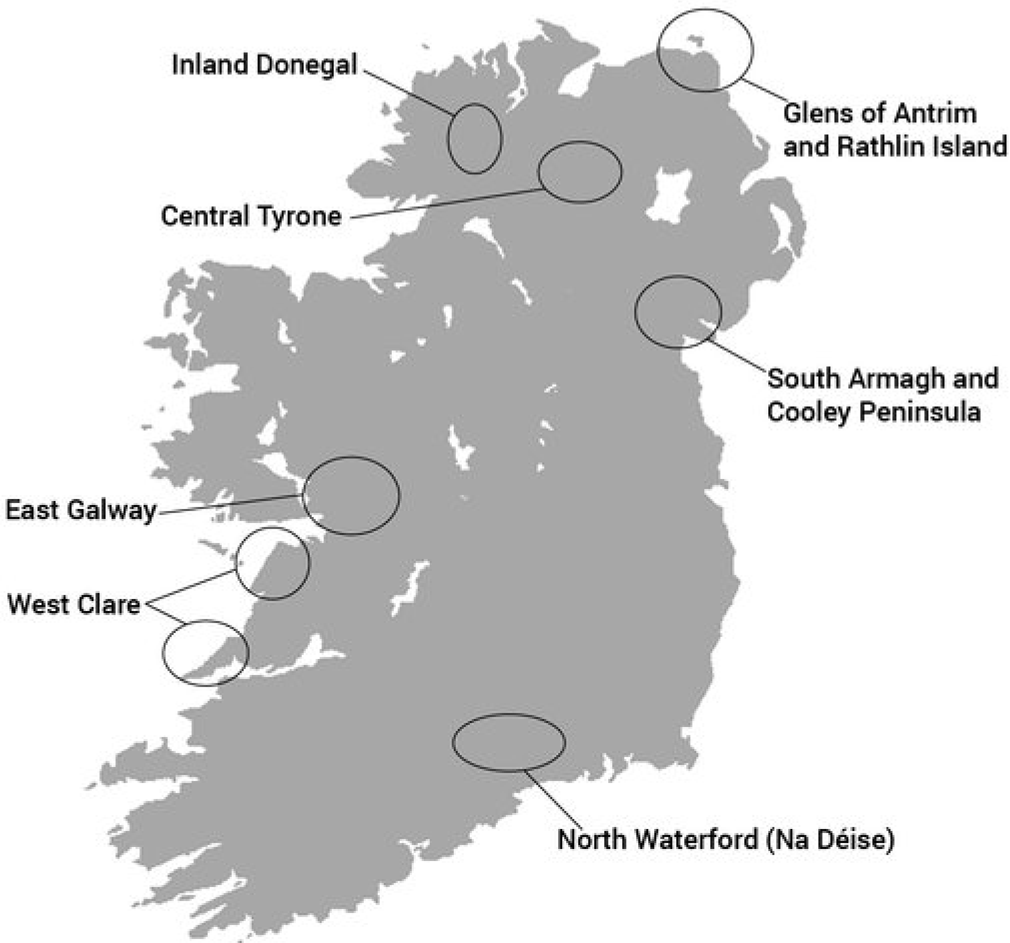
Map 1. The distribution of Irish immediately after the Great Famine (1845-1848) going on the 1851 census (after Ó Cuív 1969)



1.6. Modern Irish (1900-)

The persistent emigration of the second half of the nineteenth century reduced the numbers of native speakers of Irish significantly. But it was not the only reason for the continuing decline of the language. After Catholic emancipation (resulting from the Roman Catholic Relief Act of 1829) a system of National Schools (primary schools) for the Catholic population of Ireland was established in the 1830s. With the spread of general education for the Irish, a native middle class began to arise in Ireland and this class spoke English. In the towns, and increasingly in the countryside, the Irish language was not supported by young people and the shift to English gained in speed and intensity throughout the nineteenth century (Hindley 1990). From the following map one can see that the areas where Irish was still found as a native language at the beginning of the twentieth century, e.g. central Tyrone, West Clare or North Waterford, were all rural.

Map 2. Areas where Irish was still spoken in the early twentieth century



When the South of Ireland (26 of 32 counties) gained independence in 1922 the situation changed for the language, at least in theory. Irish became the official language and Ireland came to be referred to by the Irish form of the country's name *Éire.* Initially, the official name of the independent state was *Saorstát na hÉireann/Irish Free State* which was used from 1922 to 1937. The designation *Poblacht na hÉireann/Republic of Ireland* has been in use since 1949 when the country was declared a republic. According to Article 8 of the 1937 Constitution of Ireland (Irish: *Bunreacht na hÉireann*) Irish is the first language of the country, with English – in theory – enjoying a supplementary function. Despite this constitutional support, English is in effect the language of public and private life and over 99% of Ireland's 4.6 million Irish-born people (2011 census) speak it as a native language. Nonetheless, Irish has a special status in Ireland. Although less than 1% of the population today are native speakers, the language looms large in the minds of the Irish as the carrier of their cultural heritage, given that it was formerly the native language of the majority of the population. Many people claim that Irish is their 'native language' even though their knowledge of the language is poor. This attitude is to be found in public life as well. Government bodies all have Irish names, signposts are bilingual, official letters often contain an opening and a salutation in Irish (though the contents are generally in English).

1.6.1. Irish in the twentieth-first century

Although formerly the native language of several million people, Irish has been reduced now to some tens of thousands who use the language as their first means of communication in historically continuous communities (Ó Catháin 2012; Ó Riagáin 2007). Apart from this, there are many people in present-day Ireland with a strong interest in the language and its culture. Given that the latter group is numerically greater than that of the native speakers, it is probably their forms of Irish which will survive into the twenty-first century. Public support for the language is important in providing a framework in which the language can prosper. Such support is found in Ireland and also, since 2007, through the recognition of Irish as an official language by the European Union.

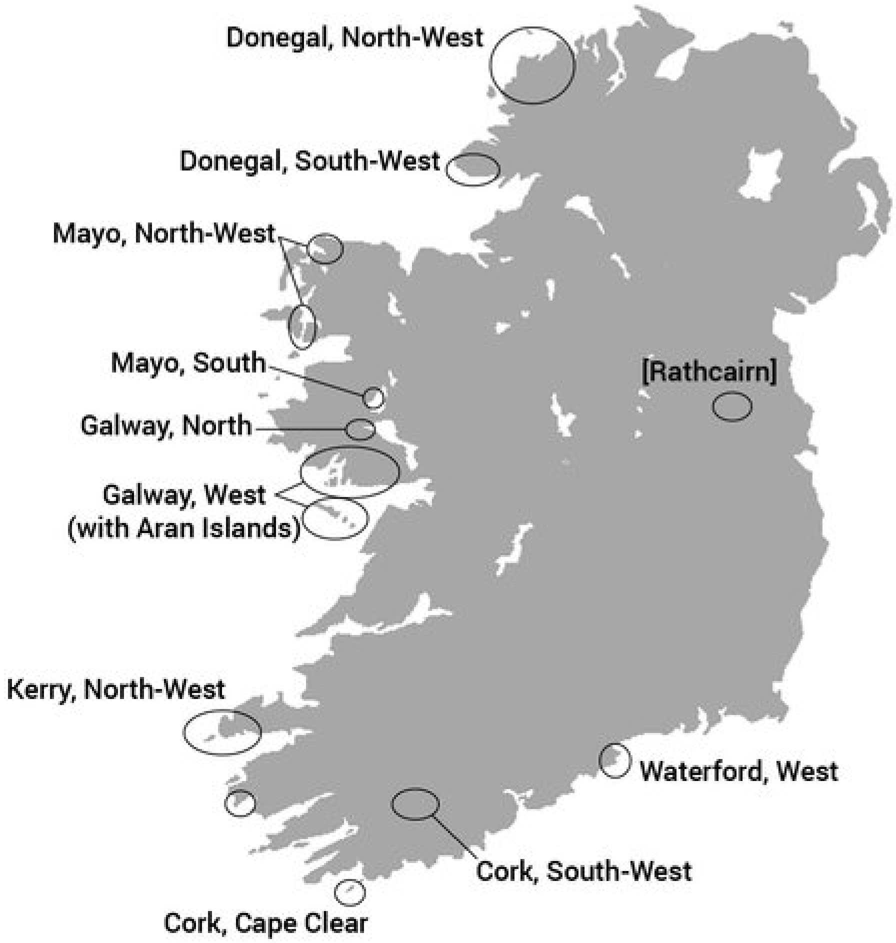
Certain issues about the language seem intractable, such as the inconsistent orthography or the question of what dialect might be taken as standard (Ó Háinle 1996). However, these issues would be surmountable if the language was viewed as fully functional for modern life. Whether the language will not only survive but perhaps spread within Ireland is a question which ultimately rests on its perception as a medium fit for use in contemporary Irish society (Cronin 2005).

2. Dialects of Irish today

Contemporary spoken Irish exists as a continuum of dialects found along the South, South-West, West and North-West coast of Ireland (Ó Curnáin 2012). Variation exists on all levels of language, but differences in pronunciation are perhaps the most striking. On a more abstract, structural level the language is fairly uniform across the different dialects. Furthermore, the differences which exist, e.g. pronunciation variants, can be accounted for by considering recent historical developments which show the different paths which the dialects have taken. This is not to imply that there is a single common underlying system, somehow part of native speakers' knowledge in each dialect, which then surfaces in varying phonetic form across the dialects. This view was supported briefly in the late 1960s and early 1970s in the wake of early generative phonology (McBrearty 1979). Studies like M. Ó Murchú (1969), Skerrett (1968) and, above all, Ó Siadhail and Wigger (1975) work on this assumption as does the monograph by Ó Siadhail (1989). These authors worked on the premise that native speakers of different dialects had a common underlying knowledge of the language. While it is true that linguists can recognise related forms across dialects and reconstruct historical antecedents for these it is quite another matter to imply that native speakers mentally store forms which represent an intersection between all possible dialectal realisations and then derive the pronunciations of just their particular dialect when speaking. There is no empirical evidence for the latter assumption: native speakers construct a phonological system by abstracting from the phonetic stream of the language as spoken around them. With related dialects it will be the case that the phonological systems constructed by speakers in early language acquisition (Ní Neachtain 2013; Owens 1992) will be similar in different geographical areas but this fact is solely due to the historical relatedness of the dialects and has nothing to do with the acquisitional process for early language learners.

In present-day Ireland, Irish is spoken as an everyday, living language in at most a dozen locations spread along the Western and Southern seaboard. Of these the largest communities are in Co. Donegal, Co. Galway and Co. Kerry. On the basis of this, the tripartite division into Northern, Western and Southern Irish is made (Doyle 2001: 10-13). In this book the three adjectives used in this division are written with capital letters to imply that in each case one is dealing with a set of varieties which are sufficiently delimited from the others to form a group. Originally there was a dialect continuum along the entire Western and Southern coast but with the geographical fragmentation of the dialect areas the separation into three large blocks became a reality.

Map 3. Locations of present-day Irish-speaking districts



The official designation for the Irish-speaking areas is the singular, collective label *Gaeltacht* 'Irish region'. A distinction used to be made between two types of Gaeltacht, depending on the numbers of Irish-speakers living there: (1) *Fíor-Ghaeltacht*, lit. 'true Irish-area', referred to those areas with a high-percentage of speakers (though the threshold for this was not officially defined) and (2) *Breac-Ghaeltacht*, lit. 'intermittent Irish-area', with considerably fewer Irish speakers. Occasionally, the English-speaking areas are referred to collectively as *Galltacht* 'region of the non-Irish' from *Gall-* meaning 'foreigner, non-Irish (speaking)'.

2.1. Research on Irish dialects

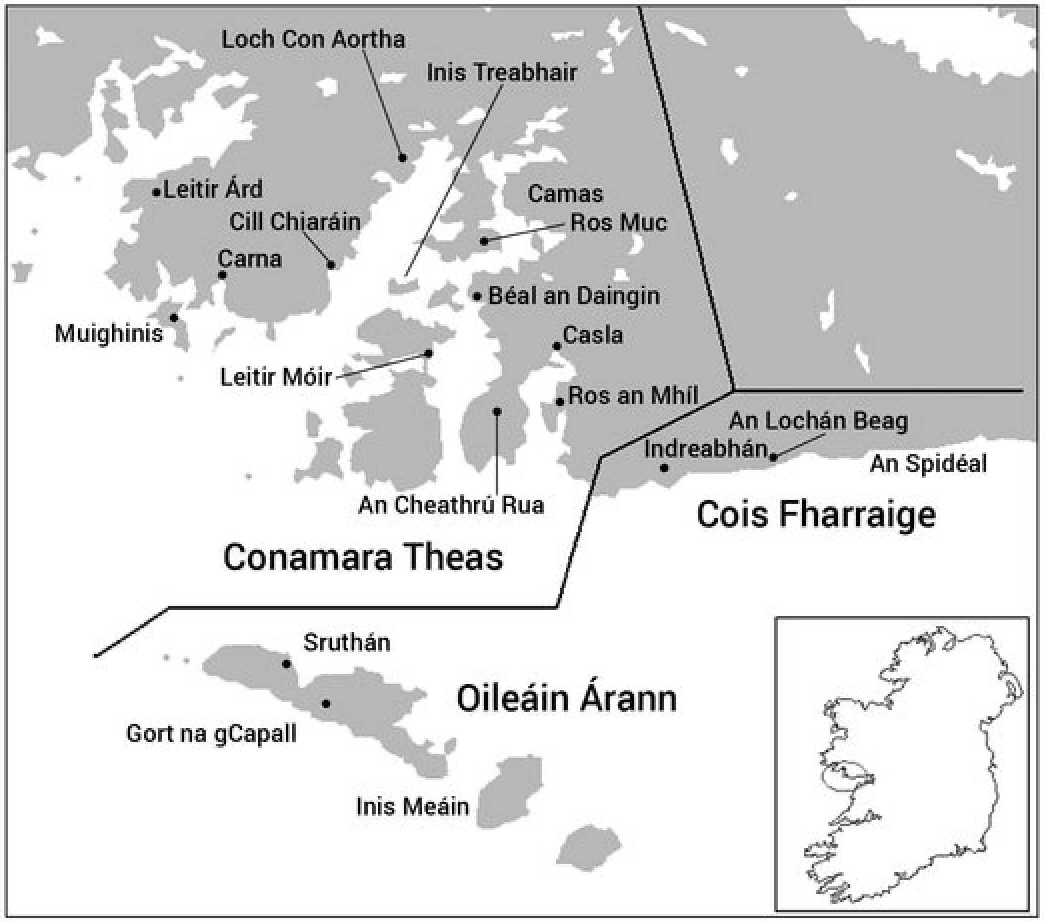
The standard dialect survey of Irish is Heinrich Wagner's comprehensive atlas (see Wagner 1958-64). But even when this was being compiled, some 60 years ago, the speakers were usually older males whose Irish was frequently moribund. The situation today is that large tracts of both halves of Ireland have no historically continuous Irish-speaking areas any more. The dialect areas were probably more or less contiguous up to the middle of the nineteenth century. Apart from famine and emigration, the language shift from Irish to English (Hickey 2007, chapter 4) meant that the geographical distribution of Irish throughout the entire island of Ireland became irreversibly intermittent. There are no historic Irish-speaking regions in Northern Ireland or in the large midland region of the country, nor are there any on the East coast, if one disregards the transplanted community of Connemara speakers in Rathchairn, Co. Meath (see Map 3. above).

The present-day dialects (see above map) show further differentiation within the tripartite division of North, West and South. In the North, in Co. Donegal (D. Ó Baoill 1996), there is a North-West dialect area including Oileán Toraigh (Tory Island, N. Hamilton 1974) along with some residual dialects in the far North of the county (Ros Goill, Lucas 1979). The second major area of the North is in the South-West which contains Teileann (Teelin) which has been investigated quite thoroughly (Wagner 1979 [1959], Ó hEochaidh 1966).

The Mayo area in the West can be subdivided into two sub-areas. The North-West comprises Iorras (Erris, Mhac an Fhailigh 1968) and the adjacent mainland, chiefly the region known as Ceathrú Thadhg (Carrowteige). North-West Mayo also includes parts of Oileán Acla (Achill Island) on the eastern part of the island, facing the mainland, notably the Corrán (Corraun) peninsula (Stockman 1974). This region shows forms of Irish which dialectally tend towards the North, given that the population there derives from Ulster immigration in the seventeenth century. The South of Mayo, the area of Túr Mhic Éadaigh (Tourmakeady, de Búrca 1958) groups with the North of Galway (the area around Corr na Móna).

The central West is the area of coastal Connemara (S. Ó Murchú 1998), West of Galway city, beginning in Bearna and extending outwards to Carna on the peninsula of Iorras Aithneach. This area has been studied intensively, starting with de Bhaldraithe (1945 and 1953a – for Cois Fharraige)[4] and continuing up to recent studies, such as S. Ó Murchú (1989), Wigger (2004 – for Ros Muc) and Ó Curnáin (1996, greatly expanded in 2007 – for Iorras Aithneach). The central West also includes the Aran Islands (J. Hughes 1952), the two minor islands of which are entirely Irish speaking: Inismeáin (Inishmaan, Finck 1899, Ó Siadhail 1978) and Inis Oírr (Inisheer, Ó Catháin 1993). The largest island, Árainn (Inishmore) has a majority of Irish-speakers outside of the main town Cill Rónáin (Kilronan).

Map 4. The Conamara Gaeltacht



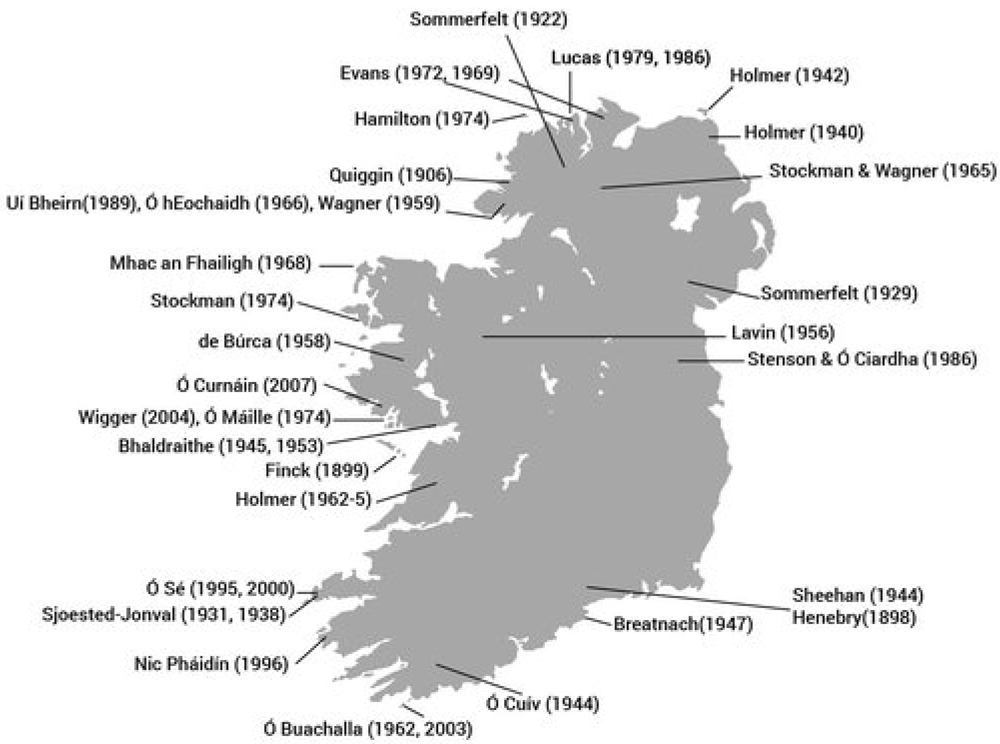
The Southern area consists of Irish-speaking regions in Co. Kerry, Co. Cork and Co. Waterford (Hickey 2011a: 129-133). The strongest of these is that of Corca Dhuibhne (somewhat inaccurately referred to in English as Dingle Peninsula). This area was the focus of studies during the 1930s by Sjoestedt (1931) and Sjoestedt-Jonval (1938) and has in the past two decades been investigated afresh by Ó Sé (1995, 2000).

There are two Irish-speaking districts in Co. Cork: (1) the area around Baile Bhúirne (Ballyvourney), called Muscraí (Muskerry, Ó Cuív 1944) in the South-West and (2) Oileán Chléire (Cape Clear or Clear Island, Ó Buachalla 1962, 2003) off the South-West coast. The numbers of speakers has declined here recently (Ní Chiosáin 2006) but there are recordings of native Irish from the area. In Co. Waterford there is the district An Rinn (Ring, R. B. Breatnach 1947) on a peninsula just South/South-West of Dungarvan (Hickey 2011a: 132-133). Irish further inland – in the Déise (the Deices, Henebry 1898) in North Co. Waterford – died out in the early twentieth century.

2.2. Linguistic research on Irish

Linguistic investigations into the sound structure of modern Irish begin at the end of the nineteenth century. The milestone among these is definitely Pedersen (1897), the author of a monumental comparative grammar (Pedersen 1909-13), an English translation of which is available as Lewis and Pedersen (1962 [1937]). Unfortunately, no English version of Pedersen (1897) – written in Danish – has been published, but it nonetheless set the tone for many investigations of Irish dialects in the twentieth century. Pedersen did fieldwork for his study in the West of Ireland and had indeed been to the Aran Islands collecting data where Franz Finck also did research (see Finck 1899). The study of Irish dialects can be said to date from both their works. Henebry's 1898 thesis on Irish in Waterford is slight in comparison to the works just mentioned. A bridge between Pedersen and Finck on the one hand and the dialect studies described in the next paragraph on the other is formed by Sommerfelt's 1922 study of a dialect of Donegal Irish (that of Torr in the inland North of the county) and by Sjoestedt-Jonval's two studies of Irish in Co. Kerry (Sjoestedt 1931 and Sjoestedt-Jonval 1938).

Map 5. Main studies of Irish dialects by location



2.2.1. Individual dialect monographs

Between the early 1940s and the late 1960s a number of dialect studies appeared which cover the major dialect areas, indeed in many cases they record varieties in communities which have all but ceased to exist. These studies are from areas where Irish was still spoken in the first half of the twentieth century from Co. Cork in the South to Co. Donegal in the North (Hickey 2011a: 88-91). The dialect studies were instigated by Thomas Francis O'Rahilly and published by the Dublin Institute for Advanced Studies (which has also produced books on Scottish Gaelic, see Dorian 1978a and M. Ó Murchú 1989a).

There was a rough blueprint for each of these books. The first half consists of a phonetic description of sounds and texts and the second half of remarks on the historical development of the dialect in question. The studies are good phonological taxonomies, but they are often based on the speech of only a few speakers, or just a single individual, and they show certain idiosyncrasies of transcription. As a rule they deal with the sound system of the dialect in question, but in the case of Cois Fharraige (immediately West of Galway city) there is both a phonetic study (de Bhaldraithe 1945) and a comprehensive treatment of the morphology of the dialect (de Bhaldraithe 1953). To this day they still serve as standard sources of information on dialects of Irish at a time when the communities were still relatively large. The main studies in this group are the following.

Bhaldraithe, Tomás de 1945. *The Irish of Chois Fhairrge,*[5] *Co. Galway*. Dublin: Dublin Institute for Advanced Studies.

Bhaldraithe, Tomás de 1953. *Gaeilge Chois Fhairrge. An deilbhíocht* . [The Irish of Cois Fharraige. The morphology] Dublin: Dublin Institute for Advanced Studies.

Breatnach, Risteard B. 1947. *The Irish of Ring, Co. Waterford.* Dublin: Dublin Institute for Advanced Studies.

Búrca, Seán de 1958. *The Irish of Tourmakeady, Co. Mayo.* Dublin: Dublin Institute for Advanced Studies.

Ó Cuív, Brian 1944. *The Irish of West Muskerry, Co. Cork.* Dublin: Dublin Institute for Advanced Studies.

Mhac an Fhailigh, Éamonn 1968. *The Irish of Erris, Co. Mayo.* Dublin: Dublin Institute for Advanced Studies.

Wagner, Heinrich 1979 [1959]. *Gaeilge Theilinn*. [The Irish of Teelin (Co. Donegal)] Second edition. Dublin: Dublin

Institute for Advanced Studies.

Some of the gaps in the coverage of historically continuous varieties of Irish were filled later with publications by the Institute of Irish Studies in Belfast, e.g. by Stockman (1974) on the Irish of Achill, N. Hamilton (1974) on the Irish of Tory Island and by Lucas (1979) on the Irish of Ros Goill, the latter two locations in North Co. Donegal. A study of East Mayo Irish based on recovered material from a mid-twentieth century study by Thomas J. Lavin (see Lavin 1956b, 1958-61a, b) is to be found in Lavin / Ó Catháin (forthcoming). A very comprehensive treatment of the Irish of Iorras Aithneach, Co. Galway (the Western edge of the Connemara Gaeltacht) is to be found in Ó Curnáin (4 vols., 2007).

2.2.2. Research on Irish phonology

From the second half of the twentieth century onwards there have been a number of linguistic analyses of modern Irish phonology, mostly PhD dissertations produced in the United States, with a few others from Germany and Poland. The earliest of these is J. Hughes (1952, PhD Columbia, New York) followed by M. Krauss (1958, PhD Harvard, Cambridge, MA). Boyle [= Dónall Ó Baoill] (1973, PhD Ann Arbor, Michigan), entitled *Generative Phonology and the Study of Irish Dialects*, is the first full-length work to consider the dialects of Irish within a theoretical framework, though Wigger 1970 had done this for the nominal area in Connemara Irish. Kelly (1978, PhD Austin, Texas) is a comparative work dealing with the interface of phonology and morphology (Ikelas 2014) and offers a comparison of Irish with the native American language Southern Paiute (see section IV.11) which has a partially comparable system of initial mutation. Nilsen (1975, PhD Harvard, Cambridge, MA) is a descriptive study of the sound structure of a dialect in West Galway. Ní Chiosáin (1991, PhD Amherst, MA) is quite different in its theoretical orientation and its application of then prevalent phonological approaches to synchronic Irish data. Cyran (1997, PhD Lublin) is in a similar vein, in this case dedicated to an analysis of Southern Irish within a government phonology framework as is Bloch-Rozmej (1998). Both these Polish dissertations have been published.

Two other monographs on the sound system of Irish appeared at the end of the last century. The first of these is Ó Siadhail and Wigger (1975). The title of this study, *Córas Fuaimeanna na Gaeilge*, literally translates as 'The sound system of Irish', an indirect reference to Chomsky and Halle (1968). One of the main aims of the authors is to posit abstract underlying forms which are taken to be those which form the basis of the different surface realisations in the various present-day dialects (see the discussion of nasalisation, Ó Siadhail and Wigger 1975: 32, as an example of this approach).

The second monograph is *Modern Irish*. *Grammatical Structure and Dialectal Variants* (Ó Siadhail 1989). The work offers a solid description of the pronunciation and grammar of Modern Irish by a writer who knows his material well. The difficulty is that Ó Siadhail adheres to a view, first proposed by him in Ó Siadhail and Wigger (1975) and explicitly continued in the current monograph (Ó Siadhail 1989: xv), that all the dialects can be linked up with each other by positing abstract underlying forms which they are supposed to share. This idea, propounded in the immediate aftermath of *The Sound Pattern of English* by such authors as Brown (1972) or Newton (1972) – for Modern Greek – had already been abandoned by the mid 1980s when this book was being written.

It should be said that generative treatments of Irish often had a specific goal. With the exception of Wigger (1970), they deal with several dialects and express the view that this method could provide a unified framework for all the dialects (M. Ó Murchú 1969), indeed that generative phonology could be a practical aid towards developing a synthesis of the dialects.

In addition to the studies listed above there is wider literature concerned with processes which happen to be central to Irish phonology. For instance, the volume by Brandão de Carvalho, Scheer and Ségéral (eds, 2008) is a broadly arranged collection of studies of lenition and fortition, some of which consider individual languages or sets of languages, e.g. Jaskula (2008) on Celtic or Brandão de Carvalho (2008) on Western Romance, and some of which look at theoretical interpretations of the two processes which form the theme of the volume, see Smith (2008). Another comparable volume is that by Andersen (ed., 1986) which offers overviews of sandhi phenomena, including initial mutation, across the languages of Europe, see the contributions by Ó Cuív (1986) and Le Dû (1986) on Irish and Breton respectively.

2.3. Main phonetic differences between dialects

The present section offers an overview of the main phonetic differences between the three chief dialect areas of Irish in the North, West and South of the country, i.e. in Co. Donegal, Co. Galway and Co. Kerry respectively. The information found here is based on published literature in the field and on recordings of native speakers from the individual areas as part of the project *Samples of Spoken Irish*, further information can be found on the DVD accompanying Hickey (2011a).

The main phonetic differences between the dialects concern the realisation of long vowels. There is some consonantal variation, most notably in the realisations of palatal /tj/ and /dj/, which ranges from affricates in the North through palatal stops in the West to alveolar stops in the South.

VOWELS

1) *Realisation of inherited <AO> vowel*

This vowel is one of a series of orthographic diphthongs in Old Irish (Greene 1976). In all dialects it has resulted in a long front monophthong[6] with an additional variant in the North. As there are many tokens of the vowel in the lexicon of Irish, it is a salient dialect indicator. Irrespective of the regional realisations, the consonant preceding the vowel is always non-palatal. This leads to a retracted quality in all cases. In citation forms of words the consonant following the vowel is generally non-palatal as well, though palatalisation in the genitive or plural of nouns can result in <AO> being followed by a palatal consonant, e.g. *saol* ‘life.NOM’ ~ *saoil* ‘life.GEN’.

vowel	N	W	S	sample
<AO>	iː	iː	eː	*baol* ‘danger’

Notes

(i) In Co. Donegal, e.g. Tory Island and the adjacent mainland, a retracted version of /iː/ is common, e.g. *aon* [ɯːnˠ] ‘one’. This is a feature which it shares with Scottish Gaelic. Sometimes this sound is transcribed as Greek lambda [λ(ː)], see N. Hamilton (1974: 131) and sometimes as inverted ‘y’, see O’Rahilly (1932: 27).

(ii) In some dialects there are lexicalised exceptions to the realisation of <AO> as a long monophthong: a few words show the diphthong /ai/ here, e.g. *faoileán* /failjɑ:nɣ/ 'seagull'. The diphthongisation may have been triggered by the following palatal /lj/. In other cases a single pronunciation applies across all dialect areas, e.g. *daoine* with /iː/.

(iii) The present-day spelling may camouflage the historical environment of the <AO> vowel. Spellings which stem from before the 1948 spelling reform show that in some cases <AO> was followed by <*gh*> which had been lost by lenition in intervocalic position. For example, *saol* 'life' was former spelled *saoghal* (Dinneen 1927: 941) which shows that the word had an intervocalic [-ɣ-] at a much earlier stage.

(iv) There is a tendency in Western Irish for <AO> to be pronounced with /e:/ in formal contexts, e.g. *Naomh Pádraig* [nɣe:v pɑ:rɪkj] 'Saint Patrick' despite the general realisation with /iː/.

(v) See O'Rahilly (1932: 27-38) for an overview in all the dialects including Scottish Gaelic.

2) *Stressed vowel reflexes before syllable-final sonorants*

A group of sonorants or sonorant plus stop which occurred in syllable-final position triggered vowel lengthening in the history of Irish. The outcome of this lengthening varies across the dialects and is a clear diagnostic of speakers' regional affiliation.

context	N	W	S	sample
<*ill*#>	ɪ	ai	iː	*maoill* ‘delay’
<*im*#>	ɪ	iː	iː	*im* ‘butter’
<*inn*(C)#>	ɪ	iː	ai	*roinnt* ‘some’
<*all*#>	a	ɑː	au	*meall* ‘pile, charm’
<*am*#>	ɑː	ɑː	au	*am* ‘time’
<*ann*#>	a	ɑː	au	*peann* ‘pen’
<*air*C(V)#>	aː	ai	iː	*airde* ‘height, attention’
<*oll*#>	ʌ, ɔː	au	au	*poll* ‘hole’
<*om*#>	ʌ	ʌ, uː	au	*trom* ‘heavy’
<*onn*#>	ʌ	ʌ, uː	au	*fonn* ‘wish’
<*ong*#>	ʌ	ʌ	ʌ, uː	*long* ‘ship’
<*or*C#>	ɔː	au	oː	*bord* ‘table’
<*orr*#>	ɔ	au	ʌ	*corr*- ‘occasional’

Notes

(i) The symbol ‘#’ indicates a word boundary.

(ii) The diphthongisation of stressed /a/ before a final nasal is greatest in East Munster, e.g. in Ring (Co. Waterford), *rang* ‘class’ is /rauŋg/. The diphthongisation is also found in unstressed, non-final position, e.g. *Breandán* /bjrjaunɣˈdɑ:nɣ/ (first name).

(iii) In Corca Dhuibhne (Co. Kerry) a final velarised /lɣ/ may be realised as [χ], e.g. *(go) mall* [mauχ] ‘slow’.

(iv) There are cases where the low vowel /a/ is retracted to /ʌ/, e.g. *ceann* [kʲʌnˠ] ‘head’ (Northern Irish). This is also found in Western Irish (in other contexts), e.g. *cat* [kʌt] ‘cat’.

3) *Long vowels and diphthongs deriving from vocalised fricatives*

In the history of Irish intervocalic voiced stops have been largely lost.. These were first fricativised, i.e. *gh, dh* became /ɣ/ and *bh, mh* became /v/ (Ó Maolalaigh 2006: 41-44). The voiced fricatives were then absorbed into the vowels which preceded them, that is they migrated from the coda of the syllable to its nucleus. This caused a lengthening of the syllable vowel because a short vowel plus a vocalised consonant led to a long vocalic element, thus maintaining the overall length of the syllable rhyme (nucleus and coda). The phonetic result could have been either a long vowel or a diphthong. The latter usually arose where the original fricatives were preceded by a low vowel. The diphthong was /ai/ where the original stop was /g/, sometimes /d/, and /au/ where it was either /b/ or /m/, this situation is most clearly seen in Southern Irish.

context	N	W	S	sample
<*agh*>	ɛi	ai	ai	*laghad* 'least'
<*abh*>	oː	au	au	*leabhar* 'book'
<*amh*>	ɔː	oː	au	*amhrán* 'song'

4) *Short vowel + long vowel in disyllabic words*

Both Western and Northern Irish have initial stress so that when the first vowel of a word is short and the second long a tension arises between the stressed short vowel and the unstressed long one. In Northern Irish the situation has been resolved by shortening the second vowel so that such words now consist of two short vowels. In Western Irish the first vowel is raised to /ʊ/, but the short + long sequence remains. In Southern Irish, which has variable stress, the long vowel of the second syllable attracts stress and remains long.

context	N	W	S	sample	
<*a*C*á*>	ˈa + a	ˈʊ + ɑː	ə + ˈɑː	*scadán*	'herring'

5) *Realisation of unconditioned long vowels*

context	N	W	S	sample
<*í*>	i:	i:, ai	i:	*luí* ‘lying’
<*é*C>	e:	e:	i:	*éan* ‘bird’
<*á*>	æ:, ɛ:	ɑ:	ɑ:	*áit* ‘place’
<*eo, ó*>	ɔ:	o:	o:	*beo* ‘alive’
<*ú*>	ʉ:	u:	u:	*fiú* ‘even’

Notes

(i) The front realisation of <*á*> in Northern Irish leads to a shift of a preceding /ɣ/ to [j], e.g. *dhá* [jæ:] ‘two’. The fronting also leads to raising with the resulting vowel often in the mid region, i.e. [ɛː] or even [e:] with some female speakers.

(ii) There is a general tendency to lower <*eo, ó*> in Northern Irish, e.g. *bróg* [brɔ:g] ‘shoe’, *beo* [bjɔ:] ‘alive’. The short mid-back vowel /ʌ/ is rounded before liquids, e.g. *fola* [fɔlɣə] ‘blood.GEN’. In the West and South there is a considerable quality difference as well, i.e. one has *moch* [mʌx] ‘early’ ~ *bróg* [bro:g] ‘shoe’.

6) *Final unstressed short vowels*[7]

N	W	S	sample
i:	ə	ə	
[ˈtʃɪn$_j$i:]	[ˈt^{j}ɪn$_j$ə]	[ˈt$_j$ɪn$_j$ə]	*tine* ‘fire’
[ˈtʃaŋi:]	[ˈt^{j}æŋ(g)ə][7]	[ˈt$_j$aŋ(g)ə]	*teanga* ‘language’

Notes

(i) The realisation of short unstressed vowels is as a schwa in the entire South and in the Connamara Gaeltacht of the West. However, there are other instances, e.g. *gadaí* [ˈgadi:] ‘thief’, where the final vowel is underlyingly long in all dialects, including those which have [ə] in words like *teanga*.

(ii) The realisation of short unstressed final vowels as [iː] is already to be found in North Galway, in Tourmakeady (de Búrca 1968: 38), and in North-West Mayo (Mhac an Fhailigh 1968: 14) and, of course, further North in the Donegal Gaeltacht.

7) *Realisation of low vowels*

North The vowel in the LÁN lexical set is considerably fronted, at least to [æ:], if not to [ɛ:], e.g. [lɣæ:nɣ ~ lɣɛ:nɣ] 'full'. The vowel in the TEACH lexical set is a short, low front vowel, i.e. [tʃæ] 'house'.

West The vowel in the LÁN lexical set is retracted and long, i.e. [1ˠɑːnˠ] 'full'. There is a general lengthening of low vowels and a very open [æ:] after palatal sounds in the TEACH lexical set (Cois Fharraige, immediately east of Galway city). This lengthening is not as salient further West towards, in An Cheathrú Rua and Carna (see relevant maps in Hickey 2011a: 119-133).

South The vowel in the LÁN lexical set is retracted and long, i.e. / 1ˠɑːnˠ/ → [lɣɑ:nɣ] 'full'. The vowel in the TEACH lexical set is a short, low front vowel, i.e. /tʲax/ → [tjæx] 'house'.

8) *Realisation of long mid and high back vowels*

North a) /u:/ is pronounced as a high mid rounded vowel, much as in the rest of Ulster (an areal feature covering both Irish and English), e.g. *cúl* [kʉ:lɣ] ‘rear’.

b) The mid back vowel /o:/ is also lowered, e.g. *pósta* [pɔ:stə] ‘married’.

West, South — Both /o:/ and /uː/ – the vowels in the SEOMRA and GÚNA lexical sets respectively – are realised without any noticeable shift in the West and South, though there can be a shortening in some cases, e.g. *seomra* [ʃʌmrə] ‘room’ for many speakers of Western Irish.

West — The raising of vowels before nasals often leads to a neutralisation of the contrast between /o:/ and /u:/, e.g. *rón* ‘seal’ and *rún* ‘secret’, both [ru:nɣ].

9) *Realisation of short mid back vowels*

The most common realisation of the short mid back vowel is as [ʌ], e.g. *turas* [ˈtʌrəs] ‘journey’. This is also found for the vowel written <*o*>, e.g. *toradh* [ˈtʌrə] ‘yield, result’. However, in Northern Irish the distinction between <*u*> [ʌ] and <*o*> [ɔ] is maintained before liquids, cf. [ˈtʌrəs] and [ˈtɔrə] for the two words just quoted.

In Western Irish there is a raised realisation of /ʌ/ before velars, e.g. *rugadh* [rʊgu:] ‘was born’, *muc* [mʊk] ‘pig’.

CONSONANTS

10) *Realisation of coronal palatals in stressed syllable onsets*

context	N	W	S	sample
<ˈ*t*-,ˈ*d*->	tʃ	t^j	t_j	*teach* ‘house’

11) *Realisation of /-x/ in word-initial position*

context	N	W	S	sample
<#*ch*->	h	x	x	*chuala* 'heard'

12) *Realisation of /-x/ in word-final position*

context	N	W	S	sample
<-*ch*#>	Ø	x	x	*cloch* 'stone'

13) *Realisation of syllable-final /-x/ in covered position*

context	N	W	S	sample
<-*ch*C>	r	x	x	*slacht* 'polish; finish'

Notes

(i) Affricates as realisations of /tʲ/ and /dj/ are also found in Western Irish, notably on the Aran Islands. The occurrence of affricates in North-West Mayo Irish (Achill Island and parts of the adjoining mainland) may be due to the Northern origin of Irish in this region. However, the affricate realisation seems to have had a wide distribution not just in the North and in Co. Mayo but reached down into North Galway as de Búrca notes (1958: 25) in his study of Tourmakeady.

(ii) Those dialects with affricates for palatal /tʲ, dj/ would seem to have a preference for [ʃ] over [s] before [r], e.g. *srúthan* [ʃrʊfɑ:nɣ] 'stream' (de Búrca 1958: 31).

(iii) The realisation of /x/ as [r] in a pre-consonantal coda position, e.g. *ocht* [art] is typical of the Northern part of the Northern Irish area, e.g. around Gaoth Dobhair and on Tory Island.

14) *Distinctions among sonorants (n- and l-sounds)*

A distinction between three types of *n* and *l* – velarised (phonologically non-palatal), alveolar and palatal – is found in Northern and Western Irish. In the South there is no systemic distinction between velarised and alveolar forms of these sonorants (Ó Sé 2000: 17-18). Hence the realisation of the nasal in a word like *bán* ‘white’ is phonetically intermediate between the fully velarised [nɣ] and the alveolar [n] found in varieties of Western and Northern Irish. For further discussion of the sonorants of Irish, see section II.1.8 below.

	$n^ɣ$ $l^ɣ$		n l	n^j l^j
North	✓		✓	✓
West	✓		✓	✓
South		✓		✓

Exx.	*naoi* ‘nine’	*duine* ‘person’	*neart* ‘strength’
	luí ‘lying’	*baile* ‘town’	*leaba* ‘bed’

N. Hamilton (1974: 139-145) has a fourfold distinction for Tory Island in North-West Co. Donegal as does Wagner (1979 [1959]: 16-32) for Teileann in the South of this county. It is uncertain whether this really exists phonologically. This issue is discussed in more detail in II.1.8 and the following subsections below.

15) *Velar stops in post-nasal position*

North	Ø		*teanga* ‘tongue’ [tʃæŋiː]
West	✓	Ø (Cois Fharraige)	[t^{j}æŋ(g)ə]
South	✓		[t$_j$æŋ(g)ə]

16) *Intervocalic /h/ after stressed vowel*

			oíche ‘night’
North	✓, Ø		[iː / iːhə]
West	✓	Ø (Cois Fharraige)	[iːhə / iː]
South	✓		[iːhə]

17) */v/ before a non-palatal vowel*

		bhog sé ‘he moved’
North	w	[wʌg ʃeː]
West	w	[wʌg ʃeː]
South	v	[vʌg ʃeː]

PROSODY

18) *Word stress*

North	Stress is on the first syllable and there is considerable shortening of post-initial long vowels (as opposed to Western Irish), e.g. *scadán* /ˈskadanɣ/ ‘herring’, see N. Hamilton (1974: 160-162) and Hickey (1997) for further discussion.
West	Initial stress applies to virtually all words with the exception of some loanwords such as *tobac* /təˈbak/ ‘tobacco’ and one or two native words, e.g. *uileag*[8] [ɪˈlʲʌg] ‘every’.
South	Long vowels in non-initial syllables attract stress, e.g. *cailín* [kaˈlji:nj] ‘girl’. This may be the result of Anglo-Norman influence (in the South-East) after the twelfth century as older authors like O’Rahilly

maintained (1932: 86-98) and certainly applied to many French loanwords, e.g. *buidéal* /bəˈdjeːlɣ/ ‘bottle’, see Hickey (1997) for further discussion of these loans. Other heavy syllables, such as /-əx(C)/ also take stress in non-initial positions, e.g. *portach* [pərˈtax] ‘bog’, *fanacht* [fəˈnɣaxt] ‘wait’. See the discussion in II.3 below for more details.

2.4. Data for former dialect areas

The Irish language has steadily declined over the past centuries with the current main dialect areas – South, West and North – the remnants of a much greater geographical distribution. Many attempts have been made to salvage and/or reconstruct what little information is available for native-speaker Irish outside the present-day Gaeltacht areas. The main source to date has been the monumental atlas of dialects by Heinrich Wagner (see Wagner 1958-64). However, of late another source has become accessible, namely the recordings of native speakers made between 1928 and 1931 by Wilhelm Albert Doegen (1877-1967).[9] These have been recovered from the original medium (poor quality vinyl records) and digitally re-mastered so that they can now be used for linguistic analysis.[10]

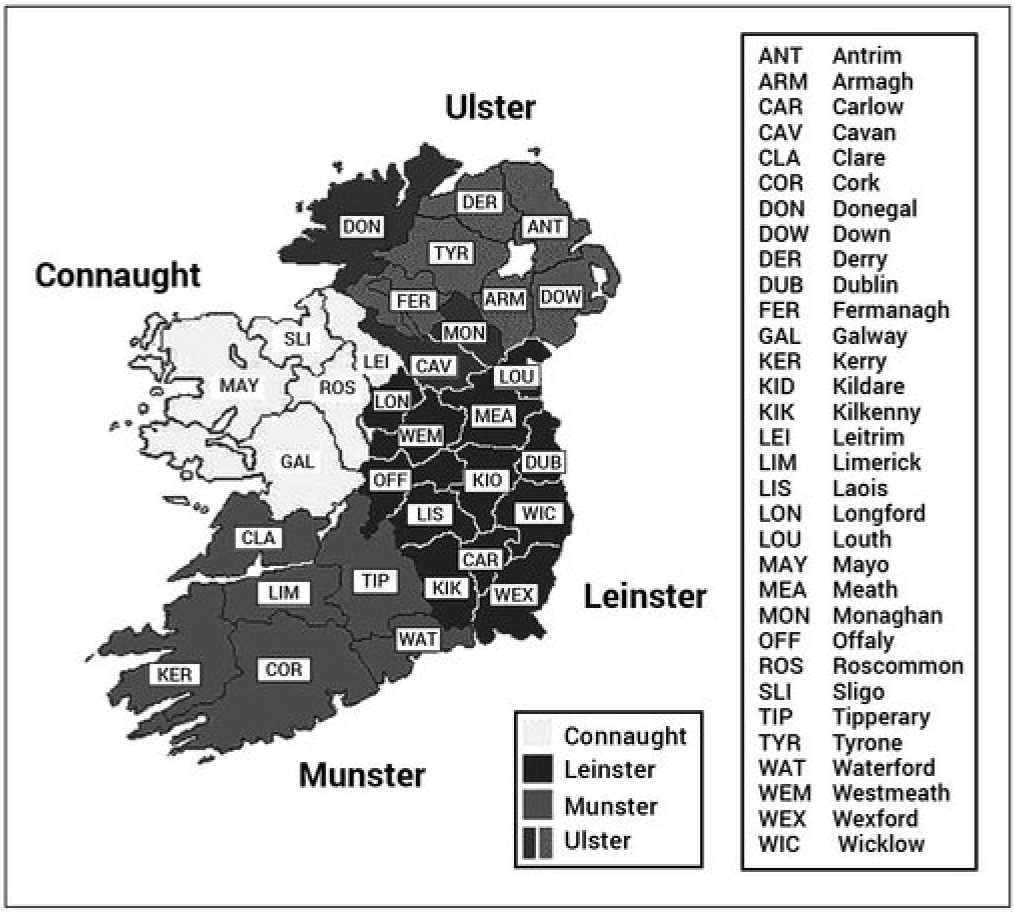
In his early career Doegen developed plans for a *Museum for the Voices of the World’s Peoples* and in 1920 he founded the Sound Archive of the Prussian State Library, a project which he continued until 1933 when he was dismissed from his post. In the context of this work, and in collaboration with the Royal Irish Academy, he collected his Irish material.[11]

The relevance of Doegen’s tapes for the phonology of modern Irish is that they contain recordings of what were some of the last native speakers of Irish from the following counties: Antrim, Derry, Tyrone, Cavan, Armagh, Louth, Sligo, Roscommon, Clare and Tipperary. The tapes from Ulster and Co. Louth were remastered by Queen's University, Belfast and those for Munster and Connaught were done by the Library of the Royal Irish Academy which has put many of these online. These recordings were examined and evaluated for both Hickey (2011a) and the present book.

2.5. Written documentation of Irish dialects

Apart from sound recordings of speakers from former Irish-speaking areas there is a body of scholarly material on Irish in these areas. This material consists of notes, wordlists and tales gathered by interested individuals, frequently folklore collectors or Irish scholars, and published to document the remnants of Irish throughout the country.

Map 6. Provinces and counties of Ireland



The main publications found here are listed below, divided by the regions concerned. The listing is in clockwise order of location, starting in the South, through the West to the North. As the labelling in Irish scholarly publications is often by county, a map is provided here which shows the locations of the counties of Ireland. The information here should be used in conjunction with that in section 2.3. above.

THE SOUTH

There is a body of literature concerned with Irish from Munster, see Cyran (1997), Sjoestedt (1931), Sjoestedt-Jonval (1938), Sommerfelt (1927) and Ua Súilleabháin (1994). Much theoretical interest has been directed at the stress system of Munster Irish (see section II.3 for a full discussion). See Blankenhorn (1981), Cyran and Rowicka (1996), Doherty (1991), Dubach Green (1996, 1997), Gussmann (1997) as representative studies.

Co. Kilkenny

County Kilkenny is to the North-East of Waterford and a very small amount of material is available, see R. A. Breatnach (1992) and C. Quin (1965). Kilkenny is outside the Déise area, a former stronghold of Munster Irish, which use to encompass west and central Waterford along with the extreme south of Tipperary. Kilkenny is furthermore part of the province of Leinster.

Ring, South-Central Co. Waterford

Ring, Co. Waterford is officially a Gaeltacht area although the number of native speakers left there is very small indeed. The standard study of recent decades is R. B. Breatnach (1947) followed by the somewhat earlier Sheehan (1944). Greater time depth is provided by Henebry (1898), the PhD thesis of the main collector of material from his native Co. Waterford. There is a certain amount of lexical material available on North-West Waterford, South Tipperary and East Cork in the following articles: Nyhan (2006, 2007a, 2007b).

Cape Clear, South-West Co. Cork

The island of Cape Clear off the south-west coast of Co. Cork officially belongs to the Gaeltacht, but with few, if any, native speakers of Irish left (Ní Chiosáin 2006). Ó Buachalla (1962) and Ó Buachalla (2003) are the two main studies of Irish on the island.

Muskerry, West Co. Cork

The main study of this official Gaeltacht area is Ó Cuív (1944). A further study appeared more than four decades later, see Ua Súilleabháin (1988).

Uíbh Ráthach, Central-West Co. Kerry

Again an official Gaeltacht area but with few if any native speakers left. Nic Pháidín (1996) is a list of words from the area.

Corca Dhuibhne, North-West Co. Kerry

The largest Gaeltacht in Munster is to be found on the tip of the Dingle Peninsula in the region called Corca Dhuibhne in Irish. Two monographs from the early twentieth century are Sjoestedt (1931) and Sjoestedt-Jonval (1938). Of a more recent date are the two studies Ó Sé (1995) and Ó Sé (2000).

Co. Limerick

Ua Súilleabháin (1997) is a collection of words from Limerick. No grammatical study is available for lack of records of Irish from this county.

THE WEST

Co. Clare

The West coast of Co. Clare (the mainland opposite the Aran Islands) had Irish-speakers up to the early twentieth century. There are a few references to Clare Irish in Wagner (1958-64). Holmer (1962-5) is a large study encompassing all the material which was still available for this county in the mid twentieth century.

Aran Islands, Galway Bay

The three Aran Islands, lying in a diagonal across Galway Bay, all have Irish speakers. General studies of Irish on the Aran Islands are J. Hughes (1952), Finck (1899), Ó Catháin (2001).

Inis Meáin, middle Aran Island

Hickey (1982), Ó Siadhail (1978)

Inis Oírr, eastern Aran Island

Ó Catháin (1993)

Connemara, Co. Galway

The term ‘Connemara’ (Irish: ‘Conamara’) is used loosely to refer to the entire Gaeltacht area west of Galway city out as far as Carna. Dialectally, there is a subdivision in this region, the area called Cois Fharraige from Bearna out to Sailearna shows a number of features, such as lengthening of short low vowels and the loss of intervocalic /-h-/ which are not normally found in Connemara proper, beginning about Indreabhán and extending out to Carna. Note that the Aran Islands tend more towards Cois Fharraige in this respect.

General

> Blankenhorn (1979), de Bhaldraithe (1985), Hartmann (1974), Ó hUiginn (1994), S. Ó Murchú (1998), Uí Bhraonáin (ed., 2008), Wigger (2000)

Rathchairn (transported variety of Western Irish in Co Meath, Leinster)

> Stenson and Ó Ciardha (1986)

Cois Fharraige, Mid-West Co. Galway

> de Bhaldraithe (1945, 1953)

Ros Muc, West Co. Galway

> T. S. Ó Máille (1974), Wigger (1996, 2004).

Iorras Aithneach, West Co. Galway

Ó Curnáin (2007), M. Ó Murchú (1989b), Wagner and McGonagle (1995)

Sraith Salach, North Co. Galway

Nilsen (1975)

An Lionán, North Co. Galway

de Búrca (1966)

Co. Mayo

The areas of North Co. Galway and South Co. Mayo were formerly contiguous from the region known as *Dúiche Sheoigheoch* 'Joyce Country' up the east coast of Lough Mask to Tourmakeady Co. Mayo and beyond. This is no longer the case and the entire region is English-speaking interspersed with some Irish speakers. Some material is available from the nineteenth century when almost the entire county of Galway was Irish-speaking, see Stenson (2003) for an edition of a text from the Clifden area.

Tourmakeady, South Co. Mayo

de Búrca (1958)

Achill, West Co. Mayo

Stockman (1974), Ó Dochartaigh (1979b)

Erris, North-West Co. Mayo

Mhac an Fhailigh (1968, 1977), Skerrett (1975-6)

East Co. Mayo

Dillon (1973), Lavin (1956a, 1956b, 1958-61), Lavin / Ó Catháin (forthcoming)

THE NORTH

Beginning with Co. Mayo, the North stretches to the Fanad peninsula in the far north of Co. Donegal. The intervening region between Achill Island and South-West Donegal, i.e. North and North-East Mayo along with Sligo is entirely English-speaking. In Donegal itself, Oileán Toraigh (Tory Island) is strongly Irish-speaking as is the region in and around Gaoth Dobhair (Gweedore) and to a lesser extent somewhat further south, around Ranna-fast and Annagary. In the south-west of Donegal, around Glencolumbcille and Teelin, Irish is in a much weaker position, despite the official status of Gaeltacht which this area has.

There is no historically continuous Irish-speaking district in the remaining counties of Ulster. Irish survived until the beginning of the twentieth century in the mountainous centre of Co. Tyrone (the Sperrin Mountains), in the Glens of Antrim (Holmer 1940), on Rathlin Island (Holmer 1942) and in very small pockets in South Armagh and on the Cooley Peninsula of Co. Louth (in the province of Leinster). Apart from the printed information on Irish in these areas, there are tapes from the Doegen collection with native speakers talking in their local dialect.

General

N. Hamilton (1971-2), A. Hughes (1994a, b), D. Ó Baoill (1996), Ó Dochartaigh (1987), Ó Searcaigh (1925, 1939)

Teileann, South-West Co. Donegal

Ó hEochaidh (1966), Uí Bheirn (1989), Wagner (1979 [1959])

Glenties, Central South Co. Donegal

A. Hughes (1986), Quiggin (1906)

Torr, North-West Co. Donegal

M. McKenna (2001), Sommerfelt (1922, 1952b)

Tory Island, North West Co. Donegal

N. Hamilton (1974)

Ros Goill, North Co. Donegal

Lucas (1979, 1986)

Inishowen and Fanad, North Co. Donegal

E. Evans (1969, 1972)

Co. Tyrone

Stockman and Wagner (1965)

Co. Antrim (with Rathlin Island)

Holmer (1940, 1942)

South Armagh

Sommerfelt (1929)

For detailed information on all of these dialects, see the discussions in Hickey (2011a).

II The phonological framework

1. Sound inventory

Irish is an Indo-European language and its affiliation is clear from most of its core vocabulary which it shares with other members of this family. However, during its development from an early form of Celtic on the European mainland (in the last millennium BCE), the language changed its grammatical type, shifting its canonical word order to VSO and establishing post-specification as the norm (Hickey 2002). The phonology of the Celtic languages also underwent major changes, notably lenition (Honeybone 2008, Jaskula 2008, Kirchner 2008; Scheer and de Lacy 2008) at the beginnings of words. This was in origin a sandhi phenomenon (Andersen 1986) but later gained systemic status in all the Celtic languages (Ternes 1990: 12-15). In the Q-Celtic sub-branch of Insular Celtic a further significant development took place. Here palatalisation[12] arose as a secondary articulation (Ternes 2010) with velar and coronal consonants in the environment of high vowels and later it spread to encompass labials as well (Jackson 1967b). In the course of time this palatalisation achieved systemic status through reanalysis by first language learners (Hickey 2003b) yielding the twin sets of changes (i) palatalisation / de-palatalisation and (ii) initial mutations. These are used to distinguish the majority of grammatical categories in the Q-Celtic languages, i.e. Irish, Scottish Gaelic and Manx (the latter two deriving historically from Irish).

The development of systemic palatalisation is something which Irish shares with Slavic (Stolz and Levkovych 2015), especially the East Slavic languages, above all Russian (Zubritskaya 1995). The main similarity with Russian is the existence of palatal and non-palatal pairs across the entire inventory of consonants (Hickey 2011a: 234-238), see section III.5 for further discussion).

Apart from the classification according to whether consonants are palatal or non-palatal (see Table 1), Irish also has a voice distinction with voiced and voiceless consonants at most common points of articulation. There are labial, labio-dental, dental/alveolar and velar pairs of segments, distinguished by voice, along with the single voiceless glottal fricative, /h/. Irish is notable, however, in not having voiced sibilants and in this respect aligns itself, for instance, with the Scandinavian languages, i.e. North Germanic (Danish, Norwegian, Swedish, Faroese, Icelandic) and Finnish. Neither does it have affricates, if one ignores one or two cases of sandhi affrication and the situation in Northern Irish and a few other points in the Irish-speaking areas which show affrication on the release of coronal palatal stops. Consonants do not have distinctive length, although this was the case in the history of the language before the Middle Irish period (900-1200) when long consonants existed word-internally and when geminates were treated differently from non-geminates in Latin loanwords (Thurneysen 1946: 85-93). The effects of long sonorants on the vowels preceding them can still be seen today. The reflexes of these sonorants are an important defining criterion for the different dialects of modern Irish (see section I.2.3 above).

Table 1. Consonants of Modern Irish

	labial	alveolar	velar	glottal
1: stops	p, p^j; b, b^j	t, t^j; d, d^j	k, k^j; g, g^j	
2: fricatives	f, f^j; v, v^j	s, s^j	x, x^j; ɣ, $ɣ^j$	h
3: nasals	m, m^j	n, n^j		
4: liquids		l, l^j; r, r^j		

The systemic distinction between long and short vowels has been maintained throughout the history of Irish. In the realisation of sound contrasts the short vowels are significant in that their quality is affected by the palatal or non-palatal quality of segments which may follow them in a syllable coda (see discussion in II.1.9.2 and II.1.9.3 below).

1.1. The palatal / non-palatal distinction

In Modern Irish this distinction applies to all consonants, with the exception of /h/, and is an essential element of both the grammar and lexical structure of the language. Phonetically, palatal consonants are produced by raising the middle of the tongue towards the palate. This provides the constriction which is the acoustic cue for such segments. Palatal sounds are indicated in transcription by placing a superscript yod [^{j}] after the sound in question, e.g. *teach* /tjax/ ‘house’. The realisation of palatal sounds, especially of coronals, varies greatly across the dialects: Northern [tʃæx], Western [tjæx], Southern [tjæx] (see section I.2.3 above). Non-palatal consonants are generally velarised with the middle of the tongue lowered and the back raised towards the velum. Acoustically, this gives a hollow sound to non-palatal segments which indicates that they are the opposite of palatal sounds with the constriction just described.

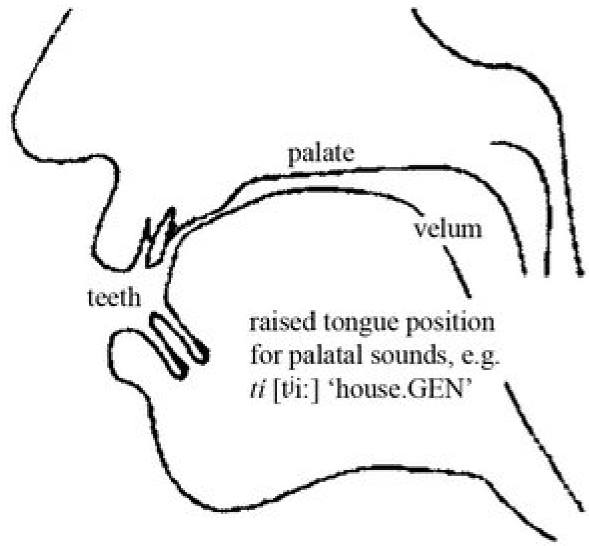


Figure 1. Articulation of palatal sounds, typical of Western Irish (just before dorso-palatal contact)

This 'hollow' quality is most noticeable with non-palatal versions of the sonorants *l* and *n*. e.g. *lá* /lɣɑː/ [ɫɑː] 'day', *ná* /nɣɑː/ [n̴ɑː] 'nor'. Perhaps for this reason, these sonorants show a three-way distinction in Western and Northern Irish, i.e. /lˠ – l – lʲ/ and /nˠ – n – nʲ/ (see section II.1.8 for further details and discussion). According to IPA practice, non-palatal sounds are indicated in transcription by placing a superscript gamma [ɣ] after the sound in question.

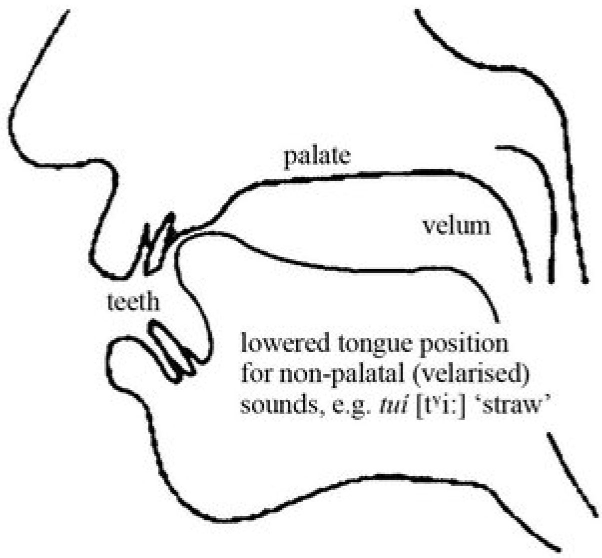


Figure 2. Articulation of non-palatal sounds, common to all dialects (just before apico-dental contact)

1.1.1. Primary and secondary palatals

In articulatory terms there is a distinction between sounds which involve the tongue as active articulator with a point of contact on the front of the palate (behind the alveolar ridge) and those sounds which do not have a palatal point of contact. The first group can be labelled primarily palatals and the second group secondary palatals as shown in the following table. [13]

Table 2. Primary and secondary palatals in Modern Irish

Primary (coronals)	**Secondary (non-coronals)**
tj, dj, sj (obstruents)	pj, bj, fj, vj, mj (labials)
nj, lj, rj (sonorants)	kj, gj, xj, ɣj (velars)

For the phonology of Irish this distinction is not a central concern because the various processes involving palatalisation and de-palatalisation apply to both primary and secondary palatals alike. But on a phonetic level the correlates of palatalisation for the secondary segments are of interest and consist of a fronting of point of articulation for velars and a tensing of the lips for labials; these articulatory features are discussed further in section II.1.7 below.

1.1.2. How phonological contrasts are realised

The contrast of palatal versus non-palatal is a binary opposition in the phonology of Irish. However, this contrast has different realisations for various sounds and shows differing degrees of manifestation depending on syllable position and the stressed or unstressed nature of syllables. The word and syllable sites which show the greatest degree of contrast are given in the following table.

Table 3. Sites of greatest palatal ~ non-palatal contrast

1) Word-initially before a stressed vowel
lá [lɣɑ:] ‘day’ versus *leá* [ljɑ:] ‘melting’
coladh [kʌlɣə] ‘sleeping’ versus *coilleadh* [kʌljə] ‘castration’
2) Intervocalically after a word-initial stressed vowel
fánach [fɑ:nɣəx] ‘wandering’ versus *fáinne* [fɑ:njə] ‘ring’
banbh [banɣəv] ‘piglet’ versus *bainne* [banjə] ‘milk’
3) Word-finally after a stressed vowel
dál [dɑ:lɣ] ‘people, tribe’ versus *dáil* [dɑ:lj] ‘assembly’

The above sites do not apply to all consonants. *R*-sounds only contrast (i) when they occur as a non-initial segment of a syllable onset, e.g. *briste* [bjrjɪʃtjə] ‘broken’, (ii) when occurring intervocalically after a stressed vowel, e.g. *boireann* [bʌrʲənˠ] ‘rocky country’ or (iii) when found in word-final position, e.g. *beoir* [bjo:rj] ‘beer’. These are the positions in which palatal [rj] occurs and hence where it can contrast with non-palatal [r].

Low-polarity coronal nasals and laterals are generally confined to an intervocalic position after a short vowel, e.g. *tine* /tjɪnjə/ [tjɪnjə] ‘fire’, *milis* /mjɪljɪsj/ [mjɪljɪʃ] ‘sweet’. In grammatical words they are found word-initially and word-finally, e.g. le [l$_j$ɛ] ‘with’ and *sin* [ʃɪnj] ‘that’. In addition one can maintain that there is no contrast between strongly palatal and non-palatal sounds before the short vowels /ɪ/ and /ɛ/ in Irish, i.e. there is nothing like /lɣɪ/ or /tɣɛ/ in Irish.

For labial segments, tensed lips and a [j]-offglide are the phonetic correlates of palatality, e.g. *mion* ‘minor’, *pioc* ‘pick’. However, in Irish there is not usually an offglide before front vowels so that a word like *mé* ‘me’ is [me:] and not [mje:], but phonologically the form is /mje:/. Although Irish does not show an [j]-offglide as the phonetic correlate of palatality before front vowels it does have an unrounded velar glide, [ɯ], after non-palatal labials and before front vowels. This gives the following distribution.

(1)

a. Palatal:

mé [me:] ‘me’ *maoin* [mˠiːnʲ] ‘wealth’

b. Non-palatal:

mion [mʲʌnˠ] ‘minor’ *mór* [moːrˠ]‘big’

This symmetrical distribution shows that glides indicating palatality or non-palatality are found before vowels which contrast with the preceding labial consonant but not otherwise.

1.1.3. How contrasts are realised: polarity and polarity reversal

The term ‘polarity’ is employed in this book to refer to the extent to which the tongue is moved away from a neutral position during the articulation of a consonant. This neutral position can be observed with nasals and liquids which in some cases do not show polarity (see below for further remarks).

In phonetic terms ‘polarity’ refers to (i) the raising of the tongue towards the palate during palatalisation and (ii) the lowering of the tongue body and raising of the back during de-palatalisation. The advantage of the term is that it simplifies the description of the morphonology of Irish: for the indication of many grammatical categories a change in polarity is often demanded. This single statement covers both the process of palatalisation and of de-palatalisation. The signal for a grammatical category, e.g. the genitive case, is often the reversal of polarity with the direction of this reversal being determined by the base form of a word. For instance, the polarity reversal which the noun *Nollaig* /nɣʌlɣəgj/ ‘Christmas’ experiences in the genitive is de-palatalisation, i.e. *am na Nollag* /nɣʌlɣəg/ ‘Christmas time’, lit. ‘time the Christmas. GEN’ whereas that which the noun *arán* /əˈrɑːnɣ/ ‘bread’ shows is palatalisation *blas an aráin* /əˈrɑːnj/ ‘the taste of the bread’, lit. ‘taste the bread. GEN’.

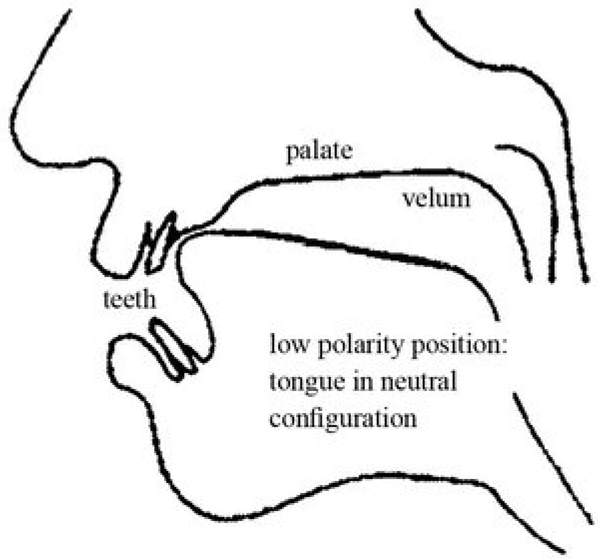


Figure 3. Articulation of coronal nasals and liquids with neutral tongue position

For most segments in the consonant inventory of Irish there are only two manifestations of polarity, palatal and non-palatal which are binary in nature, i.e. a consonant is either palatal or not, either non-palatal or not. However, among the sonorants, coronal nasals and liquids are different in that they show differing degrees of polarity. The acoustic recognition of such degrees is only possible with consonants which are primary palatals and which show a high degree of inherent sonority.[14] By this is meant that the segments in question are continuants which are clearly voiced without any frictional obstruction in the oral tract. These three phonetic factors are associated furthermore with an acoustic impression of sonority in that such segments can be perceived as maximally vowel-like, i.e. as frictionless voiced continuants. Degrees of polarity can be easily recognised for such segments as they are show a continuous articulation without any turbulence in the oral tract which could mask the perception of polarity.

The only segments in Irish which match these conditions are the coronal nasals and liquids. For them there is a cline of polarity which shows maximal palatality and maximal velarity at the two respective extremes as well as a region of low polarity in the middle where the tongue has a neutral configuration (as shown in Figure 3 above).

Table 4. Polarity values for coronal nasals and liquids in (Western and Northern) Irish

maximal palatality				**maximal velarity**
nj, lj		**n**j, lj		nɣ, lˠ
buille [-lj-]		*buile* [-lj-]		*mealladh* [-lɣ-]
'blow'		'anger'		'enticing'
neart [nj-]		*nimh* [nj-]		*nós* [nɣ-]
'strength'		'poison'		'custom'
high polarity	—	low polarity	—	high polarity

The consonants with low polarity have a subscript palatality sign which contrasts with the superscript palatality and velarity symbols which signal high polarity, in either direction. It will be noticed that there is a contrast between high polarity on the left (palatality) and low polarity in the middle (this is demonstrated by minimal pairs such as *buille* [bɪljə] and *buile* [bɪljə] in Table 4). However, there is no corresponding systemic contrast between low and high polarity segments on the velarity side to the right of Table 4.

Word pairs like *ainm* [ænjɪmj] 'name' and *anam* 'soul' might look like quasi-minimal pairs proving a contrast between /nj/ and /nɣ/ in the central, low polarity region. But although there is a single intervocalic *-n-* in the word for 'soul' this is realised as a fully velarised nasal: [ɑnɣəm] (de Bhaldraithe 1945: 13). This leaves the contrast between /nj/ and /nj/: the word for 'name' is phonetically [ænjɪmj] with a low polarity coronal nasal while the word for strength in Table 4 shows a high polarity nasal [nʲært].

The above remarks refer to the situation in Western and Northern Irish. In Southern Irish, however, non-palatal coronal nasals and liquids do not have a high degree of velarisation, compared to realisations in the other two dialect areas. Because of this scholars have used only two symbols for coronal nasals and liquids in Southern Irish, one indicating a palatal articulation and one without this indication: **l, l'; n, n'** (Ó Cuív 1944: 45-49; Ó Sé 2000: 17-18). Ó Cuív describes the non-palatal nasal and liquid respectively as a 'velarized voiced dental nasal' and a 'velarized voiced lateral consonant, formed by pressing the tip of the tongue against the upper teeth' (1944: 46, 48). Ó Sé describes the corresponding sounds in the Kerry dialect of Southern Irish which he investigated in a similar fashion. The lesser degree of velarisation of the non-palatal sonorants in Southern Irish compared to Western and Northern Irish was confirmed by the recordings which formed the primary data in Hickey (2011a).

An additional aspect of Southern Irish in respect of polarity is that there is no systemic contrast between fully palatal and slightly palatal sonorants as there is in Western and Northern Irish (see Table 4). For the word pair *buille, buile* just discussed both Ó Cuív (1944: 15) and Ó Sé (2000: 18) register no difference in pronunciation, their transcription for both being with **l'** = [lj]. De Bhaldraithe (1945: 40), on the other hand, distinguishes the two words, using **L'**and **l'** for the contrast between a fully palatal lateral, in *buille*, and a slightly palatal lateral in *buile*. The conclusion here is that the scholars dealing with Southern Irish use the symbol **l**' to denote a lateral which is not differentiated for degree of palatality. In tabular form the Southern Irish distribution can be given as follows.

Table 5. Polarity values for coronal nasals and liquids in Southern Irish

palatality	**velarity**
nj, lj	nɣ, lɣ
buille, buile	*mealladh*
‘blow’, ‘anger’	‘enticing’
neart, nimh	*nós*
‘strength’, ‘poison’	‘custom’

In the present book the phonetic symbols [nj, lj] are used to indicate the palatal coronal nasals and laterals of Southern Irish while the phonetic symbols [nɣ, lɣ] indicate the non-palatal coronal nasals and laterals which are less velarised than in Western and Northern Irish, hence the use of a subscript diacritic rather than a superscript one, i.e. [$_{ɣ}$] in the South versus [ɣ] in the West and North. The superscript diacritics are used for phonemic transcriptions no matter which dialect is being described.

1.2. Palatality and palatalisation

In the current book a strict distinction is made between the lexical property of *palatality* and its opposite *non-palatality*, i.e. the existence of palatal and non-palatal sounds in the citation form of words in the Irish lexicon, and the process of *palatalisation* and its opposite *de-palatalisation*, i.e. the reversing of polarity to indicate a certain grammatical category. This reversal can take two forms, the palatalisation of a non-palatal consonant and the de-palatalisation of a palatal consonant. This process forms the core of Irish morphonology (the interface between its phonology and syntax, Dressler 1985).

The lexical property of palatality is seen in the initial sound in *ní* /nji:/ 'wash' which distinguishes this word lexically from *naoi* /nɣi:/ 'nine'. The process of palatalisation and de-palatalisation consists of the reversal of the value for [palatal] (from a positive to a negative value or the reverse) under specific morphological conditions, for example in the change from nominative to genitive or from singular to plural with nouns (this only takes place in a syllable coda).

The term *polarity* refers to both lexical palatality and morphonological (de)palatalisation. A cover term to encompass both latter processes –palatalisation and depalatalisation – is *polarity reversal*.

Table 6. Properties and processes involving polarity in Irish

1a *Palatality*		Lexical property of words
1b *Non-Palatality*		Lexical property of words
	Lexical contrast of final palatal consonants	
	gabhar [gaur]	'goat'
	gabhair [gaurj]	'craving, strong desire'
	cuan [kuənɣ]	'cove, harbour'
	cuain [kuənj]	'litter, brood'
2a *Palatalisation*		Morphonological process
2b *De-Palatalisation*		Morphonological process

Morphonological contrast of final palatal consonants

gabhar [gaur]	'goat.NOM'
gabhair [gaurj]	'goat.GEN'
dán [dɑ:nɣ]	'poem.NOM'
dáin [dɑ:nj]	'poem.GEN'

The basic principle of palatalisation/de-palatalisation is one of alternation in the codas of syllables: the final sound or sounds in a syllable shift in value for [palatal]. All consonants in a coda – with the exception of the clusters /-xt/ and /-rt, -rd/ (see section II.2.7.4 below) – are affected by this as is the vowel preceding these, assuming that it is phonemically a short vowel, e.g. *olc* [ʌłk] 'evil.NOM' and *oilc* [ɛljkj] 'evil.GEN'.

This change in value for [palatal] is also found in unstressed syllables, as seen in *pobal, nó pobail, na hÉireann* [ˈpʌbəlɣ nɣu: ˈpʌbɪlj nɣə ˈhe:ɹjənɣ] 'the people, or peoples, of Ireland'. Perceptually, the cue for palatalisation is a somewhat retracted high front vowel [ɪ] and that for de-palatalisation is a lower schwa vowel [ə].[15] Systemically, however, the distinction is one of non-palatal versus palatal segment in the ending of the word forms just quoted, i.e. *pobal* [ˈpʌbəlɣ] versus *pobail* [ˈpʌbɪlj].

Because the change from [+palatal] to [-palatal] was triggered historically by an ending in which the vowel was non-palatal in character, de-palatalisation in modern Irish is associated with suffixation, e.g. *cáin* [kɑ:nj] 'tax.NOM' and *méid na cánach* [ˈkɑ:nɣəx] 'the amount of tax.GEN'.

1.3. Status of the feature [palatal]

Representations of palatal sounds using distinctive features, essentially stemming from the practice established by Chomsky and Halle (1968: 304-311), [16] characterise these as a combination of [+high] and [-back] . [17] However in Irish, both the property of palatality and the process of palatalisation have a particular status in the structure of the language, a status which is not adequately captured by [+high, -back] for palatal sounds, hence the use of the single feature [palatal] in this book. The value for [palatal] appears to be different from all other features in Modern Irish. [+/-palatal] can act independently from other features such as voice. It can also survive the deletion of other features of a segment which carries it. For instance, where a [+palatal] segment is deleted, e.g. as a result of lenition, the value for [palatal], which was present before the deletion, remains. The following example shows a trace of palatalisation on segment deletion:

(2)

a.	*fiúntach* [fʲu:nˠtəx]	‘worthwhile’
b.	*an-fhiúntach* [anʲu:nˠtəx] /f/ > Ø on prefixation of *an-* ‘very’	‘very worthwhile’

The trace can be discerned in the forward assimilation of the final /n/ in *an*‘very’. This type of assimilation is common in many languages and English has it in cases like *dress your* [ˈdrɛʃɔ], *what you* [ˈwɑtʃə], *said your* [ˈsədʒə] (Shockey 2003: 44-45). However, in this example one can see that in Irish the forward assimilation is present although the element which triggers it, a palatal /fj/, has been deleted. The trace of /fʲ/ changes the final consonant of an /ənɣ/ to a palatal /nj/ *although* the vowel it is immediately followed by is a back one, i.e. non-palatal. The only explanation for the forward palatal assimilation is that the feature [palatal] of the deleted /f j-/ survived this deletion and triggered the palatal assimilation of the /n/ in *an-* to [nj]. Another instance showing that the value of [palatal] is separate from the segment which carries it, involves a subset of back vowels which trigger forward assimilation, even though nothing precedes them in the base form of words they occur in.

(3)

a.	*eochair* [ˈʌxər^j]	‘key’
b.	*an eochair* [ən^j ˈʌxər^j]	‘the key’

Here the final nasal of the article *an* ‘the’ changes from a non-palatal to a palatal realisation although the following segment, the vowel [ʌ], is non-palatal (back) by nature. But as can be seen from the orthography – *eo* –this word originally had a front initial vowel. Although the vowel shifted historically to a back position, the assimilation triggered by the frontness of the original vowel has remained and is still in evidence in the forward assimilation today.

1.4. Independent and dependent segments

Not all sounds of Irish can occur initially in the citation forms of words. Some only appear as the result of applying an initial mutation, e.g. /ɣ/ only occurs in this position as the result of leniting either /g/ or /d/. The same is true of /x/ which is the outcome of leniting /k/, apart from a few grammatical words which are permanently lenited. This situation motivates the subdivision of consonants, ‘independent’ and ‘dependent’, according to whether they result from mutation or not.

Table 7. Independent and dependent consonants in Irish (i)

Independent	Those consonants which occur initially in the citation forms of words.
Dependent	Those consonants whose occurrence word-initially is dependent on the application of a mutation.

A sound can also have dual status by being both independent and dependent, e.g. /f/ in both *fadhb* /faib/ ‘problem’ (independent) and *a phota* /ə fʌtə/ ‘his pot’ (dependent on the lenition of initial /p/).

Sounds which are dependent may, however, occur in non-initial position, e.g. *fiach* /fjiəx/ ‘hunt’. In such cases the sound in question, here /x/, is part of the lexical structure of the word and is present in the citation form. The restriction of sounds to word-initial position is due to a mutation and hence such sounds cannot be found without the mutating element, e.g. *grán* /grɑ:nɣ/, */ɣrɑ:nɣ/ ‘grain’ : *a ghrán* /ə ɣrɑ:nɣ/ ‘his grain’. The only exceptions to this are a small set of words, mostly grammatical, which are permanently lenited, e.g. *duit* /ɣɪtj/ ‘to you’, *chuile* /xɪljə/ ‘every’; *dhá* /ɣɑ:/ ‘two’.

Certain consonants occur both independently and as the result of mutation leading to sounds which can be both primary and derived.

Table 8. Independent and dependent consonants in Irish (ii)

(i)	Primary nasal (independent)	*maith*	/ma/	'good'
(ii)	Derived by mutation (dependent on the mutation nasalisation)	*i mbun oibre*	/ɪ mʌnˠ aibʲrʲə/	'at work'

Morphonological and lexical contrast

Because contrasts in the feature [palatal] are found in both the lexicon and the grammar of Irish, there are words which differ in their citation forms and words which differ in some grammatical category with respect to palatality. For instance, the lexical words *cart* /kart/ 'cart' and *ceart* /kjart/ 'correct' are distinguished by the non-palatal and palatal velar stop at the beginning of each word respectively. In the words *tamall* /taməlɣ/ 'interval.NOM' and *tamaill* /taməlj/ 'interval.GEN' the difference is due to a contrast in case. Such grammatical distinctions apply at the end of word forms, i.e. in syllable codas. A morphonological contrast of palatal versus non-palatal is never found in a syllable onset in Irish.

Table 9. Polarity contrasts and syllable position in Modern Irish

	palatality (lexical)	**(de)palatalisation (morphonological)**
syllable onset	✓	✘
syllable coda	✓	✓

Because of various historical developments, the alternation between palatal and non-palatal segments may not be symmetrical. For example, the root extension /-əx/ changes to /i:/ on palatalisation as seen in *marcach* /markəx/ 'rider.NOM', *marcaigh* /marki:/ 'rider.GEN', *gaelach* /ˈge:lɣəx/

‘Irish’, *níos gaelaí* /nji:s ge:lɣi:/ ‘more Irish’. Historically, the genitive ending was /ɪç/ but the /ç/ was absorbed into the preceding vowel, lengthening it in the process. In addition the final element of the extension was voiced[18] as opposed to the voiceless segment in the nominative, see Greene (1973) for a brief survey of the historical development of palatalisation in Irish.

1.5. Pairwise notation

Because consonants in Irish come in pairs a notation is appropriate which employs a single symbol to refer to both the palatal and the non-palatal member of a consonant pair. This is not just a matter of notational convenience: there are many processes, both phonological and morphonological, which do not distinguish between the palatal or non-palatal nature of the segments involved. Hence a cover symbol to refer to both members of a pair allows a more accurate description. For instance, epenthesis occurs in heavy syllable codas (see section II.2.8 below), typically consisting of two sonorants. This applies to all such codas, irrespective of whether their constituents are palatal or non-palatal, e.g. palatal heavy coda: *ainm* /anjəmj/ ‘name’, non-palatal heavy coda: *arm* /arəm/ ‘arm’. Morphonological processes such as initial mutation (Rice and Cowper 1984) apply to pairs of segments. The result of a mutation is the same for both palatal and non-palatal members of a pair. Hence it is more economical to say that, on lenition, *M* is changed to *V* rather than writing /m/ changes to /v/ and /mj/ changes to /vj/. This notation is advantageous as it results is a more parsimonious phonological description of Irish. And there are no exceptions which require one to qualify it. Thus the feature [palatal] of consonants is ‘invisible’ to the morphonological processes of initial mutation.

What is termed in this study ‘pairwise notation’ is shown by using italicised capitals to refer to a pair of consonants which agree in all aspects except palatality, e.g. *K* refers to /kj/ and /k/. There are twelve such pairs in Modern Irish and illustrations of them can be found in the following table.

Table 10. Pairwise notation for consonants in Irish

non-palatal member			palatal member		
P	p	*post*, ‘job’	*P*	p^j	*pioc*, ‘pick’ (v.)
B	b	*bó*, ‘cow’	*B*	b^j	*beo*, ‘alive’
M	m	*mála*, ‘bag’	*M*	m^j	*meabhair*, ‘mind’
F	f	*fág*, ‘leave’	*F*	f^j	*fionn*, ‘bright, ‘fair’
V	v	*bhog*, ‘moved’, *an-mhór*, ‘very big’	*V*	v^j	*bhí*, ‘was’ *an mhí*, ‘the month’
T	t	*tóg*, ‘take’	*T*	t^j	*teach*, ‘house’
D	d	*dubh*, ‘black’	*D*	d^j	*deoch*, ‘drink’
N	n^{γ}	*naoi*, ‘nine’	*N*	n^j	*neart*, ‘strength’
L	l^{γ}	*luí*, ‘lying’	*L*	l^j	*léamh*, ‘read’
l	l	*abhaile* ‘home’	*n*	n	*tháinig* ‘came’
R	r	*roinnt*, ‘somewhat’	*R*	r^j	*trí*, ‘three’
S	s	*súil*,‘eye’	*S*	s^j	*siúl*, ‘walking’
K	k	*cá*, ‘where’	*K*	k^j	*ceart*, ‘correct’
G	g	*gach*, ‘every’	*G*	g^j	*gearr*, ‘short’
X	x	*chun*, ‘in order to’	*X*	x^j	*a cheann*, ‘his head’ (lenited)
ɣ	ɣ	*ghlac*, ‘took’	*ɣ*	γ^j	*a ghiall*, ‘his jaw’
ŋ	ŋ	*a nglór*, ‘their voice’	*ŋ*	$ŋ^j$	*a ngeall*, ‘their promise’
(h	*a hainm*, ‘her name’		h	*a hiníon*, ‘her daughter’)	

The above consonants can furthermore be grouped in pairs the members of which are distinguished by voice: *P, B; F, V; T, D; K, G; X, ɣ*. There are a few exceptions to this grouping. Firstly *S*, does not have a voiced counterpart *Z* in Irish. The prohibition on voiced sibilants is absolute. [19] The use of a voiced sibilant in Irish, e.g. in a technical term like *ozone* [o:zo:nɣ] or in personal names like *Elizabeth* [əˈlʲɪzəbət̪]], is due to code-switching to English.[20] Secondly, voiceless sonorants (nasals and liquids) do not occur on a systemic level in Irish (though they do in Welsh, see IV.3 below). Thus *M, N, L, R* are pairs of segments each member of which is voiced. However, voiceless sonorants do occur phonetically. Here they are the result of assimilation to a preceding /h/, usually as the result of a noun with an initial *SL-* or *TL-* cluster being lenited, e.g. *ar shlí* /εrʲ hlʲi:/ [εrʲ l̥ʲiː] ‘in a way’.

The phonetic realisations of the segments listed above are generally what is suggested by the symbols used, e.g. /mj/ = [mj] and /kj/ = [kj]. In some instances a consonant plus yod are written in phonological transcription although a single symbol is used for phonetic realisations, e.g. /ɣj/ = [j], /xj/ = [ç] and /sj/ = [ʃ].

The symbol *ŋ* in the above table stands for velar nasals. These can occur in word-initial position where they can contrast with dental nasals, e.g. *deich ndún* /djε nɣu:nɣ/ ‘ten forts’ versus *deich ngabhar* /djε ŋaur/ ‘ten goats’, but in initial position velar nasals are only the result of mutation. In word-internal position they can be derived from an assimilation process which causes nasals to be realised as velars before velar stops, e.g. *cúng* [ku:ŋg] ‘narrow’.

For further discussion of the different kinds of laterals and *r*-sounds, see section II.1.8 below.

Pairwise notation and morphology

Pairwise notation can also be used for morphological affixes which can have either a palatal or non-palatal realisation. Consider the following forms (the plural *crainnte* ‘trees’ is found in Connemara Irish, de Bhaldraithe 1953: 21).

(4)

a.	*crann* /krɑːnˠ/	:	*crainnte* /kriːnʲtʲə/
	‘tree’	:	‘trees’
b.	*dán* /dɑːnˠ/	:	*dánta* /dɑːnˠtə/
	‘poem’	:	‘poems’

It is obvious that the endings /-tjə/ and /-tə/ are somehow the ‘same’ and differ solely in the value for [palatal] of the initial consonant. This value is determined by the final consonant of the stem to which the ending is attached (in the case of the word for ‘trees’ the stem-final nasal is palatalised for the plural and the initial *t* of the suffix then assimilates to this). So one could capture this generalisation via the pairwise notation *-Tə*. This then refers to a single affix which begins with a palatal or non-palatal consonant depending on the stem-final consonant to which it is suffixed.

1.6. Lexical sets for Modern Irish

According to the convention introduced by J. C. Wells in *Accents of English* (3 vols, 1982), a lexical set is any group of words which show the same pronunciation for a key sound. For instance, the lexical set TRAP covers all words which show the same vowel – basically a short low front vowel – although this may be pronounced differently by different speakers. So if a speaker has [æ] in TRAP, that same speaker will have [æ] in *shall, bad, plan, cat*, etc. If a speaker has [a] in TRAP then that individual will have [a] in all words of that lexical set. Thus the sample word, here TRAP, stands for all members of the lexical set in question.

The notion of lexical set can be applied to Irish as well, both for vowels and for consonants. For instance, if one assumes a lexical set <u>TE</u>ACH then this would refer to all words which have an initial palatal /t^j/. If a speaker has a true palatal sound in this word then he/she will have the same sound in all other words which begin with palatal /tj/, e.g. *teocht* ‘heat’, *tionchar* ‘influence’. Equally, if a speaker has a phonetic affricate – [tʃ] – for palatal /tj/, then that person will have this sound for all the members of the TEACH lexical set, i.e. in all other words which begin in a palatal /tj/. This is true in the great majority of cases although there may be instances where individual speakers exhibit variation in their realisations of particular lexical sets.

The following tables contain the consonants and vowels of Irish illustrated by a series of lexical sets (Hickey 2011a: 44-52). The key sound in each lexical set is underlined.

Table 11. Lexical sets: Consonants

		Non-palatal	Palatal
Labials			
Stops			
	P	/p/ POST ‘job’	/p^{j}/ PIOC ‘pick’
	B	/b/ BUIDÉAL ‘bottle’	/b^{j}/ BEO ‘alive’
Fricatives			
	F	/f/ FADA ‘far’	/f^{j}/ FIÚ ‘even’
	V	/v/ BHOG ‘moved’ AN-MHÓR ‘very big’	/v^{j}/ BHÍ ‘was’ AN MHÍ ‘the month’
Coronals			
Stops			
	T	/t/ TÓG ‘take’	/t^{j}/ TEACH ‘house’
	D	/d/ DUBH ‘black’	/d^{j}/ DEOCH ‘drink’
Fricative			
	S	/s/ SÚIL ‘anticipation, eye’	/s^{j}/ SIÚL ‘walking’
Velars			
Stops			
	K	/k/ CÁ ‘where’	/k^{j}/ CEART ‘correct’
	G	/g/ GACH ‘every’	/g^{j}/ GEARR ‘short’
Fricatives			
	X	/x/ A CHARR ‘his car’	/x^{j}/ A CHEANN ‘his one’
	ɣ	/ɣ/ DHÁ ‘two’ A GHUTH ‘his voice’	/ɣ^{j}/ DHÍOL ‘sold’ A GHIALL ‘his jaw’
Sonorants			
Nasals			
	M	/m/ MÁLA ‘bag’	/m^{j}/ MEALL ‘pile’
	N	/n$^{\gamma}$/ NAOI ‘nine’	/n^{j}/ NEART ‘strength’
	n	/n/ THÁINIG ‘came’	
	ŋ	/ŋ/ A NGLÓR ‘their voice’	/ŋ^{j}/ A NGEALL ‘their promise’
Liquids			
	L	/l$^{\gamma}$/ LUÍ ‘lying’	/l^{j}/ LÉAMH ‘read’
	l	/l/ ABHAILE ‘at/to home’	
	R	/r/ ROINNT ‘part’	/r^{j}/ TRÍ ‘three’

In some cases two keywords are given because a sound may have two sources. This is the case with *V* which can represent the outcome of leniting *B* or *M* and with *ɣ* which results from leniting either *D* or *G*.

Table 12. Lexical sets: Vowels

	same as preceding consonant for [palatal]	different from preceding consonant for [palatal]
Short vowels		
/ɪ/	FIOS ‘knowledge’	BUIDÉAL ‘bottle’
/ɛ/	TE ‘hot’	—
/a/	SLACHT ‘neatness’	TEACHT ‘coming’
/ʌ/	COR ‘turn’	SIOC ‘frost’
	TURAS ‘journey’	FLIUCH ‘wet’
Short unstressed vowel		
/ə/	BALLA ‘wall’	
Long vowels		
/iː/	LÍON ‘fill (v.)’	BAOL ‘danger’
/eː/	MÉ ‘me’	GAELACH ‘Irish’
/ɑː/	TÁ ‘is’	BREÁ ‘good’
/oː/	BÓ ‘cow’	BEO ‘alive’
/uː/	CÚ ‘hound’	FIÚ ‘even, still’
Diphthongs		
/ai/	AGHAIDH ‘face’	BEIDH ‘will be’
/au/	CABHAIR ‘help’	LEABHAR ‘book’
/iə/	BIA ‘food’	—
/uə/	CRUA ‘hard’	—

Notes

(i) The above lexical sets for vowels are based on Western Irish pronunciations. In most cases the realisation of a vocalic lexical set is the same across the three major dialect areas, though there may be differences, e.g. TÁ shows a fronted vowel in the North: [tæ:] ‘is’. The diphthongs are a special case as the words given above do not contain diphthongs in all the dialects, e.g. *leabhar* ‘book’ is /ljo:r/ in Northern Irish.

(ii) The right-hand column in the above table contains vowels preceded by consonants with which they disagree in polarity, i.e. non-palatal consonants before front vowels and palatal ones before back vowels.

(iii) The diphthongs /iə/ and /uə/ are in complementary distribution regarding the polarity of preceding consonants so that contrast does not exist here, i.e. there is no word like **fuia* /fˠiə/ to contrast with the existing *fia* /fjiə/ 'deer' nor is there anything like **criua* /kjrjuə/ to contrast with the existing *crua* /kruə/ 'hard'. Indeed the words *bia* /bjiə/ 'food' and *bua* /buə/ 'win, success' provide a minimal pair with the two diphthongs. Given this situation it might be an option to collapse the two diphthongs phonologically to a single one, e.g. /ɨə/ with the realisation [iə] or [uə] depending on the value for [palatal] for the consonant(s) which immediately precede(s). But there are vowel-initial words which show that the contrast must be inherent to the diphthongs, e.g. *iascán* /iəskɑ:nɣ/ 'mussel' vs. *uascán* /uəskɑ:nɣ/ 'soft, sheepish person'; *iallach* /iəlɣəx/ 'constraint' vs. *ualach* /uəlɣəx/ 'load, burden'.

(iv) The diphthong /ai/ is not commonly preceded by a palatal consonant. The above example *beidh* 'will be' has /ai/ as an optional development from /ei, ɛi/ (for comments on possible sociolinguistic variation in present-day Irish, see Hickey 2011a: 374-375; for general comments, see Guy 2002). Individual forms in a dialect may show Cj + /ai/, e.g. *téann* [tjaɪnɣ] 'goes' in Connemara Irish.

(v) For the /ʌ/ vowel, two variants are posited. In unconditioned realisations in Western and Southern Irish it is generally [ʌ], but in Northern Irish the realisation before *L* and *R* is [ɔ] – C<u>O</u>R = [kɔr] – reflecting its historical source as a mid back rounded vowel (see III.3.5.2 below for further discussion). In addition, the relationship between /ʌ/ and /u:/ can be seen clearly in the variation between the two in words like *fonn* ‘desire’ in Western Irish: [fʌnˠ] ~ [fu:nɣ], i.e. lengthening /ʌ/ produces /u:/, its long counterpart before nasals.

(vi) The /i:/ vowel has two distinct variants. The first is high and front in its articulation, much like the vowel in English *tea*. The second shows considerable retraction. The /i:/ vowel after non-palatal sounds is frequently written *ao*: *sí* /sji:/ [ʃi:] ‘she’ versus *saoi* /si:/ [sɨ:] ‘expert’. This will the subject of more discussion in the relevant section, see II.1.9.9.2 below.

(vii) The long mid front vowel /e:/ can occasionally occur after non-palatal consonants, e.g. *Gael* /ge:lɣ/ ‘Irish person’, *Gaeilge* /ge:ljgjə/ ‘the Irish language’.

(viii) The short low vowel /a/ has a fronted variant after a palatal consonant, as in TE<u>A</u>CH [tjæ(:)x] ‘house’, and a central variant as in C<u>A</u>TH [ka(:)] ‘battle’. There is also a retracted and lengthened low vowel before /r/, e.g. *carr* [kɑ:r] ‘car’, *Carna* [kɑ:rnɣə] (placename).

(ix) Many speakers, especially of Western Irish, show nasalisation of vowels. This has been commented on in earlier studies, e.g. de Bhaldraithe (1945: 46), de Búrca (1958: 58-59, see his section ‘Extra-phonemic elements’) and Mhac an Fhailigh (1968: 19-20). All these authors point out the incidental nature of nasalisation, i.e. it occurs in the environment of nasal consonants and does not play a systemic role in any dialect, e.g. *naoi* [nɣĩ:] ‘nine’, *cúng* [kũ:ŋg] ‘narrow’, *múnla* [mũ:nɣlɣə] ‘form, shape’. See Ó Curnáin (2007: Vol. I., 291-361) for a detailed discussion of nasalisation in the Irish of Iorras Aithneach. Holmer (1962: 62), in his analysis of Scottish Gaelic in Kintyre, points out that nasalisation must have existed in previous centuries in the language as there are many cases where a nasalised pronunciation survives although the triggering consonant has been lost or shifted, e.g. *geimhreadh* [gjĩ:rjə] ‘winter’. However, for present-day dialects of Irish there is no robust, context-free contrast of nasalised and non-nasalised vowels.

(x) The short unstressed vowel /ə/ does not show any dialect variation beyond that which is triggered by surrounding consonants, cf. the shift of [ə] to [ɪ] which occurs, for instance, when the value for [palatal] of the following consonant changes from negative to positive as often happens between the nominative and genitive, e.g. *tobar* [tʌbər] ‘well.NOM’ : *tobair* [tʌbɪrj] ‘well.GEN’. After palatal consonants, the vowel of the BALL<u>A</u> [balɣə] lexical set is raised and fronted somewhat, e.g. *fírinne* [fʲiːrʲɪnʲɛ] (not [fji:rjɪnjə]) ‘truth’. Because these differences are contextually determined they are not of phonological significance, though they can, of course, have signal value on a phonetic level, e.g. *consan* [kʌnɣsənɣ] ‘consonant’ versus *consain* [kʌnɣsɪnj] ‘consonants’ where the fronted nature of the second, unstressed vowel may be acoustically more important to a hearer than the palatal nature of the final nasal which may not be clearly audible in connected speech.

(xi) There are a few cases of vowel contrast between a diphthong and a sequence of two vowels, e.g.

suan [suənɣ] ‘sleep, rest’ vs. *suán* [suɑːnɣ] ‘juice’, but these instances are rare.

Table 13. Lexical sets: Vowels before heavy sonorant codas

NN			
	FONN ‘desire’	GLEANN ‘valley’	BINN ‘summit’
NG			
	LONG ‘ship’	MOING ‘mane’	
M			
	TROM ‘heavy’	AM ‘time’	IM ‘butter’
RR, RD			
	CORR- ‘odd’	BORD ‘table’	AIRDE ‘height’
LL			
	MALL ‘slow’	POLL ‘hole’	MOILL ‘delay’

The realisation of vowels before heavy sonorant codas varies considerably across the dialects and will be the subject of comment at many points in this study, see section II.1.8 and the following subsections in particular.

1.7. The consonant system

Phonetically, the palatal / non-palatal pairs of consonants fall into two groups, labials and non-labials. This has to do with the manner in which palatality is realised. With the labial group, i.e. *P*, *B*, *F*, *V*, *M*, the tongue is not the active articulator but it is with the non-labial group *T*, *D*, *S*, *N*, *L*, *R*, *K*, *G*, *X*, *ɣ*. With the labials the lips, or the upper lip and lower teeth, are the active articulators but secondary palatalisation is achieved by the position of the tongue. For instance, the position during the articulation of /p^j/ is that for /i/, e.g. *pioc* [p^jʌk] ‘pick’. On release it gives rise to a very rapid glide unless the vowel which follows is also high and front in which case the tongue retains its position. During the articulation of palatalised labials the lips are spread and tense as well. With non-palatal /p/ the converse is the case: the lips are not tensed and the tongue is in the position for a high back unrounded vowel which on release into anything but a high back vowel results in a velar glide [^{ɰ}], e.g. *paor* [p^{ɰ}iːɹ] ‘grudge’.[21]

Historically, labials were palatalised after the development of palatal non-labials (Macaulay 1966). Jackson (1967b: 179-183) discusses the origins and nature of the palatals and offers an explanation for the palatalisation of labials by analogical pressure to form palatal : non-palatal pairs as the non-labials had already done.[22] Jackson’s argument is supported by Sommerfelt (1937: 276-277). Pedersen (1909-13: 345-350), while dealing with the environment in which palatalisation arose in Old Irish, does not treat the labials as a special case (see also the remarks in Greene 1973: 127-136). The pressure of a system to develop sounds which phonologically match others already present is a common phenomenon and has happened in the Slavic and Semitic languages, [23] for example.

In present-day Irish labials with lip tension and spreading can be classified as phonologically palatal. This delimits them from those labials without this tension plus spreading and links them up to the set of non-labial consonants which have a primary palatal articulation. Pattern congruity is established then, something for which there is justification, for instance in case alternations: *gob* /gʌb/ ‘beak.NOM’, *goib* /gɛbj/ ‘beak. GEN’.

Non-palatal consonants are phonetically velarised and can show a following glide. For labials this glide – [^{ɰ}] – is of particular importance as it identifies the consonant in question as being phonologically non-palatal as in *muir* [m^{ɰ}ɪrj] ‘sea’. Some Irish scholars have used this symbol, e.g. R. B. Breatnach (1947: 57) who mentions it along with [^{w}] and other possible glides. M. Ó Murchú (1969: 50) has it for the <AO> vowel of Northern Irish (found in words like *naoi* ‘nine’). Other scholars use [^{u}] or [^{w}] only (Wagner 1979

[1959]: 94, de Bhaldraithe, 1945: 43). Wagner's transcription accurately indicates the rounded off-glide from non-palatal consonants typical of Northern Irish, e.g. *amárach* [əˈmʷæːrə] 'tomorrow'.

Given the situation just described for labials and non-labials there is justification – in articulatory terms (Browman and Goldstein 1989) – of referring to the latter as 'primary palatals', i.e. coronals and velars, because the tongue makes contact with the roof of the mouth (the palatal region). This is realised by a shift forward with velars, e.g. [k] > [kʲ], or a retraction with coronals, e.g. [t] > [tj].

The articulatory correlates of palatality for coronals vary considerably across the dialects. While in Southern Irish coronal palatal stops and sonorants are only very slightly palatalised (hence the use of a subscript rather than a superscript diacritic), in Western and Northern Irish there are clear acoustic correlates of palatality. In the latter varieties palatal stops are produced with a clear arching of the tongue upwards. In the North the lips are rounded and the tongue is frequently grooved during articulation so that coronal palatal stops can be phonetically affricates. In the West this rounding and grooving is not present.

Table 14. Main dialect realisations for palatal stops

Northern	**Western**	**Southern**
teacht [tʃart]	*teacht* [tʲæ(:)xt]	*teacht* [t$_j$æxt]

Non-palatal sounds are phonetically velarised, i.e. the body of the tongue is lowered and the back raised towards the velum, giving a characteristically hollow sound to such segments. The audibility is maximal in the onset of a stressed syllable before a vowel of opposite quality: *teaghlach* [ˈtjaɪlɣəx] 'family' (palatal followed by low vowel), *tuí* [tɣɨ:] 'straw' (non-palatal followed by high, front vowel) – Western pronunciations.

Variations in palatality

The variation found between the dialects of Irish can be connected with variation in principle or with variation bound to certain lexicalised forms but not found across the board.

The first type of variation can distinguish sub-regions of larger dialect areas. For instance, in the Irish of the Aran Islands, above all Inis Meáin and Inis Oírr, there is a tendency for coronal palatal stops to be realised as affricates, e.g. *áit* /ɑ:tj/ [ɑ:tʃ] 'place'. This affrication is generally regarded as a Northern feature and is attested as far down as North-West Mayo (Achill Island).

The second kind of variation is where a certain dialect has a different value for [palatal] in a lexical word than in the other dialects. This is particularly common in Western Irish (Connemara). For instance, the word *cosain* 'defend' has an internal palatal /sj/: [ˈkʌʃɪnj]. An internal palatal segment is also found in *amárach* 'tomorrow' with /-rj-/: [əˈmɑ:rjəx]. Furthermore, /g/ is palatalised in a syllable coda when preceded by /oː/ as in *síog* [ʃi:o:gj] 'fairy'.

Southern Irish is noted for having a non-palatal /s/ in many grammatical words (deictic terms, Ó Sé 2000: 48) which in the dialects of the West and North show /sj/ [ʃ], e.g. *anso* [ənɣˈsʌ] 'here', North/West: *anseo* [ənɣˈʃʌ].

Non-palatal /s/ is also a characteristic of Irish on the Aran Islands when /sj/ occurs before /kj/ and /tj/, e.g. *scian* [skiənɣ] 'knife', *stiall* [stiəlɣ] 'strip, slice'. A more general feature of Connemara Irish is the occurrence of non-palatal /-g/ in the adverb *uilig* [ɪˈljʌg] 'all, entirely' (S. Ó Murchú 1998: 3).

1.7.1. Stops

There are seven pairs of stops in Irish. Of these six are obstruents and one is a sonorant, namely *M*. There are cogent reasons for not grouping *M* with the sonorants *N, L, R*. These have to do with the phonological behaviour of *M* which will be discussed presently (see section II.1.8.1 below). The other six pairs of stops show a symmetrical distribution: two are labial, two dental and two velar, the member of each pair being distinguished by voice: *P, B; T, D; K, G*.

Table 15. Stops in Irish

		labial	dental	velar
oral	voiceless	*P*	*T*	*K*
	voiced	*B*	*D*	*G*
nasal		*M*		

In Irish the voice distinction is phonetically robust being clearly audible among speakers of all three dialects. The only exception to this is the voiceless realisation of palatalised velars in unstressed syllables, an aspect of speech for very many speakers particularly in the West and North, see section II.1.7.1.3. below. The phonetic distinction between voiced and voiceless stops in Irish is in sharp contrast to Scottish Gaelic where the distinction on a phonetic level is between strongly aspirated stops (the equivalents of the Irish voiceless stops) and non-aspirated stops (the equivalents of the Irish voiced stops), see Ternes (2006: 9).

The labials *P* and *B* have the same realisations, bar voice which distinguishes them. The same applies to *B* and *M* which only differ in nasality: *póg* /po:g/ ‘kiss’, *bád* /bɑ:d/ ‘boat’, *maol* /mi:lj/ ‘bare, bald’. Where labials are followed by a high front vowel the non-palatal labial stops show a velar glide[24] The velar glide is particularly obvious in

Northern Irish before a fronted /ɑ:/ vowel when it tends to be rounded (see third example below).

(5)

a.	*buí* [bᶭɨː]	'yellow'
b.	*maíomh* [mᶭɨːv]	'boast'
c.	*amárach* [əˈmʷæːrə]	'tomorrow'

The palatal members of *P*, *B* and *M* are similar to each other: each shows lip tension and a very brief /j/-glide on release. The latter can be heard most clearly when a low or back vowel follows, but not before a front vowel.

(6)

a.	*bearr* /bʲɑ:r/	'cut, shave'
b.	*meabhrach* /mʲaurəx/	'clever'
c.	*piseog* /pʲɪsʲo:g/	'superstition'
d.	*béim* /bʲe:mʲ/ *mí* /mʲi:/	'pressure' 'month'

Labial stops through final closure

Especially in Western Irish, a final /ʌ/ or /o:/ can be followed by an epenthetic /b/. This is found above all among prepositional pronouns in the third person plural, e.g. *acu* [akʌb] 'at-them', *leo* [ljo:b] 'with-them'. In the case of words in final /ʊ/ or /o:/ one can speak of epenthesis because the /b/ is an additional segment in the phonological form of a word. In some cases, however, the /b/ is the result of a shift of fricative as with /v^{j}/ to /bj/ in the second person plural of prepositional pronouns, cf. *libh* /ljɪbj/ ‘with-you. PL’.

1.7.1.2. Coronal stops

The term ‘coronal’ covers all segments articulated with the front of the tongue making contact with the top of the mouth anywhere from behind the teeth to the palate. These segments are separate from both labials and velars. The latter can be grouped with the coronals to form a group of non-labials.

The non-labial stops have the tongue as an active articulator and so the palatal members are phonetically palatal. With coronal stops the palatals show a retracted point of articulation vis à vis non-palatal members. With velar stops the palatal members have a point of articulation further forward. In effect this means that there is a degree of articulatory convergence with palatal coronals and velars.

(7)

coronals			velars		
non-palatals	>	palatal	palatal	<	non-palatals
/t, d/		/t^{j}, d^{j}/	/k^{j}, g^{j}/		/k, g/

However, there is no neutralisation of the two central sets of sounds. They are kept apart by both the small difference in the point of articulation, /t^{j}/ and /dj/ being slightly more advanced on the palate than /kj/ and /gj/, and also by a (variable) degree of affrication in the case of the coronals. This affrication is greatest in Northern Irish which has [tʃ] and [dʒ] for /tj/ and /dj/ respectively (see Table 14 above).

Realisation of non-palatal coronal stops

The non-palatals /t/ and /d/ are realised as phonetically dental stops across the dialects, although the realisation of the palatals varies: *tá* [t̪ɑ:] 'is', *dó* [d̪o:] 'to-him'.

In a number of English loanwords alveolar stops are found. These were probably introduced into the language via code-switching and given that the bilingual population had, and has, no difficulty in articulating English sounds, the alveolar stops were not replaced by the corresponding dentals.

(8)

a.	*diesel* [di:sɪlʲ]	*team* [ti:m]
b.	*(an) té* [tʲe:] 'he who'	*tae* [te:] 'tea'

Affricates in Irish

Loanwords are a source of other coronal segments in Irish. The English affricates /tʃ/ and /dʒ/ occur in many recent borrowings (Hickey 1982), e.g. *chopper* [tʃapər], *jeep* [dʒi:p].

Although there are no systemic affricates in Irish, in one sandhi situation the affricate [tʃ] does arise. This is where a verb form ends in /x/ and is followed immediately by a pronoun with an initial /sʲ/. Given that the forms for 'he' and 'she' begin with /sj/, this affrication is quite frequent. It may have arisen due to the prohibition in Irish of two fricatives in sequence causing /x/ to assimilate in place of articulation to the following /sj/, yielding [tʃ]. This prohibition is in keeping with the Obligatory Contour Principle which specifies that sequences of two segments, which share manner[25] of articulation, are disfavoured crosslinguistically (Odden 2014: 22).

(9)

a.	*bhíodh sé* /vʲiːx sʲeː/ [..tʃ ..]	‘he would be’
b.	*tiocfadh sí* /tʲʌkəx sʲiː/ [..tʃ ..]	‘she would come’

1.7.1.3. Velar stops

The two velar stops of Irish are *K* and *G* which vary in realisation from a true velar position for the non-palatal member of each pair to a fronted position for the palatal members. The palatalised velars do not show any affrication so that a development as historically in many Romance languages from [k] through [kj] to [tç] to [tʃ/ʃ] has not occurred in Irish.

(10)

a.	*cúl* /kuːlˠ/ ‘back, rear’	*ceart* /kʲart/ ‘right, correct’
b.	*gá* /gɑː/ ‘necessity’	*gearr* /gʲɑːr/ ‘short, soon’

Shift of D to ɣ

The lexical incidence of *G*, *ɣ* increased greatly due to an historical development in Irish. This is the shift of *D to ɣ* which is common in Western Irish and occurs in particular with grammatical words, e.g. *mar a cheap sí don* [gənɣ] *saol nua* ‘as she thought about the new life’. This shift sometimes involves a shift to a fricative as well, i.e. from /d/ to /ɣ/. If the sound is palatal then the result is phonetically [j], e.g. *di* [jɪ] ‘to her’.

The origin of this shift is probably connected to the retraction of *D* [d, dj] to *ɣ* [ɣ, j] on lenition which occurred during the later Middle Irish period (O’Rahilly 1926: 185, 192). But although the shift of *D* to *ɣ* applies to all lenited forms in Irish, the shift of *D* to *G* is generally confined to grammatical words or a small number of lexicalised instances, such as *dul* [ɣʌlɣ] ‘going’ in Western Irish. In vernacular styles the shift is obligatory, indeed with prepositional pronouns like *duit, dó, di*, the shift is found in all registers.

Final devoicing in Irish

Irish does not show a general devoicing of word-final obstruents as do German and Russian, for example. But there is a devoicing of velar stops in unstressed syllables in Western and Northern Irish.

(11)

a. *Nollaig* [ˈnˠʌlˠɪkʲ] ‘Christmas’

b. *Pádraig* [ˈpɑ:rɪkʲ] ‘Patrick’

c. *reilig* [rɛl$_{j}$ɪkʲ][26] ‘graveyard’

The restriction of devoicing to unstressed syllables probably has to do with the reduction of phonation in these positions. However, one further feature of this devoicing cannot be accounted for in this way: it only applies to palatal velars. This leads to alternations like the following:

(12)

a. *Nollaig* [ˈnˠʌlˠɪkj] ‘Christmas.NOM’

b. *am na Nollag* [ˈnˠʌlˠəg] ‘time of Christmas.GEN’

The alternation also appears where an unstressed syllable results from the insertion of an epenthetic vowel before a palatal velar stop (see section II.2.8 below). Vowel epenthesis is a low-level operation which applies to the output of a morphonological operation, such as palatalisation in the genitive.

(13)

a. *bolg* [bʌlˠəg] ‘belly.NOM’

b. *boilg* [bɛlʲɪkʲ] ‘belly.GEN’

Palatalisation of /oːg/ to /oːgj/

A characteristic of Western Irish is the exclusive occurrence of a palatal /gʲ/ following a long mid back vowel in stem-final or word-final position (in the first example below the word is morphological *tóg* + *áil*.VERBAL-NOUN) . The syllable rhyme /o:g/ is common so that the number of tokens of /oːgʲ/ in this dialect is considerable (de Bhaldraithe 1945: 14).

(14)

a. *tógáil* [ˈto:gʲɑ:lʲ] ‘taking, building’

b. *ciotóg* [ˈkʲɪto:gʲ] ‘left-hand’

c. *sióg* [ˈʃi:o:gʲ] ‘fairy’

Final devoicing (see previous section) does not apply here although /g^{j}/ in unstressed syllables (see second example above) is normally [k^{j}]. It would seem to be that final devoicing only occurs in unstressed syllables which are also short, see examples in (11) and (12) above.

Voiced velar stops through final closure

In Southern Irish there is a strong tendency to close an historical final /j/ to /gj/ in grammatical inflections, e.g. in future verb forms. This is often extended to grammatical words as with a reflexive pronoun or adverb of direction.

(15)

a.	*inseoidh* [inˈʃo:ɪg$_{j}$]	‘will tell’
b.	*rachaidh* [ˈræçɪg$_{j}$]	‘will go’
c.	*féin* [ˈfe:n$_{j}$ɪg$_{ɣ}$]	‘self’
d.	*ó thuaidh* [o: ˈhu:ɪg^{j}]	‘northwards’

O’Rahilly (1932: 52-7), in his treatment of final palatal fricatives, suggests that the closure of /-ɪj/ to /-ɪgj/ in the South serves the function of maintaining a distinctive ending for verb forms. It is certainly true that the closure of a final short vowel (from former /-ɪj/) provides paradigmatic regularity among verb forms, e.g. with the imperative.

(16)

a.	*gearraig* [ˈg$_{j}$ærɪg$_{j}$]	‘cut.IMP.SING’ (W + N: [ˈg^{j}ærə])

b. *gearraigí* [ˈg$_j$ærɪg$_j$iː] ‘cut.IMP.PLURAL’

The /-gj/ from /-ɪj/ would appear to be of some vintage and led to the retention of syllables which were lost elsewhere. For instance, the word *trá* ‘strand’ comes from an earlier disyllabic *tráigh* which in the West and North lost the second syllable. Due to the closure of /-ɪj/ to /-ɪgj/ this second syllable has been retained in the South: [ˈtrɑːɪg$_j$].

Voiced velar stops through borrowing

It is common for English words to be used in Irish with little if any phonological adaptation to the sound system of Irish, i.e. to code-switch from Irish to English (Hickey 1982, Stenson 1991). However, in previous centuries borrowing did involve phonological adaptation and in the present context this is relevant. Voiceless velar stops in word-final position in such borrowings were replaced by the corresponding voiced ones, increasing the tokens of voiced segments at the ends of words. This adaptation may have been motivated by patterns already present in the language. For example /-ʌg/ and /-o:g/ were already established syllable rhymes in Irish, cf. *clog* /klʌg/ ‘bell’ (← Latin *clocca*) and *bileog* /bɪljo:g/ ‘leaf’.

(17)

a. *cnag* /kn^{ɣ}ag/ ‘knock, tap’ (← OE *cnocian* / ON *knoka*)[27]

b. *pancóg* /pan^{ɣ}ko:g/ ‘pancake’

c. *bológ* /ˈbʌlˠo:g/ ‘bullock’ (E. G. Quin ed., 1990: 91)

The original reason for the voiced stops in words like *clog* and *cnag* was intervocalic voicing which remained after the loss of the final syllable in each word. This along with the Irish suffix *-óg* /-o:g/, a common ending with locative and diminutive meanings, cf. *crannóg* ‘lake-dwelling’, *daróg* ‘little oak’ (← *dair* ‘oak’), provided a pattern to which the English words were adapted on borrowing.

Final voicing on borrowing also applied to coronal stops, especially in unstressed position, e.g. *buiséad* /ˈbʌsje:d/ ‘budget’, *cairpéad* /ˈkarjpje:d/ ‘carpet’. See McCone (1981) for a historical treatment of final voicing.

1.7.2. Fricatives

There are five pairs of fricatives in Irish with the glottal fricative outside the palatal ~ non-palatal grouping.

Table 16. Fricatives in Irish

	labial	dental	velar	glottal
voiceless	*F*	*S*	*X*	/h/
voiced	*V*	—	ɣ	

The fricatives are more irregular than stops, both in occurrence and distribution. In occurrence because there is a significant gap among the sets of fricatives, viz. there is no voiced sibilant in Irish. In distribution because the voiced members of the labial and velar sets are dependent phonemes (see I.1.5 above) and hence do not occur in the lexical forms of words in initial position. Furthermore, voiced fricatives do not occur before stops and independent voiced fricatives are not to be found word-initially at all: ɣ only occurs word-initially as the result of leniting *D* or *G* or in some few permanently lenited forms. *V* occurs initially through the lenition of *B* or *M*. Intervocalically and word-finally, due to internal lenition of *B* or *M*, *V* is also found.

The distribution of fricatives depends crucially on whether they occur as the result of applying an initial mutation or whether they are also to be found in the lexical citation form of words. Only *S* and *F* occur in the latter in word-initial position, e.g. *sásta* ‘satisfied’, *fuar* ‘cold’.

The absence of a voiced sibilant is something which Irish shares with other languages such as Finnish and the North Germanic languages. This absence is all the more remarkable as Irish experienced considerable intervocalic voicing during its history and the shift of /s, s^j/ to /z, zj/ in this context would have paralleled the other instances of such internal voicing. One possible reason for the non-voicing of sibilants could be the acoustic prominence of friction with these sounds which would mask voicing and hence render the latter non-salient.

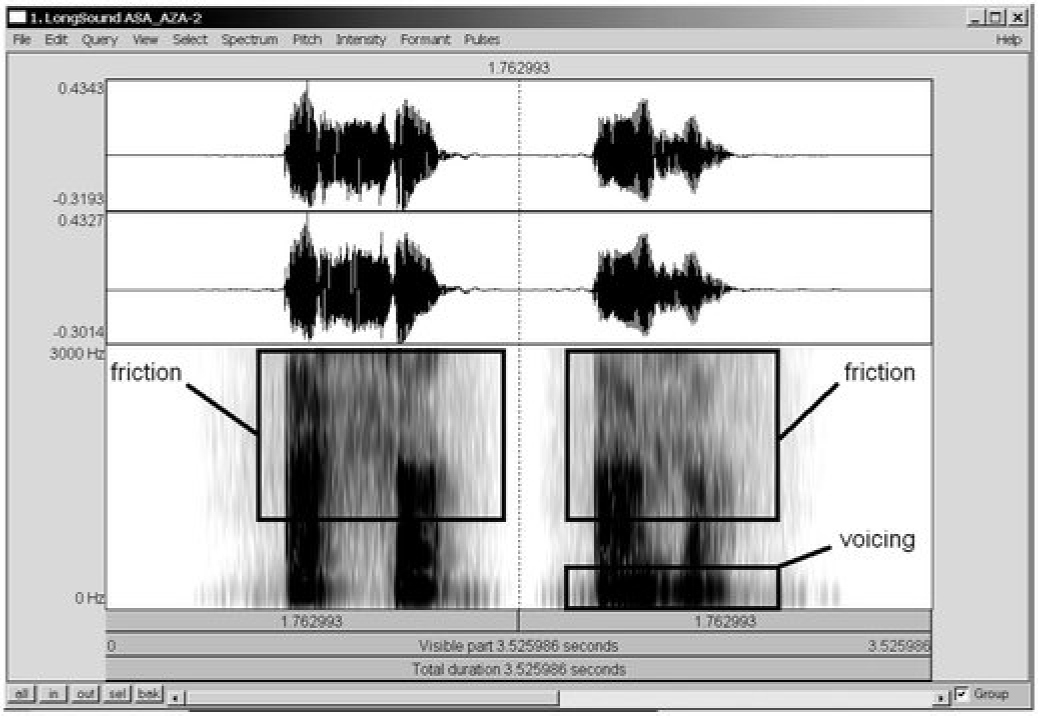


Figure 4 Spectrogram showing friction (square boxes) for voiceless and voiced sibilants (reading is of [asa] and [aza]).

Consider the preceding spectrogram: although the voiced fricative of [aza] (on right) is shorter than its voiceless counterpart [asa] (on left) the friction is still prominent compared to the voicing (on the bottom right) which it masks acoustically.

Despite the non-occurrence of analogical intervocalic sibilant voicing in Irish there is one instance in which such voicing is marginally attested: when preceded by a preposition which demands initial voicing some few varieties do show sibilant voicing (see section II.1.7.2.2 below).

1.7.2.1. Labial fricatives

The voiceless labial fricatives /f/ and /fj/, i.e. *F*, are generally realised as labio-dentals, though a bilabial articulation for /f/ is found occasionally. The palatal member of *F* is realised with lip tension plus a short [^{j}] release much as are the stops /pj/ and /bj/. For shifts in the articulation of *F*, see section II.2.7.6 below).

The voiced labial fricatives /v/ and /vj/, i.e. *V*, show some variation in realisation across the dialects. In Southern Irish both are labiodentals, i.e. [v] and [vj]. But in the West and North, the non-palatal member has an open articulation. This may be either a bilabial fricative [ß] or an approximant [w]. However, if the sound following /v/ is non-vocalic then a fricative pronunciation, [v], is more common (second set of forms below).

(18)

a.	*a bhua* [ə wuə]	‘his victory’
	faoin bhalla [walˠə]	‘under the wall’
	a mhála [wɑ:lˠə]	‘his bag’
b.	*brón* /bro:nˠ/	‘sorrow’
	a bhrón [ə vɹo:nˠ]	‘his sorrow’
	blas /blˠas/	‘taste’
	a bhlas [ə vlˠas]	‘his taste’

The distribution of realisations in the North and West can be formulated as follows.

(19)

$$
/\text{v}/ \rightarrow \begin{array}{ll} [\beta, \text{w}] & / \_\_\_ \text{V}, \# \\ [\text{v}] & / \_\_\_ \text{C} \end{array}
$$

In many cases, word-final position triggers further weakening of articulation which may lead to vocalisation (in

Western and Northern Irish). This demands that the segment preceding the labial fricative is a vowel, the fricative being absorbed into the nucleus of the syllable, lengthening the vowel in the process. The source of the vowel is irrelevant, it may in some cases be the result of epenthesis, e.g. *garbh* [ˈgarəv] ~ [ˈgaru:] ‘rough’.

This word-final variation is not confined to [-əv] and [-u:]. There is a further option which is common in Western and Northern Irish which involves a slight closure to a bilabial fricative, e.g. with the autonomous past form of many verbs. This closure of final /uː/ often has iconic value for the past autonomous and can be seen with verbs in which the word-final /uː/ has not arisen from the vocalisation of a labial fricative historically (second example below).

(20)

a.	*scríobh*	W: [sʲkʲrʲiu:ß], N: [sʲkʲrʲiʉ:ß]	‘was written’
b.	*frítheadh*	W: [frʲiu:ß], N: [frʲiʉ:ß]	‘was found’

Certain historical developments with sequences of a vowel plus a voiced fricative in a syllable rhyme can result in variation across the dialects. For instance, <*amh*> (/a/ followed by a non-palatal labial fricative) has the following realisations in Connemara with the second favoured in the westernmost parts of the region.

(21)

a.	*samhradh*	[ˈsaurə] ~ [ˈsavrə]	‘summer’
b.	*amhras*	[ˈaurəs] ~ [ˈavrəs]	‘doubt’

There is also a degree of morphological motivation for the retention of post-vocalic /v/ in unstressed syllables. This is to be found with the past autonomous form which shows [u:] ~ [əv] variation in Connemara, especially for second conjugation verbs (see third example below).

(22)

a. *an bhliain a rugadh mé* [ˈrʊgu:] ~ [ˈrʊgəv] ‘the year I was born’
b. *an áit ina tógadh mé* [ˈt̪o:gu:] ~ [ˈt̪o:gəv] ‘the place I was reared’
c. *foilsíodh graiméar nua* [ˈfailjʃi:əv] ‘a new grammar was published’

Southern Irish, in contradistinction to the other dialect areas, shows a vocalisation of palatal /vj/ which can render certain morphological paradigms opaque as the palatal sound only appears at a certain point.

(23)

a.	*sliabh*	[slʲiəv]	‘mountain’
b.	*barr an tsléibhe*	[bɑ:r ənʲ tʲlʲe:]	‘the top of the mountain’

Because the vocalisation of /vʲ/ can lead to a long vowel through absorption into the syllable nucleus in some cases this long vowel furthermore attracts stress in Southern Irish as in the following instances.

(24)

a.	*Gaillimh* ‘Galway’	[ˈgaljɪvj]
b.	*cathair na Gaillimhe* ‘the city of Galway’	[gaˈlji:]

Apart from the variation just discussed there are lexicalised examples of loss of a labial fricative, for instance, word-medial /-f-/ is lost in the word *uafásach* 'terrible, dire' in Connemara Irish: [ˈuːɑːsəx].

1.7.2.2. Sibilants

The phonology of Irish shows certain gaps, the most notable of which is the absence of voiced fricatives in the central area, i.e. it has no /z/ and no /ʒ/. Words like *ózón* 'ozone' are really examples of code-switching which have been given orthographical recognition. Native speakers generally use voiceless sibilants in such cases as in, e.g. [oːsoːnˠ lɣɛːɹ] for 'ozone layer'.

For Irish dialects the only case of a voiced sibilant being clearly recorded [28] is from Oileán Chléire (Cape Clear) off the coast of Co. Cork (Munster) where voiced sibilants were reported by Ó Buachalla (1962: 105) as occurring in intervocalic sandhi position in the environment where the initial mutation nasalisation (manifested as voicing) occurs, e.g. *tigh na sagart* [tjɪgj nɣə zagərt] 'the house of the priests', *i Sasana* [ə zasənɣə] 'in England' (Ó Buachalla's examples). The last case shows clearly that the voicing of /s/ in Cape Clear is by analogy with other examples of voicing on nasalisation, e.g. *F* to *V*. The word-internal /-s/, which is part of the lexical structure of the word for 'England', is not affected in this case.

Voiced sibilants (/z/ and /ʒ/) are reported for Scottish Gaelic by Dorian (1973: 41-42) who sees these as the result of phonemic borrowing. Breton is the only Celtic language which has established regular voiced fricatives in the coronal area (Ternes 1979: 217).

Realisations of sibilants

S stands for the pair /s – sj/. The non-palatal /s/ is a thin-groove laminal sibilant, much as in English, which before a stressed front vowel has velarisation as secondary articulation.

(25)

a. *suí* /si:/ [sˠɨ:] ‘sit’ *saoi* /si:/ [sˠɨ:] ‘expert’

b. *sé* /sʲe:/ [ʃe:] ‘he’ *sí* /sʲi:/ [ʃi:] ‘she’

The palatal /sj/ has a realisation in pre-vocalic or final position which is very similar to the [ʃ] of English. [29] When it occurs in homorganic clusters, i.e. before sonorants, it tends to be less broad-grooved and has a truly palatal realisation as a result of assimilation to the following palatal sound.

(26)

a. *Séamus* /sʲe:məs/ [ʃeːməs] ‘James’

b. *sneachta* /sʲnʲaxtə/ [ʃʲnʲæ:xtə] ‘snow’

Non-grooved fricatives are not found in present-day Irish (or Scottish Gaelic), although Holmer (1962: 25) has [θ][30] as an allophone of /s/ in Kintyre in the south-west of Scotland opposite the Antrim coast.

Subdivision of S

On a phonological level the sibilant *S* requires a subdivision into two types, S_1 and S_2, as follows.[31]

(27)

a.	S_1	=	*S* before *N, L, R*	(coronal sonorants)
			S before vowels	
b.	S_2	=	*S* before *M, P, T, K*	

This subdivision derives from the grammatical behaviour of the sonorants: S_1 can be lenited before *N, L*, *R* or a vowel whereas S_2 cannot.

(28)

a.	*a shnaois* /ə hnˠiːsʲ/	'his snuff'
b.	*a shlabhra* /ə hlˠaurə/	'his chain'
c.	*a shrón* /ə hruːnˠ/ /	'his nose'
d.	*á sheachaint* /ɑː haxənʲtʲ/	'avoiding him'
e.	*a smaoineamh* /ə smiːnjəv/[32]	'his thought'
f.	*faoin spéir* /fiːnj sʲpʲeːrʲ/[33]	'under the sky, in the open'
g.	*a staid* /ə stadʲ/	'his/her/their state'
h.	*a scéal* /ə sʲkʲeːlˠ/	'his/her/their story'

This behaviour is one of the main reasons for regarding *M* as a stop in Irish. The non-lenition of *S* before *M* is something which it shares with the stops *P, T, K* but not with the coronal sonorants which allow lenition in this context.

An additional feature of S_2 is that it is not realised as [ʃ] (the general realisation[34] of /sj/ in Irish) when preceding a palatal *M* or *P*. This behaviour appears to be determined by place of articulation: S_2 followed by a labial does not palatalise whereas S_2 followed by any non-labial does. Because of this S_2 can be subdivided into S_{2a} and S_{2b}.

(29) S_{2a} (does not palatalise when followed by a palatal labial)[35]

a.	*sméar*	/sʲmʲeːr/	[smʲeːr][24]	‘berry’
b.	*taispeáin*	/taˈsʲpʲɑːnʲ/	[təˈspʲɑːnʲ]	‘show’

S_{2b} (palatalises when followed by palatal)[36]

c.	*aisteach*	/asʲtʲəx/	[æʃtʲəx]	‘strange’
d.	*stíl*	/sʲtʲiːlʲ/	[ʃtʲiːlʲ]	‘style’
c.	*scian*	/sʲkʲiənˠ/	[ʃkʲiənˠ][25]	‘knife’

The non-labial sounds which can follow S_{2b}, i.e. *T* and *K*, do not palatalise in all dialects. For instance, a notable feature of Irish on the Aran Islands is the lack of palatalisation of S_{2b}, e.g. *scian* [skʲiənˠ] *[ʃkʲiənˠ], *scéal* [skʲeːlˠ] *[ʃkʲeːlˠ] ‘story’. This lack of palatalisation does not apply to S_1, e.g. *sneachta* ‘snow’ is [ʃʲnʲæxtə] and not *[snʲæxtə], *sioc* ‘frost’ is [ʃʌk], not *[sʌk], even in those dialects which do not have palatalisation for S_{2b}.

The following table summarises the behaviour of the different types of *S* sound. Where *s* is not followed by *h* in the table there is no lenition, e.g. *a smig* ‘his/her chin’[37] does not show lenition, that is the pronunciation is [ə smjɪgj] and not *[ə hmʲɪgʲ].[38]

Table 17. Types of *S*-sound in syllable-initial position

***S*-type**	**allows lenition**	**allows palatalisation**

S_1	yes: *a shrón* [ə hruːnˠ] 'his nose'	yes: *stíl* [ʃtʲiːlʲ] 'style'
S_{2a}	no: *a smig* [ə smʲɪgʲ] 'his/her chin'	no: *sméar* [smʲeːr] 'berry'
S_{2b}	no: *a stíl* [ə ʃtʲiːlʲ] 'his style'	yes (no): *scian* [ʃkʲiənˠ] (variant: [skʲiənˠ])

1.7.2.3. Velar fricatives

The velar fricatives of Irish, *X* and *ɣ*, are both dependent phonemes, i.e. they only occur word-initially as the result of mutating a velar stop. There are very few exceptions to this and all of these involve grammatical words of common occurrence which have permanent lenition, e.g. *chun* /xʌn/ 'to', *chomh* /xoː/ 'as'. In word-final or morpheme-final position *X* occurs independently.

(30)

a.	*sách* /sɑːx/	'fairly'
b.	*bacach* /bakəx/	'lame'
c.	*bacachán* /bakəxɑːnˠ/	'beggar'

This highlights a difference between *X* and *ɣ*: the voiced velar fricative only results from leniting *D* or *G* and so cannot occur in word-medial or word-final position. Historically, it did occur in these positions as the orthography suggests. In final position it has been vocalised or lost, while in word-medial position it was absorbed into the nucleus of the syllable containing it, frequently resulting in a diphthong in the West or South or a long vowel in the North.

(31)

		North	West + South	
a.	*laghad*	/lˠɛ:d/	/lˠaɪd/	‘least’
b.	*gadhar*	/gɛ:r/	/gaɪr/	‘dog, beagle’

The palatal member of *X* – /xʲ/ – is a fronted variant of /x/, phonetically [ç], e.g. *cheapaim* /xjapɪmj/ [çæpɪmj] ‘I think’, the palatal member of *ɣ* is also a palatal fricative, e.g. *an-ghearr* /anɣ ɣjɑ:r/ [anj jɑ:r] ‘very short, near’.

The distribution of /x/ is similar to that of /xʲ/. However, word-finally /xj/ is rare as the point in paradigms which demand palatalisation have /-i :/ (from earlier /-ɪj/) where /x/ was present at the velar point. This is true of nominal paradigms so that the genitive of *bacach* is *bacaigh* /baki:/ from a much earlier /bakɪj/.

Historically, there were also oblique case forms with the voiceless velar fricative, e.g. *tossuch* ‘beginning.DAT’ (Thurneysen 1909: 77) but clearly the voiced variant won out as only this could have led, on vocalisation, to the present-day /i:/ found in place of the palatal fricative. Furthermore, the spelling <*(a)igh*>, found in so many genitive endings today, implies that the ending formerly contained a voiced fricative.

As mentioned in II.1.7.1.3 above, Southern Irish /-j/ was strengthened to /-gj/ and is now found finally in nominal and verbal forms: *cheannuigh* ‘bought’, Southern: [çanɪgj] (Ó Cuív 1944: 77), Western: [çæ:nɣə] (de Bhaldraithe 1945: 101).

The status of X in Northern Irish

In Northern Irish (Donegal) the voiceless velar fricatives /x/ and /xʲ/ have an uncertain status. The reason for this is that they have been shifted or deleted in specific phonotactic environments. In fact for Donegal speakers, especially in the far north of the Gaeltacht (Gort an Choirce, Croithlí, An Fál Carrach), <*ch*> [x] may hardly exist at all. In the recordings for *Samples of Spoken Irish* (Hickey 2011a: 411-430) there was one sentence in which /x/ in the onset of a lexical stem occurred, i.e. *Tá a charr briste*. ‘His car is broken’. This is the position in which [x] is most likely to occur. But some speakers only had [h-], despite the fact that the sentences were presented to them in printed form which would trigger a reading style pronunciation.

In word-medial position, <*ch*> /x/ tends to be realised as [-h-] as well, especially with grammatical words. In word final position, /x/ disappears completely. Where it is followed by a further segment, in effect by /t/, it is realised as [r]. The following table summarises the phonotactic environments for *X* and the realisations it shows in these.

Table 18. Environments for *X* <ch> in Northern Irish

	Position	Realisation	Example
1)	Word-final	-Ø	*iontach* [i:nˠtə] ‘very’
			cláirseach [klˠæ:rʃə] ‘harp’
2)	Pre-consonantal	/-rC/	*slacht* [slˠart] ‘polish; finish’
3)	Remaining positions	/(-)h-/	*cha* [ha] ‘not, none’
			grácha [ˈgræ:hə] ‘loving.PL’
		/xʲ/: ç, h, Ø	*oíche* [i:, i:hə] ‘night’
			ar bith [ər bʲi:(ç, h)] ‘at all’

The first position also includes syllable-endings, e.g. *claochlú* [ˈklɣi:lɣʉ:] ‘transformation’. The second position is always a syllable coda and because of the possible coda clusters in Irish, C is always /t/ because clusters like /-xd, -xh, -xg, -xp, -xb/ do not exist. Furthermore, because /x/ does not assimilate to palatal /tj/ in syllable-codas (see II.2.1 below), there is no [ç] before /tj/. The /x/ in this position is realised as [r] just as it is in non-palatal clusters, e.g. *níos beaichte* [njiːs bʲærtʃə] 'more precisely'. In the third position there is variation between /h/ and zero. Furthermore, the loss of /-x/ has sometimes led to a specific vowel arising through coalescence with the preceding syllable nucleus, e.g. *bheadh* [vjeʉː] ← [vjeəx] 'would be'.

However, most speakers have a velar fricative in their sound repertoire because they would be aware of its use in more standard varieties of Irish. The preconsonantal realisation of /x/ as [r] is almost a shibboleth for Northern Irish and many speakers attempt to avoid it in careful speech, saying [ɒxt] for their vernacular [ɑrt], *ocht* 'eight'.

Voiced velar fricative

The ɣ set – /ɣ/ and /ɣʲ/ – is confined to word-initial position in Modern Irish, the orthographic sequences *-gh(-)* and *-dh(-)*, indicating word-medial and final ɣ, having long since been vocalised (see Ó Maolalaigh 2006 for a detailed discussion of these developments, especially pp. 42-66). They are furthermore always the result of applying an initial mutation.

(32)

a.	*ró-ghearr* /roː ɣʲɑːr/	'too short'
b.	*a ghluaisteán* /ə ɣlˠuːsʲtʲɑːnˠ/	'his car'
c.	*an-ghlic* /anɣ ɣʲlʲɪkʲ/	'very clever'

d. *leathgheal* /lʲaɣʲal/ ‘half-bright’

The quantity of voiced velar fricatives is quite high in Irish as it represents the lenited forms of both *G* and *D*: *a dhóthain* /ə ɣo:nj/ ‘his fill, portion’, *á dhéanamh* /ɑ: ɣji:nɣə/ ‘doing it.MASC’.

In all dialects the realisation of /ɣj/ is [j], e.g. *an ghirseach* [ənj ˈjɪrʃə(x)] ‘the young girl’. However, the non-palatal member of the set, /ɣ/, has a different realisation in the North from the West and South. In the former area /ɣ/ is fronted to [j] due to the fact that back vowels are frequently fronted in Northern Irish.

(33)

		North	South, West	
a.	*dhá*	[jæ:, jɛ:]	[ɣɑ:]	‘two’
b.	*do ghúngaí*	[də ˈjʉ:ŋi:]	[də ˈɣu:ŋgi:]	‘your hunkers’

These realisations contribute to the lack of velar fricatives in Northern Irish. Together with the realisations of palatal /tʲ/ and /dj/ as [tʃ] and [dʒ] respectively, they contribute significantly to the ‘soft’ acoustic impression conveyed by Northern Irish, rendering it quite different from Western and Southern Irish in this respect.

1.7.2.4. The glottal fricative

The glottal fricative [h] is has a defective distribution in Irish: in general it does not occur word-initially in native words though some common and long-established loans from English have [h] in this position, e.g. *halla* ‘hall’, *hata* ‘hat’. It is also an exception to the rule that all Irish consonants can appear in a velar and palatal form. Its place of articulation, the glottis, lies oral cavity where consonantal distinctions are made. [h] occurs frequently, apparently as a hiatus element between two vowels, for instance after the article or negator and before a following vowel-initial word.

(34)

a.	*na heaglaisí* /nˠə haglˠɪsʲiː/	‘the churches’
b.	*muintear na hÉireann*	‘the people of Ireland’
c.	*ní hé sin a rá*	‘that is not to say’

However, it is not obligatory in all situations where two vowels adjoin each other, e.g. *Bhí siad á athrú* ‘They were changing it’, so that it cannot be regarded as the automatic insertion of [h] between any two vowels. Furthermore, there is no doubt that /h/ has systemic status in Irish because it is involved in grammatical contrast, cf. its use as a marker of the third person singular feminine possessive pronoun. This /h/ is not present with the masculine possessive pronoun.

(35)

a.	*a hainm* /ə hanʲəmʲ/	‘her name’
	a hiníon /ə hɪnʲɪːnˠ/	‘her daughter’
b.	*a ainm* /ə anʲəmʲ/	‘his name’
	a iníon /ə ɪnʲɪːnˠ/	‘his daughter’

The glottal fricative also occurs as a result of lenition in Irish, i.e. as a dependent segment. Both *S* and *T* lenite to /h/.

(36)

a. *shnámh sé* /hnˠɑ:v sʲe:/ ‘he swam’

b. *shíl mé* /hi:lʲ mʲe:/ ‘I thought’

c. *thóg sé* /ho:g sʲe:/ ‘he took’

d. *thiocfadh tú* /hʌkəx tu:/ ‘you would come’

When /h/ occurs before a sonorant the phonetic effect is to devoice this segment so that /h/ is perceived as voicelessness with sonorants, e.g. *ar shlí* /ɛr hlʲiː/ [ɛr l̥ʲiː] ‘in a way’.

The realisation of /h/ is furthermore sensitive to the relative frontness or backness of the following vowel. When /h/ derives from a palatal segment, i.e. /sj/ or /tj/, it is always phonetically [h] despite the palatal articulation of the sound from which it is derived by lenition: *trí thine* [tjrji: hɪnə] ‘on fire’, cf. *tine* /tjɪnə/ ‘fire’; *shiúl mé* [çu:lɣ me:] ‘I walked’, *siúl* [ʃu:lɣ] ‘walk’. However, when the following vowel is back /h/ is realised as [ç] which increases the acoustic contrast between the consonant and the vowel: *thiocfadh tú* [çʌkəx tu:] ‘you would come’, *ar sheol* [ər ço:lɣ] ‘on the way, moving’. With the low vowels the palatality of the input segment is only retained before the low back vowel /ɑ:/.

(37)

a. *tais* [ta:ʃ] ‘damp’

b. *aimsir thais* [a:mʲʃɪrʲ ha:ʃ] ‘damp weather’

c. *teach* /tjax/ [tʲæ:x] ‘house’

d. *a theach* /ə hax/ [ə hæ:x] ‘his house’

e. *Seán* /sʲɑːnˠ/ [ʃɑːnˠ] ‘John’

f. *a Sheáin* /ə hɑːnʲ/ [ə çɑːnʲ] ‘John.VOCATIVE’

The conclusion from the above forms is that dependent /h/ which derives from leniting *S* and *T* contrasts in polarity with the following vowel: [h] occurs before front vowels and the low short vowel, whereas [ç] is found before back vowels, including the long low back vowel /ɑː/.

A further source of [h]

Established English loans in Irish show changes in their onsets. Sometimes the words were treated as mutated and the supposed initial mutation was reversed. This accounts for *balla* from *wall* where the initial [w-] was assumed to have derived from leniting /b/. *Seabhac* /sjauk/ [ʃauk] from English *hawk* is another instance. In this case the initial /h-/ was assumed to have derived from leniting /sj/ and this was ‘restored’, hence the Irish form of the English word. In a handful of English loans with initial /h/ this was not changed. These words are not subject to initial mutation.

(38)

a.	*haiste* ‘hatch’	*halla* /halˠə/ ‘hall’	*hata* /hatə/ ‘hat’
b.	*a hata* ‘his hat’	*a hata* ‘her hat’	(**a ata* ‘his hat’)

Word-medial [h]

The positions which [h] can occupy in a word varies both across and within the dialect areas. Within Western Irish a major difference between Irish in Cois Fharraige (immediately West of Galway city) and Irish in South Connemara (the area from An Cheathrú Rua to Carna, including the coastal islands) is that the former region shows a deletion of word-medial [h] with attendant vowel lengthening. The region of Cois Fharraige also shows a phonetic lengthening of the low vowel /a/ (de Bhaldraithe 1945: 12-14).

(39)

		Cois Fharraige	South Connemara	
a.	*bóthar*	[bo:r]	[bo:hər]	‘road’
b.	*oíche*	[i:]	[i:hə]	‘night’
c.	*athair*	[a:rʲ]	[ahərʲ]	‘father’

This deletion of intervocalic /-h-/ can also lead to homophony as with *oíche* /i:/ ‘night’ and *aoi* /i:/ ‘guest’. In this respect the Aran Islands would appear to align itself with Cois Fharraige rather than with South Connemara and shows a similar deletion of intervocalic /h/.

The presence of intervocalic /-h-/ in South Connemara has led to a phonetic shortening of vowels which precede it, even if these are systemically long. In addition, the preceding vowel receives strong stress, stronger than in other instances where it is an open syllable or in one not closed by /h/.

(40)

		South Connemara	
a.	*faoi láthair*	[fʷi: ˈˈlˠɑˑhɪrʲ]	‘at the moment’
b.	*lá*	[ˈlˠɑ:]	‘day’
c.	*lán*	[ˈlˠɑ:nˠ]	‘full’

1.7.2.5. The glottal stop

Irish does not have any glottalisation of oral stops. The lenition of stops involves frication. This is usually to a homorganic fricative as with *P* and *K* but can be to /h/ as with *S* and *T.* Lenition can also lead to deletion as with *F*. Nonetheless, a glottal stop (Ní Dhomhnaill 1977) can be recognised in Irish at the beginning of stressed words beginning with a vowel (as in German, for example, Kohler 1995: 168) and gives a clipped quality to such words. There is never any contrast between the glottal stop and its absence in this environment so that it cannot be accorded systemic status. This is different from the glottal fricative (see previous section) which can be present or absent in one and the same phonotactic position.

(41)

a. *Caithfimid na rudaí sin a athrú*
[... ˈʔæ ˈʔæru:]
'We have to change these things'

b. *D'iarraigh mé cruiniú a eagrú*
[... ˈʔæ ˈʔægru:]
'I tried to organise a meeting'

1.7.3. Shifts in articulation

Various shifts in articulation are found in Irish with variation across the dialects. These shifts can be divided into three basic types:

(42)

1) shift due to lenition, often as a step towards deletion
2) shifts resulting from the fortition of fricatives
3) shifts which are unrelated to either lenition or fortition and only involve a change in place of articulation

Before looking at attestations of shifts, it should be said that some shifts are instances of assimilation, e.g. *dromchla* [drʌmplə] ‘top, ridge, surface’ shows a shift of /x/ to /p/ under the influence of the preceding labial nasal. Other shifts are confined to a particular grammatical category. For instance, in Western Irish a final /-b/ is found in second and third personal plural prepositional pronouns. The closure appears to have been originally phonotactically determined, i.e. it arose before a fricative as in *acu* + *san* > /ˈʌkəbsən/ ‘at them’ (emphatic form) or *sibh* + *se* /ˈsjɪbsjə/ ‘you.PL’ (emphatic form) but came to be generalised. Evidence for this interpretation comes from similar cases of closure outside the prepositional pronoun paradigms e.g. *An raibh sé* [rʌbsje:] *thall riamh*? ‘Was he ever over there?’

All the shifts discussed here are separate from those which are the result of the mutations in present-day Irish, e.g. the shift of *K* to *X* with morphonological lenition as in *ceann* /kj-/ ‘head’ ~ *a cheann* /xj-/ ‘his head’. Perhaps because of the initial mutations, the other shifts involve, in the main, medial and final fricatives, but not stops.

1.7. 3.1. Lenition-related shifts

Shifts related to phonetic lenition, not the initial mutation of lenition, can involve the removal of the oral gesture with fricatives, e.g. the shift of *X* to /h/, found regularly in Northern Irish in word-initial position with grammatical words, e.g. *cha* [ha] ‘not’. Some fricatives do not shift to /h/ but are deleted. For instance, word-internal <*mh*> /vj/ can disappear, lengthening the vowel before it or triggering a diphthong, a common process in Southern Irish (see II.2.3 above), e.g. *go deimhin* ‘for sure’, Western Irish: [gə djɪvjɪnj], Southern Irish: [gə daɪnj].

1.7.3.2. Fortition-related shifts

A shift is regarded as an instance of fortition if it involves a shift from a glottal to an oral articulation or a shift from a vowel to a fricative. For example, the glottal fricative is shifted with adjectives and some grammatical words in Connemara Irish and that of the Aran Islands. The shift is from /h/ to /x/ and is found in cases like *ó shin* /o: xɪnj/ and *(níos) breátha* /bjrjɑ:xə/ 'finer'. Analogical extensions are also found, e.g. *crua* /kruəxə/ 'hard' (Ó Catháin 2001: 254).

The fortition may also involve the strengthening of a syllable onset. The word *breátha* just quoted also has the form /bjrjɑ:x.tjə/ in which a stop is inserted after the fricative to strengthen the onset of the second syllable.

The historical reduction of fricatives in final, unstressed syllables may not be present in some cases where the fricative (sometimes shifted in point of articulation) has iconic value for a certain grammatical category. For instance, in Western Irish the past autonomous of second conjugation verbs shows a word-final /-v/ (from a very early /-ð/) as in *Foilsíodh* [ˈfaɪljʃi:əv] *leabhar nua le gairid* 'A new book was published a short while ago'.

Occasionally a palatal fricative, which was vocalised in most contexts, can be retained or indeed inserted, cf. *-th* to [ç] as in *go maith* [gʌ mæˈç] 'as well' (above all in Northern Irish).

1.7.3.3. Shifts in place

The shift of a velar fricative to a labial (or labio-dental) one is a widespread occurrence (Hickey 1984a) which is amply attested in Germanic, cf. German *lachen* /laxən/ and English *laugh* /lɑ:f/. The change is also common in other branches of Indo-European, for instance in Slavic, cf. the Russian genitive ending *-ogo* /əvə/.

In Irish this is a common shift and is attested both historically and across the present-day dialects. The shift to the velar fricative is found historically in Irish, cf. *lochta* 'loft' (the Irish and English words are both from Old Norse *lopt* 'air, sky') which appeared in Old Irish as *lo(p)ta, labhta* so that one can posit a language-internal development for the shift from /-ft/ to /-xt/. Another example of this internal shift is *rachta* ‘rafter’ from Old Irish *rafta* (de Bhaldraithe 1981: 60) implying the shift of /-ft/ to /-xt/ happened within Irish rather than on borrowing. For some words the shift involves the palatal members of *X*, cf. Western and Northern Irish, *cluiche* [klɪfjə] ‘game’.

A related type of shift involves (mute) <*-th*> which can optionally be realised as /-f/, e.g. *go ceann leathuaire* [gə kjɑ:nɣ ljæfu:rjə] ‘for half and hour’. This is attested widely in Northern Irish and in the Irish of NorthWest Mayo, which is historically related to the former.

On the other hand, the appearance of *X* in word-final position, occasionally word-internally, is a common feature of Southern Irish. Here it is probably a retention of an earlier realisation for <*th*> : *luath* [lɣuəx] ‘early’, *tráthnóna* [trɑ:xnɣu:nɣə] ‘afternoon’, which goes back to earlier forms with word-final <*-th*> (Dinneen 1927: 681). These were carried forward from Old Irish where the pronunciation was originally [θ] (E. G. Quin ed., 1990: 443, col. 225-226). The dental fricative was shifted to a glottal fricative in the Middle Irish period but existed in the South-East as a voiceless velar fricative, hence the pronunciation in the South, especially in Ring (Ua Súilleabháin 1994: 487).

Range of reflexes of common input forms

The situations just described show what diverse reflexes are found across the dialects for a certain uniform input, here <*th*>. The common word *sruthán* ‘stream’ clearly illustrates this situation.

(43) Reflexes of historical <*th*> in Irish dialects (*sruthán* ‘stream’)

a. [srɑ:nˠ] (Cois Fharraige, de Bhaldraithe 1945: 104)

b. [ˈsrʊhɑ:nˠ] (South Connemara, de Bhaldraithe 1945: 104)

c. [ˈsrʊfɑ:nˠ] (Tourmakeady, de Búrca 1958: 31)

d. [srʊˈxɑ:nˠ] (East Mayo, Lavin 1956b: 310)[39]

e. [ˈsrʊhæ/ɛnˠ] (Donegal, Sommerfelt 1922: 12-13, Wagner 1979 [1959]: 66)

The pronunciation in Ulster without a phonetic reflex of the earlier word-internal /-θ-/ <*-th-*> is implied by anglicisations of placenames in East Ulster such as *Stranmillis, Co. Antrim* < *An Sruthán Milis* ‘sweet stream’ (McKay 2007: 137). See O’Rahilly (1932a: 175) for further comments on intervocalic <*-th-*> in Ulster. On more general issues of dialect relationships within Ulster, see C. Ó Dochartaigh (1987: 8-12).

Anglicisations of Irish placenames also provide evidence for these types of shifts, e.g. *Knocktopher* [-to:fər] in Co. Kilkenny which comes from Irish *Cnoc an Tóchair* [-to:xərj] ‘hill of the causeway’.

Dental to velar retraction

During the Middle Irish period (900-1200), a voiced velar fricative came to be used for lenited *D* (O’Rahilly 1932: 56 + 211). This shift in place of articulation became obligatory and in the contemporary language *D* always lenites to *ɣ*. This shift exercises a certain analogical attraction on *D* in non-lenited contexts, especially with grammatical forms, e.g. *do* /gə/ ‘for’, *de* /gə/ ‘from’ and the prepositional pronouns deriving from them (de Bhaldraithe 1953: 142). Homophony can result from the dental to velar retraction, especially in Western Irish where it is widespread.

(44) *Bhí chuile duine ag dul / gol* (both: [gʌlɣ]) *inné*
'Everyone was going / crying yesterday'

Lateral to uvular fricative
A shift which is found to the Kerry Gaeltacht on the Dingle Peninsula (Corca Dhuibhne) involves velar /lɣ/ [ɫ]. In word-final position this can shift to a voiceless uvular fricative [χ]. The shift was also found (in *Samples of Spoken Irish*) with an older, traditional speaker from Cill Chiaráin (Iorras Aithneach) in the West of the Galway Gaeltacht which would suggest that [χ] for [ɫ], as in *go mall* /gə mɑ:lɣ/ [gə mɑ:uχ] 'slowly', was more widespread previously.

1.8. The sonorant system

In Irish sonorants share patterning and processes not found with obstruents (plosive and fricatives). There are many phenomena such as allophony, grammatical alternation, pathways of language change, indeed a degree of interchangeability which justify treating sonorants as a class, cf. the exchange of sonorants in non-initial position both in native words, such as *mná* [mɹɑ:] 'women', and in loanwords, such as *feirm* [fɛljɪmj] 'farm'.

In a narrower sense the term 'sonorant' refers to nasals and laterals. In a wider sense it would include *r*-sounds and possibly glides.[40] For the current analysis, the term 'sonorant' will be used in the narrower, consonantal sense. The question also arises, how cohesive is this group of segments? At first sight sonorants can be subdivided into nasal and non-nasal sonorants. The various nasals – /m, n, nj, ŋ/ – have in common that they are oral stops with nasal continuance during their articulation due to the lowered velum. Non-nasal sonorants are more difficult. These consist of laterals and to a certain extent these sounds pair with *r*-sounds to give the subclass of liquids. But what have laterals and *r*-sounds in common which they do not share with nasals and, conversely, what do they have in common with nasals which they do not share with obstruents? From the articulatory point of view laterals and *r*-sounds have in common that they are non-fricative continuants (approximants) without nasal participation. This hives them off from other segments. Evidence for their similarity can be found in diachronic developments, such as the development of /r/ to /l/ (cf. Latin *peregrinus* and Italian *pellegrino* 'pilgrim') or in their non-distinctiveness in some Asian languages, such as Japanese and Sinitic languages. The articulatory similarity between /r/ and /l/ (both non-nasal, non-fricative continuants) is matched by an acoustic similarity, both having vowel-like formants, *r*-sounds being distinguished by a much lower third formant (Fry 1979: 120). These remarks apply to a (non-trill) apical *r* ([ɹ]) of course; different realisations of phonological /r/, such as alveolar taps or uvular fricatives, represent special cases.

Formant structure represents a similarity between nasal and non-nasal sonorants. Both groups are characterised by a clear formant structure during their articulation. This structure is determined by the preceding vowel (and the following, if there is one). Evidence for the coupling of nasal and non-nasal sonorants in Irish phonology can be found in their interchangeability or in such synchronic processes as vowel-lengthening before sonorants or stress placement on vowels preceding sonorants as will be shown presently.

The vowel-like nature of sonorants is obvious from their recognisable formant structure during articulation. This separates sonorants from fricatives which are characterised by a stochastic acoustic pattern during articulation and of course from stops which simply have closure. The acoustic proximity to vowels is stressed in standard definitions of sonorants. Like vowels, sonorants are normally voiced, a consequence of an unhindered air-flow (nasal or oral) during their articulation: Sonorants are characterised by a 'vocal tract cavity configuration in which spontaneous voicing is possible' Chomsky and Halle 1968: 302).

The similarity of sonorants and vowels is also obvious from the frequent vocalisation of sonorants, for example the nasalisation of vowels with loss of nasal consonants (French), vocalisation of syllable-final /r/ to a central vowel (Received Pronunciation, Standard German), vocalisation of /lj/ to /j/ and /lɣ/ to /u/ (various Romance languages, e.g. Spanish, French, Brazilian Portuguese, Polish, vernacular London English, Catalán and historical stages of English). With regard to Irish, however, the vocalisation of sonorants is not of relevance. There are (practically) no instances of vocalisation of /r/[41], /l/ or nasals. Furthermore, the present-day language has no phonemic glides: /j/ and /w/ do not exist; although phonetically [j] and [w] are present, the former deriving from palatal /gj/ on lenition and the latter from non-palatal /v/ through loss of friction. What is characteristic of Irish sonorants are their internal relationships and their effect on preceding vowels.

1.8.1. Nasals

In Irish three coronal sonorants and one labial sonorant exist: *N, L, R* and *M*. However, there is internal evidence from Irish that *M* is not to be recognised as a sonorant with the same undoubted status as the coronals *N, L, R*. Five items of evidence can be put forward here.

1) Firstly, *M* changes to *V* on morphonological lenition. The latter is a process of frication in Irish which occurs under specific grammatical conditions, e.g. *cat* /kat/ 'cat', *a chat* /ə xat/ 'his cat', *madra*[42] /madrə/ 'dog', *a mhadra* /ə vadrə/ 'his dog'. None of the coronal sonorants change their primary

articulation to a fricative on lenition.[43] In this respect *M* behaves like an obstruent.

2) Secondly, the presence of *M* after *S*, for instance in the genitive, blocks the prefixation of *T* found with words without *M* when they are masculine.

(45)

a. *gluaisteán an tsagairt* (← *sagairt*) /t- / 'the priest's car'
b. *bonn an tsléibhe* (← *sliabh*) /tjlj- / 'the bottom of the mountain'
c. *mí-usáid an smachta* (← *smacht*) 'the misuse of power'

3) Thirdly, clusters consisting of *S* and a sonorant are usually lenitable but not when the sonorant is *M*, consider the following instances *a sméar* (**a shméar*) 'his finger', *gan sprioc* (**gan shprioc*) 'without a goal' but *a shlacht* 'his tidiness', *a shrón* 'his nose'. The generalisation here is that *S* can only be lenited in a syllable onset if it is followed by a segment which shows higher sonority than lenited *S*, i.e. the glottal fricative. As sonorants count as more sonorous than fricatives a cluster like *shr*-/hr-/ or *shl*- /hl-/ is permissible. The illegality of lenited clusters like *shp*- /hp-/ and *sht*- /ht-/ can be explained by the decrease in sonority which a fricative + stop would represent a syllable onset. The non-occurrence of *shm*- /hm-/ could thus be accounted for by regarding M as having stop qualities which would block the lenition of *S* before it in a syllable onset.

4) Fourthly, *S* does not palatalise to [ʃ] before a palatal labial sonorant whereas it does before a palatal coronal sonorant cf. *sméar* [smje:ɹ] 'blackberry', but *sneachta* [ʃnjæ:xtə] 'snow'. In this respect *M* couples with the labial stops which do not show palatalisation of /s/ before a palatal labial, cf. *spéir* [spje:rj] 'sky'.

5) Fifthly, *N* after *M* in a syllable onset often shifted to *R*. This resulted in a type of cluster which *M* shares with stops: *mná* /mnˠɑ:/ ➜ /mrɑ:/ ‘woman. GEN’, *cnoc* /knˠʌk/ ➜ /krʌk/ ‘hill’, *Luimneach* /-mʲnʲ-/ ➜ *Limerick* /-mr-/, cf. *troid* /trɛdj/ ‘fight’, *prátaí* /prɑ:ti:/ ‘potatoes’.

Although certain reservations concerning the exclusive sonorant character of *M* are called for in light of the above facts it can nonetheless be regarded as a sonorant for the current analysis. Vowel alternation is attested before *M* as it is before *N, L, R*, e.g. *am* /ɑ:m/ ‘time’, *ama* /amə/ ‘time.GEN’ (see I.2.3 and II.1.8.5 for details).

Status of the velar nasal [ŋ]

As in many other languages, including English, there is a strong case for deriving the velar nasal from an assimilation process which causes a dental/alveolar nasal to be retracted to a velar position immediately before a nasal stop. While in English this requires positing a stop which is only sometimes realised, as in *long : longer* but not in *sing : singer*, in Irish the alternation between /ŋ/ and /g/ is always present. A form like *a nglór* /ə ŋlɣo:r/ ‘their voice’ can only occur if an independent form *glór* /glɣo:r/ ‘voice’ already exists. The same is true of a palatal version of /ŋ/ as in *a ngiolla*, /ə ŋjʌlɣə/ ‘their servant’, cf. the simple form *giolla* /gjʌlɣə/ ‘servant’. For this reason no segment pair *ŋ*, which would consist of /ŋ/ and /ŋj/, is posited here on a systemic level. Instead it is assumed that all instances of velar nasals derive from alveolar nasals via a low-level assimilation in place of articulation to a following velar stop.

The dialects of Irish vary considerably in the realisation of /ng/ clusters. All have assimilation of the nasal to the following stop, i.e. /n/ ➜ [ŋ] /__ /g/ but some have deletion of the stop so that the velar nasal is all that is present phonetically. This is another distinguishing feature of the Cois Fharraige area (de Bhaldraithe 1945: 39) vis à vis the South Connemara area in Western Irish, though it is variable here. The bare nasal was also found in South-West Cork in the recordings of *Samples of Spoken Irish* (Hickey 2011a: 411-430).

Syllabic nasals

In Southern Irish a nasal following a homorganic stop is not released but realised as a syllabic nasal. Because of the structure of stop + nasal clusters, only coronals occur here: *maidin* [ˈmad$_j$n̩$_j$] ‘morning’ (Hickey 2011a: 73).

1.8.2. Laterals

Laterals in Irish dialects vary from a palatalised [l^j] through an alveolar [l] to a velarised [lɣ] (the conditions for these segments will be the subject of section II.1.8 and its subsections below). Some variation in the realisation of laterals is found across the dialects. In Southern Irish, in the Corca Dhuibhne Gaeltacht in North Co. Kerry, and in Western Irish, in the Carna area, a shift is found where the strongly velarised lateral appears as a uvular [χ] in simple syllable codas (this shift is not accompanied by a parallel shift of palatalised /lj/ to [ç]).

1.8.3. R-sounds

There are a number of realisations of *R* in Irish dialects depending on palatality and the phonetic environment in which it occurs. The first is as a frictionless continuant, much as in Irish English (Hickey 2007: 320), when *R* is non-palatal, e.g. *barr* [bɑːɹ], *rún* [ɹuːnɣ] ‘secret’. The palatal /rj/ is a continuant but the raising of the body of the tongue towards the palate means that friction can occur, e.g. in a word like *dréimire* [djrjeːmjɪrjɛ] ‘ladder’. There may be a flap realisation of /r/, following a labial or velar stop in the onset of a stressed syllable, that is where the consonant preceding /r/ is not homorganic with it, e.g. *brón* [bɾo:nɣ] ‘sorrow’, *crann* [kɾɑ:nɣ] ‘tree’, or intervocalically after a short vowel, e.g. before an epenthetic vowel as in *gorm* /gʌrm/ ➔ [gʌɾəm] ‘blue’. After dental stops a slightly trilled realisation of /r/ may occur, e.g. in *trom* [trʌm] ‘heavy’. This trilled realisation of /r/ after dental stops is also attested for Irish English as well (Hickey 1989).

These realisations apply to present-day Irish. The evidence that the language had a non-trill # trill distinction at a previous stage is partly orthographic and partly based on dialectal evidence. D. Ó Baoill (1979: 81) reports that older speakers of Northern Irish have a trill for those words which are written with *rr*. As literacy in Irish is low among older speakers in rural Irish-speaking districts one can take it that this is not due to orthographic influence and so postulate for previous stages of Irish a [r] # [ɹ] distinction similar, for example, to that which is found in Spanish (Malmberg 1962: 81-86) in words like *perro* [pero] ‘dog’ and *pero* [peɹo] ‘but’. The trill was then the phonetic exponence of a phonological geminate /rr/ and a contrast may well have existed in a word pair like *urraim* [ʌrɪmj] ‘esteem’ and *orm* [ʌɹəm] ‘on-me’, the shorter [ɹ] corresponding historically to a non-geminate or lenited geminate (under specific morphonological conditions). This distinction is tenuous and certainly recessive, if it exists at all, and only for Donegal Irish. In the case of former geminates, Wagner (1979 [1959]: 27), in his study of South Donegal Irish, has a de-palatalised trill as in *farraige* [fɑrɪgjə] ‘sea’.

R in word-initial position

The distinction between palatal and non-palatal *R* is generally neutralised in word-initial position in Modern Irish. In this respect *R* is different from the remaining sonorants *N, L, M* all of which maintain the palatal # non-palatal distinction in this position. The non-occurrence of /rj/ in word-initial position has led to a lowering and retraction of high vowels after /rʲ-/ [r-] in varieties of Connemara Irish (last example in the following).

(46)

a.	*reo* /rʲo:/ [ro:]	‘frost, freezing’
b.	*riamh* /rʲiəv/ [riəv]	‘ever, never’
c.	*rith* /rʲɪ/ [rʌ]	‘run’

Because of initial mutation there are instances where a word with an initial cluster containing /rj/ as a second element is altered such that the /rj/ becomes word-initial. In these cases the /rʲ/ is generally depalatalised.

(47)

a.	*freagra* [fʲrʲægrə]	‘answer’
b.	*ní bhfuaireamar aon fhreagra* [e:nˠ rægrə]	‘we didn’t get any answer’
c.	*freasúra* [fʲrʲæsu:rə]	‘opposition’
d.	*san fhreasúra* [sənˠ ræsu:rə]	‘in the opposition’
e.	*friotaíocht* [fʲrʲʌti:əxt]	‘resistance’
f.	*an fhriotaíocht* [ənˠ rʌti:əxt]	‘the resistance’

The lack of palatal /r/ in word-initial position in a single-segment syllable onset means that there are instances where the lack of contrast leads to (near)homophony, compare the second and third examples in the following.

(48)

a. *raon* [riːənˠ] ‘way, route’

b. *rian* [riːənˠ] ‘trace’

c. *trian* [tʲrʲiːənˠ] ‘third; town quarter’

The distinction between /r/ and /rj/ is also neutralised when the latter is the first element of a cluster (de Bhaldraithe 1945: 38, 42). The non-palatal /r/ furthermore may have the effect of retracting a preceding vowel or leading to a preference for /aʊ/ over /aɪ/ where a diphthong is involved, see first example below.

(49)

a. *toirneach* /taʊrnʲəx/ ‘thunder’

b. *ceird* /kaɪrdʲ/ ‘trade, craft’

c. *comhairle* /ku:rlʲə/ ‘advice, council’

That the neutralisation of the palatalisation contrast applies not just to word-initial but to stressed syllable-initial position can be seen in the word *aréir* [əˈre:rj] ‘last night’ where the first /r/ is non-palatal.

Uvular r in Irish

The occurrence of uvular *r* in Europe is generally confined to a band which stretches from Northern France to Southern Sweden and which is found in languages such as standard French, standard German, Danish and in Southern dialects of Swedish (Malmberg 1971: 89-91). In the British Isles, uvular *r* is generally thought to be confined to traditional varieties of English in Northumbria where it is referred to as a ‘burr’ (Beal 2008: 139-140).

However, uvular *r* is also quite widespread in Ireland. In the recordings for *A Sound Atlas of Irish English*, the present author not only found actual instances of uvular *r* in North Leinster (for instance in the town of Drogheda, Hickey 2004a: 77-79) but also recorded vowel retraction to the uvular area as a reflex of previous uvular *r* among speakers spread across a much larger region of north-central Ireland. The recordings for *Samples of Spoken Irish* revealed uvular *r* with speakers from the Western and Southern Irish areas. Uvular *r* is known from speakers from other parts of the country, for instance rural Co. Waterford in the South-East.

The conclusion to be drawn from these facts is that previously uvular *r* had a much wider distribution across all parts of Ireland, both in Irish and in English. That uvular *r* is strongly recessive can be seen from its occurrence in a confined area for English (North Leinster, Hickey *loc. cit.*) and only with some older speakers in Western and Southern Irish. From the limited distribution in the latter it would seem that uvular *r* only occurs for non-palatal /r/and has a retracting effect on a preceding vowel.

Assibilation of R

Palatal /rj/ in word-final position is often articulated as a voiceless fricative. This is found in all dialects but is most obvious in Western Irish. The devoicing is a consequence of a reduction in phonation and is most salient in unstressed syllables. The non-palatal /r/ is not generally affected by this final devoicing.

(50)

a.	*b’fhéidir* /ˈbʲe:dʲərʲ/ [ˈbje:djɪʂ]	‘perhaps’
b.	*láidir* /ˈlˠɑ:dʲərʲ/ [ˈlˠɑ:dʲɪʂ]	‘strong’
c.	*foclóir* /ˈfʌklo:rʲ/ [ˈfʌklo:ˈʂ]	‘dictionary’

A voiced fricative realisation – [ʐ] – can be found for /rj/ when it follows a palatal labial in the onset of a stressed syllable.[44] In this case, the lip spreading combined with the raised tongue position for palatal labials leads to an assibilation of /rʲ/.

(51)

a.	*breá* [bʲʐɑ:]	‘good, fine’
b.	*breis* [bʲʐeʃ]	‘addition, excess’

This assibilation[45] is found chiefly in Western and Northern Irish and is mentioned briefly in the descriptive studies by Mhac an Fhailigh (1968: 44), de Bhaldraithe (1945: 41) and D. Ó Baoill (1973: 37). It has a parallel in certain developments in other languages, for instance in Czech (Kučera 1961: 31-32) which also experienced the assibilation of /r/. In a sonogram investigation Romportl (1973: 88) established the intermediary character of this sound between [r] and [ž] (= IPA /ʒ/). This shows parallels with Irish where /rj/ lies between [ɹ] and [zj]. The essential difference is that the [ʐ] sound of Irish /rj/ is not trilled. If the tip of the tongue were lowered and the blade grooved then [ʐ] would become [ʒ]. This similarity is paralleled by a development in Polish where /ř/ and /ž/ had fallen together by the eighteenth century (Stieber 1973: 69-70).[46]

Influence of English on the pronunciation of R

In Irish English during the 1990s a shift in the pronunciation of /r/ occurred: the traditional velarised [ɹ̴] came to be replaced by a retroflex [ɻ],[47] first in the Dublin area among young females, then spreading out to the rest of the country (Hickey 2003a, 2005: 76-77, 2007: 358-359). As all young speakers of Irish are bilingual, features of recent Irish English are being transferred to their Irish, both in grammar and pronunciation, not to mention vocabulary and phraseology. One obvious feature of the speech of young female Irish speakers is the use of a retroflex [ɻ], especially in word-final position after a vowel, a position where it is prominent in Irish English, e.g. *north* [no:ɻt̪], *sore* [so:ɻ]. In instances where the *R* of Irish is palatal, the retroflex [ɻ] is not acoustically obvious, but it is where *R* is non-palatal as in *leor* /ljo:r/ ‘enough’, realised as [ljo:ɻ].

1.8.4. Three-way distinctions among sonorants

In the discussion of obstruents so far, two-member pairs were assumed for each, e.g. /t/ and /tj/, represented in the present study, and Hickey (2011a), as *T*. But there is strong evidence from the dialects of Irish that there was, and is, a three-way distinction among coronal nasals and laterals in the West and North (Ní Chasaide 1979).

Because of the continuous and resonant nature of sonorants, palatality and velarity are particularly clear. There is no difficulty in acoustically separating not only a palatal and a velarised nasal or lateral, for instance, but also of recognising an intermediate, ‘neutral’ value. Transcriptionally, these can be represented as follows.

Table 19. Division of coronal nasals and laterals in Western Irish

	palatal	neutral	velarised
	n^j	n_j	$n^{\text{ɣ}}$
	l^j	l_j	$l^{\text{ɣ}}$

Such a tripartite acoustic division exists for Western and Northern Irish. A palatal nasal occurs in a word like *neart* /njart/ ‘strength’ and a non-palatal, velarised sound is found in *naomh* /nɣi:v/ [n̴i:v] ‘saint’, for instance. Neutral versions of both nasals and laterals, i.e. without any significant polarisation along a palatal-velar axis, occur after a short stressed vowel and before another, e.g. *duine* [dɪnjə] ‘person’, *baile* [baljə] ‘town’, *eile* [ɛljə] ‘other’ (note the absence of off-glides or on-glides in the environment of these sonorants). The sonorants appear most often immediately after a stressed syllable, as in the examples just given. The polarised variants also occur in this position providing contrast, cf. *codladh* [kʌlɣə] ‘sleep’ and *bainne* [bæ:njə] ‘milk’.

Neutral sonorants do not occur normally at the beginning of a word,[48] i.e. they are excluded from the onset of a stressed syllable but not from the coda of stressed syllables. However, due to the fact that palatal sonorants can also occur in non-initial position there is genuine systemic contrast as in the following examples.

(52)

a.	*céile* /k^je:lə/	‘spouse’
	céille /k^je:l^jə/	‘sense.GEN’
b.	*fine* /f^jɪnə/	‘race, tribe’

finne /fʲɪnʲə/ ‘fairness, brightness’

Furthermore, the neutral sonorants would seem to be examples of depalatalised or non-palatalised palatals, i.e. there are no neutral sonorants in a velar environment. Any word with a single sonorant in an orthographically velar environment (between back or low vowels) has a phonetically velarised realisation.

(53)

a.	*dona* [dʌnˠə] ‘bad’	*dúnadh* [du:nˠu:] ‘was closed’
b.	*solas* [sʌlˠəs] ‘light’	*codladh* [kʌlˠə] ‘sleep’

This situation applies to Western and Northern Irish. But for Southern Irish a distinction between neutral and velarised sonorants is tenuous (Ó Sé 2000: 17-19). If a transcription was being used for all dialects then divisions such as the following would be required (this is in fact those used for the comprehensive dialect study in Hickey 2011a).

Table 20. Polarity cline for *N*- and *L*-sounds

maximal palatality				maximal velarity
n^j, l^j		n_j, l_j	n_γ, l_γ	n^γ, l^γ
high polarity	—	low polarity	—	high polarity

Because the dialects have sonorants which show varying degrees of palatalisation and velarisation, the fourway system would be necessary to refer to individuals sounds.

The high polarity sonorants are indicated by a *superscript* diacritic and the low polarity are shown by a *subscript* diacritic as can be seen in Table 20. But no one dialect appears to have a robust form of the maximum fourway distinction.

The greatest degree of phonetic contrast is between the two extremes of palatality and velarity. The sounds in the centre of the continuum cannot show high phonetic contrast as they are much more similar phonetically than those at either end of this continuum. In articulatory terms the four distinctions on the polarity cline can be described as follows.

(54)

[n^j]	The body of the tongue is arched upwards towards the palate with dorsal contact; the tip of the tongue is behind the lower teeth (convex tongue configuration).
[$n^ɣ$]	The body of the tongue is arched downwards away from the palate; the tip of the tongue is behind the upper teeth (concave tongue configuration).
[n_j]	There is apico-alveolar contact with slight raising of the body of the tongue towards the palate.
[$n_ɣ$]	There is apico-alveolar contact with slight lowering of the body of the tongue away from the palate.

1.8.5. Polarised and non-polarised sonorants

In terms of polarity there are two types of sonorant in Irish: one with low polarity, i.e. without audible palatalisation or velarisation, and one with high polarity in which the palatalisation or velarisation is indeed audible. Type one sonorants are similar to those in English and do not trigger any effect on other segments. Type two, however, induces a change in a preceding vowel: the nucleus vowel is long in monosyllables and short in polysyllables.[49] Because of the nature of Irish morphology, a change in grammatical category can involve the addition of an inflection, rendering monosyllabic stems polysyllabic thus triggering a shortening of the stem vowel. Consider the following phonetic forms.

(55)

a.	*carr*	[kɑ:r]	'car'
b.	*carrannaí*	[karənˠi:]	'car.PL'
d.	*gearr*	[gʲɑ:r]	'cut.VERB'
e.	*gearradh*	[gʲarə]	'cut.VERBAL NOUN'

Vowel alternation in pre-sonorant position

Here there is an alternation of vowels, i.e. before the sonorant both [ɑː] and [a] are found.[50] This alternation can be seen even more clearly where the low vowel follows a palatal consonant. In such cases the short /a/ is realised as [æ] which is phonetically very distinct from the long back realisation of /a:/, i.e. [ɑ:].

(56)

a.	*geal* [gʲælɣ]	'bright'
b.	*geala* [gʲæ$l_{ɣ}$ə]	'bright.PL'
c.	*geall* [gʲɑ:lˠ]	'promise.VERB'

d. *gealladh* [gʲælˠə] ‘promise. VERBAL NOUN’

Vowel changes do not just affect low vowels: high vowels are equally altered. There are forms like *im* /iːm/ ‘butter.NOM’ and *ime* /ɪmə/ ‘butter.GEN’ with stem-vowel shortening in the inflected form. These alternations do not seem to correlate with any appreciable length difference among the sonorants in question.[51]

Variation in vowel alternation

The quality of vowels before length-inducing sonorants varies greatly across dialects of Irish[52] and it provides one of the easiest ways of distinguishing varieties of the language (see Table 66 in Hickey 2011a: 265). A word like *am* ‘time’ varies across all dialects: /am/ in the North, /ɑːm/ in the West and /aum/ in the South (as in Northern Scottish Gaelic, O’Rahilly 1932: 51).

The diphthongal forms in the South alternate with short monophthongs across grammatical categories, e.g. *am* ‘time.NOM’ [aum] ~ *ama* [amə] ‘time.GEN’. This applies to Ring in County Waterford (mid South), R. B. Breatnach (1947: 142), to Muskerry, Co. Cork (South West), Ó Cuív (1944: 121) and to the Corca Dhuibhne area (tip of the Dingle peninsula; north of the South West), Ó Sé (2000: 110).

In the North of the country (North Co. Mayo and Co. Donegal) the original short vowels are retained in monosyllabic forms, e.g. *aill* ‘cliff’ [alj], *geall* [gjalɣ] ‘promise’ (de Búrca 1958: 132) with occasional phonetic lengthening of low vowels (Mhac an Fhailigh 1968: 113). This means that, for Northern Irish, there is no robust alternation of long and short vowels before length-inducing sonorants between monosyllables and polysyllables.

In Western Irish there is differential output among the vowels which provide the input to the vowel alternation process. The vowel which shows this most is the short back /ʌ/-vowel, written <*o*> or <*u*>, depending on the historical source (see II.1.9.5 *Short vowels* for further discussion). For example, <*orr, oir*> is /au/ in South Connemara but /ʌ/ in North Galway, cf. *toirneach* ‘thunder’, [taurnjəx] versus [tʌrnjəx] (de Búrca 1958: 41). Another case in Western Irish is the singular form *tonn* ‘wave’ which occurs as both [tu:nɣ] and [tʌn$^{\gamma}$] (because the plural form is disyllabic the short vowel is always used, i.e. *tonnta* ‘waves’ is /tʌnɣtə/). The long vowel in monosyllables is most common in the Cois Fharraige area and is the only pronunciation recorded by de Bhaldraithe (1953a: 27) though in the earlier phonetic study he remarks on variation among speakers, especially in those of the eastern part of the Cois Fharraige area (de Bhaldraithe 1945: 122-123). In the recordings of *Samples of Spoken Irish*, there was a great deal of variation with words like *fonn* ‘wish, desire’ pronounced [fʌnɣ] or [fu:nɣ] not just by different, but also by the same speakers (Hickey 2011a: 268). Going on their point of articulation, the vowels of Western Irish can be ordered according to the likelihood of their appearing long in pre-sonorant position. Lengthening is categorical with low vowels, almost so with high front vowels and variable with mid-high back vowels.

(57) Decreasing likelihood of pre-sonorant vowel lengthening in Western Irish

(i)	low vowels	*crann* [krɑ:n$^{\gamma}$] ‘tree’
(ii)	high front vowels	*im* [i:m^{j}] ‘butter’
(iii)	mid-high back vowels	*tonn* [tʌn$^{\gamma}$] ~ [tu:n$^{\gamma}$] ‘wave’

In Northern Irish vowel lengthening does not occur before the historical geminate, now length-inducing sonorants (Hickey 2011a: 263-266). Hence words like *fonn, tonn* are pronounced with [ʌ].[53] Vowel lengthening does not appear with low vowels either, e.g. *gleann* 'valley' is [gjlʲanɣ] in the North but [gjlʲɑ:nɣ] in the West.

In Southern Irish the short back vowel /ʌ/, due to generalised diphthongisation before length-inducing sonorants, has /au/ before non-palatal sonorants and /ai/ before palatal ones, cf. *ann* [aunɣ] 'there' and *tinn* [tjainj] 'ill' respectively. When lengthening has produced a diphthong there is no variation, that is /au/ or /ai/ does not vary with /ʌ/ or /ɛ, ɪ/ with one and the same word among speakers. Hence a word like *gleann* is [gjlʲaʊnɣ] (Ó Sé 2000: 24).

For more details on phonetic realisations across the dialects, see section I.2.3 above.

Development of mid-vowels before length-inducing sonorants

Long mid-vowels are not found in monosyllables before sonorants as these have been diphthongised, i.e. /e:/ has become /ai/ and /o:/ /au/. This is responsible for the alternation between diphthongs and short vowels as in *moill* /mailj/ 'delay' and *moille* /mɛljə/ 'delay.GEN'. An alternation of /au/ with /ɛ/ is generally not found as those forms with the diphthong in the uninflected stem show /ai/ in the genitive, e.g. *poll* /paulɣ/ 'hole' and *poill* /pailj/ 'hole.GEN'.

Ambisyllabic and tautosyllabic sonorants

Words with former geminate sonorants can have plurals in the dialects which differ from the standard. For instance, *crann* 'tree' has the plural *crainnte* 'trees' in Connemara

Irish (de Bhaldraithe 1953: 21). The singular has a long vowel before <*nn*>, i.e. [krɑ:nɣ], but the plural also has: *crainnte* [kri:njtjə]. The plurals of words such as *peann* [pʲɑːnˠ] ‘pen’, i.e. *peanna* [pjanɣə] ‘pens’, seem to suggest that the long vowel shortens once it no longer occurs in a monosyllabic form (this is the case when the plural suffix [-ə] is added). If this is the case then how does one reconcile the pronunciations *peanna* [pʲanˠə] with a short and *crainnte* [kri:njtjə] with a long vowel? The answer would seem to lie in the syllabification of these and similar words.

Consider first that vowel lengthening is triggered in monosyllabic forms due to the presence of a former geminate sonorant in the coda, e.g. *crann* [krɑ:nɣ]. If this sonorant remains entirely in the coda then it still triggers lengthening, i.e. *crainnte* [kri:nj.tjə]. Where the sonorant is ambisyllabic, it is spread across the coda of the first syllable and the onset of the next. This happens when a vowel-initial suffix is added to a base but not if this suffix has an initial consonant. This means that the sonorant in *peanna* [pjanɣə] is ambisyllabic and hence no longer triggers vowel lengthening. The assumption here is that there are ambisyllabic segments and that the effects they have on their phonetic surroundings are different from tautosyllabic segments.

1.8.6. Historical development

The origin of the vowel alternation exemplified by forms like *am* /ɑ:m/ ‘time.NOM’, *ama* /amə/ ‘time.GEN’ lies in Old Irish (see Greene 1952: 218f., and de Búrca 1981: 263-264, for remarks on the historical development). It is assumed that Old Irish had a distinction in length among its consonants (Thurneysen 1946: 89-96) even though the writing of double consonants was irregular and thus not always a safe guide to regarding a particular sound as a geminate (Lewis and Pedersen 1962 [1937]: 148). The geminates arose mostly from the assimilation of consonant clusters to the most sonorous element, sometimes within the borders of a lexical stem and sometimes across the boundary of prefix plus stem, e.g. *proinn* 'meal' (cf. Latin *prandium*) or *ammus* Old Irish 'attempt' from an earlier *ad* + *mess* (Thurneysen 1946: 93-94). These developments indicate that a length distinction existed on a phonological level at the oldest stage of the language.

There is a further argument for sonorant length in Old Irish. A phenomenon existed at this stage of the language known as MacNeill's Law[54] named after Eoin MacNeill who first described it (MacNeill 1909, Hamp 1974) and which specifies that in an immediately post-tonic syllable a sonorant (*l, n, r*) is 'strengthened' to a double sonorant, i.e. a geminate, if the syllable opens with a non-geminate sonorant.

(58)

a. *Érenn* 'Ireland.GEN.SG', cf. *Ériu* 'Ireland.NOM'
b. *Domnall* [dovnɣəlɣlɣ], but: *Túathal* [tuaθəlɣ] (proper names)

This is a type of dissimilation which prohibits two non-geminate sonorants in the same (unstressed) syllable and is reminiscent of Grassmann's Law in Sanskrit and Classical Greek which blocks the occurrence of two aspirated stops in succession. The Irish situation can only have arisen if phonological length for sonorants was already an established feature of the language.

The question of syllable quantity

Syllable quantity in Old Irish showed greater variation than in the present-day language as consonant length was distinctive. The original distribution of phonemically long segments in Old Irish, going on the orthography, prohibited long vowels before long consonants in monosyllables. But in the course of Old Irish hypercharacterised syllables (Lass 1984: 256) would seem to have developed in which original short vowels were lengthened before geminates, judging by the right-slanting stroke, the length-mark (Thurneysen 1946: 20), which at a later stage was occasionally written on the vowel preceding geminate sonorants (O'Rahilly 1932: 51). Such pre-sonorant vowels were later regarded by teachers of Irish poetry as half-long (*meadhónach*, i.e. intermediate, O'Rahilly, *loc. cit.*), perhaps because of their alternation with short vowels on inflection.

This development in Irish can be compared with the lengthening of vowels before clusters of sonorant and homorganic stop in the late Old English period (Lass 1994: 259-260) which produced realisations like [mi:nd] from earlier [mɪnd] 'mind' or [bu:nd] from earlier [bʊnd] 'bound'. This situation is quantitatively similar to that in Old Irish because an original heavy syllable rhyme VCC became superheavy through the lengthening of the nucleus V:CC. A possible reason for such lengthening may be the anticipation of the long cluster of the coda during the articulation of the preceding vowel, something which is phonetically plausible but which led to a phonologically unbalanced system, at least for Old English which had an embryonic complementary distribution of short vowels with heavy codas on the one hand and long vowels with light codas on the other (as in present-day Swedish or Norwegian, for example).

The sonorant geminates which arose in Irish differed from non-sonorant geminates inasmuch as the latter simplified to single consonants when, in the course of the history of Irish, phonological length was lost among obstruents, cf. Old Irish *attach* (← **ad* + *tech*, Thurneysen 1946: 91), Modern Irish *atach* 'beseeching'. The difference lies in the trace of former gemination which these special sonorants left behind. In

Modern Irish they cause diphthongisation in the South and vowel lengthening in the West and North. The reflex of this situation in the lexicon of present-day Irish is that some lexical items with a word-final sonorant have lengthening of the preceding vowel and some do not have this. It is normal for Irish phonologists to remark on the orthographic correlation between the shift before sonorants and the occurrence of the graphemes *-ll, -nn, -m* and to note the optionality of shift before clusters of sonorant and stop such as *-nt* (de Bhaldraithe 1945: 109). The words written with these digraphs do indeed contain the reflexes of former geminates. For example, Old Irish *all, cland, lomm* (Thurneysen 1946: 93-96) all have reflexes in Modern Irish with a long vowel when monosyllabic: *aill* /ɑ:lj/ 'cliff' (with palatalisation of the original final /l/), *clann* /klɣɑ:nɣ/ 'family' (with assimilation of the stop to preceding sonorant), *lom* /lɣu:m/ 'bare' (with a single *m* in present-day Irish as vowel lengthening is taken to have occurred before all instances of final /m/).

Loss of phonological length

Scholarly opinion, e.g. O'Rahilly (1926: 183), assumes that phonological length in Irish[55] existed until at least the end of the Middle Irish period (900-1200) and survived longest in Northern dialects. Polarised segments, i.e. phonetically velarised or palatalised sounds, were geminates: a word like *tinn* 'sick' was originally [tjɪnjnj], later [tji:njnj], then [tji:nj]. The orthography of Modern Irish, which is essentially that of Early Modern Irish, indicates the former geminates by double sonorant graphemes.

The development of long vowels before former geminates only holds where the geminate sonorant was in the coda of a stressed syllable but not before a further syllable beginning with a vowel, hence words like Modern Irish *corraigh* ‘move’, *corrach* ‘bog, marsh’, *corrán* ‘hook, sickle’ do not show long vowel reflexes in the first syllable.

In the subsequent course of Irish, consonantal length was lost, this redressing the syllable quantity in words like *gann* ‘scarce’, leaving them with a long vowel and a short (or rather quantitatively non-distinctive) sonorant, i.e. [gɑ:nɣ]. Side by side with these forms others existed in which the sonorant was never long. Thus a word like *gan* ‘without’ always had a vowel followed by a short sonorant. The present-day difference is the following: *gann* [gɑ:nɣ] with a long low back vowel versus *gan* [ganɣ] (full form) ‘without’ with a short low central vowel. There are a number of such pairs, e.g. *tinn* /tji:nj/ ‘sick’ and *tine* /tjɪnjə/ ‘fire’. This type of alternation contrasts with the lack of vowel alternation with words which contain systemically long vowels which are long in all contexts, e.g. *bán* /bɑ:nɣ/ ‘white.SG’, *bána* /bɑ:nɣə/ ‘white.PL’.

The geminate ~ non-geminate contrast and later developments

In earlier Irish non-geminate palatal sonorants in post-stress position were realised without polarisation and the reflex of this in present-day Irish is a neutral pronunciation, i.e. without audible secondary articulation, cf. *tine* /tjɪnjə/ ‘fire’, *duine* /dɪnjə/ ‘person’. These neutral segments were confined to a given phonotactic position, after stressed vowels. However, there is a complication which results from the initial mutation, lenition. This would seem to have entailed the reduction of polarisation for sonorants, e.g. the use of [nj] for [nj] in a word like *neart* [njæɹt] ‘strength’ when it occurred in a leniting environment, e.g. after the masculine third person possessive pronoun *a*. If one assumes that, phonetically, lenition is a reduction in articulatory gesture, e.g. using a fricative for a stop as in *a* *cheann* [ə çɑ:nɣ] ‘his head’, then the neutralisation of a polarised sonorant on lenition is in keeping with this phonological directive.

Studies of Irish dialects postulate that, for strong monoglot speakers, de-polarisation on lenition still applies to palatal nasals and laterals, e.g. de Bhaldraithe (1945: 114) has /n’/ [nj] as the lenited form of /N’/ [nj] but not of /N/ [nɣ]. Equally, he has /l’/ [lj] as the lenited form of /L’/ [lj] but not of /L/ [lɣ]. This lenition of sonorants would then occur, for instance, in the change from present to past tense, a shift which requires lenition of the verb stem as in *coigil* /kɪgjɪlj/ ‘save, spare’, *choigil* /xɪgjɪlj/ ‘saved, spared’.

(59)

Present	**Past**
nigh [nʲɪ] ‘wash’	*nigh* [n$_j$i] ‘washed’
náirigh [nˠɑ:ɹʲɪ] ‘shame’	*náirigh* [nɣɑ:ɹʲɪ] ‘shamed’
líon [lʲiːnˠ] ‘fill’	*líon* [l$_j$iːnˠ] ‘filled’
láidrigh [lˠɑ:dʲɹʲɪ] ‘strengthen’	*láidrigh* [l$_ɣ$ɑ:dʲɹʲɪ] ‘strengthened’

Numbers of distinctions in the dialects

Donegal Irish would appear to have the largest number of distinctions among sonorants. Wagner (1979 [1959]: 16) has four symbols. For /l/ and /l’/ he remarks that these are the sounds which were written with a single *l* in Old Irish. By this he seems to mean that they were not geminated which would imply that the distinction between /L, L’/ and /l, l’/ was formerly one of length. But there are contradictions in

Wagner's treatment, for instance when he classifies (*loc. cit.*) /L/ and /L'/ (and also /N'/ and /R/) as double consonants and /LL, L'L'/ (and also /NN/, /N'N'/ and /RR/) as geminates. In a later study, M. Ó Múrchú (1969) also uses a fourfold division with the same symbols, /L, L', l, l'; N, N', n, n'/ and does not explain his uppercase symbols beyond maintaining that they are tense (Irish *teann*), a term which has existed for centuries in Irish scholarship (Hickey 2001a).

Mhac an Fhailigh (1968) is a description of the Irish of Erris (intermediate between Connemara and Donegal Irish) and has a fourfold distinction for sonorants. The author admits that it is very tenuous and not always upheld by younger speakers. He furthermore says that when he was conducting his research he was expecting the distinction and states that he found it where it is 'historically justified' (Mhac an Fhailigh 1968: 41). As with Wagner, the distinction in Erris between /l, n/ and /l', n'/ seems to derive from the symbols and not from phonetic reality. De Búrca (1958: 38-39) maintains that the distinction between (his) /L', N'/ on the one hand and /l', n'/ on the other is that the former has laminal contact but the latter apical contact. But de Búrca does not seem to recognise any difference in degree of polarisation between his two sets of consonants.

The uppercase symbols used for polarised nasals and laterals in traditional Irish dialects studies (see Appendix 1) are also found in Holmer, for example, to denote dental /t, d/, i.e. /T, D/ (Holmer 1962: 16). He refers to the correspondence between them and /N, L/ by which he probably means the velarisation which is found with all the segments. This is in keeping with traditional works on Irish where capitals indicate a polarised pronunciation, either fully velar or fully palatal.[56] Additional support for this is found with some scholars, such as Henry (1958), who also used /T, D/ to represent /t̪, d̪/ (with laminal contact immediately behind the teeth), this time as the Irish English equivalents of Standard English /θ/ and /ð/.

Describing ‘tense’ sonorants

The effect of geminate sonorants on preceding vowels survived the loss of phonological length in the Middle Irish period. These reflexes were then given the label ‘tense’ – Irish *teann* – which goes back at least to the Bardic Tracts (L. McKenna 1979 [1944]), a series of treatises for instructing professional writers in the grammar of Irish and which belong to the period from 1200 – 1600 (Classical Modern Irish, Ó Cuív 1965:141).

‘Tense’ sonorants cover both palatal and non-palatal segments and early analyses of these segments are found in Pedersen (1897) and Henebry (1898) and mentioned by even earlier authors like O’Donovan (see O’Donovan 1845: 31-36, for example). Such discussions are also to be found in the early twentieth century, e.g. with Sommerfelt (1922: 48+73). These scholars attempt a close description of the sounds’ articulation. Sommerfelt specifies that the sonorants, which he writes with capital letters to indicate ‘tenseness’, are produced by pressing the tongue tip or blade strongly against the upper teeth ridge. He states that in the case of his /N/ the ‘nasal resonance is very strong’, which could be interpreted as meaning that the velarisation of this and similar segments is clearly audible. Pedersen (1897: 21) describes the primary dental articulation of the non-palatal nasals and laterals, comparing them with the stops /d/ and /t/ ‘but combined in these two cases with subsidiary action of the back of the tongue’ (translation RH), i.e. combined with secondary velarisation. Equally accurate descriptions are to be found in Finck (1899: 72) where he describes /L/ as a ‘plosive formation between the tongue tip and the upper teeth in combination with a raising of the back of the tongue’ (translation RH). Henebry (1898: 5) specifies that for the same sound ‘the throat organs are held in the position of a *u* vowel’. In the case of the palatalised sonorants, written /L’/ and /N’/ by Pedersen and all Irish scholars since, the main feature would seem to be distinct palatal quality.

> L’ and N’ have a much stronger palatalisation, so strong that with a transition from these sounds to a (back-tongue) vowel one believes one hears a transitional sound *j*’ (Pedersen 1897: 21, translation RH).

A point to stress here is that these scholars do not speak of consonantal length but of clear secondary articulation, palatalisation or velarisation. All later scholars, from O’Rahilly (1932: 49) and de Bhaldraithe (1945: 37) to Ó Siadhail (1989: 9 + 92), avoid any phonetic discussion and simply refer to these sounds as ‘tense’. If ‘tenseness’ is to be interpreted as phonological length then the term is not accurate but if it implies audible secondary articulation then it is. One author who has made the necessary distinction between length and secondary articulation is D. Ó Baoill (1979), who offers both a good sketch of the origin of sonorants in Modern Irish as well as a survey of the distribution of the reflexes of Old Irish in the present-day dialects.

1.8.7. The position of the orthography

In attempting to retrace what sonorants in Modern Irish stem from original geminates one can use the orthography as a guide. Bearing its practices in mind one can reconstruct the system of sonorant geminates in Old Irish with reasonable certainty.

In Irish long vowels are indicated orthographically by a right-slanting accent over them. Phonetically long vowels which do not show an accent in the orthography are the result of 1) vowel lengthening before former geminate sonorants or 2) the vocalisation of voiced fricatives. When the vowels have resulted from the first process the following sonorant is doubled in writing, but only when the sonorant is coronal, i.e. *L, N* or *R*. The only velar sonorant in Modern Irish is [ŋ] and is the realisation of /n/ before /g/. A sonorant and a homorganic voiced stop induced lengthening, going on the realisation of monosyllables with such clusters in Modern (Western) Irish, cf. the long vowels in *long* [lɣu:ŋg] 'ship', *moing* [mi:ŋjgj] 'mane'. The lengthening would also appear to have occurred before a coronal sonorant and a voiced stop as the sequence /-rd/ in monosyllables is preceded by a diphthong in Western Irish today, e.g. *bord* [baʊrd] 'table.NOM', *boird* [baɪrd] 'table.GEN'.

Where a long vowel has resulted from pre-sonorant lengthening and where the sonorant is labial only a single grapheme is found in Modern Irish. While the final sequence -*mm* was permissible in Old Irish it is not in Modern Irish nor is the doubling of sonorant graphemes word-initially or the doubling of obstruent graphemes in any position.

The long vowels which arose through pre-sonorant lengthening are always one of three vowels: /i:/, /ɑ:/ or /u:/. The two mid-vowels of Modern Irish, /e:/ and /o:/, do not occur in the forms which exhibit pre-sonorant lengthening (in Western Irish). The orthography of Modern Irish suggests that many of the long vowels here derive from original mid vowels which through raising (above all in pre-nasal position) and lengthening led to either /i:/ or /u:/: *trom* /tru:m/ 'heavy', *tonn* /tu:nɣ/ 'wave'.

There are furthermore words which apparently had a geminate sonorant previously but where the word does not occur as a monosyllable and so never exhibits a phonemically long vowel before it, e.g. *bainne* /banjə/ 'milk' which is always disyllabic. Still other words show vowel alternation, but where one of the forms does not necessarily have to be monosyllabic. In such instances, the long vowel occurs when a previous geminate sonorant comes together – because of syncope – with a further (nongeminate) sonorant to yield a lengthening environment, e.g. *coinneal* /kɪnjəlɣ/ ‘candle.NOM’ and *coinnle* /ki:njljə/ ‘candle.GEN’ (with lengthening of /ɪ/ before /njlj/).

1.8.8. Possible phonological analyses

The effect of Old Irish geminate sonorants on the vowels which preceded them has led to a situation in Modern Irish which has drawn the interest of many scholars over the past few decades. The occurrence of long and short vowels before certain sonorants is the subject of a number of studies or sections of monographs of which the following is a small selection: Bloch-Rozmej (1998: 107-50), Carnie (2002), Cyran (1997: 109-116),[57] Ní Chiosáin (1991: 188-211), D. Ó Baoill (1979), Ó Siadhail and Wigger (1977: 89-93), Ó Siadhail (1989: 48-55).

The essential question for phonologists considering this situation is the following: how do native speakers of Irish know which sonorants in present-day Irish show long-short vowel alternation between monosyllables and polysyllables and which sonorants do not? Speakers know that long vowels before certain sonorants shorten when the words are no longer monosyllabic, e.g. after adding a grammatical inflection. But which sonorants have this effect is not derivable from the realisation of the sonorants as can be seen from the following instances.

(60)
vowel shortening on suffixation?

no   yes

bán [bɑːnɣ] ‘white’ *peann* [pʲɑːnˠ] ‘pen’

The fact that vowel shortening on suffixation cannot be predicted from the phonetic shape of sonorants has engendered many interpretations which can basically be divided into two types (i) abstract and (ii) lexicalist. The former favour a marking of the underlying forms of sonorants at an abstract level which will account for the observed surface effects. The latter view assumes that language learners store all words in their mental lexicon which have vowel-alternation before sonorants with something like a tag which indicates that they trigger this. Such tagging of lexical forms would be similar to that for nouns in languages with grammatical gender, including Irish itself (Hickey 2011b).

Since the 1960s the question of abstractness has been of considerable concern to phonologists. The basic issue behind abstract analyses is how well-motivated they are. In the heydey of generative phonology an analysis was motivated if it fulfilled the following criteria.

(61)
Evaluation metric for phonological analysis

1) simple, i.e. parsimonious in the assumption of categories
2) where possible, motivated by several phenomena

Sometimes there was a degree of synchronically unjustified reference to historical forms of a language, e.g. in treatments of the English long vowel shift which referred to the historical outset values in the underlying forms of present-day words showing this shift, see the discussions in Chomsky and Halle (1968: 187-190), in Lass (1976) and the detailed treatment in Wolfe (1972). A supposed advantage of appealing to historic conditions was that this rendered possible, in the synchronic phonology of a language, the linking together of morphonologically alternating forms which had superficially drifted apart due to independent phonetic developments. This position was transferred to Irish and lies behind analyses such as M. Ó Murchú (1969) and those in Ó Siadhail (1989).

1.8.9. Abstract phonological derivations

An abstract phonological analysis of the pre-sonorant vowel alternations might postulate the existence of underlying geminates (reflecting the diachrony of the language).[58] Such geminates would account for the phonemic lengthening of vowels in monosyllables. In disyllables the geminates would not have this effect. Words like *bán* /bɑːnˠ/ 'white', which always have a long vowel, would be assumed to contain non-geminate sonorants underlyingly as would a word like *glan* /glanˠ/ 'clean' which only has a short vowel in all forms.

(62)

a. derivation of *peann* 'pen'

/pʲann/		
/pʲɑːnn/	1)	pre-geminate vowel lengthening
/pʲɑːn/	2)	geminate simplification
[pʲɑːnˠ]		(surface form)

b. derivation of *peanna* 'pens'

/pʲannə/		
/pʲannə/	1)	pre-geminate vowel lengthening inapplicable due to disyllable
/pʲanə/	2)	geminate simplification
[pʲænˠə]		(surface form, front vowel after palatal)

It is important here not to confuse diachronic developments with the synchronic phonology of the language, assumed to be common to all native speakers of Irish, at least in three main forms for the chief dialect areas. While the above derivations recapitulate the diachronic development of pre-geminate sonorant vowels in Irish there is no evidence to support the assumption that this also represents native speakers' underlying knowledge of the present-day phonology. The main criticism of derivations such as those in (62) is that they postulate geminates which do not exist in Irish today. Placing them in synchronic derivations would be a case of absolute neutralisation, a postulation which does not have support among phonologists today.

Other kinds of derivation have been postulated in the literature on Irish sonorants. One is that underlyingly sonorants can carry a mora – a unit of phonological length – and that this can be transferred to another segment during a derivation. This is essentially the position offered in Ní Chiosáin (1991). She assumes that the vowels in monosyllables like *peann* are underlyingly short (Ní Chiosáin 1991: 189) and gain additional length, i.e. a mora, from the following sonorant, making the stem vowel long, hence the surface form [p^{j}ɑːn^{ɣ}]. However, this account does not explain how native speakers know which sonorants trigger vowel alternation, in Ní Chiosáin's terms what sonorants carry an extra mora which they can transfer to a preceding vowel in monosyllables.

In the case of vowel length-inducing sonorants there are exceptions to the lengthening environment specified. There is no shortening in cases where a consonant-initial suffix is added to a word.

(63)

a. *gleannta* /gjljɑ:nɣtə/ 'valleys' ← *gleann* + *ta*

b. *crinnte* /kjrji:njtjə/ ‘trees’ ← *crann* + palatalisation + *ta* (plural form in Connemara Irish)

Indeed there are cases where a singular nominative noun has a short vowel in the stem and a long one in the genitive when syncope leads to a consonant immediately following the sonorant.

(64)

a. *coinneal* /kɪnjəlɣ/ ‘candle.NOM’
b. *coinnle* /ki:njljə/ ‘candle.GEN’

An explanation for this might be that in *gleannta* the syllabification is /gjljɑ:nɣ.tə/, i.e. the sonorant belongs to the first syllable and hence still triggers vowel lengthening. This would also hold for *coinnle* /ki:nj.ljə/ with a long vowel but not for *coinneal* /kɪ.njəlɣ/ where the medial /nj/ can be seen as forming the onset of the following syllable (see Ní Chiosáin 1991: 188 for a similar interpretation of the phonetic facts).

The derivation of *peann* in (62) above assumes that schwa suffixation is grammatical and applies prior to the phonological steps of the derivation. If the order were reversed, suffixation would apply to /pʲɑːnˠ/, yielding the unacceptable form *[pjɑ:nɣə]. It is furthermore necessary to distinguish here between inflectional and word-formational morphology. Pre-sonorant vowel shortening only applies to the former type where the inflectional endings are more closely bound to their lexical stems. Where two lexical stems come together, i.e. in compounding, pre-sonorant vowel shortening is not triggered, but it is in the case of a stem plus suffix, consider the following forms.

(65)

Vowel shortening with suffixation

a.	*ceann*	/kʲɑːnˠ/	'head'
b.	*ceannas* (stem + suffix)	/kʲanˠəs/	'authority'
c.	*ceannasach* (stem + adjectival inflection)	/kʲanˠəsəx/	'sovereign' (adj.)

No vowel shortening with compounding

d.	*ceannchathair* ('head' + 'city')	/kʲɑːnˠxahərʲ/	'capital city'
e.	*ceanníseal* ('head' + 'low')	/kʲɑːnˠiːsʲəlˠ/	'dejected'

Returning to the derivation of *peann* 'pen' : *peanna* 'pens' discussed above, one can list the arguments for and against it. In favour of underlying geminates is the fact that they provide a mechanism for predicting phonetic vowel lengthening in monosyllables. However, as derivations are postulates, one must in turn find arguments for them from other areas. The fact that this analysis uses entities (geminate sonorants) which once existed in the language is not a valid argument in synchronic derivations. Equally, the fact that the orthography uses double sonorant graphemes is irrelevant; the orthography itself is not even consistent, cf. the lack of final *-mm* in Modern Irish. Furthermore, the first and crucial stage in the above derivations, pre-geminate vowel lengthening, is phonetically unusual. On the contrary, the hypercharacterised syllables, which would result on the surface, are prohibited in those languages (such as Swedish, see Árnason 1980, in particular pp. 70-76) which have surface geminates and strict quantity rules for monosyllables (i.e. either VCC or VVC, but not *VVCC). Above all, one can doubt the hypothesis that underlying geminates exist as they do not occur on the surface elsewhere in Western Irish, i.e. they would be a case of absolute neutralisation, as mentioned above.

The abstract view would see the cause of pre-sonorant vowel lengthening within the phonological derivation of surface forms. By implication the forms stored in the mental lexicon of speakers contain geminates. But in a language which does not have phonetic geminates postulating them underlyingly is simply to use the term ‘geminate’ metaphorically, i.e. it is a way of saying that speakers have marked the forms with vowel-lengthening sonorants in some way so as to always produce the correct forms phonetically. Underlying geminates thus remain a theoretical construct to explain the effect of a certain subset of sonorants in the phonetics of Modern Irish.

Not only do many abstract analyses appeal to the diachrony of a language, but they often use evidence from other dialects to support an analysis in a particular dialect. The classical study which employs this procedure is Newton (1972) which attempted to link up the dialects of Greek on an abstract underlying level. But there are difficulties in using evidence from one dialect in the analysis of another. For instance, in her analysis of sonorants in Western Irish Ní Chiosáin (1991: 204-206) uses the situation in Northern Irish for support. There geminate sonorants preceded by short vowels are claimed to occur (Sommerfelt 1922, Wagner 1979 [1959], D. Ó Baoill 1979) in the same environments as non-geminate sonorants in Western Irish.

(66)

gleann	Western Irish	Northern Irish (putatively)
‘valley’	/glʲɑːnˠ/	/glʲanˠnˠ/

However, the dialects are no longer contiguous and young language learners in the West do not normally have any significant exposure to Northern Irish, so that they would not hear the (postulated) geminates of the North and hence could not analyse their own non-geminates as underlyingly geminates.

1.8.10. Lexical tagging

An alternative to an abstract phonological derivation is provided by a lexicalist analysis of vowel-alternation before sonorants. The lexicalist position does not posit any underlying phonological segments (here: geminate sonorants) which do not appear on the surface, i.e. it does not involve absolute neutralisation. Instead it assumes that words in the mental lexicon can show tagging for non-derivable phonological patterning. In the present context this would mean that certain sonorants in Modern Irish are marked as having a lengthening effect on the vowels before them in monosyllables while other sonorants would lack such marking. Those sonorants with marking would incidentally be equivalent to those which were geminates in Old Irish, this fact, however, is irrelevant for synchronic analysis.

In the present discussion this marking is represented by the notation [+long] ← [+son]. This simply formulates the fact that a certain sonorant causes the preceding vowel to be lengthened phonetically when occurring in a stressed monosyllable. Length induction by sonorants refers to vowels only. However, this does not need to be specified as obstruents do not occur before sonorants in word-final position in Modern Irish. If the environment for lengthening is a monosyllable, then the lack of lengthening in polysyllables does not have to be specified.[59]

Only length-inducing sonorants are tagged. The default case is for there to be none, hence words which always have long vowels, e.g. *fán* /fɑːnˠ/ 'wandering', *fánaí* /fɑːnˠiː/

‘wanderer’ and words which always have short vowels, e.g. *fan* ‘wait.IMPERATIVE’, *fanacht* ‘waiting’, carry no tagging.

The implied consequence of the notation [+long] ← [+son] for language acquisition is that when speakers of Irish acquire certain word forms they note in their mental lexicon that the final sonorant has a lengthening effect on the preceding vowel (or not as the case may be). This view of lexical storage gains support from a number of areas in Irish, for instance with nouns there is great variation with case inflection and plural formation (Carnie 2008) so that tagging is required to flag nouns as belonging to one of a large number of types. As has been pointed out above similar tagging is also required for gender assignment in Irish.

In the evaluation metric given in (61) above it is specified that an analysis should be motivated by encompassing several phenomena in a language. When one examines the phonology of Irish further one finds that lexical tagging provides an account for the behaviour of other segments which show non-derivable phonological effects.

1.8.11. Vowel tagging and assimilation

In Modern Irish sonorants show progressive assimilation towards the vowels which follow them in terms of the value for [palatal] which the latter show. This type of assimilation can be seen when a palatal consonant is followed (across a word boundary) by a low or back vowel, or when a non-palatal consonant is followed by a high vowel.

(67)

a.	*in* /in_j/ ‘in’	*in aghaidh na gaoithe* /ɪn_j ai/ [ɪnˠai] ‘in the face of the wind’

b. *aon* /e:n^{ɣ}/ ‘one’[60] — *aon teach amháin* /e:n^{ɣ} t^{j}ax/ [e:n^{j}t^{j}æ:x] ‘one house only’

The final sonorant assimilates to the following segment (consonant or vowel). This assimilation may of course be vacuous if the sonorant and the following segment agree in their value for palatality, e.g. *aon dún amháin* /e:nɣ du:nɣ.../ [e:nɣdu:nɣ...] ‘one castle only’ where both the final consonant of the first word and the first of the second are non-palatal.

There are, however, a large number of cases in which unexpected assimilations take place. The form of the article in Irish is *an* /ənɣ/; it ends in a non-palatal consonant but can, because of assimilation, be palatalised before a word starting with a front vowel: *an íobairt* [ənj i:bərtj] ‘the sacrifice’. In this context consider two further words: /asərj/ ‘litter’ and /ʌmərkə/ ‘excess’. According to the assimilation rule just described there should be no change in the articulation of the final /nɣ/ of the article, the assimilation being vacuous. This is not the case, however, as the /n/ palatalises.

(68)

a.	*an easair* [ən^{j} asər^{j}]	‘the litter’
b.	*an iomarca* [ən^{j} ʌmərkə]	‘excess, large amount’

Here the orthography provides the clue to what has happened in the development of Irish. Originally, there was a front vowel with a back offglide to the following non-palatal consonant. But in the course of the development of Irish[61] the accent shifted to this off-glide syllabifying it with the consequent loss of the original stressed vowel, see Greene (1976: 39-41) for details of the changes involved. The development of the forms just given (neglecting unnecessary phonetic detail) was as follows: /eas ɪrʲ/ ➜ /asɪrj/, /iomərkə/ ➜ /omərkə/ ➜ /ʌmərkə/. This change is similar to the so-called ‘Akzentumsprung’ of Old English (Lass 1988) which is responsible for the shift of *lēosan* with /e:o/ to Modern English *lose* with /u:/ (from an earlier /o:/).

Although there was a change from front to back with the stressed vowel of the Irish words the palatal assimilation of preceding consonants caused by the original front vowel was retained. The shift was widespread in Irish so that many words differ phonologically solely in the assimilatory effect they have on a preceding consonant.

(69)

a.	*iúl* /u:lˠ/ (with palatal assimilation)	‘knowledge’
b.	*úll* /u:lˠ/ (with non-palatal assimilation)	‘apple’
c.	*iúr* /u:r/ (with palatal assimilation)	‘yew’
d.	*úr* /u:r/ (with non-palatal assimilation)	‘fresh’

The question here is: how do native speakers of Irish know which vowel-initial words cause assimilation contrary to that which one would expect from the initial vowel? One could postulate that children learning Irish store, with the form of certain (vowel-initial) words, the additional information that the assimilation caused by this vowel is contrary to that expected from the normal assimilation rule of Irish, e.g. *an iúr* [ənj u:r] 'the yew'. The marking of a vowel which causes unexpected assimilation could be indicated by a notation like that used for length-inducing sonorants:

(70)
[+pal] ← /u:lɣ/
gan iúl /gənɣ u:lɣ/ [gənj u:lɣ] ‘without notice, heedless’

An example of a word which has an initial front vowel and yet causes non-palatal assimilation (of the prefixed *t* accompanying the definite article) would be *uisce* ‘water’; this could be indicated similarly:

(71)
[-pal] ← /ɪsjkjə/
an t-uisce /ənɣ tɪsjkjə/ [ənɣ tɪʃkjə] ‘the water’

The phenomenon of lexicalised palatalisation/velarisation is incidentally also found in the morphology of Irish.[62] There are certain inflectional suffixes in Irish which demand palatalisation or velarisation of the final consonant of the stem to which they are appended although the vowel the suffix begins with has opposite polarity.

(72)

a.	*uasal* /uəsəl^{ɣ}/	‘noble’
b.	*uaisleacht* /uɪs^{j}l^{j}əxt/	‘nobility’
c.	*duiniúil* /dɪn$_{j}$u:l^{j}/	‘human, natural’
d.	*duiniúlacht* /dɪn$_{j}$u:l^{ɣ}əxt/	‘humanity, kindliness’

Because the stem-final change for [palatal] on /əxt/-suffixation is not predictable from the phonetic environment, it would be necessary to specify which version of the suffix produces which type of effect on the final stem coda as follows (i) [+pal] ← /əxt/, (ii) [-pal] ← /əxt/.

1.8.12. Parallels in other languages

The examination of Irish dialects shows that there is a three-way distinction among nasals and laterals: (i) palatal, (ii) alveolar and (iii) velarised (at least for varieties of Western and Northern Irish). There are other languages in which similar three-way distinctions are to be found, consider the following two examples.

Romanian

Distinctions similar to those in Irish were discussed in Petrovici (1959) for Romanian. He remarks (Petrovici 1959: 185) that three-way distinctions in Slavic were of short duration but that for Romanian these have existed for a long time. These observations were supported by Halle and Jakobson (1959) in their supplementary note to Petrovici's article. The segments he recognises are: palatal, palatalised and non-palatalised, in his words: 'palatal, diesés et non-diesés' (lit. 'palatal, sharp and non-sharp' – RH). A number of questions arise in this connection: are his non-palatals phonetically velarised? If this were the case it would increase their acoustic separation from neutral or palatal segments. Petrovici is also unclear about whether the palatalised segments are neutral, corresponding in articulation to the alveolar segments as in Irish *baile* [baːljə] 'town'. Another issue which arises is what the functional load of the distinction between palatalised and palatal is for the Romanian which Petrovici was describing. The answers to these questions are not forthcoming in the short article by Petrovici but the very discussion does support the contention that sonorants can carry such a three-way distinction.

Polish sibilants

A parallel to three-way distinctions among the sonorants *l* and *n* in Irish is provided by Polish sibilants and affricates (Stone 1990: 83; Gussmann 2007: 3-7). The language realises the differences between the different types by means of tongue body height much as in Irish.

(73)

a. *ci, si, zi* /tɕ, ɕ, ʑ/
b. *c, s, z* /ts, s, z/
c. *cz, sz, rz* /tʃ, ʃ, ʒ/

(Padgett and Zygis 2003: 157, Stone 1990: 86-92)

The first series consists of true palatal fricatives, voiced and voiceless and, in combination with stops, palatal affricates. The second series is alveolar and the sounds are much as in English or German, for example. The third series may appear to show the fricatives found in English *ship*, *chip* and *rouge*, respectively. But phonetically they are velarised with the body of the tongue arched downwards to give a hollow secondary articulation which contrasts clearly with the palatal articulation of the first series, much as the non-palatal sonorants of Irish, [nɣ] and [lɣ], contrast in their secondary articulation with the palatal sonorants [nʲ] and [lj].

1.9. The vowel system

Height levels in vowel space

There are three distinct levels of height for vowels in Irish. [63] Differences in the phonetic quality of certain vowels, e.g. [iː] versus [ɪ], can be adequately accounted for in terms of length and tenseness.

(74)

	Front		Mid	Back	
Level 1	iː	ɪ			uː
Level 2	eː	ɛ	ə	ʌ	oː
Level 3			a		ɑː

Long vowels	*bí* /bʲiː/ ‘be’, *cé* /kʲeː/ ‘although’, *tá* /tʲɑː/ ‘is’, *bó* /boː/ ‘cow’, *tú* /tuː/ ‘you.SG’[64]
Short vowels	
(i) stressed:	*ar bith* /bʲɪ/ ‘at all’, *te* /tʲɛ/ ‘hot’, *cath* /ka/ ‘battle’, *scoth* /skʌ/ ‘choice (adj.)’, *luch* /lˠʌx/ ‘mouse’
(ii) unstressed:	*domhanda* /daunˠdə/ ‘global’, *arís* /əˈrʲiːsʲ/ ‘again’

Interrelationship between short mid vowels

The vowel inventory of Irish consists of four short vowels, five long vowels and four diphthongs. The vowels are distributed fairly evenly across phonological space and the diphthongs are regular consisting of two rising and two falling diphthongs. The vowel length distinctions are phonological, a difference between Irish and Slavic languages which show a systemic palatal/non-palatal distinction but not a vowel length distinction.[65] Among the Indo-European languages, Lithuanian is closest in this respect: systemically, it has both a vowel length and palatal/non-palatal distinction.

Going on the above division of vowel space in (74), the short vowels /ɛ/ and /ʌ/ are located on Level 2 and differ only in relative frontness. Words with alternative realisations involving these vowels can be seen to manifest either progressive or regressive assimilation to the polarity of a flanking consonant.

(75)

	V ➔ +P	-P ← V	
coileach	/kɛlʲəx/	/kʌlʲəx/	‘cock’

This front to back switch, or vice versa, can only apply to vowels which are systemically short and on the same height level. [66] As Level 2 is the only one with more than one short vowel, the possibility of a switch from front to back does not apply to either Level 1 or Level 3.

Long vowels do not show this influence of the polarity of flanking consonants to anything like the same extent. Furthermore, it is much more common to find a following consonant influencing a preceding syllabic vowel. Hence on the palatalisation or de-palatalisation of final consonants, the preceding vowel within the rhyme assimilates in polarity to the consonant(s) in the coda.

(76)

a.	*glas* /glˠas/	‘green’
b.	*níos glaise* /nʲiːs glˠɛsʲə/	‘greener’
c.	*troid* /trɛdʲ/	‘fight.NOM’

d. *troda* /trʌdə/ 'fight.GEN'

The assignment of vowels to certain levels is justified by alternations which are found throughout Irish. For instance, a change in word class can often involve a change in vowel. The pair below shows that /ɛ/ and /o:/ are short and long mid vowels respectively which furthermore differ in front versus back. Variation in forms of the future for the verb 'get' shows that /ʌ/ can be regarded as the short counterpart of /oː/ (not before nasals).

(77)

a. *te* /tʲɛ/ 'hot'

b. *teocht* /tʲo:xt/ 'heat'

Loans from English (de Bhaldraithe 1953b) also provide evidence for regarding /ɛ/ and /ʌ/ as mid vowels (Level 2 above) which differ solely in front versus back position. In the case of the following two words the coronal fricative at the end of the word determined the nature of the vowel used in Irish on borrowing. From the Irish vantage point, English [ʃ] is regarded as a palatal fricative and English [s] as non-palatal.

(78)

a. Front mid short vowel before a palatal coda /ɛ/ before [ʃ] English *brush* > Irish *bresh* [brɛʃ]
b. Back mid short vowel before a non-palatal coda /ʌ/ before [s] English *stress* > Irish *struss* [strʌs]

Mid-vowel diphthongisation

Apart from the relationship between front and back mid vowels just discussed, variation between a long mid front /e:/ and the front rising diphthong /ai/ is also found. For instance, de Bhaldraithe (1945: 11) remarks on 'by-forms' in [aɪ] for words which normally have [eː].[67]

(79)

a.	*droichead*	[dre:d] ~ [draɪd]	'bridge'
b.	*soitheach*	[se:x] ~ [saɪx]	'ship'
c.	*meitheal*	[m^{j}e:l$^{\gamma}$] ~ [m^{j}aɪl$^{\gamma}$]	'party, group'

Lexical sets for Irish vowels

The distinction between long and short vowels can be demonstrated clearly by a number of keywords which illustrate the vowels in typical contexts. The following table shows the lexical sets of Irish vowels devised by the author and first presented in Hickey (2011a: 44-69).

Table 21. Vowels of Irish

Short vowels				
	/ɪ/	FIOS 'knowledge'		BUIDÉAL 'bottle'
	/ɛ/	TE 'hot'		
	/ʌ/	SIOPA 'shop'		TURAS 'journey'
	/a/	TEACH 'house'		SLACHT 'polish'
Long vowels				
	/i:/	LÍON 'fill'		BAOL 'danger'
	/e:/	ÉAN 'bird'	/a:/	ÁIT 'place'
	/o:/	ÓL 'drinking'	/u:/	GÚNA 'dress'
Diphthongs				
	/ai/	LAGHAD 'least'	/au/	LEABHAR 'book'
	/iə/	BIA 'food'	/uə/	CRUA 'hard'

In the following the vowels are arranged in twos each with a short and a long member illustrated by words which form minimal pairs.

Table 22. Vowel contrasts

/a/ and /ɑ:/	
tairg [tarʲɪkʲ] 'offer, attempt'	*táirg* [tɑ:rʲɪkʲ] 'produce'
airigh [arʲɪ] 'perceive, hear'	*áirigh* [ɑ:rʲɪ] 'count, reckon'
cainteach [kanʲtʲəx] 'talkative'	*cáinteach* [kɑ:nʲtʲəx] 'critical'
ciste [kɪʃtʲə] 'fund, chest'	*císte* [ki:ʃtʲə] 'cake'
/ɪ/ and /i:/	
frith- [fʲrʲɪ] 'counter-, anti-'	*fríth* [fʲrʲi:] 'find'
/ɛ/ and /e:/	
te [tʲɛ] 'hot'	*(an) té* [tʲe:] 'he who'
/ʌ/ and /o:/	
solas [sʌlˠəs] 'light'	*sólas* [so:lˠəs] 'refreshments (food)'
ros [rʌs] 'headland'	*rós* [ro:s] 'rose'
/ʌ/ and /u:/	
tur [tʌr] 'dull, dim'	*túr* [tu:r] 'tower'
/o:/ ~ /ʌ/ variation	
seomra [ˈʃo:mrə] ~ [ˈʃʌmrə] 'room'	

gheofar [jo:fər] ~ [jʌfər] 'will get'

Notes

(i) The distinction between phonemically long and short low vowels is one of place: the long vowels are further back. The only exception to this is Northern Irish where there is frequently a movement of back vowels to a front position. The short vowels are furthermore sensitive to the polarity of the preceding consonant: after palatals /a/ is realised as [æ], after non-palatals it is [a] (both realisations with possible lengthening, especially in the Cois Fharraige area of the Galway Gaeltacht).

(ii) In Western and Southern Irish the back vowels /uː/ and /o:/ have one short equivalent which can be realised as [ʊ] or [ʌ] depending on the following segment: the high back vowel occurring before velars, e.g. *rug sé* [rʊg ʃe:] 'he was born'. In some words, such as those quoted above, there is variation between a long and a short vowel. The long vowel /o:/ varies with /ʌ/ which confirms that the latter is the short equivalent of /o:/; /ʌ/ can also vary with /ɛ/ on the back-front axis.

1.9.1. Long vowels

The vowels of Irish can be characterised via a combination of three parameters (1) height, (2) frontness / backness and (3) length. There is a front/back distinction for both long and short vowels and there are three height levels, low, high and mid, as discussed above. In the mid area there is one systemic level although vowels may vary somewhat in their relative height here. For instance, vowels before nasals show raising. In Western Irish this raising can lead to a merger with systemic high vowels. The realisation of mid back vowels can vary, e.g. through vowel lowering in Northern Irish (see following section). For Western Irish one can postulate five long vowels as follows:

(80)

1:	[i:] ——————	[u:]
2:	[e:] ————	[o:]
3:		[ɑ:]

/i:/	*íoc* /i:k/ ‘pay’	/u:/	*úll* /u:lˠ/ ‘apple’
/e:/	*éan* /e:nˠ/ ‘bird’	/o:/	*ceo* /kʲo:/ ‘fog’
		/ɑ:/	*dán* /dɑ:nˠ/ ‘poem’

Frontness or backness is generally sufficient to predict roundness: there are neither front rounded nor back unrounded vowels, bar the low back unrounded [ɑ:]. Phonetically, there may be some exceptions to this, e.g. with the retracted [ɯː], for the <AO> vowel found in the North of the Northern Irish area (see section II.1.9.9.2 below) or with the high central [ʉ:] found throughout Northern Irish for /u:/, cf. *dún* [dʉnɣ] ‘castle’.

1.9.1.1. Variation with long vowels

The long vowel system of Irish is stable, but nonetheless there is variation across the dialects which is important in delimiting them. In Northern Irish there is a tendency for vowels in the back area to move forward. This can be seen with the fronting of /u:/ to [ʉ:] just mentioned, but it is also noticeable with two further instances of fronting. The first is a shift forward of long /ɑ:/. This is normally to a low front vowel, [æ:], or slightly raised to [ɛ:], cf. *dá* [ɣɑ:] > [jæ:] > [jɛ:] ‘two’. With some female speakers the raising is considerable, i.e. to [e:] (see examples in Hickey 2011a: 375).

A further instance of unconditioned vowel shifting is the lowering of mid back /oː/ to [ɔ:] in Northern Irish, cf. *pósta* /po:stə/ [pɔ:stə] ‘married’. Together with the fronting of /uː/ this has led to a migration away from the high back vowel area in Northern Irish, something which is in stark contrast with the vowel realisations in Western and Southern Irish.

North-Mayo Irish is the result of an historical merger of North Connaught Irish with Northern Irish through immigrants from Ulster who settled in the region in previous centuries (Stockman 1974: ii and 351). Along with established features of Northern Irish, such as affricates as the realisations of /tj/ and /dj/, it shows features of its own which help to delimit it from other varieties. One salient feature is the retraction of /eː/ to /ɒ:/, seen in a word like *déanta* [ˈdʒɒ:ntə] ‘done’.

Variation in the realisation of /eː/ is found in Southern Irish, especially in the Corca Dhuibhne Gaeltacht (Dingle Peninsula). The long mid vowel is raised to /i:/ with a perceptual offglide to a following non-palatal consonant. [68] This raising is found in Western Irish in lexicalised instances, e.g. in the word *féasóg* [fji:so:gj] ‘beard’ (de Bhaldraithe 1945: 9).

In the South-West Cork Gaeltacht (the area of Muskerry, Ó Cuív 1944), a fronting and raising of long /ɑː/ can be found which is similar to that in general Northern Irish. The raising is just to [ɛː] as opposed to the North where speakers often show an [e:] vowel for <*á*>.

Especially in Western Irish, mid or high front vowels may diphthongise to /aɪ/. This is seen with the verb ‘come’ where *téann (sé)* ‘(he) comes’ is pronounced [tjaɪnɣ]. Another instance is the verb ‘to lie’ where the long /i:/ has shifted to /aɪ/: *luí* [lɣaɪ].

1.9.2. Relative status of vowels

Whether vowels or consonants should be regarded as more central in phonological analyses is an issue in all languages with a palatal/non-palatal distinction among its consonants. The situation in Irish is comparable to that in Russian or indeed to that of certain Caucasian languages which may, phonologically considered, be regarded as having only a pair of vowels, or indeed none[69] if all vowel occurrences are entirely predictable from the consonantal environment.

In Irish morphology consonants normally alter their quality with a change in grammatical category. An example would be palatalisation in the genitive of masculine nouns where the change in the consonant is the phonetic signal for this category. The change in quality with a preceding short vowel is of no grammatical relevance.

(81)

a.	*blas* /blˠas/	[blˠas]	‘taste.NOM’
b.	*blais* /blˠasʲ/	[blˠɛʃ]	‘taste.GEN’

Palatalisation or de-palatalisation of consonants are exponents of grammatical categories and as such part of the language’s morphology. However, from an acoustic point of view, vowel quality may be equally, if not more salient than the value for palatality. This applies in particular to final consonants in unstressed syllables where the acoustic difference between [ə] and [ɪ] is at least as salient as that between the non-palatal and palatal quality of the final consonant between the singular and plural in the following instances.

(82)

a. *capall* /kapəlɣ/ [kapəlɣ] ‘horse’
b. *capaill* /kapəlj/ [kapɪlj] ‘horses’

1.9.3. Weighting of consonants and vowels

Some authors, such as Fife (1993: 15), would seem to interpret vowel changes as carrying grammatical weight. Examples like *fear* /fjar/ 'man' : *fir* /fjɪrj/ 'men' are cited to support this stance. However, such vowel changes are attendant on the change for [palatal] in the syllable coda. The question is what arguments there are for assigning grammatical significance to the vowel rather than the consonant change? Consider the following forms in this light.

(83)

a. *beag* /bʲʌg/ 'small'
b. *teach an fhir bhig* /tjax ənɣ ɪrj vjɪgj/ 'the small man's house'
c. *an bheoir* /ənɣ vjo:rj/ 'the beer'
d. *blas na beorach* /blɣas nɣə bjo:rəx/ 'the taste of the beer'
e. *éan* /e:nɣ/ 'bird'
f. *éin mhara* /e:nj varə/ 'birds of the sea'

The adjective *beag* /bjʌg/ shows an alternation of /ʌ/ and /ɪ/ between nominative and genitive. The remaining forms show no such alternation, the vowel remaining largely constant across the change in grammatical case. This situation would suggest that the vowel quality changes are only found with short vowels. Assigning grammatical significance to vowel changes would then imply that with phonologically long vowels no exponents of grammatical categories like the genitive are to be found. The alternative interpretation which would see the change in consonant quality in the coda of stressed syllables as primary and changes with short vowels as secondary would lead to a consistent exponence for grammatical categories, irrespective of vowel length.

Another argument in favour of the primacy of consonant change can be seen in epenthesis. Here a vowel is inserted between two segments in the coda of a stressed syllable (see discussion in II.2.8 below).

(84)

a. *tarbh* /tarəv/ [tarəv] 'bull.NOM'

b. *tairbh* /tɛrʲəvʲ/ [tɛrʲɪvʲ] 'bull.GEN'[70]

The epenthetic vowel adopts the polarity of the consonant cluster it is inserted into. A palatal cluster triggers [ɪ]-insertion while a non-palatal causes [ə]-insertion. As it is the consonants which determine vowel quality, this is an additional argument in favour of consonants over vowels.

1.9.4. Vowel gradation

In this section the effect of a polarity change with syllable coda consonants on the preceding nucleus vowel is examined.[71] The initial consonant or consonants are not affected by polarity reversal in the coda of the syllable of which they form the onset. The effect which polarity reversal in a coda can have on a preceding nucleus vowel is labelled 'vowel gradation' in the current discussion.[72] This term has been deliberately chosen here rather than say 'umlaut' (used by Wigger 1973: 70 and, to a lesser extent, by Ó Cuív 1972) for two reasons. Firstly, 'umlaut' might wrongly suggest that the changes are due to high vowels or /j/ in a following syllable as in the history of Germanic (Cercignani 1980). Secondly, and more importantly, the term 'vowel gradation' allows one to subsume changes in a nucleus vowel from either palatalisation or velarisation in the consonants of a syllable coda, i.e. gradation refers to changes in a syllable nucleus where the coda consonants have undergone polarity reversal.

(85)

	vowel gradation	polarity reversal	
syllable onset	syllable nucleus	syllable coda	
/fʲ/	/a/	/r/	*fear* 'man.NOM'
/fʲ/	/i/	/rʲ/	*fir* 'man.GEN'
/r/	/ʌ/	/k/	*roc* 'wrinkle.NOM'
/r/	/ɛ/	/kʲ/	*roic* 'wrinkle.GEN'

The manner and amount of gradation in Irish varies across the dialects and here the situation in colloquial Western Irish is taken as the basis for the discussion. But even within what seems to be a single variety there is variation in speakers' usage. Especially the gradation of /a/ or /ʌ/ on palatalisation of the coda is subject to variation. While some speakers have /a/ ➜ /ɛ/ and /ʌ/ ➜ /ɛ/ as in *blas* /blɣas/ 'taste.NOM' ~ *blais* /blɣɛsj/ 'taste.GEN' and *stoc* /stʌk/ 'stock.NOM' ~ *stoic* /stɛkj/ 'stock.GEN' respectively, others retain the vowel of the citation form or only raise/front it slightly, i.e. they phonetically have pronunciations like [blɣæʃ] and [stʌ́kʲ] respectively. The clearest cases are those where the gradation is recognised orthographically and with these it is found in all dialects, e.g. *fios* /fjɪs/ 'knowledge.NOM' ~ *feasa* /fjasə/ 'knowledge.GEN'.

Because of the nature of inflectional morphology in Irish, nouns which experience vowel gradation may also have suffixation, e.g. the example just given (*fios* ~ *feasa*) or *sprioc* /sjpjrjʌk/ 'goal.NOM' and *sprice* /sjpjrjɪkjə/ 'goal.GEN'.[73] Gradation and suffixation have various possibilities of realisation and the combinations of these over the categories of case and number result in the five traditional nominal declensions. These have been the subject of much comment by linguists. Ó Cuív (1956: 86 +

114) comments on the lack of justification for grouping nouns in this way and discusses both Sjoestedt-Jonval's (1938) and Finck's (1899) different systems. He mentions the ten groups (in fact there are only nine) which Finck (1899: 160-185) derives from Zeuss (1871) and other grammarians of the nineteenth century whose classes are ultimately based on the continuation of inherited Indo-European classes with certain stems, however vaguely these may still be reflected in Irish.

For Old Irish Thurneysen has 12 classes (6 vocalic and 6 consonantal) and a group of irregular and indeclinable nouns (Thurneysen 1946: 154-216) which are even more directly based on inherited features. Dottin (1913: 45-71) reduced this number to five for Middle Irish but they are not the same as for Modern Irish. The latter are found in modern works, for example, in the *Caighdeán Oifigiúil* 'Official Standard' (Government of Ireland 1958: 5-24), the Christian Brothers' grammar (Christian Brothers 1960: 57-80) and in de Bhaldraithe's morphology of Cois Fharraige Irish (de Bhaldraithe 1953a: 16-41). By far the most vigorous reformulation of the declensional system is to be found in Carnie (2008: 19-39) and Wigger (1970: 14-45 + 61). Both Wigger's treatment and that of Skerrett (1968) are in the style of early generative phonology. This is also true of Wigger (1973) which has some inconsistencies, for instance, when he, while establishing putative major and minor rules in Irish, separates palatalisation (with velarisation as a sub-rule) from 'umlaut', that is vowel alternation between various grammatical categories. This prevents him from connecting the two and seeing them as related aspects of a general process triggered by polarity reversal.

When dealing with the changes in phonetic shape of nouns across grammatical categories, such as case and number, it is essential to distinguish automatic changes due to phonetic environment and those which are triggered by the mechanism used to indicate the category in question.

This consideration is particularly important when viewing the stressed vowels of nouns in Irish.

There are a number of vowel changes here. The first involves vowel lengthening before former geminate sonorants:[74] /a/ ➜ /ɑ:/ *crann* /krɑ:nɣ/ 'tree.NOM', /ɪ/ ➜ /i:/ *crainn* /kri:nj/ 'tree.GEN'. The second is the diphthongisation of short vowels. This is a fairly simple process which involves the shift of mid vowels: /ʌ/ ➜ /au/ *poll* /paulɣ/ 'hole.NOM', /ɛ/ ➜ /ai/ *poill* /pailj/ 'hole.GEN'. The third change is one in which a back or low vowel changes to a front position. This shift is attendant on a change in palatality for the consonant(s) in the coda of the syllable in which this vowel occurs: *blas* /blɣas/ 'taste.NOM', *blais* /blɣɛsj/ 'taste.GEN'. This latter type of vowel gradation only affects short vowels.

(86)

/ʌ/	➜	/ɪ/
/a/	➜	/ɪ/
/ʌ/	➜	/ɛ/
/a/	➜	/ɛ/

a. *cnoc* /krʌk/ 'a hill'
b. *barr an cnoic* /xrɪkj/ 'the top of the hill'
c. *an t-olc* /ənɣ tʌlɣk/ 'evil'
d. *ag troid an oilc* /ɛljkj/ 'fighting evil'
e. *fear* /fjar/ 'man'; *geal* /gjalɣ/ 'bright'
f. *fir* /fjɪrj/ 'men'; *níos/is gile* /gjɪljə/ 'brighter/brighest'
g. *glac* /glɣak/ 'handgrasp'
h. *lán glaice* /glɣɛkjə/ 'a handful'

When making generalisations about this vowel gradation, /a/ and /ʌ/ cannot be collapsed as both occur in a purely palatal environment. But one can predict the front vowel used in gradation with short stressed vowels.

(87)

/a/ → /ɛ/ / C^{ɣ} ___ C^{j}
/ɪ/ / C^{j} ___ C^{j}

a.	*glas* /gl^{ɣ}as/	‘lock.NOM’
b.	*glais* /gl^{ɣ}ɛsj/	‘lock.GEN’
c.	*neart* /n^{j}art/	‘strength.NOM’
d.	*nirt* /n^{j}ɪr^{j}t^{j}/	‘strength.GEN’

The upshot of this is that if the base form has /a/ then the result of gradation is predictable. The difficulty in determining gradation with /ʌ/ is that the nouns which show the structure C^{j} /ʌ/ C^{ɣ} do not palatalise for the genitive singular or the plural but have a vowel change and suffix without any alteration of palatality.

(88)

a. *sioc, seaca* /sjʌk/ /sjakə/ ‘frost.NOM’, ‘frost.GEN’
b. *cion, ceana* /kjʌnɣ/ /kjanɣə/ ‘fondness.NOM’, ‘fondness. GEN’

But there are a few instances where structures of the kind C^{ɣ} /ʌ/ C^{ɣ} occur and on palatalisation of the final velar segment the vowel changes to / ɪ/:

(89)

a. *muc, mice* /mʌk/ /mɪkjə/ ‘pig.NOM’, ‘pig. GEN’
b. *cur, cuir* /kʌr/ /kɪrj/ ‘putting’, ‘put’

The second example is taken from the verb ‘put’ to show that if the consonant following a stressed /ʌ/ is palatalised then the vowel shifts to /ɪ/. What is not possible, given the structure of the modern Irish lexicon, is to predict whether a coda of the kind /ʌ/ C$^{\gamma}$ will palatalise for the genitive or plural as both suffixation and zero change are valid options. There are also cases where alternative pronunciations of a word have the vowels which are involved in this change: *goid* /gʌd^{j}/ ~ /gɛdj/ ‘steal’.[75] In this example one can see that both the orthographic vowels – <*oi*> – serve the function of consonant quality indicators so that it cannot be said for definite which vowel is the syllable indicator. If this is *o* then /ʌ/ results, if it is *i* (after *o*) then /ɛ/ is the result. But the syllable must have a vowel which is somehow specifiable, after all the word cannot be pronounced arbitrarily as /giːd^{j}/ or /gaudj/. One solution is to regard the syllable as containing a vowel which is [-long] and [-high]. Allowing for the moment only two divisions on the height continuum will delimit both /ʌ/ and /ɛ/ sufficiently from /ɪ/. This vowel is, however, unspecified for palatality. The value it obtains is determined by assimilation to the following consonant, i.e. to the coda of the syllable rhyme in which it is located. According to this, cases such as /gʌdj/ are exceptional in that the value of [palatal] for the vowel comes from the consonant in the syllable onset. As would be expected forms such as /gʌdj/ are fairly rare as they constitute instances of assimilation to segments outside a syllable rhyme.

The above explanation for vowel values has the advantage of rendering superfluous any recourse to grammatical category, for example, by specifying /ɛ/ for the genitive and /ʌ/ for the nominative. This would not predict the attested forms in instances such as the following which are best handled by reference to rhyme-internal (regressive) assimilation.

(90)

Skeleton	**Example**
Cˠ /ɛ/ Cʲ	*troid* /trɛdʲ/ ‘fight.NOM’
Cˠ /ʌ/ Cˠ	*troda* /trʌdə/ ‘fight.GEN’
Cʲ /ɪ/ Cˠ	*fios* /fʲɪs/ ‘knowledge.NOM’
Cʲ /a/ Cˠ/ə/	*feasa* /fʲasə/ ‘knowledge.GEN’
Cʲ /ɪ/	*rith* /rʲɪ/ ‘run.NOM’
Cˠ /a/ (h) /ə/	*reatha* /rʲahə/ ‘run.GEN’
Cˠ /ɪ/ Cʲ	*muir* /mɪrʲ/ ‘sea.NOM’
Cˠ /a/ Cˠ /ə/	*mara* /marə/ ‘sea.GEN’
Cʲ /a/ Cˠ	*fear* /fʲar/ ‘man.NOM’
Cʲ /ɪ/ Cʲ	*fir* /fʲɪrʲ/ ‘man.GEN’
Cˠ /a/ Cˠ	*glas* /glˠas/ ‘lock.NOM’
Cˠ /ɛ/ Cʲ	*glais* /glˠɛsʲ/ ‘lock.GEN’[76]
Cʲ /ʌ/ Cˠ	*sioc* /sʲʌk/ ‘frost.NOM’
Cʲ /a/ Cˠ /ə/	*seaca* /sʲakə/ ‘frost.GEN’
Cˠ /ʌ/ Cˠ	*roc* /rʌk/ ‘wrinkle.NOM’
Cˠ /ɛ/ Cʲ	*roic* /rɛkʲ/ ‘wrinkle.GEN’
Cˠ /ʌ/ Cˠ	*muc* /mʌk/ ‘pig.NOM’
Cˠ /ɪ/ Cʲ /ə/	*muice* /mɪkʲə/ ‘pig.GEN’

C^{j} /a/ $C^{ɣ}$	*leac* /l^{j}ak/ ‘stone slab.NOM’
C^{j} /ɛ/C^{j} /ə/	*leice* /l^{j}ɛk^{j}ə/ ‘stone slab.GEN’
$C^{ɣ}$ /ɪ/ C^{j}	*fuil* /fɪl^{j}/ ‘blood.NOM’
$C^{ɣ}$ /ʌ/ $C^{ɣ}$ /ə/	*fola* /fʌl^{ɣ}ə/ ‘blood.GEN’

Several generalisations can be made about the above patterns: in genitive forms, /a/ only occurs if the following segment is non-palatal and in inflected forms only if there is schwa suffixation. / ɪ/ only occurs if one of the flanking consonants is palatal. The environment C^{j}_ shows least predictability as all short vowels can occur in monosyllables.

(91) Vowels occurring in monosyllables after a palatal consonant

a. *fios* C^{j}/ɪ-/ ‘knowledge’
b. *feic* C^{j}/ɛ-/ ‘see’
c. *leac* C^{j}/a-/ ‘slab’
d. *sioc* C^{j}/ʌ-/ ‘frost’

If both flanking consonants are palatal the vowel is nearly always /ɪ/ (*fir*). In vowel-initial words with a following palatal consonant the stem vowel is /ɛ/ not /ɪ/ (*oilc*). /aɪ/ derives regularly from /ɛ/ in syncopated inflectional forms (*oibre*, *soilse*). Conversely, this diphthongisation never occurs with words which have epenthetic vowels in their coda clusters (*boilg* /bɛljəgj/ ‘stomach.GEN’).

(92)

a.	*olc* /ʌl^{ɣ}k/ ‘evil.NOM’	*oilc* /ɛl^{j}k^{j}/ ‘evil.GEN’
b.	*alt* /al^{ɣ}t/ ‘article.NOM’	*ailt* /ɛl^{j}t^{j}/ ‘article.GEN’

c.	*obair* /ʌbər^{j}/ ‘work.NOM’	*oibre* /aɪb^{j}r^{j}ə/ ‘work.GEN’
d.	*solas* /sʌl^{ɣ}əs/ ‘light’	*soilse* /saɪl^{j}s^{j}ə/ ‘lights’

These vowel alternation patterns are not mutually exclusive. There are stems which show different patterns (including no change) and are associated with different meanings as can be seen from the following forms.

(93)

a.	*cion* /k^{j}ʌn^{ɣ}/ ~ *ciona* /kjʌnɣə/	‘offence.NOM’,‘offence. GEN’
b.	*cion* /k^{j}ʌn^{ɣ}/ ~ *ceana* /kjanɣə/	‘love.NOM’, ‘love.GEN’
c.	*cion* /k^{j}ʌn^{ɣ}/ ~*cion* /kjʌnɣ/	‘share.NOM’, ‘share. GEN’

That the vowel alternations discussed above are caused by the change in polarity for the consonant(s) in the coda can be demonstrated quite simply by considering the following fact. In the coda /-xt/ the velar fricative cannot be palatalised, i.e. the palatalisation of /-xt/ results in /-xt^{j}/, not /-x^{j}t^{j}/. When this occurs the vowel of the syllable rhyme remains unchanged as can be seen in the following example.

(94)

a.	*locht* /l^{ɣ}ʌxt/ ‘fault.NOM’	*loicht* /l^{ɣ}ʌxt^{j}/ ‘fault. GEN’

b. *bocht* /bʌxt/ ‘poor’ *(níos) boichte* /bʌxtʲə/ ‘poorer’

Finally it should be mentioned that while vowel gradation is most commonly a process of alternation across the grammatical categories of number and case there are also instances where it exists without such alternation. For instance, in the noun *oiliúint* ‘nutrition; training’ the first syllable is [ɛlʲ-] with the shift of <*o*> /ʌ/ to [ɛ] because of the following palatal lateral. In the genitive, *oiliúna*, this lateral remains palatal and so there is no change of [ɛ] to [ʌ] as would be expected if the lateral underwent polarity reversal, i.e. if it were velarised.

1.9.4.1. Vow el-altering processes

The cases of vowel gradation discussed so far all illustrate the effect which consonant polarity has on adjacent vowels. There are, however, a number of further vowel alternations which are independent of consonant polarity.

Short vowel raising before a closed syllable with a long low vowel

A connection between short and long vowels in successive syllables is evident in Western Irish in an alternation which involves the raising of a low or mid vowel when this is short and followed in the second syllable of a word by a long low vowel. Examples illustrating this are the following.

(95)

a. *snasán* [snˠʊsɑːnˠ] ‘polish’

b. *coláiste* [kʊlˠɑːsʲtʲə] ‘college’

c. *caisleán* [kɪsʲlʲɑːnˠ] ~ [kʊsʲlʲɑ:nˠ] ‘castle’

d. *gearán* [gʲɪrɑ:nˠ] ‘complain’

Here the low vowel of the stressed initial syllable – phonologically /a/ or /ʌ/ – is shifted upwards to a high vowel. This vowel normally agrees in palatality with the following consonant, but there is some fluctuation as the penultimate example just given shows.

The trigger for this raising is a following long low vowel in a closed syllable. The initial syllable need not be closed as pronunciations like *anáil* [ˈʊnˠɑ:lʲ] ‘breath’ show. Whether the syllable containing the long vowel must always be closed is difficult to ascertain given that there are very few words which have this type of phonotactic structure. An example is the word *tochrá* ‘distress, anguish’ which is pronounced [ˈtʌxrɑ:]. However, this is not a common word in present-day spoken Irish and may be excluded from the raising of /ʌ/ to [ʊ]. Another example is *antráth* ‘inopportune time, late hour’ which does not have raising either, cf. de Bhaldraithe (1953a: 255) where [ˈɑ:ntrɑ:] is the transcription given. Because of the dearth of words with the structure C(CC) + /a, ʌ/ + C(CC) + /ɑ:/ it cannot be decided whether an open syllable per se blocks the raising of /ʌ/ to [ʊ] . What is certain, though, is that the vowel of the closed second syllable must be low. Mid and high vowels do not trigger the change in the first syllable as can be seen from the following examples: *gasóg* [ˈgaso:g] ‘boy scout’, *lasúil* [ˈlˠasu:lʲ] ‘ardent’ (de Bhaldraithe 1945: 87) or *botún* [ˈbʌtu:nˠ] ‘mistake’.

R-lowering

In the environment of /r/ there is a frequent lowering of vowels which is most noticeable with short vowels in Western Irish as seen in the following forms which show [ɛ] rather than [ɪ].

(96)

a.	*cruimh* /krɛv^{j}/	'maggot'
b.	*tirim* /t^{j}ɛr^{j}ɪm^{j}/	'dry'

This lowering can occur between non-palatal and palatal segments (or a combination of these). The outcome of *R*-lowering may involve more than one process. For instance, the word *nuair* 'when' has the formal realisation /nɣuərj/ [nɣuɪɹj]. However, in casual speech one finds [nɣɛrj]. This has arisen through a shift of the syllable nucleus from [u] to the palatal on-glide to the following [rj], [ɪ]. This element became syllabic and was affected by *R*-lowering, yielding [ɛ].

R-lowering only effects the highest vowels. There is no lowering of /ɛ/ to /a/. A phonetic reason can be offered here: acoustically /r/ has the effect of lowering a vowel's third formant[77] which leads over time to a substitution of lower vowels for previous higher ones. R-lowering is by no means uncommon. In Old Norse, for example, /i/ was lowered to /e/ followed by /r/: **miR* ➜ *mer* 'me.DAT' (Hanssen, Mundal and Skadberg 1975: 24); /u/ was also lowered to /o/.

Nasal raising

A process of wide occurrence in Western Irish is nasal raising (cf. O'Rahilly 1932: 33). By this is meant the raising of a vowel[78] in the environment of a nasal vowel which usually follows the vowel.

(97)

a. *seomra* [ʃu:mɹə] ‘room’

b. *trom* [tru:m] ‘heavy’

c. *bonn* [bu:nˠ] ‘bottom’

d. *inneóin* [ɪnʲu:nʲ] ‘anvil’

e. *déanamh* [dʲi:nˠə] ‘doing’

f. *pension* [pɪnʃən]

g. *nós* [nˠu:s] ‘custom’

h. *(ar) meisce* [mʲɪʃkʲə] ‘drunk’

i. *ar ndóigh* [a:ɹ nˠuː] ‘of course’

Phonetically, this is a process with an auditive motivation. Nasals have resonance below 800 Hz and above 2000 Hz with anti-resonance in between (Fry 1979: 117) as the following formant values for four English vowels show (Catford 1977a: 58-62).

(98)

	F1	F2	
/e/	570	1970	as in *head*
/i/	300	2300	as in *heed*
/o/	450	740	as in RP *hoard*
/u/	300	940	as in RP *who’d*

Nasal raising of /e/ to /i/ and of /o/ to /u/ can be seen as a kind of assimilation maximalising the distance between the first and second formants in anticipation of the distance between the two with nasals. In each case of nasal raising the distance between F1 and F2 increases.

Nasal raising can lead to homophony where a vowel is raised to a value which makes it coalesce with an existing word, consider the placename *Cloch na Rón* ‘Seals’ Rock’ (English: ‘Roundstone’) in Western Connemara. With raising of the mid back vowel in the word *rón*, i.e. [roːnˠ] ➔ [ruːnɣ], it becomes homophonous with *rún* ‘secret’.

Vocalisation of non-palatal /v/ after central/back vowels

In Western and Northern Irish non-palatal /v/ tends to have little or no friction in open positions, e.g. before a vowel or in final, unchecked position, e.g. *a bhád* [ə wɑːd] ‘his boat’. When this sound follows /ʌ/ or a schwa in an unstressed syllable, it merges with the vowel to yield /uː/, e.g. *subh* [su:] ← /sʌv/ ‘jam’, *marbh* [maɹu:] ← /marəv/ ‘dead’. This development means that with some words which undergo epenthesis due to a heavy coda, there is an alternation of [u:] with [ɪvj].

(99)

a.	*tarbh* [taɹu:]	‘bull.NOM’
b.	*tairbh* [tɛɹʲɪvʲ]	‘bull.GEN’
c.	*garbh* [garu:]	‘rough’
d.	*níos gairbhe* [gɛrɪvʲə]	‘rougher’, lit. ‘more rough’

1.9.5. Short vowels

Systemically, Irish has four short vowels in stressed syllables. In unstressed position the number is reduced to just one with two variants depending on the polarity of flanking consonants.

(100)

Short vowels

Stressed		*Unstressed*
ɪ		ɪ/ə
ɛ	ʌ	
a		

Dialectal variation among short mid vowels

Short stressed vowels show considerable variation across dialects of Irish as well as in Scottish Gaelic and Manx. In particular the environment $C^{ɣ}$__C^{j} shows fluctuation for vowel realisation in all dialects (on Erris, see Mhac an Fhailigh (1968: 18); on Tourmakeady, see de Búrca (1958: 59-60 + 63); see also M. Ó Múrchú (1969: 67) and recordings contained on the DVD in Hickey (2011a). Any monosyllable which has consonants differing in palatality on either side of the vowel will either have (i) a glide plus a vowel or (ii) a vowel plus a glide depending on what is chosen as the syllable bearing vowel of the word (de Búrca 1977/8: 398). To take the case of Tourmakeady Irish and Scottish Gaelic, a monosyllable of the $C^{ɣ}$__C^{j} type with a short vowel can have the following vocalic structure.[79]

(101)

a.	$C^{ɣ}$	^{u}ɪ	C^{j}
b.	$C^{ɣ}$	ʊ^{ɪ}	C^{j}

luibh ‘herb’	[$l^{ɣ}$ɪv^{j}]	Tourmakeady Irish
	[$l^{ɣ}$ʊ^{ɪ}]	Scottish Gaelic

Scottish Gaelic and Manx would appear to have syllabified the velar glide of Irish in general (O’Rahilly 1932: 141):

duine ‘people’ [dɪnə] Irish, [dʊnə] Scottish Gaelic, *cuid* ‘part’ [$k^{ɯ}$ɪd^{j}] Irish, [kʊdj] Scottish Gaelic. This also happened with many forms between Old and Modern Irish (see Pedersen (1909-13: 343-345), who deals with this and with modern variants as well).

Short front vowels

The two stressed short front vowels, /ɪ/ and /ɛ/, are clearly distinguished in Irish, cf. /ɪ/ *tit* /tjɪtj/ ‘fall’, /ɛ/ *feis* /fjɛsj/ ‘festival’. It is true that via *R*-lowering or *N*-raising (see above) either can be shifted to the value of the other. Nonetheless, there are minimal pairs which show clearly that the vowels are systemically separate.

(102)

a.	*bith* /b^{j}ɪ/	‘ever, at all’
b.	*beith* /b^{j}ɛ/	‘being; birch’

The behaviour of the two vowels in the morphonology of Irish is quite different, however. The mid short front vowel /ɛ/ alternates with the mid short back vowel /ʌ/ as discussed above. The high short front vowel /ɪ/ does not partake in any back – front alternation across grammatical forms. Instead it alternates with /a/ when, through a change in polarity in the consonants of a coda or because of suffixation, a vowel shift occurs.

(103)

a.	*fios* /f^{j}ɪs/	‘knowledge.NOM’

b.	*feasa* /fʲasə/	‘knowledge.GEN’
c.	*fear* /fʲar/	‘man’
d.	*fir* /fʲɪrʲ/	‘men’

In Northern Irish there is a tendency to lower and retract high front vowels. This is a feature which it shares with different forms of English in Ulster and with both Scottish Gaelic and English in Scotland. Hence this lowering and retraction can be regarded as an areal feature (Hickey 1999) of Ulster and (Western) Scotland, as can the high mid realisation of/uː/ as [ʉ:] in both Irish and English in this large area.

Low vowels

There is one low short vowel in Irish which can be transcribed phonologically as /a/. This vowel has two main realisations, [a] and [æ] after non-palatal and palatal segments respectively. Variation across and within the dialects can be found. In particular in Cois Fharraige Irish there is an almost automatic lengthening of all phonologically short low vowels.[80] Consider the following forms which all show phonetically long vowels which are, however, qualitatively different.

(104)

a.	*teanga* [tʲæ:ŋgə]	‘language’	(Cois Fharraige)
b.	*baile* [ba:l$_j$ə]	‘town’	(Cois Fharraige)
c.	*bán* [bɑ:nˠ]	‘white’	
d.	*deán* [dʲɑ:nˠ]	‘channel in strand’	

The vowels in the above forms can be organised into two groups. The first two belong to the short vowel /a/ and the latter to the long vowel /ɑː/.[81] In keeping with the other long vowels, /ɑ:/ does not show any appreciable difference in quality despite varying polarity with preceding consonants (see third and fourth forms above). Hence the unaffected realisation of /ɑː/ as [ɑ:] can be taken as evidence for its status as a systemically long vowel. Because this vowel is always a low back vowel, and not low central, the transcription /ɑ:/ for the systemically long low vowel of (Western and Southern) Irish is preferred. The situation in Northern Irish is different as this vowel can be considerably fronted, see I.2.3. above.

The fact that /a/ shows two realisations is connected to its status as a short vowel: it is affected by the polarity of the preceding vowel with [æː] as a front realisation after palatals. This is a phonetic effect which only applies to systemically short vowels.

(105)

/a/	➜	[æ] /Cj ___	*ceathach* [k^{j}æhəx] ‘showery’
		[a] / C$^{\gamma}$ ___	*cathach* [kahəx] ‘warlike’

The status of [æ] as the realisation of /a/ in post-palatal position can be demonstrated by considering recent loans from English which contain the [æ] vowel. The consonants which precede this English vowel are generally palatalised because in Irish only such segments can occur before [æ] (but after this vowel a non-palatal consonant is possible, e.g. *fear* [fjær] ‘man’). Loans from English with [æ] are written with <*e*> preceding <*a*> (and possibly <*i*> following it) in accordance with Irish orthographical practice (see Hickey 2011a: 392-404 for an explanation of the principles of Irish spellings), e.g. *plean* [pljænɣ] for *plan*, *veain* [vjænj] for *van*.

The variation among low vowels just discussed can be accounted for by the polarity of ambient consonants and/or general features of a variety, such as the ‘drawl’ of Cois Fharraige Irish. Nowhere are more than two low vowels justified on a phonological level, one long and one short, i.e. /a/ and /ɑː/, and these are the segments responsible for the distinction in minimal pairs such as the following.

(106)

a.	*camas* /kaməs/	‘small bay’
b.	*cámas* /kɑːməs/	‘finding fault’

The early phonetic studies of Irish dialects produced during the 1940s assume that each phonetically recognisable variant corresponds to a separate phoneme. De Bhaldraithe (1945: 12-13) has no less than five low vowel ‘phonemes’: [æː], [æ], [aː], [a] and [ɑː]. Ó Cuív (1944: 17-20) has four low vowels which he terms ‘phonemes’: [a], [aː], [ɑ] and [ɑː]. Later studies assume a smaller number of systemic units and discuss variation within the context of one or other of these units. Both de Búrca (1958: 13-14) and Mhac an Fhailigh (1968: 13-16) have a set of two vowel ‘phonemes’ [a] and [aː].

Low vowels before R

A length contrast is found in Irish before tautosyllabic *R* among mid and high vowels. There are some (near) minimal pairs which illustrate this contrast clearly as in the following.

(107)

a.	/ɪ/ *fir* /fɪr^j/	‘men’
b.	/iː/ *fíor* /fiːr/	‘true’
c.	/ɛ/ *deir* /d^{j}ɛr^j/	‘say.IMPERATIVE’
d.	/eː/ *déarfainn* /d^{j}eːr.hɪn^j/	‘say.CONDITIONAL’
e.	/ʌ/ *cor* /kʌr/	‘move, development’
f.	/oː/ *cór* /koːr/	‘choir; corps’

However, with low vowels before tautosyllabic *R*, no length contrast is found. There are basically two realisations here which depend on the weight of the syllable coda. Where this consists of just *R* [rj] or *R* [r, rj] and a voiceless consonant [t, k] the realisation is [æ] (this type of syllable is phonologically light in Irish).

(108) Realisation of /a/ before light codas

a.	*stair* [stær^j]	‘history’
b.	*neart* [n^{j}ært]	‘strength’
c.	*cearc* [k^{j}ærk]	‘hen’

Where the coda consists of *R* followed by a voiced consonant, in effect by /d/ or /n/, then a long vowel is found.[82] For those dialects which have [ɑː] (Western and Southern Irish) this vowel is found before /-r/ + /n, d/. In Northern Irish a tautosyllabic *R* has the effect of blocking /ɑː/-fronting (see section I.2.3 above), i.e. a word like *carr* ‘car’ is [ka(ː)r], contrasting with a word like *fáth* [fwæː] ‘reason, cause’. The retracting effect of /-r/ (+ /n, d/) is seen most clearly in Connemara where /a/ in this environment is lengthened and strongly retracted with a degree of lip rounding.

(109) Realisation of /a/ before /-r/ (+ /n, d/)

a.	*carr* /kar/ [kɑːr]	‘car’
b.	*Carna* /karna/ [kɑːr.nˠə]	(placename)
c.	*nach ndearna muid* [ˈnʲɑːr.nˠə]	‘which we didn’t do’
d.	*garda* [ˈgɑːrdə]	‘policeman’

Where the post-vocalic *R* forms the onset of the following syllable, or is ambisyllabic (depending on phonological interpretation), the low vowel is not lengthened. This also applies to words in which epenthesis occurs in the coda of the syllable containing the low vowel in questions.

(110)

a.	*garach* [ˈga.rəx]	‘obliging’
b.	*garbh* [ˈga.rəv] ~ [ˈga.ruː]	‘rough’
c.	*gairm* [ˈga.rɪmj]	‘profession’

Lexicalised variation among low vowels

The possible variation among low vowels in Irish is not exhausted by the discussion above. There are other variants in the different dialects. These are lexicalised and thus confined to certain words which are then a diagnostic feature for just this region.

In the Ring dialect of West Waterford, the diphthongs /aɪ/ and /aʊ/ show many more tokens than in other dialects. The diphthongs appear before former geminate sonorants and often spread to long vowels which did not stem from original short vowels before geminates, e.g. *Brendán* /brjɛnˈdaʊnɣ/ ‘Brendan’. The /aɪ/ diphthong occurs due to the absorption of a former fricative – *ch* or *dh* – into the nucleus of the syllable of which it forms the coda, hence *rachaidh* ‘will go’ is [raɪgj]. The far North of the Donegal Gaeltacht has a high unrounded back/central vowel [ɯ:, ʉ:], which it shares with Scottish Gaelic, and this can be found in *feadh* ‘during’ which in other dialects has /a/ [æ]. An example for a more sporadic lexicalised form can be seen with [tji:əxt] for *teacht* (*tíocht*) in Western Irish which is not part of a general shift upwards of low vowels.

In both Western and Northern Irish before a non-palatal consonant a short low vowel /a/ can be shifted to a short back vowel as a result of place assimilation. There is only one such vowel so that the result is always /ʌ/, e.g. *cat* [kʌt] ‘cat’. There may be a degree of variation here: the retraction is more common in the Western part of Connemara, for example *agus* ‘and’ is [ˈʌgəs] in the Rosmuc-Camas and Carna areas, but [ˈagəs] is found in Cois Fharraige. Between the West and the North there can be different preferences, e.g. *eaglais* ‘church’ is [aglɣɪʃ] in the West but [ʌglɣɪʃ] in the North.

(111)

a.	*agus*	/agəs/ (S)	/agəs/ ~ /ʌgəs/ (W+N)	‘and’
b.	*acub*	/akə/ (S)	/akəb/ ~ /ʌkəb/ (W+N)	‘with-them’

Short back vowels

For the discussion of short vowels which are non-low and non-front one can begin with the situation in Southern and Western Irish. These two dialect areas agree in having an unrounded mid-open vowel as the unconditioned, and hence most common, realisation of vowels which are written <*o*> and <*u*>.

(112)

a.	*siopa* [ʃʌpə]	‘shop’
b.	*doras* [dʌrəs]	‘door’
c.	*fliuch* [fʲlʲʌx]	‘wet’
d.	*rud* [rʌd]	‘thing’

Historically, the two vowels /u/ and /o/ collapsed to an unrounded low back vowel. The syncretism can be seen quite clearly with pairs of homophones: *loch* ‘hole’, *luch* ‘mouse’ both: [lɣʌx] or *ocht* ‘eight’, *ucht* ‘breast, bosom’ both: [ʌxt]. The vowel values in Northern Irish are different: here a short back rounded vowel for <*o*> has been maintained before liquids, e.g. *olc* [ɔlɣk] ‘evil’, *corr-* [kɔr] ‘occasional’, and an unrounded low back vowel as a reflex of <*u*>, e.g. *rud* [rɑd] ‘thing’.

The situation in present-day Irish dialects suggests that a lowering of /u/ occurred in Early Modern Irish, similar to that which led to the lowering of Early Modern English /ʊ/ to /ʌ/ during the seventeenth century in South-East England (Lass 1987: 131). Dating the shift in Irish is difficult as the corresponding sound in Irish is not referred to at all in the relevant literature. Irish scholars consistently use /u/ to transcribe <*u*> and /o/ for <*o*>. Outside of the North both sounds merged in non-conditioned contexts and continue as /ʌ/. This sound is not commented on in the ‘Historical Development’ sections of the dialect studies (de Bhaldraithe 1945: 84-96; Mhac an Fhailigh 1968: 134-164; and Ó Cuív 1944: 96-123). Wagner (1979 [1959]: 5-6) gives [ɨ], i.e. the

Northern Irish variant of /ʌ/, in forms where it would be expected in Southern and Western Irish /bɨg/ for *bog* 'soft', /hɨg/ for *thug* 'gave'.[83]

(113) Western and Southern Irish

short <*o*>	/ʌ/	*cor* /kʌr/ [kʌr]	'move'
short <*u*>	/ʌ/	*luch* /l^{ɣ}ʌx/ [l^{ɣ}ʌx]	'mouse'

Northern Irish

short <*o*>	/ʌ/	*cor* /kʌr/ [kɔr]	'move'
short <*u*>	/ʌ/	*luch* /l^{ɣ}ʌx/ [l^{ɣ}ʌx]	'mouse'

Raised realisation before velar stops and labial fricatives (all dialects)[84]

short <*u*>	/ʌ/	*rug* /rʌg/ [rʊg]	'was born'
short <*u*>	/ʌ/	*dubh* /dʌv/ [dʊv][73]	'was born'

Even as careful a writer as de Bhaldraithe uses this transcription as in *fliuch* /fl′ox/ 'wet' (de Bhaldraithe 1945: 14) whereas what native speakers say in the dialect which he investigated is [fjljʌx]. A curious paradox of the dialect studies is that the description of the sounds which is given at the beginning of each section is often quite accurate. De Bhaldraithe describes his /o/ as a 'half-open vowel, with neutral lip position, which is very slightly advanced from the back line' (de Bhaldraithe *loc. cit.*). Such a description would call for the use of IPA [ʌ] in phonetic transcriptions, or [ʌ̈] to indicate a slightly centralised realisation of the unrounded vowel.

In the historical development of <*u*> /u/ to /ʌ/ it is not known whether unrounding or lowering occurred first. The possible developments with both orders are given below.

(114)

(i) Unrounding before lowering of <*u*>

cur /kur/ ‘putting’ <*u*> /u/ ➜ /ɤ/ ➜ /ʌ/ *cur* /kʌr/

cor /kor/ ‘turn’ <*o*> /o/ ➜ /ʌ/ *cor* /kʌr/

(ii) Lowering of <*u*> and merger with <*o*>

cur /kur/ ‘putting’ <*u*> /u/ ↘

cor /kor/ ‘turn’ <*o*> /o/ ➜ /ʌ/ *cur, cor* /kʌr/

Sources of [ʊ]

Although one can posit an unconditioned short mid-back unrounded vowel /ʌ/ on the grounds just outlined, there are many instances of [ʊ] in Irish all of these are, however, contextually determined.

1) *Blocking of lowering before labial fricatives*

In Irish, just as in English, there are phonetic contexts in which the shift of /u/ to /ʌ/ does not take place. In English it is before /ʃ/ and /ɫ/, cf. words like *push* and *pull*, in Irish it is before labial fricatives. The lowering of historic /u/ seems to have been blocked by a following /v/ which in its turn may be lost or vocalised, lengthening the preceding vowel in the process: *dubh* ‘black’ /duv/ > [dʊv], [dʊw], [du:], *subh* ‘jam’ /suv/ > [sʊv], [sʊw], [su:].

Because of the structure of Irish words, it is the voiced labial fricative which occurs abundantly in word-final position. A voiceless labial fricative is only found in word-final position with English loans and the lowered realisation of earlier /u/ is used in such instances, e.g. *stuff* [stʌf] ‘stuff’.

2) *Blocking of lowering before velar stops*

When it occurs before a tautosyllabic velar stop, /ʌ/ has a raised variant [ʊ]. Historically, it is most probable that the original /u/ was not lowered in pre-velar position, given that velar stops are close to the high back vowel in terms of tongue position. However, earlier /u/ did lower to /ʌ/ before velar fricatives (see third example below).

(115)

a.	*muc* /mʌk/ [mʊk]	'pig'
b.	*thug* /hʌg/ [hʊg]	'gave'
c.	*lucht* /lˠʌxt/	'capacity, load'

The occurrence of /ʊ/ before velar stops means that this is an environment where historic /o/ and /u/ are still distinguished, despite their merger in unconditioned contexts.

(116)

a.	*rug* /rʌg/ [rʊg]	'took; was born'
b.	*bog* /bʌg/ [bʌg]	'move'

3) *Raising before nasals*

In Irish nasals have a raising influence on preceding vowels (see discussion in II.1.9.4.1 above).[85] The raising applies to both <*o*> /ʌ/ and <*u*> /ʌ/. It is most common before a coda nasal but occasionally an onset nasal can have this effect (see last example below).

(117)

a. *dona* [dʊnˠə] ‘bad’

b. *nós* [nˠu:s] ‘custom’

c. *Conamara* [kʊnˠəma:ɹə] ‘Connemara’

d. *muise* [mʊʃə] ‘indeed’ (exclamation)

4) *Raising before low back vowels*
The realisation of low vowels in Western Irish can be subject to polarisation on a height axis. The situation which gives rise to this has been discussed above (see section II.2.3) and leads to tokens of [ʊ] where the preceding onset is non-palatal and the following syllable contains [ɑː], for instance, *cupán* [ˈkʊpɑ:nɣ] ‘cup’, *cabáiste* [ˈkʊbɑ:ʃtjə] ‘cabbage’, *pacáiste* [ˈpʊkɑ:ʃtjə] ‘package’.

Realisations of <o>

The situation with historic <*o*> /o/ is more straightforward: this vowel is always realised as [ʌ] in the West and South and does not show the raising before velars which is found with short <*u*>, e.g. *cogar* [ˈkʌgər] ‘whisper’, *ogam* [ˈʌgəm] ‘ogam’. This realisation is also found in open syllables where there is no conditioning due to a following consonant, e.g. *scoth* [skʌ] ‘choice, pick’, *do* [dʌ] (stressed form) ‘for’. The only exception to this distribution is Northern Irish which has retained [ɔ] for <*o*> before liquids, e.g. *olcas* [ɔlɣəs] ‘evil’
Summing up, one can note that for the historic input of /o/ and /u/, in words like *loch* ‘hole’ and *luch* ‘mouse’, there has been a merger in all environments except before velar stops and labial fricatives. Synchronically, this means that there is a distinction between [ʌ] and [u] in this context as seen in the following forms.

(118)

a. *log* [lɣʌg] ‘place’, e.g. in the compound *logainm* ‘placename’
b. *lug* [lɣʊg] ‘lug’, e.g. in *lug seoil* ‘lugsail’
c. *ubh* [ʊv] ‘egg’
d. *gob* [gʌb] ‘beak’
e. *rubar* [rʊbər] ‘rubber’

The development of <*u*> /u/ in Irish demands that a decision be made as to what vowel realisation is regarded as primary and hence used in phonological transcriptions. There are basically two option as follows:

(119)

(i) There is a phoneme /ʊ/ which only occurs before velar stops and labial fricatives (having shifted to [ʌ] in all other environments).
(ii) There is a phoneme /ʌ/ which has a realisation [ʊ] before velar stops and labial fricatives, otherwise [ʌ].

For the present study, short <*u*> is transcribed as /ʌ/ because [ʌ] is the majority realisation (option (ii) above). Furthermore, it is the realisation of the vowel in a word-final open syllable where conditioning by a following consonant is excluded, e.g. *guth* [gʌ] ‘voice’. This interpretation does, however, imply that there is a split of /ʌ/ before voiced velars stops and voiced labial fricatives, a trace of the two units [ɔ] and [ʊ] before the historic merger of <*o*> and <*u*>.

Further reflexes offormer/o/and/u/

Although both these vowels have merged to /ʌ/ in unconditioned environments there are cases where both historic vowels have resulted in diphthongs. These occur before /r/ followed by a further segment, either another sonorant or /d/. The diphthong is /aʊ/ and is found from Cois Fharraige (de Bhaldraithe 1945: 22) across to Carna. In North Galway this diphthongisation did not occur and there /ʌ/ is found in a word like *urlár* [ʌrlɣɑ:r] (de Búrca 1958: 134). The lack of diphthongisation can be seen immediately North of coastal Connemara in the placename *Corr na Móna* [kʌr nɣə mo:nɣə].

(120)

a.	*bord* /baurd/	'board'
b.	*corr-* /kaur/	'odd, occasional'
c.	*urlár* /ˈaʊrlˠɑ:r/	'floor'

Alternation of /ʌ/ and /ai/

In general a consonant cluster after a short vowel can cause diphthongisation. Where the consonant cluster is palatal the diphthong which can arise is /ai/. This is found in Western Irish in cases like the following where the cluster arises through syncope triggered by the addition of a genitive ending, here a final schwa. Southern Irish *oibre* /ɛbjrjə/ (Ó Sé 2000: 107) shows the same steps as Western Irish but without the diphthongisation of the stem vowel /ɛ/ to [aɪ].

(121)

a.	*obair* /ʌbər/	'work.NOM'
b.	*oibre* /aibʲrʲə/	'work.GEN'

/ʌbər/ + palatalisation + /ə/ ➜

/ʌbʲərʲə/ + assimilation of vowel ➜

/ɛbʲərʲə/ + syncope ➜

/ɛbʲrʲə/ + diphthongisation [aɪbʲrʲə] ➜

There can be further instances of /aɪ/ from /ʌ/ which are linked to a common source of diphthongs in Irish. This is where an original short vowel was followed by a lenited word-medial consonant. Frequently, this consonant was absorbed into the nucleus, lengthening and usually diphthongising it in the process. This is the development which lies behind words like *gadhar* /gair/ 'beagle', *slaghdán* /ˈslɣaidɑ:nɣ/ 'cold, infection' (in Western and Southern Irish, Northern Irish shows a long monophthong here, i.e. *slaghdán* /ˈslɣɛ:dɛnɣ/).

/ʌ/ in Irish and Irish English

The unrounded mid back vowel of Modern Irish has the same quality as the /ʌ/ in the STRUT lexical set of (Southern) Irish English, itself a conservative pronunciation of the English /ʌ/-vowel.[86] The following words are homophones for speakers of Irish and (Southern) Irish English, allowing for the velarisation of the final segment in Irish: *gol* /gʌl/ [gʌlɣ] 'crying', *gull* /gʌl/ [gʌl]. What cannot be determined for lack of historical data is whether there is a causal relationship here. It may well be that the lowering of Irish /u/ was furthered, if not initiated, by the /ʌ/ of English. However, this vowel cannot have been imported into Irish English earlier than the seventeenth century (in the second period of Irish English, Hickey 2007: 30-48) because before that English /u/ had not been lowered in Southern England anyway. But even the late seventeenth century would be an early date for /ʌ/ in Irish English, given that [ʊ] for the STRUT vowel is still characteristic of local Dublin English (a variety reflecting English in Ireland from the period before 1600, Hickey 2005: 35-45). The /ʌ/ vowel is known from comments by the elocutionist and prescriptivist Thomas Sheridan to have occurred among educated speakers in Ireland in the late eighteenth century (Hickey 2008a, 2009). This may have spread throughout the country subsequently, especially given that during the large-scale shift of the nineteenth century the Irish-speaking population were striving to acquire a variety of English which was publicly acceptable in Irish society.

The upshot of these considerations is that, if the lowering of Irish /u/ is connected to Irish English /ʌ/, then it cannot be dated to much earlier than 1800. However, there is a cogent argument against a link between the two languages despite the identity of [ʌ]-realisations in Irish and Irish English today. In Irish, inherited /o/ (mostly like [ɔ] phonetically), as in *cor* /kor/ 'turn', was unrounded and merged with /ʌ/ from earlier /u/. Nothing like this is attested in Irish English. Words with the LOT vowel (historically / ɔ/ in English) show variation, chiefly lowering with possible unrounding to /ɑ/ or /a/. This is an established phenomenon and is reflected in older loans from English into Irish where /a/ corresponds to /ɒ/, both on its own and in the diphthong /ɒɪ/, e.g. *caife* /ˈkafjə/ 'coffee' [87] and *baghcat* /ˈbaikət/ 'boycott'. At no time in Irish English was there a merger of the STRUT and LOT vowels, but this did happen in Irish with the RUD and OLC vowels, at least in the Western and Southern dialects.

Quality of short and long vowels

The developments among short back vowels outlined above have meant that there has been a clear qualitative dissociation between short and long vowels which is seen with many pairs of words in the present-day language.

(122)

a.	*post* /pʌst/	‘post’
b.	*pósta* /po:stə/	‘married’
c.	*cur* /kʌr/	‘putting’
d.	*cúr* /ku:r/	‘foam; kite’

1.9.6. Unstressed vowels

Irish is clearly a stress-timed language (Roach 1983), even more so than English. Within the set of Irish dialects, Connemara Irish has the most pronounced stress timing. Here there is an obvious qualitative difference between stressed and unstressed vowels. This applies to the other dialects as well, including Southern Irish which shows non-initial stress in many instances (see the detailed discussion in II.3 below). For the long vowels, the difference is not so great as to demand either a different transcription or a different systemic allocation vis à vis stressed vowel counterparts. The only dialect where this is, however, the case is Northern Irish, which shows considerable shortening of long low vowels in unstressed position. Furthermore, the shift of long low vowels to a front position in this dialect does not occur when these do not carry primary stress.

(123) *Long unstressed vowels*

	open syllables	closed syllables
a.	*ionsaí* /ˈʌnˠsi:/ ‘attack’,	*cailín* /ˈkalʲi:nʲ/ ‘girl’

b.	*contae* /ˈku:nˠte:/ ‘county’	*cáipéis* /kɑ:pʲe:sʲ/ ‘document’
c.	*íomhá* /ˈi:vɑ:/ ‘image’	*scannán* /ˈskʊnˠɑ:nˠ/ ‘film’
d.	*barrdhó* /ˈbɑ:rˠo:/ ‘singe’	*bábóg* /ˈbɑ:bo:g/ ‘doll’
e.	*laghdú* /ˈlˠaidu:/ ‘decrease’	*bunús* /ˈbʌnˠu:s/ ‘basis; origin’

Long vowels can contrast with short ones in both open and closed syllables as can be seen from the following examples.

(124) *Short-long vowel contrast in unstressed syllables*

	short vowel	**long vowel**
a.	*comhartha* /ˈko:rə/ ‘sign’	*comhrá* /ˈko:rɑ:/ ‘conversation’
b.	*foras* /ˈfʌrəs/ ‘foundation’	*forás* /ˈfʌrɑ:s/ ‘development’
c.	*cathair* /ˈkahərʲ/ ‘city’	*cathaoir* /ˈkahi:rʲ/ ‘chair’

Furthermore, a short unstressed vowel can be contrastive through its absence or presence.

(125) *Zero-short vowel contrast in unstressed syllables*

Bare stem	Stem + /ə/
a. *seas* /ˈsʲas/ ‘stand’	*seasamh* /ˈsʲasə/ ‘standing’
b. *stop* /stʌp/ ‘stop’	*stopadh* /ˈstʌpə/ ‘stopping’
c. *líon* /lʲiːnˠ/ ‘fill’	*líonadh* /lʲiːnˠə/ ‘filling’

Quality of short unstressed vowels
Phonetically, there are two unstressed short vowels in Irish but the difference between them is determined by consonant quality. In closed syllables the higher vowel occurs before palatal consonants and the more central one before non-palatal segments. In open syllables it is the preceding consonant which determines the vowel quality, see examples (c) and (d) in the following.

(126) unstressed

$$\text{/ə/} \rightarrow \begin{array}{l} \text{[ə]} \, / \, \_\_\_ \, C^{ɣ} \\ \text{[ɪ]} \, / \, \_\_\_ \, C^{j} \end{array}$$

a.	*ocras* /ʌkrəs/ [ˈʌkrəs]	‘hunger.NOM’
b.	*ocrais* /ʌkrəsʲ/ [ˈʌkrɪʃ]	‘hunger.GEN’
c.	*misneach* /mʲɪsʲnʲəx/ [ˈmʲɪʃnʲəx][88]	‘courage.NOM’

unstressed

$$\text{/ə/} \rightarrow \begin{array}{l} \text{[ə]} \, / \, C^{ɣ} \, \_\_\_ \\ \text{[ɪ]} \, / \, C^{j} \, \_\_\_ \end{array}$$

c. *feasta* /fʲastə/ [fʲæstə] ‘anymore’

d. *rince* /rɪŋʲkʲə/ [rɪŋjkjɪ] ‘dance’

As the quality of short unstressed vowels depends on the polarity of the preceding consonant both of them can be assigned on a systemic level to the schwa vowel /ə/. It would be phonologically redundant to posit both /ə/ and /ɪ/ in Irish as they can never contrast with one another. A minimal pair such as /ˈsʌdə/ and /ˈsʌdɪ/ is not possible in Irish.

Occurrence of final /ə/ and /iː/

Across the dialects of Irish there is a correspondence of final /ə/ and final /i:/ with many nouns in the singular. From North Galway northwards, i.e. covering all of Co. Mayo and Co. Donegal, final unstressed /i:/ can correspond with final /ə/ elsewhere, e.g. *teanga* ‘tongue; language’ [tʃæŋgi:] (North-West and North), [tjæŋgə] (West [Connemara] and South). Furthermore, in Connemara Irish and in Southern Irish, a final /i:/ is frequently the acoustic signal for a plural noun.

(127)

a. *fataí* /fatiː/ (SG: *fata* /fatə/) ‘potatoes’ (West)

b. *prátaí* /prɑ:ti:/ (SG: *práta* /prɑ:tə/) ‘potatoes’ (South)

Indeed Connemara Irish is known for an extension of plural marking through the use of long suffixes (Hickey 1985a) as in *áit* /ɑ:tj/ ‘place’, *áiteachaí* /ɑ:tjəxi:/ ‘places’.

In Western and Northern Irish a final unstressed /-iː/ is also taken as a signal of the verbal adjective (roughly equivalent to the English past participle).

(128)

a.	*fághta* /fɑːkiː/	‘left’
b.	*tógtha* /toːkiː/[89]	‘taken’
c.	*bailithe* /balʲiː/	‘gathered’
d.	*sáinnithe* /sɑːnʲiː/	‘trapped’

This also holds for those cases where the form has become an independent adjective as in *sciobtha* /sjkjʌpiː/ ‘quick, nimble’ (from the past participle of the verb *sciob* ‘snatch’).

Signal value of [ə] and [i]

While [ə] and [ɪ] can be collapsed phonologically as /ə/, on the phonetic level the difference between these vowels is important. Because the difference correlates with the palatal ~ non-palatal distinction it is often the vowel which is taken as a perceptual cue when recognising the polarity of a consonant.

(129)

a.	*carball* /karbəlˠ/ [karəbəlˠ]	‘gum, palate.NOM’
b.	*carbaill* /karbəlʲ/ [karəbɪlʲ]	‘gum, palate.GEN’
c.	*scamall* /skaməlˠ/ [skaməlˠ]	‘cloud.NOM’

d. *scamaill* /skaməlʲ/ [skamɪlʲ] ‘cloud.GEN’

This applies to short vowels only. Where the vowel preceding the consonant which changes polarity is long, the vowel quality does not change but there is a short high glide towards a following palatal consonant. This is particularly clear when the long vowel in question is the low vowel /ɑː/.

(130)

a. *caisleán* /ˈkasʲlʲɑ:nɣ/ [ˈkʊʃlʲɑ:nˠ] ‘castle.NOM’

b. *caisleáin* /ˈkasjljɑ:nj/ [ˈkʊʃlʲɑ:ˈnʲ] ‘castle.GEN’

Front rounded vowels

There are no front rounded vowels in Irish and in the remaining Celtic dialects they are the results of borrowing or secondary developments. The former case is seen in Breton which has front rounded vowels, most probably from French, in the standard dialect based on that of Léon (Ternes 1979: 215-216). Late Manx (nineteenth and to mid-twentieth centuries) may have had rounded realisations of front vowels, see comments in Broderick (2009: 307-310) and a low front vowel was apparently found before /r/, cf. *myr* /mœr/ ‘as’ (Jackson 1955). The paucity or absence of front rounded vowels in Q-Celtic languages may well have to do the fact that languages with a series of palatal and non-palatal consonants do not show rounding with front vowels as this clashes phonetically with the unrounded, spread-lipped nature of palatal consonants.

1.9.7. Diphthongs

On a systemic level there are four diphthongs in Irish, [90] two of which are centring and two of which are rising. The latter are the result of a variety of historical processes, chiefly the vocalisation of lenited intervocalic fricatives and of diphthongisation before former geminate sonorants. The centring diphthongs have been in the language from the earliest records, that is they were written as <*ia*> and <*ua*> already in the Old Irish period.

(131) Centring diphthongs

a.	/iə/ :	*Dia*	/dʲiə/	‘God’
		cliath	/kʲlʲiə/	‘harrow; hurdle’
b.	/uə/ :	*bua*	/buə/	‘victory’
		tuath	/tuə/	‘country’

Rising diphthongs

c.	/ai/ :	*claí*	/klˠai/	‘stone wall’
		poill	/pailʲ/	‘hole.GEN’
d.	/au/ :	*cleamhnas*	/kʲlʲaunˠəs/	‘marriage match’
		poll	/paulˠ/	‘hole.NOM’

The phonological diphthongs, e.g. *bia* /bjiə/ ‘food’, need to be distinguished from frequently identical sequences which are, however, the result of an off-glide to a following consonant with opposite polarity to a vowel.[91]

(132)

a. *gréas* [gʲrʲe:əs] ‘ornamentation’

b. *cíos* [kʲi:ᵊs] ‘rent’

The above forms have counterparts with back vowels followed by front glides, e.g. *báid* [bɑ:ˈdj] ‘boats’. The diphthongal character of the vowels in such words is lost when the following consonant agrees with the vowel in polarity, e.g. *bád* [bɑ:d] ‘boat’. On the other hand, the phonological diphthongs remain phonetically diphthongal in word-final position, where it is not possible to interpret them as arising from off-glides to consonants of opposite polarity, e.g. *dua* /duə/ ‘toil, hardship’.

The four diphthongs listed above are furthermore distinct from sequences of two vowels which are found in many words, especially where the second vowel is the realisation of an inflectional suffix.

(133)

a.	*báigh* /bɑːiː/	← /bɑː/ + /iː/	‘drown’
b.	*sáigh* /sɑːiː/	← /sɑː/ + /iː/	‘thrust, stab’

These vowel sequences are reduced to a single vowel when the lexical stem occurs on its own, e.g. with third person singular present forms of verbs as in *sádh sé* ‘he thrusts’.

1.9.7.1. The diphthongs /ai/ and /au/

The most common source of the diphthongs /ai/ and /au/ are the following.

(134)

(i) a vowel and a former fricative where this has been absorbed in to the nucleus of its syllable

(ii) a vowel and a former geminate sonorant or a vowel and a cluster consisting of a sonorant and an obstruent

With (i) the clue to pronunciation is the voiced fricative which still appears in writing. The realisation of sequences of vowel + voiced fricative varies greatly across the dialects and the result of the historical development is not always a diphthong. But as a source of diphthongs in principle these sequences are found in all dialects.

(135)[92]

		North	West	South	
a.	*laghdú*	/lˠɛ:du:/	/lˠaidu:/	/lˠaidu:/	'reduction'
b.	*gadhar*	/gɛ:r/	/gair/	/gair/	'beagle'
c.	*leabhar*	/lʲo:r/	/lʲaur/	/lʲaur/	'book'
d.	*amhrán*	/o:rənˠ/[81]	/o:rɑ:nˠ/	/auˈrɑ:nˠ/	'song'

In Northern Irish the reflexes of the sounds originally represented by <*amh*>, i.e. /av/, can be either /o:/ or /au/. Furthermore, the actual realisation of the /au/ diphthong varies, for instance in Teileann it is [o ʊ] or [ɔʊ] (Wagner 1979 [1959]: 77-78). The reflex of <*agh, adh*>, both /aɣ/, is generally a low front long vowel, i.e. [ɛ:].

In general it is true that Northern Irish has fewer tokens of the diphthongs /ai/ and /au/. This may well be due to the smaller number of phonetically long vowels in this dialect given that it is the long vowels which historically are the input to diphthongisation to either /ai/ or /au/.

Because /ai/ has arisen before palatal consonants there are many paradigms (nominal and adjectival) where there is an alternation between a monophthong and a diphthong or two diphthongs at different points as seen in the following examples.

(136)

a.	*obair*/ ʌbərʲ/'work.NOM'	*oibre* /aibʲrʲə/ 'work.GEN'

b. *reamhar* /r^{j}aur/ ‘fat’ *reimhre* /r^{j}aiv^{j}r^{j}ə/ ‘fatter’

The second most common source of diphthongs, vowels before former geminate sonorants, shows a good deal of variation across the dialects. In general, Southern Irish[93] has diphthongs before the sonorants *M* and *N* in base forms (Ó Cuív 1944: 28-29, 121) whereas this is not the case in Western and Northern Irish.

(137)[94]

		South	West	North	
a.	*trom*	/traum/	/tru:m/[83]	/tru:m/	‘heavy’
b.	*ann*	/aun/	/ɑ:n/	/an/	‘there, in-it’
c.	*roinnt*	/raɪn^{j}t^{j}/	/ri:n^{j}t^{j}/	/rɪn^{j}t^{j}/	‘some, part of’

Before a former geminate lateral *L* or *R* diphthongs are common in both the West and the South. If the sonorant does not belong to the group of former geminates then there is no lengthening, compare *gal* /galɣ/ ‘valour, fury; steam’ (no lengthening) and *gall* /gɑ:lɣ/ ‘foreigner’ (with lengthening).

(138)

		South	West	North	
a.	*poll*	/paul^{ɣ}/	/paul^{ɣ}/	/pʌl^{ɣ}, po:l^{ɣ}/	‘hole’
b.	*bord*	/bo:rd/	/baurd/	/bɔ:rd/	‘table’

Lengthening before former geminate sonorants only takes place in stressed syllables. There are a number of instances in unstressed syllables which show short vowels in present-day Irish.

(139)

a.	*ólann sé* /ˈoːlˠənˠ sʲeː/	‘he drinks’
b.	*cumann* /ˈkʌmənˠ/	‘association’
c.	*inneall* /ˈinʲəlˠ/	‘engine, machine’

However, the condition that lengthening only occurs before sonorants in word final position is not absolute. With disyllables both short and long vowels are attested: *iompar* /ʌmpər/ ~ /uːmpər/ ‘transport’. This applies when the second syllable is part of the lexeme. When it is an inflectional suffix, however, the stem vowel is usually short:[95] *am* /ɑːm/ ‘time.NOM’, *ama* /amə/ ‘time.GEN’.

Lack of diphthongisation

Diphthongs are not always found in the environments in which they might be expected. There are a number of reasons why this is the case.

1) In some dialects diphthongisation of former sequences of low vowel and voiced fricatives occurred but the latter also survived as optional realisations. For instance, in Connemara Irish (especially in the West of this area) words like the following shows two main variants.

(140)

a.	*samhradh*	/saurə/	/savrə/	‘summer’
b.	*samhnas*	/saunˠəs/	/savnˠəs/	‘disgust’
c.	*amhras*	/aurəs/	/avrəs/	‘doubt’
d.	*amhlaidh*	/aulˠi/	/avlˠi/	‘like, the same’

While the diphthong is probably the more common of the two realisations the retention of the /av/ sequence may have gained support from the existence of the sequence /ɑ:v/ as in *sámhnas* [sɑ:vnɣəs] ‘respite, lull’.

2) If the diphthongisation of mid front and back vowels frequently yields /aɪ/ and /aʊ/ before palatal and non-palatal codas respectively then one must account for instances like the following where the shift takes place in the second and fourth forms but not in the others.

(141)

a.	*saibhir*	/sɛvʲɪrʲ/	‘rich’
b.	*saibhreas*	/saivʲrʲəs/	‘richness’
c.	*solas*	/sʌlˠəs/	‘light’
d.	*soilse*	/sailʲsʲə/	‘lights’

With the first and third forms above a short epenthetic[96] vowel is present which separates the sonorant and obstruent or vice versa, i.e. /-vjɪrj/ and /-lɣəs/ prevent the clusters /-vjrj/ and /-lɣs/ from arising, neither of which would be phonotactically acceptable in Irish. The epenthetic nature of the vowel is clear from (i) its short unstressed character, (ii) the fact that the consonant to its left and right agree in polarity and (iii) from its disappearance on suffixation which triggers syncope (see second and fourth forms respectively).

It is a valid observation across the lexis of Irish that the short stressed vowels which precede an epenthetic cluster never diphthongise. The reason may be that the vowels in a form which is disyllabic because of epenthesis, e.g. *bolg* /bʌlˠg/ → [ˈbʌlɣəg] ‘belly’, are particularly short phonetically and this then prevents the stressed vowel from forming the input to diphthongisation.

When syncope is triggered by suffixation in forms like *saibhreas* and *soilse*, a resyllabification also takes place, i.e. one has /saiv^{j}.r^{j}əs/ and /sail^{j}.s^{j}ə/ respectively. The single segment light codas in the first syllable of each word do not block mid short vowel diphthongisation, hence the /ai/ in both these words.

1.9.7.2. The centring diphthongs /iə/ and /uə/

There are two diphthongs in present-day Irish with /ə/ as a second element: /iə/ and /uə/.[97] Their origin lies in the off-glide which arose between a high vowel and a following consonant. Where this was a fricative, voicing arose and the segment was eventually lost. The off-glide remained, however, so that it could no longer be interpreted as a glide but associated with the preceding vowel to form an unconditioned diphthong. In some cases the former fricative is still indicated orthographically depending on the extent to which the Early Modern Irish spelling is used: *bia* (← *biadh*, E. G. Quin ed., 1990: 93) /biə/ 'food', but: *tuath* /tuə/ 'country'.

The centring diphthongs show a complementary distribution with regard to the polarity of the immediately preceding segments (syllable onset). /iə/ only occurs after palatals and /uə/ only after non-palatals.

(142)

	/iə/ / C^{j} ___			/uə/ / C$^{\gamma}$ ___	
a.	*siar* /s^{j}iər/	'West'	c.	*cuas* /kuəs/	'hollow'
b.	*riamh* /riəv/	'ever'	d.	*suan* /suən$^{\gamma}$/	'slumber'

These are falling diphthongs so that their second elements are not phonetically prominent. This element – /ə/ – is also a typical vocalic offglide from a syllable nucleus to a following non-palatal consonant, e.g. *téad* /tje:d/ [tje:əd] 'rope'.[98] The upshot of this is that for many pairs of words the pronunciation may be the same, even though phonologically a contrast between a diphthong and a long vowel before a non-palatal segment might be present.

(143)

a.	*iatacht* /iətəxt/ [iətəxt]	‘constipiation’
b.	*íotacht* /i:təxt/ [iətəxt]	‘avidity’
c.	*cuairt* /kuərt^j/ [kuɪrtj]	‘visit’
d.	*cúirt* /ku:rt^j/ [kuɪrtj]	‘court’

Diphthongs and vowel dyads

In present-day Irish there are a number of forms which show either /i/ or /u/ followed immediately by /ɑ:/.

(144)

a. *suán* /suɑ:nɣ/ ‘juice’
b. *suáilce* /suɑ:ljkjə/ ‘virtue’
c. *lián* /ljiɑ:nɣ/ ‘trowel’
d. *sián* /sjiɑ:nɣ/ ‘fairy mound’

Here two long vowels are found within a consonantal frame. There is a restriction on the occurrence of these vowel sequences. They are only found before sonorants, mainly *N*. One could postulate that, synchronically, both represent realisations of two basic underlying diphthongs.[99]

(145)

a. /IA/ → /iə/ / ___ #
/iɑ:/ / ___ sonorant

b. /UA/ → /uə/ / ___ #
/uɑ:/ / ___ sonorant

However, this interpretation is untenable as there are instances where /iə/ and /uə/ occur before *N* or *L*, e.g. *sian* /sjiənɣ/ 'whistling, hum of voices' or *buan* /buənɣ/ 'enduring, permanent'. Hence the following sonorant is not a sufficient trigger for /ɑ:/ in the sequences /iɑ:/ and /uɑ:/.

Although the vowel dyads cannot be subsumed under the diphthongs /iə/ and /uə/ they do share with them the fact that they occur within a lexical stem. Other combinations of two vowels between consonants such as *bóín* /bo:i:nj/ 'little cow' are not directly comparable as there is a morpheme boundary between the two vowels, /bo:/ + /i:nj/ (= diminutive suffix).

It is perhaps remarkable that the vowels /i, u, a/ are the components of the centring diphthongs and vowel dyads. These vowels occupy the periphery of vowel space and tend to group together (Kenstowicz and Kisseberth 1977: 26-27; 1979: 199-204). They also have a certain primacy in many languages (Anttila 1989 [1972]: 112, 186), for instance in Russian the five vowel phonemes are reduced to these three in unstressed syllables (Romportl 1973: 37-58).

Lastly it should be remarked that there is considerable variation in the realisation of the centring diphthongs. There is a tendency for them to be smoothed to monophthongs, especially in word-final position, often with lowering one level to a mid vowel in the process.

The morphology of diphthongal forms

Centring diphthongs are lowered to mid vowels for grammatical category changes in Modern Irish. The most common alternation is that between /i ə/ in the nominative and /e:/ in the genitive as is seen in the following.

(146)

a. *ciall* /kʲiəlˠ/ ‘sense.NOM’ ~ *céile* /kʲe:l$_j$ə/ ‘sense.GEN’
b. *iasc* /iəsk/ ‘fish.NOM’ ~ *éisc* /e:sʲkʲ/ ‘fish.GEN’
c. *grian* /gʲrʲiənˠ/ ‘sun.NOM’ ~ *gréine* /gʲrʲe:n$_j$ə/ ‘sun.GEN’

The /e:/ of the genitive appears to correlate with the palatalisation of the coda of the syllable containing this vowel. However, coda palatalisation is not a sufficient condition for the appearance of /eː/ because there are nouns which on palatalisation show a different type of alternation (first example below). Other nouns which have /e:/ in the genitive have this vowel in the nominative as well (second example below with long stem vowel). There are also cases where /iə/ alternates with /e:/ in an open syllable so that there is no question of this being triggered by the polarity of the syllable coda (third example below). Finally, there are instances of /iə/ which show no change across different cases and/or number (last two examples).

(147)

a. *scian* /sʲkʲiənˠ/ ‘knife.NOM’ ~ *scine* /sʲkʲɪn$_j$ə/ ‘knife.GEN’
b. *spéir* /sʲpʲe:rʲ/ ‘sky.GEN’ ~ *spéire* /sʲpʲe:rʲə/ ‘sky.GEN’
c. *Dia* /dʲiə/ ‘God.NOM’ ~ *Dé* /dʲe:/ ‘God.GEN’
d. *bia* /biə/ ‘food.NOM’ ~ *bia* /biə/ ‘food.GEN’
e. *fia* /fiə/ ‘deer.SG’ ~ *fianna* /ˈfiənə/ ‘deer.PL’

1.9.8. Glides

On a phonetic level there are two glides in Irish. The first is a palatal continuant [j] and the second a labial approximant [w]. The former is the realisation of /ɣj/, a voiced fricative and the palatal counterpart of /ɣ/. The latter is the realisation of /v/ in Western and Northern Irish where the non-palatal voiced labial fricatives lost all friction in pre-vocalic positions. In word-final position, typically after an epenthetic vowel, /v/ can appear as [w] or can merge with the preceding short vowel to form [u:] (fourth example below). There are also instances where word-final /v/ <*mh*> after /ɑ:/ varies between [v] and [w] (last example below). When /v/ is followed by a sonorant (in the West and North) and in all environments in Southern Irish it is realised with audible friction as [v].

(148)

a. *cuid dheas* /kɪdj ɣjas/ [k^{ɯ}ɪd^{j} jæs] ‘a nice part’
b. *bhog sé* /vʌg sje:/ ‘he moved’
[wʌg ʃeː] (West + North) [vʌg ʃeː] (South)
c. *béilí bhlasta* /bje:lji: vlɣastə/ [bje:lji: vlɣastə] ‘tasty meals’
d. *marbh* /marv/ [marəv, marəw, maru:] ‘dead’
e. *snámh* /snɑ:v/ [snɑ:v ~ snɑ:w] ‘swim’

In the past autonomous of second declension verbs the final /-uː/ is often realised as [əv] in Western Irish. The contrast between first and second declension verbs in this respect can be seen in the following examples.

(149)

a.	*briseadh* [ˈb^{j}r^{j}ɪʃu:] *an gloine*	‘the glass was broken’
b.	*gortaíodh* [ˈgʌrti:əv] *í*	‘she was injured’

In Western Irish, a further source of [w] is where the /u/ of the centring diphthong /uə/ loses its syllabic character to become a glide when the diphthong occurs in word-initial position, for instance in the set of prepositional pronouns based on *ó* ‘from’.

(150)

a. *uaim* /uəm^{j}/ [wɛm^{j}] ‘from-me’

b. *uaidh* /uəi/ [waɪ] ‘from-him’

Where /uə/ occurs before a long vowel in the following syllable it cannot be reduced to [w]: *uafásach*[100] [ˈu:(f)ɑ:səx] (*[wɑ:səx]) ‘terrible’.

1.9.9. Historical developments

A glance at the orthography of Irish shows that many letters are still written which have not the phonetic value expected of them, if any at all. This is still true although the orthographical reform undertaken in the late 1940s eliminated many instances of ‘silent letters’. The spelling before the reform can be found in the dictionary by Dinneen (1927 [1924]) while the dictionary by Ó Dónaill (1977) uses the reformed orthography. The letters which were removed from writing generally were of no use as an indication of present-day pronunciation. For instance, the <*gh*> in *saoghal* ‘life’ (Dinneen 1927 [1924]: 941) was entirely redundant because the pronunciation /si:lɣ/ (North and West) or /se:lɣ/ (South) is sufficiently indicated by the <AO> vowel in the present-day spelling *saol*.

In other instances, the reform masked the diachronic development of certain sounds. For instance, the centring diphthongs /iə/ and /uə/ are regarded as independent of following segments, indeed many occur in final open syllables, e.g. *crua* ‘hard’. For the phonological analysis of present-day Irish it is right to view them as such, but pre-reform spelling betrays their origin in closed syllables, cf. *cruaidh* (Dinneen 1927 [1924]: 941) from a very much earlier /kruəɣj/. Similar cases can be found with the other centring diphthong, /iə/, e.g. *bia* /bjiə/ (< *biadh*) ‘food’.

The spelling of present-day Irish retains the letters for vocalised fricatives where these have resulted in diphthongs or long vowels by absorption into the nucleus of the syllable which contained them. If the syllable in question was unstressed the outcome was usually /ə/.

The development of unstressed final /ə/ is historically due to 1) vowel reduction because of lack of stress (O'Rahilly 1932: 125-128) and 2) disappearance of certain word-final fricatives.

(151)

a.	*dada* (← *dadamh*)	/dadə, tadə/	'nothing'
b.	*togradh*	/tʌgrə/	'will, inclination'

However, in the grammatical system of Irish, final schwa has been reanalysed with verbs, either as a marker of the verbal noun or of the past autonomous form.

(152)

a.	*líonadh* /lʲi:nˠə/	'filling'
b.	*líonadh* /lʲi:nˠu:/	'was filled'
c.	*dúnadh* /du:nˠə/	'closing'
d.	*dúnadh* /du:nˠu:/	'was closed'

This is a synchronic interpretation which views /ə/ and /u:/ as productive markers of grammatical categories. Historically, the /u:/ can be traced through a series of shifts (O'Rahilly 1932: 66-67 and 70)

(153)

adh /əð/ → /əɣ/ → /əv/ →	/u:/ in the West and North
	/ə/ in the South

This analysis requires two shifts, dental to velar and velar to labial. It is supported by the dental to velar shift, common in Irish anyway, and the velar to labial shift found in such words as *aniodh* (originally /əˈnjɔɣ/) /(ə)nʲʊv/ → /njʊ/ 'today' (de Bhaldraithe 1945: 97).

1.9.9.1. Vocalisation of fricatives

Vocalised fricatives are evidence of phonetic change in Irish. In postvocalic position in stressed syllables there are many instances and they are the cue to the pronunciation of the vowels they follow.

The fricatives in question were usually voiced. This stands to reason as vocalisation is a diachronic process which extends lenition beyond the area of consonants into that of vowels. The stages of lenition can be represented as follows.

(154)

voiceless Stop	→	voiced Stop	→	voiced Fricative	→	Vowel
		voiceless Fricative	→	voiced Fricative	→	Vowel

Once a positive value has been introduced for voice it cannot be reversed. The sound which stands at the end of the progressions above is orthographically a voiced fricative preceded by a vowel. The four possibilities are as follows (O'Rahilly 1932: 24-26): *bh, mh; dh, gh.*

In Western Irish the vocalisation of *bh, mh; dh, gh* resulted in either a long vowel or a diphthong, either /ai/ or /au/. Certain former consonants tend to be responsible for one or other of the diphthongs so that patterns of orthographic-phonetic correspondence emerged. Thus the first two – *bh*,[101] *mh* – if they lead to a diphthong, result in /aʊ/. The latter two – *dh, gh* – can result in either /ai/ or /au/. The preceding vowel is the determining factor.

(155)

a.	*-abh, -amh, -odh, -ogh* /au/	
	samhradh /saurə/	‘summer’
	cabhair /kaur^{j}/	‘help’
	foghail /faul^{j}/	‘trepassing’
	foghraíocht /fauri:xt/	‘phonetics’
b.	*-adh, -agh* /ai/	
	radharc /rairk/	‘sight’
	aghaidh /aɪ/	‘face’

The large degree of predictability in phonetic realisation of the orthographic sequences just discussed has led to their being used in giving English loanwords written forms such as the following.

(156)

a. *foghlaeir* /faule:rj/ ‘fowler’
b. *babhta* /bautə/ ‘bout’, *dabht* /daut/ ‘doubt’
c. *seabhac* /sjauk/ ‘hawk’

These forms provide additional evidence for the vowel values which the English source words must have had in Ireland at the time of borrowing. Because <*abh*> was used it is clear that the English words must have been pronounced with /au/. In addition the word *seabhac* shows that the initial

/h-/ in English was interpreted as the lenited form of /sj/ which was ‘restored’ in Irish.

The orthographic forms above contain back vowels followed by fricatives. In these cases only *a* and *o* occur. If *u* precedes then the result is a long monophthong: *dubh* /du:/ ‘dark’ (← /dʌv/). When a sequence of back vowel and front vowel (always *i*) occurs the resulting diphthong is /ai/, if the fricative is *dh* or *gh.*

(157)

a. *leigheas* /l^{j}ais/ ‘cure’

b. *staighre* /stair^{j}ə/ ‘stair’

This is understandable as the original sound in both these cases would have been /j/ (← *idh, igh*) which resulted in /ai/ (with lowering of the nucleus vowel to /a/ in the first word above).

Because palatalisation is a productive process in present-day Irish there are alternations between /au/ and /ai/ in grammatical paradigms which involve a change in polarity. In the following example the <*bh*> was absorbed fully into the nucleus of the first syllable and the form became monosyllabic. In the plural of this word syncope occurred and the original vowel in <*-bhainn*> of the singular was deleted. As the <*bh*> was not just palatalised but no longer intervocalic it was retained and triggered diphthongisation of the stressed initial vowel to /ai/.

(158)

a. *abhainn* /aun^{j}/ ‘river’

b. *aibhneachaí* /aivʲnʲəxi:/ ‘rivers’ (Western Irish)

Intervocalic voiceless fricatives

In immediately post-stress position voiceless fricatives can occur as well. The two most common are <*th*> /h/ and <*ch*> /x, xj/ and these can vocalise along the following trajectory.

(159)
voiceless oral Fricative → voiceless glottal Fricative → Vowel

The combination *ph* does not occur word-internally in Irish so there are no diphthongs deriving from it. Word-internal <*sh*> is a signal that a word is a lexical compound and the part beginning with <*sh*> is the head which is lenited by the first part, the modifier. This is true even in cases like the second example below where the modifier (*clap*) does not exist as an independent word.

(160)

a. *dháshiollach* (← *dhá* + *siollach*) ‘disyllabic’

b. *clapsholas* (← *clap* + *solas*) ‘twilight’

Word-internal <*f*> is not common in Irish and is historically the result of a voicing assimilation of /v/ to a preceding voiceless fricative. For instance, the word *uafásach* ‘horrible, enormous’ has an internal /-f-/ (which may be deleted) through a very early assimilation of one fricative to another: Old Irish *uath* + *bás* = *uathbhás* /-θ.v-/ → /-θ.f-/ → /-f-/ ‘death from terror’ (E. G. Quin ed. 1990: 623). Internal <*f*> is also found in the so-called *f*-future of Irish with first conjugation verbs (Ó Buachalla 1985, D. Ó Baoill 2001, C. Watkins 1966) as in *tiocfaidh* ‘come’ (future). However, it is now only pronounced in the autonomous future form, cf. *tiocfar* /ˈtjʌkfər/ ‘will come’.

Non-initial <ch>

In word-initial position *<ch> X* is the result of leniting the onset of a stressed syllable. *X* can occur, however, with a small number of mostly grammatical words which show permanent lenition (examples two and three below).

(161)

a.	*chaith sé* /xa sʲe:/	‘he spent’
b.	*chun* /xʌnˠ/	‘in order to’
c.	*chomh* /xo:/	‘as (much as)’

X also occurs in non-initial positions where it is part of the lexical structure of a word. The non-palatal member of *X*, /x/ is always realised as [x] in Western and Southern Irish. Northern Irish has either [r] or zero depending on whether /x/ is pre-consonantal or word-final (see discussion in II.2.3 above).

(162)

a.	*achar* /ˈaxər/	‘distance, period’
b.	*go brách* /gə ˈbrɑ:x/	‘forever’

The palatal member of *X*, /xj/ can behave somewhat differently. When word-medial it may be debuccalised to /h/ or deleted, lengthening a preceding vowel if this is short.

(163)

a.	*fiche* /fʲixʲə/	[fʲɪhə]	'twenty'	(general)
b.	*oíche* /i:xʲə/	[i:hə, i:]	'night'	(general, Cois Fharraige)

Non-initial <th>

What is written <*th*> is the result of leniting *t* is pre-Old Irish. Until the Middle Irish period, the outcome was /θ/ which then was reduced to a glottal fricative, the value it still has in all the dialects when it is pronounced. In word-final position <*th*> was lost and after the mid twentieth century spelling reform it ceased to be written in many words. [102]

(164) Word-medial <*th*> = /h/

a.	*leithéid* /ˈlʲɛhe:dʲ/	'like, equal'
b.	*mothúchán* /ˈmʌhu:xɑ:nˠ/	'feeling'
c.	*othar* /ˈʌhər/	'patient' (noun)

Word-final <*th*> = zero

d.	*brath* /bra/	'depend (on)'
e.	*meath* /mʲa/	'decline'
f.	*scíth* /sʲkʲi:/	'tiredness, rest'

Dialects vary is their realisations of both <*ch*> and <*th*>. A common development is for either or both of these to have shifted to a labial fricative. Another option is for <*th*> /h/ to be deleted, especially in Cois Fharraige and on the Aran Islands where the second form below is [srɑːnˠ].

(165)[103]

a.	*cluiche* /klɪfʲə/	'sports match'	
b.	*sruthán* /srʌhɑːnˠ/[92]	'stream'	
c.	*ithe* /ɪhə/ → [iː]	'eating'	(Cois Fharraige)
d.	*mothaigh* /mʌhə/ → [muː]	'to feel'	(Cois Fharraige)
e.	*bóthar* /boːhər/ → [boːr]	'road'	(Cois Fharraige)

The range of variation for <*ch*> and <*th*> is considerable. A shift of <*th*> [h] to [ç] is attested as is a shift to [ʃ] in North-West Mayo.

Fricative vocalisation in unstressed syllables

The above discussion of diphthongisation refers to word-initial, stressed syllables. Although fricatives have disappeared in unstressed syllables no diphthongisation takes place. Instead the vowel of the preceding syllable remains as [ə], the unstressed realisation of all short low vowels. Where the unstressed syllable contains a long vowel this remains long, unaffected by the lost fricative.

(166)

a.	*súgradh* /suːgrə/	'playing'
b.	*folamh* /fʌlˠə/	'empty'
c.	*éalódh*[104] /eːlˠuː/	'escape'

1.9.9.2. The case of the <AO> vowel

There are a large number of words in Modern Irish which can be traced back directly to Old Irish and which were written at this stage of the language with digraphs, representing, it is assumed, phonetic diphthongs (Greene 1976, Shaw 1968-71). There is a class of words in Modern Irish which contain a long, stressed vowel, now written <AO>, and which has two basic realisations: /e:/ in Southern Irish and /i:/ in Western and Northern Irish. Both vowels have a retracted pronunciation reflecting their non-palatal status.

(167)

		South	West	
a.	*baol*	[be:-l_{γ}]	[bɨ:l^{γ}]	‘danger’
b.	*taobh*	[te:-v]	[tɨ:v]	‘side’
c.	*saol*	[se:-l^{γ}]	[sɨ:l^{γ}]	‘life’
d.	*saoi*	[se:-]	[sɨ:]	‘expert’

Whatever the pronunciation of the lexical precedents of these words in Old Irish may have been, in present-day Irish they all share a clear characteristic: consonants which precede them are always non-palatal. Phonetically, <AO> is pronounced as a retracted vowel which can and does cause non-palatal assimilation if preceded by a high- or low-polarity palatal consonant as the word-final segment of a preceding word.

(168)

a. *in aon lá* /ɪnj e:nɣ lɣɑ:/ [ɪnɣ e:-nɣ lɣɑ:] ‘in one day’
b. *sin aonach beag* ‘that’s a small fair’ /sɪnj i:nɣəx bjʌg/ [sɪnɣ ɨ:nɣəx b^{j}ʌg]

Going on the non-palatal assimilation which <*ao*> causes in Irish today, it was a back vowel in Old Irish[105] and if the orthography is basically accurate it was furthermore a diphthong with values which all began with a low vowel, viz. /ae, ao, au/. It later shifted[106] to /e:/ (O'Rahilly 1932: 27-38). This mid vowel was further raised to /iː/ in Western and Northern Irish. To some extent there is variation in the realisation still: *aon* /e:n/ ~ /i:n/ 'one', *glaoch* /gle:x/ ~ /gli:x/ 'call', *faraor* /ˈfariːr ~ ˈfare:r/ in Western Irish. In Western Irish the mid vowel is also found as a general pronunciation in more formal contexts, e.g. in the pronunciation [nɣe:-v] for *naomh* 'saint'. In all dialects, i.e. including the South, *daoine* 'people' has /i:/.

In one or two cases <*ao*> has resulted in a front diphthong /ai/ (de Bhaldraithe 1945: 84) though this may be a case of mid front vowel diphthongisation, which is a separate process in Western Irish and which can been seen, for instance, in the alternative realisations for *glaodh (glae)* /glɣe:/ ~ /glɣai/ 'slime'. Words in which <*ao*> stands for /ai/ are *slaodán /*slɣaidɑ:nɣ/ 'cold (illness)' and *faoláin* /failɣɑ:nj/ 'seagull'. In the first of these the diphthongised pronunciation has received orthographic recognition and in accordance with the phonetic realisation of vocalised fricatives has the present-day written form *slaghdán* (Ó Dónaill 1977: 1108). But the source of the /ai/ pronunciation is unshifted /e:/ which in turn goes back to Old Irish <*ae*>, cf. Old Irish *slaetán* 'disease, illness' (E. G. Quin ed, 1990: 547). In keeping with similar shifts of mid level vowels, <*ae*> /e:/ was probably diphthongised rather than further raised to /iː/. The occurrence of /ai/ for <*ao*> is quite prevalent in the Irish of Ring, Co. Waterford, e.g. *naoi* [nɣaɪ] 'nine'.

Because the <*ao*>-vowel never occurs after a palatal consonant (though it can be found before one) there are grounds for regarding it synchronically as the non-palatal counterpart of /iː/ in the West and North and of /e:/ in the South. The vowel would then be an allophone of a long front vowel – /e:/ or /i:/ – determined by the polarity of the syllable onset.

(169)
Western and Northern Irish

/i:/ → [ɨ:] / Cˠ ___
[i:] / Cʲ ___

a. *saol* [sɨ:ɫ] ‘life.NOM’ *saoil* [sɨ:lʲ] ‘life.GEN’
b. *síol* [ʃi:ɫ] ‘seed.NOM’ *síl* [ʃi:lj] ‘seed.GEN’

Southern Irish[107]

/e:/ → [e:-] / Cˠ ___
[e:] / Cʲ ___ [96]

a. *saol* [se:-ɫ] ‘life.NOM’ *saoil* [se:lʲ] ‘life.GEN’
d. *sé* [ʃeː] ‘he’, *séimh* [ʃe:v_j] ‘mild, gentle’

The retraction of /i:/ to [ɨ:] is most pronounced in Northern Irish. In the North of Co. Donegal, in the area around Fál Carrach and on Oileán Toraigh (Tory Island), but also in the extreme South-West (Teileann), the realisation is a high back unrounded vowel [ɯː] which Northern Irish shares with Scottish Gaelic.

1.10. The syllable in Modern Irish

Throughout the present book there are many references to the role of syllables in Irish phonology so it is appropriate to consider the status of the syllable in Irish in a dedicated subsection of the present study. In phonological literature from the early 1970s onwards the syllable was moved

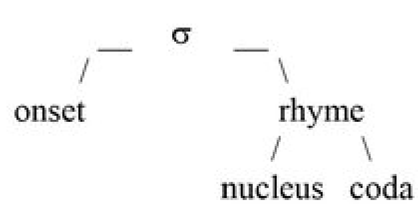
centre-stage in the analysis of sound systems, not least as a reaction to its relative neglect in classical generative phonology of the preceding decade. At least two major volumes have been dedicated to the role of the syllable in phonology, see van der Hulst and Ritter (eds, 1999) and Cairns and Raimy (eds, 2011) while in both editions of the *Handbook of Phonology* (Goldsmith ed, 1995, 2014) there are central chapters on the syllable, see Blevins (1995) and Goldsmith (2014) as well as monograph studies such as Duanmu (2009, see pp. 36-71).
A discussion of the syllable involves clarifying two central questions to begin with (cf. Goldsmith 2014: 168-171).

(170)

(i) What constituents are assumed for syllables
(ii) Can the internal composition of these constituents be optimally characterised by the relative sonority of their segments

Characterisations of the syllable (Fudge 1969: 273, 1987; Hooper 1976: 199; Fallows 1981; Selkirk 1984; Maddieson 1985; Picard 1987a; Vennemann 1988; van der Hulst and Ritter 1999: 22-29) generally assume two major constituents with a further subdivision of the second, i.e. 1) an onset and 2) a rhyme with the latter divided into 2a) nucleus and 2b) coda. Both the onset and the coda can be internally complex with a maximum of three segments in an initial and of two segments in a coda in Modern Irish.

(171)



There are many aspects of Irish phonology and morphonology which support this division. Consider the domains of the two chief morphonological processes in Modern Irish 1) initial mutation and 2) polarity reversal (palatalisation or velarisation depending on the nature of the input). Here there is an absolute division of syllables into onset on the one hand and rhyme on the other, for instance, polarity reversal never affects the onset and initial mutation never the coda of a syllable.

Table 23. Domains of morphonological processes in Modern Irish by syllable position

initial mutation	vowel gradation	polarity reversal
	rhyme	
onset	nucleus	coda

The question of the composition of syllable constituents primarily concerns onsets and codas. These constituents are consonantal while the nucleus is usually vocalic but can also show a syllabic nasal as in the second syllable of English *button* [bʌtn̩] or Munster Irish *maidin* [madʲn̩] 'morning'.

Sonority-based analyses of onsets and codas (Zec 1995) assume that, preferentially, a syllable consists of a maximum onset[108] and a minimum coda with an increase in sonority within an onset and a decrease in a coda,[109] should these positions be segmentally complex in a given syllable (Dogil 1988, 1992; Dogil and Luschützky 1990; Daland et al. 2011: 198). 'Maximum'/'increase' and 'minimum'/ 'decrease' refer in this context to the value of segments on a scale of sonority (vowel-like quality, the opposite of consonantality). Such scales have been an established instrument in analyses in recent decades, see Hooper (1976) and Vennemann (1982) as representative examples for the re-establishment of sonority scales in phonology. These scales go back at least to Jespersen: his scale (Jespersen 1913: 191) is only slightly different from that in Hooper (1976: 196+206), and in his use of the term *Schallfülle* ‘sonority’ Jespersen (1913: 190) is close to late twentieth-century understandings of sonority (Nathan 1989: 60-62; Rice 1992: 65). A maximally differentiated sonority scale could be arranged as follows, with a few contestable positions, e.g. those of voiced stops and voiceless fricatives.

(172)

vowels	0	greatest sonority
semi-vowels	1	
/r/	2	
/l/	3	
nasals	4	
voiced fricatives	5	
voiceless fricatives	6	
voiced stops	7	
voiceless stops	8	greatest consonantality

In phonological literature of the 2000s there has been much criticism of sonority scales. This criticism is largely centred around the lack of predictability which sonority shows (see Harris 2006 as a typical example of such criticism). It is quite justified to reject sonority should any predictive power be attributed to it or should explanations of phonological change be regarded as always deriving from considerations of sonority. However, as a heuristic tool by means of which one can explain tendencies across languages and local distributions of segments within syllables for individual languages, the notion of sonority is useful. Epenthesis in Modern Irish (see II.2.8 below) and phonotactic distributions (see following section) can be accounted for satisfactorily by appealing to the sonority values of segments (Fudge 1976). Nonetheless, bilingual speakers in Ireland with comparable competence in Irish and English will produce *banbh* ‘piglet’ as [banəv] or *tarbh* ‘bull’ as [tarəv] and have *swerve* [swɚ:v] when speaking English and not necessarily [swɛrəv]. So the sonority-motivated epenthesis of Irish does not necessarily carry over into Irish English, i.e. it lacks predictive power for Irish English, though historically, due to the much stronger position of Irish, there was transfer into Irish English, cf. pronunciations like *term* [tɛrəm] and *girl* [gɛrəl] which are well attested both diachronically and in many rural varieties today.

2. Phonotactics

Phonotactics refers to patterns of segments which occur within the boundaries of the phonological word. Sounds form typical combinations which are in keeping with restrictions which apply within a language (Ní Chiosáin 1999). Apart from the sounds which Irish has, for instance *ɣ* and *X*, it is the arrangement of the sounds which makes them immediately recognisable as Irish: *cnuasaigh* /knɣuəsə/ ‘gather food’, *a mbláthannaí* /ə mlɣɑ:hənɣi:/ ‘their flowers’, *dlúthcheangal* /dlu:xjaŋgəlɣ/ ‘close connection’, *a ghrá* /ə ɣrɑ:/ ‘his love’. Added to this is the frequency with which sounds occur, e.g. a high frequency of /ə/, /əx/, /əxt/ and /i:/ in unstressed final syllables is typical of Irish.[110]

In the phonotactics of a language one can distinguish between what is theoretically possible and what is actually attested. For example, * *blick* is legal in English phonotactics but not attested as opposed to * *bnick*, which is neither. In Irish a similar situation is found, for instance the word * *blón* /blɣo:nɣ/ does not exist but it is phonotactically possible. The form * *bnón* /bnɣo:nɣ/ is illegal in principle as the combination of labial stop and nasal in syllable-initial position does not occur.

The distribution of segments is not affected by their value for palatality. For instance, among monosyllables which end in a long vowel one finds *teo-* /tjo:/ 'heat, warm' (prefix) but there is no corresponding syllable with a non-palatal consonant **tó* /to:/. This is accidental as one can conclude from the pairs of other consonants which occur in the same environment as /tj/ above: *ceo* /kjo:/ 'fog', *comh* /ko:/ 'co-' (prefix), *(go) deo* /djo:/ 'forever', *(a) dó* /do:/ 'two'.

Phonotactics encompasses not just sequences of sounds but the preferred structures of words. This can be shown by considering the forms of English loanwords which entered Irish during its history. In general one can say that English monosyllables were rendered disyllabic by the addition of a final schwa as the forms in the following table illustrate.

Table 24. Canonical disyllabic form of English loans

ápa 'ape'	*babhta* 'bout'	*balla* 'wall'
banda 'band'	*bosca* 'box'	*gúna* 'dress'
sluma 'slum'	*slúpa* 'sloop'	

The restructuring of English words which is in evidence here does not apply to contemporary loans. These maintain their structure, even if monosyllables, and also have segments, such as voiced affricates, which historically would have been devoiced and metathesised.

Table 25. Phonological adaptation with English loans [111]

Devoicing and metathesis with Anglo-Norman loans in Irish[2]					
college	/kʌlˠɑːʃtʲə/	/-dʒ-/ >	/-tʃ-/ >	/-ʃtʲ-/	*coláiste*
page	/pɑːʃtʲə/	/-dʒ-/ >	/-tʃ-/ >	/-ʃtʲ-/	*páiste*
Contemporary loans without affricate adjustment (devoicing / metathesis)					
jeep	/dʒiːp/	*job*	/dʒaːb/		

2.1. Permissible clusters

Syllable-initial consonant clusters can have up to three elements, much as in English though the combinations of sounds are quite different. Two-item initial clusters show considerable flexibility because sequences of velar stop and coronal nasal, /kn-/ and /gn-/, are legal as are clusters of two sonorants (nasal + liquid), e.g. /mr-, ml-/. When two non-sonorants follow on each other initially then the sequence is always a fricative plus a stop.

(173)
C_1C_2V- C_1 = fricative; C_2 = stop

The only fricative which can precede a stop is *S* which automatically excludes *F, V, X, ɣ* and /h/. As all clusters agree on the value for [voice], *B, D, G* are not found after *S* either. But in Irish, as in other languages like English and German, there is a lack of aspiration of voiceless stops after *S*. This has led many Irish authors to transcribe the stops in this position as voiced stops which is phonetically inaccurate, e.g. *scéal* /ʃg’eːL/ ‘story’ (de Bhaldraithe 1945: 29, his transcription). The origin of this practice may lie in the older spelling which indicated the lack of aspiration by means of an orthographic voiced stop, i.e. *sgéal.*

Three-term initial clusters must show the sequence: *S*, stop and liquid. All but one of the six possibilities are attested: *SPR-, STR-, SKR-, SPL-, SKL-,*STL-.*

(174)
C_1C_2SV- C_1 = *S*; C_2 = *P, T, K*; S = *R, L*
spreagadh /sjpjrjagə/ 'excitement'
stráice /strɑ:kjə/ 'strip'
scread /sjkjrjad/ 'scream'
spleách /sjpjljɑ:x/ 'dependent'
sclog /sklɣʌg/ 'gulp, gasp'

Due to mutations a number of initial clusters are possible in Irish which are not attested in the lexicon of the language, see the section *Phonotactics and the initial mutations* below.

Final clusters

Final sequences in Irish tend to mirror initial sequences: with final clusters the first consonant is either a sonorant or a fricative with the second a stop.

(175)
-VC_1C_2 C_1 = sonorant/fricative; C_2 = stop

In Western and Northern Irish there are no syllabic sonorants as in English *table* /teɪbl̩/ or *button* /bʌtn̩/ so that sonorants cannot follow stops in codas. In comparable cases an intervening vowel is found, e.g. *maidin* /madjɪnj/ 'morning'. However, in Munster Irish such syllable sonorants are indeed found and this is a significant delimiting feature vis à vis other dialects. Here the pronunciation of *maidin* is /madʲn̩ʲ/.

Voiceless and voiced stops contrast after sonorants in this position (first and second examples below). If C_1 is a fricative then it may be either *S* or *X*, the latter being far more common (third and fourth examples).

(176)

a.	*ceart*	/k^{j}art/	‘correct’
b.	*ceird*	/k^{j}air^{j}d^{j}/	‘trade, occupation’
c.	*blaosc*	/bl^{ɣ}iːsk/	‘eggshell’
d.	*nuacht*	/n^{ɣ}uəxt/	‘message, news’

Voiced stops after sonorants are in fact restricted to those after *R*. The final sequences *-ND* and *-LD* have long since lost their stop element through assimilation, leading to geminates and later simplification due to loss of distinctive consonant length by the end of the Middle Irish period, e.g. *cland* (< Latin *planta*) → *clann* ‘children’.

Final clusters in Irish can have a maximum of two elements. Mitigating against complex clusters is the fact that Irish does not have any consonantal suffixes, or if it does then these can only be appended to single-segment final clusters (second example below).

(177)

a.	*gealt* + *-a* plural suffix	*gealta* /g^{j}al^{ɣ}tə/	‘lunatics’
b.	*dán* + *-ta* plural suffix	*dánta* /dɑːn^{ɣ}tə/	‘poems’

What is common to both the plural formations above is that they result in a vowel final, disyllabic form. Where plural formation does not increase the number of syllables in a word then it must always be realised by a change in polarity. With this type of plural formation the number of elements in the stem-final cluster remains unchanged.

(178)

a.	*bord* [-palatal] + reversal	*boird* [+palatal]	‘tables’
b.	*poll* [-palatal] + reversal	*poill* [+palatal]	‘holes’

A further restriction in Irish is that it does not allow sequences of two fricatives even when a suffix is added to a stem. This means that none of the following combinations can occur.[112]

(179)

fs	vs	sf, sv	xf, xv	ɣf, ɣv
fx	vx	sx, sɣ	xs	ɣs
fh	vh	sh	xh	ɣh
fɣ	vɣ			

Prohibition on sequences of similar segments

The statement just made with reference to sequences of two fricatives can be broadened to include any two segments of the same type, i.e. plosives, fricatives or sonorants. None of these are allowed in immediate succession within the boundaries of a non-compounded word.

Table 26. Impermissible and permissible segment sequences in Irish

	Impermissible		Permissible			
1:	F + F	/-sx-/	F + S	/-sn-/	*trasna*	‘across’
2:	P + P	/-kt-/	F + P	/-xt-/	*seachtú*	‘seventh’
3:	S + S	/-mn-/	S + P	/-lk-/	*folcadh*	‘bathing’
	F=fricative, P=plosive, S=sonorant					

The above situation is that of Modern Irish and is the result of a long period of phonological development in which the phonotactic distributions found today became established, these allowing the generalisation given in Table 26. Why the patterning is as observed has been considered by Gussmann (2002: 92-95) who gives an analysis in terms of government phonology without, however, recognising that the patterning is more general. A single constraint, which will account for the distribution in Table 26, is the following.

(180) Consonants in sequence always show an alternation in type.

Not only will this constraint explain the above distribution, it also provides a principled explanation for diachronically attested shifts such as /mn-/ > [mr-] as in *mná* [mrɑː] 'women' or *Luimneach* > *Limerick* in English.[113] In such instances a sequences of two nasals is resolved by shifting the second sonorant in the cluster from a nasal to a liquid thus rendering the segments less similar (allowing for the subdivision of sonorants into nasals and liquids).[114] At a much greater time depth there are instances like *seacht* (Old Irish *secht*) 'seven' from an Indo-European root **sept* (cf. Latin *septem*) which probably first had a coda /-kt/ from a retraction of /p/ to /k/ before /t/ and then fricativisation of the stop occurred yielding the sequence /xt/ which is very common in Irish, either as a single coda, as in *bocht* /bʌxt/ 'poor', or a coda+onset when in word internal position, e.g. *crochta* /krʌx.tə/ 'hung'.

Phonologists have recognised, when observing other phenomena, such as tone, that a continuous alternation in type is preferred in many languages. This has been labelled the 'obligatory contour principle' (Odden 1986) and has been applied to the analysis of segmental variation by Guy and Boberg (1998) who have shown that consonantal cluster reduction, such as *fact* / fækt/ > [fæk], is motivated by the avoidance of codas of like segments. In Irish there is a continuous changing contour in segment type (fricative ~ plosive ~ sonorant) which excludes sequences of like segments already in the phonological forms of words. This patterning has an additional advantage in that, by maximising contrast between adjacent sound segments, it renders perception of phonological distinctions and hence the recognition of words easier for the hearer. In this respect the Irish patterning supports the dispersion theory of contrast as developed by Edward Flemming (see, for example, Flemming 2001, especially p. 24-26) and applied to Russian and Irish by Jaye Padgett and Máire Ní Chiosáin (see Padgett 2001; Ní Chiosáin and Padgett 2009).

The distribution of segments in consonantal sequences discussed above holds primarily for lexically simple words. Compounds need not comply to the phonotactic restrictions on like segments, cf. *seachghalar* /sʲaxɣalˠər/ 'complication' which shows /-xɣ-/. Indeed the second fricative /ɣ/ results from the lenition of /g/ due to the higher-order requirement that the second element of a compound be lenited, this overriding the phonotactic restriction on sequences of like segments. The same is true on a phrasal level, consider a sentence like *Táimid sách sásta leis an toradh* 'We are fairly satisfied with the result' where *sách* is followed by *sásta* yielding /sɑ:x sɑ:stə/.

Palatality agreement in clusters

/x/ has the peculiarity in Irish of not assimilating in place of articulation to a following palatal consonant (O'Rahilly 1932: 192). The retention of /x/ in the environment of __ Cʲ means that there is no alternation before /x/ with short vowels in a syllable nucleus which would have a vowel change otherwise: *bocht* [bʌxt] 'poor', *boichte* [bʌxtjə] 'poorer' (not *[bɪxʲtʲə]). /rj/ is not palatal in a syllable coda if followed by a stop. This means that the clusters /-rjtj/ and /-rjdj/ are phonetically [-rtj] and [-rdj] respectively, cf. *cúirt* [ku:rtj] 'court', *ceird* [kjairdj] 'trade, craft'. However, when /rj/ is the only coda element then it is also phonetically palatal, cf. *cóir* /ko:rj/ [ko:rj] 'share, due; justice'.

Phonotactics and the initial mutations

Due to the change in consonant articulation on initial mutation combinations of sounds occur in initial clusters which are not found in the lexical forms of words. For instance, the sequence *ML* is common when it results from nasalizing *B* which is followed by *L* as in *a mblas* ‘their tastes, accents’, *i mbliana* ‘this year’. The occurrence of *N* as C_2 is somewhat more restricted, four of nine consonants are permissible.[115] Where the second element of an initial cluster is *R* or *N* there are many more attestations in the lexical forms of words.

(181)

a.	MN-	*mná*	‘women’
b.	TN-	*tnúth*	‘envy’
c.	KN-	*cnagach*	‘tough, hardy’
d.	GN-	*gnó*	‘business, affair’
e.	SN-	*snaidhm*	‘knot’

The application of mutations is furthermore subject to certain restrictions. Only in those cases where the second consonant is a sonorant can lenition apply: *a bhláthanna* ‘his flowers’, *an-chnagach* ‘very tough’, *a threise mheabhrach* ‘his intellectual power’. Thus no consonant cluster CCCV can have lenition applied to its initial element because the second element is not a sonorant: *an-spleách* ‘very dependent’, *Scrios sí a gúna*. ‘She ruined her dress.’

The phonotactics of S

The single sibilant of Irish is also the most flexible segment (in both its palatal and non-palatal versions). It is subject to least restrictions regarding cluster formation, showing the greatest number of combinations with following consonants. It can also be lenited when followed by a vowel or any sonorant. The complete distribution of *S* in CCV structures with reference to lenitability is as follows.

(182) Lenitable combinations of *S* and a further segment

S + vowel /h/	*a shaol*	‘his life’
S + sonorant /h/	*a shláinte*	‘his health’
	a shnua	‘his complexion’
	a shrón	‘his nose’

Non-lenitable combinations of *S* and a further segment

S + /m, p/	*smacht*	‘power, control’
	spéis	‘affection’
S + /t, k/	*staid*	‘condition’
	scoil	‘school’
S+ /f/	*sféar*	‘sphere’

2.2. Cluster simplification

The most common reduction of clusters involves homorganic sequences of a stop and sonorant. In these instances it is the sonorant which is lost: [116] *Pádraig* /pɑ:rɪkj/ ‘Patrick’, *Breatnach* /bjrjanɣəx/ ‘Walsh/Welsh’ (surname), *codladh* /kʌlˠə/[117] ‘sleep’, *taitneamh* /tanjəv/ ‘shining’. The reverse of this development is also attested. Here an epenthetic stop has arisen to provide a strong onset for a second syllable,[118] e.g. *banrán* /bɑ:nɣtrɑ:nɣ/ ‘complaining’, *anraith* /ɑ:nɣtrə/ ‘soup’, *malrach* /malɣdrəx/ ‘young boy’ (de Bhaldraithe 1945: 36-37). There is also a strengthening of word-final syllable codas by introducing a homorganic voiceless stop: *ceangal* /kjaŋgəlɣt/ ‘bond’, *oiliúint* /ɛlju:njtj/ < *oileamhain* ‘nutrition; training’, *bagairt* ← *bagair* /bagərjtj/ ‘threatening’ (de Bhaldraithe 1945: 115).

2.3. Sonorant shift

There is a historically attested shift of *N* to *R* in clusters where the sonorant is the second element: *mná* /mnˠɑː/ → /mrɑ:/ ‘women’, *cnoc* /knˠʌk/ → /krʌk/ ‘hill’, *imní* /iːmʲnʲiː/ → /i:mjrji:/ ‘worry’, *cneasta* /kjnjastə/ → /kjrjastə/ ‘honest’, *gníomh* /gʲnʲiːv/ → /gjrji:v/, ‘act, action’, *gnó* /gno:/ /gro:/ ‘business’. It is obvious that this change is quite old. It must at least date back to the Middle Irish period as it is present in Scottish Gaelic, as is the shift of former interdental fricatives (the result of leniting *T* and *D*) to the velar/glottal area.

Sonorant shift may also involve *R* and *N*, often with loans from English: *N* → *L*: *kniotáil* /kjljɪtɑ:lj/ ‘knitting’, *R* → *L*: *feirmeoir* /fjɛljɪmjo:rj/ ‘farmer’. The shift of *N* to *R* is also attested onomastically, e.g. in *Luimneach* /lˠɪmʲnʲəx/ ‘Limerick’.

The [ŋ] in *teagmháil* ‘contact’ is an example of nasal insertion before a voiced stop similar to that which occurred in Middle English, cf. *nightingale, messenger* from earlier forms without the word-internal nasal (Luick 1940: 992; Jespersen 1909: 36).

(183)
teagmháil ~ *teangmháil* ‘contact’

2.4. Interpretation of [h]

In Irish /h/ is most often the result of mutating both *S* and *T*. Furthermore, it occurs for zero mutation when the host lexical stem is vowel-initial, e.g. *a hainm* /ə hanjmj/ ‘her name’. With regard to palatality, /h/ shows the following distribution.

(184)

$$
/\mathrm{h}/ \;\rightarrow\; \begin{array}{l} [\text{ç}] \;/\; \_\_\_\; V^{\text{ɣ}} \\ [\mathrm{h}] \;/\; \_\_\_\; V^{j} \end{array}
$$

a.	*a Sheáin* /ə hɑ:n^{j}/ [ə çɑ:n^{j}]	‘John.VOC’
b.	*a theach* /ə hax/ [ə hæ:x]	‘his house’

The distribution applies irrespective of the source of /h/. There results from this an overlap with the voiceless palatal fricative which is the lenited form of the palatal velar /kj/: /kj/ + lenition → /xj/ [ç], e.g. *ceann* /kjɑ:nɣ/ [kjɑ:nɣ] ‘head’, *a cheann* /ə xjɑ:nɣ/ [ə çɑ:nɣ] ‘his head’.

The most frequent occurrence of /h/ is in the following environment: /h/ / # ___ V. The boundary here can refer to either a morpheme or a word. But as /h/ is also the result of leniting *S* it can also occur in the following environment: /h/ / # ___ *N, L, R,* e.g. *a shrón* /ə hru:nɣ/ ‘his nose’, *a shlat* /ə hlɣat/ ‘his stick’, *a shnáth* /ə hnɣɑ:/ ‘his thread’. On a phonetic level /h/ before a sonorant is realised as a voiceless sonorant, i.e. the realisations for the three words just given would be [ə r̥u:nˠ], [ə l̥ˠat], [ə n̥ˠɑ:].

For a small number of loans from English, some of which are quite early, an initial /h-/ is found in the citation form of the words, e.g. *hata* /hatə/ ‘hat’, *halla* /halɣə/ ‘hall’, *haca* /hakə/ ‘hockey’.[119] In addition, there are neoclassical words from English which have equivalents to Irish, especially in formal usage, e.g. *hipitéis* ‘hypothesis’, *histéire* ‘hysteria’, *homalóg* ‘homologue’.

Lastly mention should be made of cases where initial /h-/ appears in expressive phrases such as *Ní raibh hob nor hé as* ‘There was not a budge out of him’, *Huga leat!* ‘Get out of here!’, *Bhí húirte háirte ar siúl* ‘There was a hubub going on’, *Hois, hois* ‘Shoo!’ (used with animals).

2.5. Vowel phonotactics

All vowels may occur initially and finally and not infrequently on their own, e.g. /ai/, /oː/, /ɑː/, /a/, /iː/, etc. are lexical or grammatical words in Irish: *aoi* /iː/ ‘guest’, *é* /eː/ ‘him’, *ea* /a/ ‘is’, *aghaidh* /ai/ ‘face’, *ádh* /ɑː/ ‘luck’, *ó* /oː/ ‘from’, *uaigh* /uə/ ‘grave’. Short final vowels, e.g. *crith* /kjrjɪ/ ‘shake, tremble’, *te* /tjɛ/ ‘hot’, can only occur in monosyllabic words as they would automatically be reduced to /ə/ with polysyllabic forms where they would not carry primary stress. The high incidence of final /-iː/ can be seen from the range of contexts in which it occurs. This is particularly true of Northern Irish.

(185)

(i) in comparative, superlative of adjectives *suntasach* ‘strange, unusual’ *níos, is suntasaí* ‘more, most unusual’
(ii) as marker of the verbal adjective *tógtha* /to:ghə/ > [toːk$^{(j)}$iː] ‘taken’
(iii) due to the vocalisation of intervocalic /h/ *imithe* /ɪm^{j}ɪhə/ > [ɪm^{j}iː] *oíche* /iːhə/ > [iː]
(iv) in many plural formations /ə/ versus /iː/ in the standard and the dialects *laethanta* / *laethantaí* ‘days’ *amanna* / *amantaí* ‘times’

The use of long plurals, see (iv) above, may be in part motivated by the wish to signal plurals in a manner which is different from other inflections involving /i:/.

2.6. Prefixed consonants

F and P in Irish It is undoubtedly part of the phonological awareness of native speakers of Irish that a vowel-initial word can potentially contain a retrievable *F* which was lenited to zero. This has led to many cases in the various dialects (de Bhaldraithe 1945: 114; Ó Cuív 1944: 125-126; Mhac an Fhailigh 1968: 165-167) of an *F* being 'restored' to words where it never existed historically: *uar* → *fuar* 'cold', *ás* → *fás* 'growth', *adhb* → *fadhb* 'problem'.[120] There are also cases of fortition where *F* is strengthened to *P*, e.g. *feircín* → *peircín* /pjɛrjkji:nj/ 'firkin'. This is a genuine case of fortition as the noun is a loanword from Middle English *ferkyn* where the /f/ of the original form of the Irish word case be seen in the source of the loan. There are also cases where *P* is prefixed to a word becoming part of its lexical form, e.g. *leadhb* → *pleadhb* /pjljaib/ 'listless fool' (de Bhaldraithe 1945: 114). The historical development in this case might have been Ø → *F* → *P* but as there is no attested intermediary stage with *F* it is not possible to say whether there was such a stage. Indeed there are several other words with a prefixed *P* which are related in meaning – *(p)leib* 'dolt', *(p)leidhce* 'fool', *(p)leota* 'clumsy fool', *plobaire* 'babbler, excessive talker', *plucaire* 'impertinent, cheeky person' – so that the prefixation may well have been semantically motivated (de Bhaldraithe *loc. cit.*).

Prefix T The *T* which is found word-initially without etymological justification is different from the cases of *F* and *P* just discussed. It only occurs pre-vocalically, e.g. *aiséadach* → *taiséadach* /tasje:dəx/ 'shroud'. What is involved here is probably a false segmentation –

metanalysis – of the kind which led to such instances in English as *an ekename* → *a nickname* or *an apron* from French *naperon* ‘apron’ (Onions 1966: 609 + 46). The prefix *T* of the nominative singular with the definite article, i.e. *an t-aiséadach* ‘the shroud’, came to be regarded as belonging to the word independently of the presence of the article.

Prefix S The comments on prefix *P* imply that the sequence *PL-* has a certain phonaesthetic value (Firth 1964 [1930]: 190-215; Hinton, Nichols and Ohala eds, 1994). A similar motivation may be responsible for the prefixation of *S* to the beginning of a word. Consider the following selection of such items; where <*s*> is in brackets the forms are attested without the initial sibilant.

(186)

(i) human referents

(s)crománach	‘a tall crooked person’
(s)laibéara	‘a garrulous person’
(s)glugaí	‘soft, flabby person’
(s)gluitéara	‘glutton’
(s)mairtíneach	‘cripple’
scabhaitéir	‘blackguard’
scadhrach	‘chattering woman’
slabóg	‘slovenly woman’
slaimiceálaí	‘untidy person, messer’
scanróir	‘miser, skinflint’

(ii) inanimate referents

(s)giobal	‘rag’
(s)liobar	‘tatter’
(s)grúnlach	‘dregs’
(s)meadráil	‘act of messing’
scadarnach	‘inferior things, refuse’
scáinteacht	‘thinness, sparseness’
slabáil	‘puddling, sloppy work’
scaits	‘whopping lie’
scáfar	‘frightful, dreadful’

Group (i) denotes individuals with unpleasant or unaesthetic characteristics and the items in group (ii) also have negative connotations. However, it is difficult to definitively associate a certain sound or sound sequence with a given meaning or meaning component. There are examples of words in SC(C)- with a positive connotation or occasionally some which are both pejorative and ameliorative: *scafánta* ‘strapping, tall and vigorous’, *scuabaire* ‘dashing person’, *slachtmhaireacht* ‘neatness, tidiness’, *scailleagántacht* ‘lankiness; liveliness, breeziness’.

2.7. Minor processes

2.7.1. Internal lenition and voicing

This involves the lenition of the initial consonant of the second element of a compound (the head in Irish), e.g. *foclóir* /fʌklo:rj/ ‘dictionary’, *gearr-fhoclóir* /gjɑ:rʌklo:rj/ ‘short dictionary’, and is very widely attested in the present-day lexicon of Irish.

(187)

Internal lenition in compounds

a. *mathshlua* (← *maith* 'good' + *slua* 'crowd) 'large crowd'
b. *géarfhocal* (← *géar* 'sharp' + *focal* 'word') 'nasty remark'
c. *dubhfhluich* (← *dubh* 'black' + *fluich* 'wet') 'very wet'
d. *comhbhrón* (← *comh* 'co-' + *brón* 'sorry') 'sympathy'
e. *ildaite* (← *il* 'many' + *daite* 'coloured') 'colourful'
f. *ilteangach* (← *il* 'many' + *teangach* 'tongued') 'multilingual'

This kind of sandhi is obligatory in the word-formational type 'modifier + head'.[121]

(188)
Modifier + Head compounding

a.	*ceannchluiche* (ch = /x/)	'final match'
	'head' + 'game'	
b.	*tromchúiseach* (ch = /x/)	'serious, grave'
	'heavy' + 'cause.ADJ'	
c.	*neamhbheartaithe* (bh = /v/)	'unintentional'
	'not' + 'planned'	

Where the final consonant of the modifier is homorganic with the initial one of the head there is no lenition, e.g. *ceanndána* (← 'head' + 'bold', not *ceanndhána*) 'wilful, stubborn'.

There are other cases of sandhi processes which are, or were, phonetically motivated and which play no role in the morphonology of the present-day language. Two such processes are prominent. The first is voicing and the second fricativisation. While the latter is phonetically identical to morphonological lenition, the former process – voicing – is not an initial mutation of its own (unlike Welsh), but a part of the nasalisation mutation.

Throughout the history of Irish many cases of internal voicing occurred but for the modern language those which took place on compounding are still recognisable because the elements of the compounds, which do not have this voicing or lenition, are still present.

(189)
Internal voicing in compounds

	K <c> → G <g>	
a.	*iargúlta* (← *iar* ‘back’ + *cúlta* ‘cornered’)	‘remote’
b.	*aingiallta* (← *ain* ‘not’ + *ciallta* ‘sensible)	‘irrational’
c.	*éigiontach* (← *éi* ‘un-’ + *ciontach* ‘guilty’)	‘innocent’
d.	*éiginnte* (← *éi* ‘un-’ + *cínnte* ‘certain’)	‘uncertain’
d.	*éagóir* (← *éa* ‘un-’ + *cóir* ‘just’)	‘injustice’
e.	*éigeart* (← *éa* ‘un-’ + *ceart* ‘correct’)	‘injustice’
	T <t> → D <d>	
c.	*éadrom* (← *éa* ‘un-’ + *trom* ‘heavy’)	‘light’

F <f> → V <bh>

b. *ainbhios* (← *ain* ‘not’ + *fios* ‘knowledge’) ‘ignorance’

The internal voicing, exemplified above, is now fossilised and no new instances arise in strong contrast strongly with the Modifier + Head type of compound.

2.7.2. Final devoicing

A further type of sandhi which can occur in Irish and which it has in common with languages such as Russian, German or Sanskrit (Thumb and Hauschild 1958: 316-333) is final devoicing. This has not, however, spread to cover all instances of final voiced obstruents in Irish. It is found most frequently with final velar stops in unstressed syllables, e.g. *Pádraig* /pɑ:rɪkj/ ‘Patrick’, and can occur in more words such as *Nollaig* /nʌlɣɪkj/ ‘Christmas’. The non-palatal counterpart of /g^j/, i.e. /g/, is not devoiced finally. The result of this is that, for many speakers, there is variation in voice for the final consonant of words between the nominative and genitive forms: *bolg* /bʌlg/ [bʌlɣəg] ‘belly.NOM’, *boilg* /bʌljgj/ [bɛljɪkj] ‘belly. GEN’. There is much onomastic evidence that this final devoicing extends back in the history of Irish: nearly all the anglicisations of the common Irish placename element *carraig* ‘rock’ are rendered as ‘Carrick’, e.g. *Carraig-na-Súire* ‘Carrick-on-Suir’; in addition the word *reilig* /rɛlɪg/ ‘graveyard’ is often anglicised in placenames as [rɛlɪk] suggesting final devoicing, e.g. *Poll na Reilige* [hollow of the graveyard] ‘Pollnarelick’.

2.7.3. Assibilation of glottal fricatives

A development (O’Rahilly 1932: 206) which is the reverse of the lenition of *S* is found in Erris (Mhac an Fhailigh 1968: 156-157) and consists of a shift of /h/ to /ʃ/ after /r/, cf.

gearrtha [gja:rʃi:] ‘cut’, *threibh* [ʃrjevj] ‘tribe’ (lenited form). These examples show the influence of /r/ causing retraction of /s/ yielding [ʃ] phonetically. For Northern Irish Wagner (1979 [1959]:
29) also has examples of /s/ → [ʃ] / __/r/__ which again show /r/-retraction. De Bhaldraithe (1945: 51) remarks that /rj/ becomes [ɹ] and that [ʃ] is retracted and velarised.

2.7.4. Depalatalisation of /r^j/

This is common in all dialects, de Bhaldraithe (1945: 111), Mhac an Fhailigh (1968: 44), Wagner (1979 [1959]: 30), but it is confined to certain phonotactic environments. The most common position for depalatalisation is (1) word-initially, *rian* /riənɣ/ ‘path, course; trace’, *rince* /rɪŋjkjə/ ‘dance’, and (2) in a cluster with a coronal obstruent, *ceird* /kjairdj/ ‘trade’, *srian* /sri:nɣ/ ‘bridle, rein’.

The general tendency to depalatalise /rj/ initially is matched by the opposite tendency in Connemara Irish to palatalise /r/ when in an intervocalic position. Thus the forms *amáireach* ‘tomorrow’ and *aimhreas* ‘doubt’ have the official forms *amárach* and *amhras* where the medial non-palatal /r/ can be seen.

2.7.5. Assimilation across word boundaries

The most common type here is that concerning the value for palatality which is changed due to regressive assimilation, *Más é do thoil é* /mɑ:sj e: də hʌlje:/ [mɑ:ʃe:…] ‘(if you) please’, *Níor chas sí air* /nji:r xa sji: ɛrj/ […xaʃe:…] ‘she did not meet him’. Geminates do not arise due to assimilation in Irish. What assimilation causes is simplification with the latter element usually prevailing, e.g. /s + s^j/ → /sj/ [ʃ].

(190)

a. *’S é* [ʃe:] *an chaoi cheart.*
‘It’s the right way.’

b. *An tusa an runaí nua? ’S mé* [sme:]
‘Are you the new secretary? Yes, that’s me.’

Forward assimilation can only operate if the resulting cluster is phonotactically acceptable, hence [sme:] and not [ʃme:] for *’S mé* as [ʃm-] is not a permissible phonotactic cluster in Irish.

2.7.6. F in Irish

The reflexive pronoun *féin* /he:nj/ contains neither /fj/ as an initial consonant nor Ø (zero) which would be found if the form was permanently lenited. The development of /h/ in this position has been the subject of many comments (O’Rahilly 1932: 81). Gleasure (1968: 79-87) devotes an article to the *féin* = /he:nj/ problem and contests O’Rahilly’s explanation of its origin. The typical position for *féin* is after a pronoun which it qualifies: *Tá siad féin anseo* [is-them-self-here] ‘They themselves are here’.

The weakening of /f^{j}/ and the devoicing it triggers in both pronominal and verbal forms has led some scholars to connect the two. Pedersen (1897: 19) sees the development of /f^{j}/ → /h/ as normal and quotes the forms of *fuair* ‘cold’ in some varieties of Scottish Gaelic and of Manx, which have initial /h/, as support for this argument.

Another account of the origin of /h/ is to take the third person plural of certain prepositional pronouns and to link up another development which occurred with these forms. Here a stop has developed: *fúthu* /fu:b/ ‘under-them’, *leo* /ljo:b/ ‘with-them’, *acu* /akəb/ ‘at-them’. The original forms had /v/, e.g. *acabh*, and O’Rahilly (1932: 81) assumes that on contact with *féin* a stop developed (an assumption supported by the presence of this stop in the third person plural forms). The analyses of both O’Rahilly and Gleasure stand or fall on the assumption that /v/ + /f^{j}/ yielded /p/ by dissimilation of two fricatives. The full chain of development according to O’Rahilly would be as follows.

(191)

$V/v/\#/f^j/V$	$\rightarrow$	$V\#/p^j/V$	$\rightarrow$	$V/b/\#/h/V$
féin		*péin*		*héin*

The intermediate orthographic form is actually attested with the reanalysis of /pj/ [phj] as /b/ and /h/. But alternatively the form *péin* could be explained via a false analysis of final /p/ of the prepositional pronouns (devoiced by the /fj/ which disappeared) as belonging to the following reflexive pronoun. Such an analysis is conceivable as *féin* functions almost as an enclitic to the prepositional pronoun and so no articulatory pause need arise between the two forms which would indicate which of these /p/ belonged to.

In each of the proposed analyses the final /v/ must have been fortified to /b/[122] but in the present proposal this occurred because of an independently motivated phonotactic restriction on sequences of two fricatives. The stop which arose in this way would then have spread to the forms of the second person plural personal pronoun where it was palatal (deriving from a former palatal fricative), e.g. *fúibh* /fu:bj/ ‘under-you.PL’, *libh* /ljɪbj/ ‘with-you.PL’, etc.

The operation of analogy is also to be seen with *féin*. The /fj/-less form became regular and holds for all positions, even post-vocalically. But why does the /fj/-less form here have an initial /h/? To account for this one must postulate a fixed order and a re-analysis. Assume also that morpheme-initial, as opposed to word-initial, /f/ has a tendency to lenite just as the initial element of a second morpheme does in word formation. The form /ljo:b/ (*leo* ‘with-them’) is taken as a typical environment and could have developed historically as follows.

(192)
Development of *féin* = [he:nj]

[lʲo:b fʲe:nʲ]	
[ljo:ph fʲe:nʲ]	devoicing by regressive assimilation
[ljo:ph e:nʲ]	loss of [fʲ] by lenition
[lʲo:b he:nʲ]	reanalysis of [pʰ] as [b] and [h]
[he:nʲ]	reinterpretation with analogical spread

2.7.7. Elision phenomena

All dialects of Irish are strongly stress-timed with considerable reduction in the phonetic quality of unstressed syllables. Stress is on the first syllable in Western and Northern Irish, with virtually no exception, while in Southern Irish there are complex conditions which determine where stress is placed in words (see section II.3. below for details). The reduction of unstressed syllables also results in the elision of vowels,[123] not just in fast speech, as can be seen from the following examples.

(193)

CV # VC →	**CVC**	
arú anuraidh	[aru:(ə)nˠʌrə]	‘the year before last’
bua abhí acu	[buə(ə)vʲi:akʌb]	‘they were successful’

V # V → V

ach an oiread	[a:x(ə)ɛrʲəd]	‘either’
ba iontach an lá	[b(ə)i:nˠtəx...]	‘the day was marvellous’
ag crith le ocras	[ə kʲrʲɪl$_{j}$(ə)ʌkrəs]	‘shaking with hunger’
lán de aire	[lɣɑ:nˠ dʲ(ə)arʲə]	‘full of care’
gach uile > *chuile*	[xɪl$_{j}$ə]	‘each one’

The susceptibility of vowels to elision can be expressed in terms of an elision hierarchy as in the following, where (1) represents the greatest tendency to elide and (3) the least.

(194)

1 /ə/
2 other short vowels
3 long vowels

The number of monosyllabic words with a final short vowel other than /ə/ is very small and if one finds examples of /a/ or /ʌ/ on both sides of a word boundary then there may well be a natural pause between them as in the following instance: *’s ea, olann a bhí aige.* [sja _ ʌlɣənɣəvji: ɛgjə] ‘that’s right, it’s wool he had with him.’. Where long and short vowels come into contact it is the short one which is elided: In some cases (see fourth example below) the contraction may lead to a new sound arising, here the diphthong /au/.

(195)

a.	*Cá bhfuil mo pheann luaidhe?*	‘Where is my pencil?’
	[kɑ:l$_j$ ...] ← [kɑ: wɪl$_j$ ...]	
b.	*Cá fhad a mbeidh tú ann?*	‘How long will you be there?’
	[kɑ:d ...] ← [kɑːad ...]	
c.	*Ba í sin an chúis.*	‘That was the reason’
	[bi: ʃɪn$_j$ …] ← [bə i: ʃɪn$_j$ …]	
d.	*Ní raibh a fhios agam faoi sin.*	‘I didn’t know about that’
	[n^{j}i: raus …] ← [n^{j}i: rə ə ɪs …]	

These contractions have not as yet received orthographic recognition though some instances have: *céard* /kje:rd/ ‘what’ (← *cad é an rud* ‘what is the thing’), *ch’uns* /xu:ns/ ‘while’ (← *chomh fada is* ‘as far as’).

Elision and phonetic reduction has led to many cases of homophony which can only be disambiguated by contextualisation. For instance, in spoken Western Irish the full form of the definite article *an* /ənɣ/ is reduced to a schwa: *an leabhar* /ə ljaur/ ‘the book’, *a leabhar /*ə ljaur/ ‘her book’. A similar case of homophony is in the following where the syntax is, however, sufficient to disambiguate the forms: *is* /ɪs/ ‘is’, *agus* /agəs/ → [ɪs] ‘and’.

Although Irish has initial stress there are a few forms to which this does not apply. In such words the first syllable is liable to elision. These are mostly adverbs with [ə] and stress on the following syllable: *anois* /əˈnɣɪsʲ/ → [nˠɪʃ] 'now', *anseo* /ənˈsʲʌ/ → [n̩ˈʃʌ] 'here', *anuas* /əˈnˠuːs/ → [nˠuːs] 'down from'.

In the form for 'here' the syllabic nasal arises from the phonotactic impossibility of having /nsj/ syllable-initially. An intermediate stage between the full and the elided form may be present by pre-glottalisation of the nasal: *anois* [ʔnɣɪʃ]. This is not infrequent when a syllable is lost from the beginning of a word. With the verb *taispeáin* 'show' the first syllable is often lost as it is unstressed (an exception to the verbal stress pattern) and leaves behind a trace as a glottal stop: *taispeáin* [ʔspjɑːnj] 'show'.

This procope frequently affects the interrogative marker *an* which is placed before a dependent form of a verb and where it is not always phonetically present, e.g. *An bhfuil Máire tinn?* [wɪlj mɑːrjə tjiːnj] 'Is Mary sick?'. This means that the dependent form of the verb then takes on the function of an interrogative verb form as it would be the only segmental difference between the question just quoted and the corresponding declarative sentence: *Tá Máire tinn* [tɣɑː mɑːrjə tjiːnj] 'Mary is sick'.

As a diachronic process final elision is attested throughout the whole history of Irish. The reason for this is the strong initial stress which leads to considerable reduction in the phonetic distinctiveness of succeeding syllables. Among the consonants which disappear in this position two can be singled out, *X* and *V*, e.g. *salach* /salə/ 'dirty', *talamh* /talə/ 'land, soil'. Of these two consonants, /x/ disappears less often, perhaps because it is frequently part of a morphological suffix and its loss with certain words would be in conflict with its functionally significant presence with others.

V can also disappear in a few unstressed monosyllabic forms, often with a change in vowel quality: *raibh* /rʌv/ ~

/rə/ ‘was’, *taobh* /ti:v/ ~ /ti:w/ ~ /ti:/ ‘by the side of’. The reduction may be accompanied by elision with a vowel at the beginning of the lexical word which follows as in *taobh istigh* [ti:ˈʃtjɪ] < /ti: ˈɪsjtjɪ/ ‘inside’ (de Bhaldraithe 1945: 58).

2.7.8. Articulatory shifts

There are a few shifts in place of articulation which are found in Irish. The shift from velar to labial with voiceless fricatives, /x/ → /f/,[124] is attested in a few words where the orthography still indicates the former pronunciation, cf. *cluiche* /klɪfjə/ ‘game’. This type of shift is found in many other languages, cf. German *Kachelofen* → Polish *kaflowy piec* ‘tile stove’, German *lachen* ~ English *laugh*, Dutch *kopen* ‘to buy’ ~ *gekocht* ‘bought’.

In Western Irish the shift from glottal to velar place of articulation is occasionally attested as in *ó shin* [o: xɪnj] ← /o: hɪnj/ ‘since’. The shift of a glottal to a palatal articulation is frequent after high vowels: *ar bith* [ər bjɪç] ‘at all’, *ag ithe* [ɛgj ɪçə] ‘eating’. A reverse shift from a front point to a back one is found with the verb form *dul* → *goil* ‘going’ (de Bhaldraithe 1945: 99) and the forms of the pronoun *do* ‘for’, including the prepositional pronouns, e.g. *dom* [ɣʌm] ‘for-me’, *duit* [ɣɪtj] ‘for-you’, *dó* [ɣo:] ‘for-him’, as well as some of the compound forms of *de* ‘from’: *de* [jɛ], *di* [jɪ].

2.8. Epenthesis

Epenthesis is the addition of phonetic substance to a word, either vocalic or consonantal. It may be either obligatory – as in the case of vowel epenthesis in Irish (Ní Chiosáin 1997) – or it may be merely probable, i.e. it need not of necessity occur when the conditions for epenthesis are met. This is generally the case with consonantal epenthesis in Irish.

The following account of epenthesis appeals to the notion of coda sonority. This should not be understood as something which is absolute across all languages but which has local significance in Irish and which furthermore has varying degrees of significance in the different dialects of the language. How the sonority-based restrictions on codas in Irish arose historically is not an issue here and the criticism of attributing predictive power to sonority considerations and/or regarding them as the cause of phonological phenomena, as outlined by Blevins (2006: 159-164) or Harris (2006), is justified.[125] But for present-day Irish, sonority is a useful heuristic tool for accurately describing the distribution of epenthesis. [126]

Vowel epenthesis is known from pronunciations of Irish English, e.g. *film* [fɪləm] (Hickey 2007: 307-308), and from languages like Dutch, e.g. *melk* [mɛlək] 'milk' (Hickey 1985c). Essentially, it is similar in Irish but here it is necessary to distinguish epenthesis from a related but distinct phenomenon, syncope. The latter is the loss of a vowel between two consonants when a vowel follows the second, e.g. *cosain* [kʌsənj] 'defend', *cosnaím* [kʌsnɣi:mj] 'I defend'. Epenthesis on the other hand is a general restriction which is due to phonotactic rules which apply globally for the language. Certain statements can be made for epenthesis as a general phenomenon in Irish.

Table 27. Conditions for vowel epenthesis in Irish

1)	An epenthetic vowel never carries lexical information.
2)	An epenthetic vowel never indicates a grammatical category.
3)	The values for [palatal] with both the preceding and the following consonant always agree.
4)	An epenthetic vowel is never stressed.

The likely purpose of an epenthetic vowel is to break up a phonotactically inadmissible cluster by re-syllabifying its members. The clusters which are broken up by an epenthetic vowel show a characteristic segmental structure as seen in the following examples.

(196)

a.	*borb* [bʌrəb]	'fierce, violent'
b.	*corb* [kʌrəb]	'to corrupt'
c.	*lorg* [lʌrəg]	'search'
d.	*bolg* [bʌlɣəg]	'belly'

Epenthesis involves two-item clusters in the coda of a stressed syllable. If the segments of the coda cluster are homorganic then no epenthesis takes place. In essence, this means that the second element must be labial or velar as in the above examples. The first element must furthermore be a sonorant and due to the distribution of sonorants in syllable codas in Irish, these are always coronal. This means that a coronal sonorant followed by a voiced non-coronal obstruent (velar or labial) induces epenthesis: *seilbh* [ʃɛljɪvj] 'possession', *banbh* [banɣəv] 'piglet'.

The conditions for epenthesis can be deduced from the attested forms which show it. For instance, the failure of epenthesis with clusters showing voiceless or homorganic obstruents is confirmed by forms such as the following: *bord* [baurd] 'table', *gort* [gʌrt] 'field', *feilt* [fɛljtj] 'felt', *corp* [kʌrp] 'body', *folc* [fʌlɣk] 'downpour', *olc* [ʌlɣk] 'evil', *port* [pʌrt] 'port, harbour; tune'.

It is obvious from monosyllables without epenthesis that a coda cluster must have a sonorant as first element for this to occur: *toisc* [tʌsjkj] 'journey, purpose', *tost* [tʌst] 'silence'.

However, the required nature of the second element is not as obvious. Words such as the following show epenthesis regularly:[127] *ainm* [anjɪmj] ‘name’, *arm* [arəm] ‘army’, *feilm* [fjɛljɪmj] ‘farm’. These instances would seem to indicate that epenthesis is induced where the second element is at least as sonorous as a voiced obstruent. This would include sonorants as second element, a postulation which is vindicated by the forms just quoted.

Epenthesis does not occur before voiceless stops.[128] As epenthesis is the development of a short unstressed vowel then this can arise preferentially in an environment which is voiced and where the cluster-final consonant is lenis. With stops voicelessness sets in immediately at the end of the sonorant or slightly before this thus reducing the likelihood of epenthesis, cf. *olc* [ʌlɣk], *[ʌlɣək] ‘evil’. However, epenthesis is attested before voiceless fricatives, e.g. *seilf* [ʃɛl$_j$ɪf^j] ‘shelf’. This phonetic nature of voiceless stops also explains their position at the lowest point on a sonority scale.[129]

Table 28. Preferential environments for epenthesis

a)	sonorants and sonorants
b)	sonorants and voiced fricatives
c)	sonorants and voiceless fricatives
d)	sonorants and voiced stops
e)	[sonorants and voiceless stops]

There are two further conditions on epenthesis apart from the relative sonority of the segments of the coda in which it occurs: 1) the cluster must consist of heterorganic segments and 2) there must be no morpheme boundary between the elements of a cluster. This holds for Cois Fharraige Irish (although Southern Irish allows slight leakages, R. B. Breatnach 1947: 127), so that a form like *cionmhar* /kjʌnɣ#vər/ ‘proportional’ does not show epenthesis.

If one now orders the consonants of Irish on a hierarchy according to their relative sonority[130] then one can recognise that there is a cut-off point for epenthesis between voiced and voiceless plosives so that the latter cannot trigger epenthesis in a syllable coda with a preceding sonorant.

Table 29. Sonority hierarchy for Irish

(vowels		**0, greatest degree)**
liquids	*R, L*	1
nasals	*M, N*	2
fricatives, voiced	*V*, *ɣ*	3
fricatives, voiceless	*F, S, X*, /h/	4
plosives, voiced	*B, D, G*	5
plosives, voiceless	*P, T, K*	6

2.8.1. Motivation for epenthesis

Epenthesis is governed by the phonotactics of syllable codas in Irish. All the clusters which show epenthesis contain a sonorant and a further sonorous element. These are instances of heavy clusters in Irish. If the preceding stressed vowel is short then the weight of the coda cluster is resolved by inserting an epenthetic vowel which resyllabifies the phonological word. The final element of the input coda cluster is shifted to the second syllable created by epenthetic vowel insertion forming a single-element coda in this new syllable.

(197)

Coda resyllabification on epenthesis *bolg* [.bʌlg.] → [.bʌl.əg.] ‘belly’

There could be a case for the sonorant after a short stem vowel being ambisyllabic in the new epenthetic structure. Whether or not this is the case is irrelevant. The main point is that the second element of the coda in the lexical form is no longer in the stem syllable after epenthesis.

Evidence for the status of epenthetic vowels as syllable nuclei is found in Irish where these vowels, before non-palatal /v/, can vocalise to /u:/ which is most definitely a separate syllable.

(198)

Vocalisation of epenthetic syllable coda[131] *garbh* [.garv.] → [.gar.əv.] → [.gar.u:.] ‘rough’

Seeing as how vowel epenthesis serves the function of resyllabification its lack of lexical or grammatical significance is to be expected. The resyllabification interpretation gains credence if one examines those instances in which epenthesis does not occur. Consider the following instances where the elements of the central clusters belong to different syllables.

(199)

a.	*céimnigh* /kʲe:mʲ.nʲə/	‘to step’
b.	*téarma* /tʲe:r.mə/	‘term’
c.	*corrdhuine* /kaur.ɣɪn$_j$ə/	‘occasional person’
d.	*fíordhrochlá* /fʲi:r.ɣrʌxlˠɑ:/	‘very bad day’

e. *cúlchaint* /ku:lˠ.xanʲtʲ/ ‘bad talk’

f. *fuarchúis* /fu:r.xu:sʲ/ ‘apathy’

But there is a further condition required to explain the attestations just given. Epenthesis is blocked in polysyllabic forms if the vowel before the potential epenthetic cluster is long. In the following forms this vowel is short triggering epenthesis.

(200)

dorcha /dʌr.xə/	→	[dʌrəxə]	‘dark’
Donncha /dʌnˠxə/	→	[dʌnˠəxə]	‘Denis’
Murchú /mʌrxu:/	→	[mʌrəxu:]	‘Murphy’
simléar /sʲɪmʲ.lʲe:r/	→	[ʃɪmʲɪlʲe:r]	‘chimney’

The stipulation that epenthesis does not occur after long vowels is upheld by compounds. There are a large number of these with the qualifying element *mór-* ‘big’. None of them show epenthesis though in the first word epenthesis may be found in the head of the compound: *mórbholgach* [mo:rvʌlɣəgəx] ‘big-bellied’, *mórchuid* [mo:rxɪdj] ‘large amount’, *mórga* [mo:rgə] ‘exalted, majestic’, but: *morgadh* [mʌrəgə] ‘corruption, putrefaction’. In addition if the lexical form of a word is monosyllabic and the nucleus vowel is long then epenthesis is indeed present in the phonetic realisation, e.g. *táirg* [tɑ:rjɪgj] ‘manufacture’ with epenthesis just like *tairg* [tarjɪgj] ‘offer’.

Unstressed central vowels between segments which would form heavy clusters in Irish are found across the board in the lexicon of the language. The status of these vowels would on closer inspection seem to show differences despite superficial similarities. Consider the following forms which show [-lɣv] (a heavy cluster) separated by [ə]: *folamh* [fʌlɣəv] ‘empty’,[132] *ollamh* [ʌlɣəv] ‘professor’. These words are different from epenthesis-triggering forms like *bolg* [bʌlɣəg] ‘belly’, *banbh* [banɣəv] ‘piglet’ in one essential respect: when the final consonant is palatalised, e.g. in the genitive case, then the sonorant before this is not palatalised, e.g. *Leabhar nua an ollaimh*. [... ʌlɣɪvj] ‘The professor’s new book’, *Dath an bhuidéil fholaimh* [... ʌlɣɪvj] ‘The colour of the empty bottle’. An absolute rule of Irish phonology is that all clusters – except /-xtʲ/, /-rdj/ and /-rtj/ – agree in the value for [palatal]. Seeing as how the [lɣ] and the [vj] of the second syllable of the forms just quoted differ in this respect, palatalisation must have operated on an input which already contained a vowel between [lɣ] and [vj], otherwise the outcome would have been /ʌljvj/, then *[ʌljɪvj] by epenthesis which is not the case. The second vowel in the form *folamh* is thus lexical. It may be deleted via syncope when a palatalising suffix is added to the base form, e.g. in the comparative *níos foilmhe* ‘emptier’. Syncope, however, brings the lateral and the voiced fricative together, palatalising the former and then satisfying the input condition for epenthesis on phonetic realisation.

(201)

a.	*folamh* /fʌlˠəv/ + palatalising suffix *e*	‘empty’
b.	*foilmhe* /fɛlʲvʲə/ → [fɛljɪvʲə]	‘emptier’

2.8.2. Epenthesis and syncope

In Modern Irish syncope always occurs where the input for resyllabification is satisfied by suffixation. In instances like the following, the /-dl-/ must be split between two syllables as such a sequence cannot occur in a monosyllable in Irish.

It violates the sonority cline for codas which demands that sonorants come before obstruents word-finally in monosyllables.

(202)

a. *codail* /kʌdɪl^{j}/ ‘sleep’

b. *codlaím* /kʌd.l$^{\gamma}$i:m^{j}/ ‘I sleep’

After syncope applies to /kʌdɪlj/ the liquid is non-palatal (/kʌdlɣi:mj/ not */kʌdjlji:mj/). The reason why the /vj/ in the word *foilmhe* ‘emptier’ remains palatal is that the palatalisation is a morphonological process which cannot be reversed by a low-level rule like assimilation. In the case of the verb for ‘sleep’ one can see, looking at a whole series of verbs, that the stem is /kʌd/ and the /-ɪlj/ is an ending, one of a group of sonorant endings, which characterises the imperative and the past (with lenition): *codail* /kʌd/ +/ɪlj/ ‘sleep’, *oscail* /ʌsk/ + /ɪlj/ ‘open’, *iomair* /ʌm/ + /ɪrj/ ‘row’, *cosain* /kʌs/ + /ɪnj/ ‘defend’, *cigil* /kjɪgj/ + /ɪlj/ ‘tickle’, *imir* /ɪmj/ + /ɪrj/ ‘play’, *taitin* /tatj/ + /ɪnj/ ‘please’.

Any low-level rule such as assimilation cannot alter the phonological composition of a lexical stem. Thus the sonorant stem endings do not alter the value for [palatal] with the stem final segment when syncope deletes the vowel of the ending. This principle applies vacuously to the last three verbs just listed as their stems end in palatal consonants anyway.

2.8.3. Epenthesis and areality

Epenthesis is an areal phenomenon: the Q-Celtic languages, Irish, Scottish Gaelic and previously Manx, all show epenthesis.[133] Furthermore, it is present in the varieties of

English which are either in contact with or stem from these forms of Celtic. Granted the manifestation of epenthesis may differ within the dialect continuum of the Q-Celtic languages. Scottish Gaelic generally reduplicates the stressed vowel in epenthetic clusters[134] whereas Irish has a central vowel which ranges from /ə/ to /ɪ/ depending on the value for [palatal] of the surrounding consonants. The extent of epenthesis can vary also. In Munster Irish it has a greater scope than in Western Irish as it is found in the former dialect across morpheme and even word boundaries (see following section). Nonetheless the assertion is valid that epenthesis is found in geographically contiguous areas. Nor are there any 'pockets' without epenthesis within an area for which it is characteristic. There is no variety of Irish or English in Ireland which does not have epenthesis in the most susceptible clusters, such as double non-homorganic sonorants after short vowels, e.g. Irish *ainm* [ænjɪmj] 'soul', Irish English *film* /fɪləm/.

The areal view of epenthesis is supported by evidence from other languages. In the West Germanic group one finds epenthesis in Dutch (van den Toorn 1973: 144) and Flemish and also in the contiguous dialects of German in the lower and middle Rhine region, at least down as far as Cologne and Bonn (Hickey 1985c). The presence of features like epenthesis in structurally unrelated languages, Irish and English, would point to its status as a low-level phenomenon. Contrast it with palatalisation in Irish. This is a central morphonological feature of the language and despite the centuries of contact with English it was not transferred to the latter.

2.8.4. Epenthesis in Southern Irish

The view that vowel epenthesis is triggered by heavy codas must be expanded to include instances of heavy clusters whose elements are allocated to separate syllables. This wider conception of epenthesis holds for Southern Irish in particular. Consider its occurrence across word boundaries[135] as in compounds where the final consonant of the first element and the first of the second element together form a cluster which requires epenthesis (here: a sonorant plus voiced fricative): *an-mhaith* /anɣ.vah/ [anɣəvah] ‘very good’ (Southern Irish, Ó Cuív 1944: 106). This situation has further phonetic ramifications: the prefix *an-* takes lenition, but generally lenition does not apply to a lexical word whose initial consonant is homorganic with the final consonant of the mutating element, thus one has *an-deas* /anɣdjas/ ‘very nice’ and not **an-dheas* /anɣɣjas/ in Western and Northern Irish. But in Southern Irish because of extended epenthesis the initial consonant of the stem is no longer in contact with the final one of the leniting particle and so one has indeed /an$_{\text{ɣ}}$ɣ^{j}as/ → [anɣəjas]. This situation implies that in the generation of the phonetic output vowel epenthesis removes the contact between the stem-initial and the particle-final consonants thus allowing the lenition of the stem-initial consonant.

2.8.5. Epenthesis and other processes

D. Ó Baoill (1980) sought to demonstrate the relatedness of epenthesis, preaspiration (in some varieties of Northern Irish and in Scottish Gaelic) and the vowel lengthening which occurred before former geminate sonorants in the history of Irish (see section II.1.8.6 above).

Preaspiration consists of a period of voicelessness before a voiceless stop. The configuration of the supra-glottal tract is entirely automatic: the oral cavity retains the position which it has adopted for the vowel which precedes when preaspiration begins. Behind D. Ó Baoill’s treatment of preaspiration, epenthesis and vowel lengthening is the view that syllable length in all the forms which have come to manifest these phenomena has remained constant. D. Ó

Baoill (1980: 80) talks of a conspiracy to maintain syllable length. The other process in D. Ó Baoill's triad is vowel lengthening and the cases he considers (D. Ó Baoill 1980: 103) all involve vowels before former geminate sonorants. He maintains that the length of the sonorant was transferred to the vowel which preceded it before consonantal length was lost in Irish. In this respect linking preaspiration and lengthening before geminate sonorants (D. Ó Baoill 1980: 81+104) is valid seen in terms of syllable rhyme quantity. Of course this view only applies to those Northern dialects of Q-Celtic in which preaspiration arose.

(203)

Old Irish	*macc*	'son'	/makk/	→	/mahk/
	geall	'bright'	/g^{j}all/	→	/g^{j}a:l/

The most crucial process for the testing of D. Ó Baoill's hypothesis (1980: 80) that the innovating factor in all three processes was the existence of geminate consonants in Old Irish is epenthesis. The cases of epenthesis which he deals with (D. Ó Baoill 1980: 94) are similar to those already viewed in this section with the exception of a few examples of weak epenthesis after long vowels – e.g. *seomra* [ʃo:mərə] 'room' – which he himself says is confined to Munster Irish and arose by extension from the original triggering environment disregarding the condition which still holds in all other dialects apart from Munster Irish[136] that the premier site for epenthesis is in syllable codas with a short vowel as nucleus. Not only does it occur after short vowels but also after sonorants. This fact leads D. Ó Baoill to attempt a link with vowel lengthening. His argumentation on epenthesis is compelling with close observation of the phonetic facts. D. Ó Baoill claims that epenthesis arose through the vocalisation of the first part of a geminate consonant after a sonorant (D. Ó Baoill 1980: 97). He assumes, after outlining certain arguments, that the double graphemes in a form like Old Irish *gargg* represented a geminate. This is essential to his thesis. If there were no post-sonorant word-final geminates then not only can he not link up epenthesis with preaspiration and vowel lengthening before geminate sonorants but he has no means within his framework of explaining epenthesis at all. However, granted his assumption about Old Irish phonology, all the processes he examines can be viewed as quantity preserving, i.e. *gargg* /gargg/ →/garəg/ 'bitter, rude, harsh' on the loss of geminates.

2.8.6. Consonant epenthesis

There are two basic types of consonant epenthesis in Irish. Both have to do with the structure of syllable margins. The first strengthens a syllable onset by introducing a homorganic coronal stop before a sonorant. A closer look at the attestations shows that the trigger for epenthesis in this environment has to do with syllable contact.[137] If a liquid, effectively /r/, forms the onset of a syllable and the coda of the previous syllable ends in either a nasal or liquid, effectively /l/, then stop epenthesis takes place: *anró* /anɣ.tro:/ 'hardship due to bad weather', *banrán* /banɣ.trɑ:nɣ/ 'grumbling', *malrach* /malɣ.drəx/ 'youngster', *lonradh* /lɣʌnɣ.trə/ 'brightness'. Put in broader terms one can maintain that syllable contact does not permit two sonorants to adjoin, if they do a stop intervenes to form a consonantal peak between coda and onset. In a few cases this principle has been extended to cover cases where a sonorant and a vowel adjoin at a syllable boundary. Epenthesis is common here in non-standard plural forms (Hickey 1985a), the suffix starting with a coronal stop: *scéal : scéalta* 'story' : 'stories', *traein : traentachaí* 'train' : 'trains'. Indeed the association of *T* epenthesis with plural markers has led to analogical epenthesis in instances where syllable contact can hardly have been the motivation for it: *rása : rastaí* ‘race’ : ‘races’, *rí : rítí* ‘king’ : ‘kings’.

The second kind of epenthesis is one which strengthens the coda of a word-final syllable by the addition of a consonant.[138] For this to be triggered the coda-final element must be either a nasal or /s/.[139] This is relatively rare in Irish and is only widely attested in the form *arís* /əˈri:sjtj/ ‘again’.

2.9. Metathesis

Metathesis is well attested throughout the history of many languages, not least in English. Its manifestations and effects have been investigated quite often (Bailey, 1970; Hogg 1977; Ultan 1978; Hock 1985; Hume) and it has been postulated as a stage in underlying structures to account for surface forms. Its manifestations are often presented traditionally as taxonomies, for example in the various traditional grammars of Old English, Campbell (1959: 184-186), Luick (1940: 917-920). It is widely documented for the varieties of Irish and all the dialect studies which were published during the twentieth century (see section I.2.1 above) have dedicated sections on metathesis.

Metathesis has no obvious cause and some linguists admit to the difficulties in determining its possible motivation (Hogg 1977: 174; Alexander 1985) while others ascribe it to a *lapsus linguae* (see the summary of views on metathesis in Thompson and Thompson 1969: 213). Another opinion, outlined in Ultan (1978: 373, 383-394), is that metathesis has the function of preserving phonological segments from deletion. The view that metathesis is due to a *lapsus linguae* links up with research done on slips of the tongue (Fromkin (ed.) 1973) which have been shown not to be irregular, although in any given instance they would not be predictable. Slips of the tongue, it would seem, occur on the basis of syllable organisation (Crompton 1981: 689-690). The same would appear to be true of metathesis which has the syllable as its domain.

Below an attempt is made to account for metathesis in terms of syllable structure. It does not account for all cases of metathesis but it can provide a framework for describing attested instances of metathesis in Irish with considerable consistency and regularity.

In order to deal with metathesis a sufficient number of instances of it are required. With modern standard languages, as a result of regularised orthography and universal education, metathesis is not in evidence. If it occurs then it is confined in its nature and extent, cf. the simple *r*-vowel metathesis in Irish English *modern* [mɒdrən] which is the only type found in supraregional Irish English.[140] However, placenames are a fruitful source of metathesis in Ireland, e.g. *Ballytruckle* from Irish *Baile an Turcail* with [...tʌrk...] appearing as [...trʌk...], in turn from Scandinavian 'Torkell's town' (Sommerfelt 1952a: 228).

Dialect material is useful for the study of metathesis as it shows forms within a single language which vary geographically and which are independent of orthographical or standardising influences. One of the few detailed treatments of metathesis is that of Grammont (1971 [1933]: 339-357) which takes as its material dialect realisations, mainly of standard French word forms and of words from other Romance languages.

2.9.1. Delimiting metathesis

Attestations of metathesis point to it being a phonological phenomenon which affects the ordering of segments in words. It is essentially non-phonetic as can be seen from the most common case of metathesis, that of *r* and a vowel, e.g. Irish English V+/r/ → /r/+V, *pattern* /pætərn/ → [pætrən]. The metathesised form has an alveolar fricative (Cruttenden

2008: 181-185) for the sequence /tr/ while the non-metathesised form has a rhotacised central vowel [ɚ]: phonetically this is not a sequence of a vowel followed by an /r/ articulation.

The recognition of metathesis may sometimes present difficulties, particularly as it may co-occur with further processes or indeed one of these can be misinterpreted as metathesis. For instance, with the word *spioraid* ‘spirit’ the first vowel may be deleted with the /r/ forming a cluster with the initial /sp-/: /sʲpʲɪrədʲ/ → [sjpjrjɪdj]. This is not a case of *r*-vowel metathesis (note that the /r/ palatalises when it joins the palatal onset and that the second vowel /ə/ when stressed become /ɪ/ between palatal consonants). There are many similar instances in Irish, e.g. *furasta* ‘simple’ with /fʌrəstə/ → [frʌstə] or *turas* ‘journey’ /tʌrəs/ → [trʌs] (Southern Irish).

2.9.2. Metathesis and other processes

Metathesis consists of a linear reordering of segments. It may occur between various syllable positions and be found twice within a single word. Although it is true that metathesis never involves the deletion of a phonological segment, phonetic substance may be deleted as a consequence of metathesis having applied. Irish has a system of epenthesis (see previous section) which introduces an epenthetic vowel /ə/ in a cluster of /r/ and voiced stop (among many other types of clusters. Thus one has /krɪɣʲarg/ ‘blood-red’ (*croidhearg*) (Ó Cuív 1944: 128), where the phonetic realisation shows an epenthetic vowel which splits up and thus resyllabifies the final cluster: /krɪ.ɣʲarg/ → [krɪ.jarək] (the phonotactic restriction states that a sequence of sonorant and heterorganic voiced stop cannot share the same syllable coda). However, with *r*-vowel metathesis the surface form is disyllabic as the underlying form after metathesis no longer meets the

structural description for epenthesis: /krɪɣʲarg/ → /krɪɣjrak/, /krɪɣʲ.rag/ → [krɪj.rak].
Metathesis may also block another process which fortifies voiced stops to voiceless ones in homorganic post-sonorant word-final position. If the sonorant is removed from its immediately pre-stop position through metathesis this fortition (McCone 1981) can be reversed: /malˠərʲtʲ/ → /malɣrədj/ ‘opposite’ (*malairt*) (de Bhaldraithe 1945: 116).

2.9.3. The domain of metathesis

Metathesis appears to be only phonological. There would seem to be no cases where it has been co-opted for grammatical purposes, though there may be one language in which vowel-vowel metathesis seems to indicate grammatical categories. This is Kasem (also Kasim, Kasena), a language of the Niger-Congo phylum spoken by about 250,000 mostly in northern Ghana and Burkina Faso, see the discussion in S. Anderson (1974: 152-160) and in Phelps (1975), who refutes the analysis as metathesis. One case where it co-occurs with, but is not determined by, a grammatical category is found in Lithuanian where an alveolar or palatal fricative and a velar stop are reversed in sequence before a further obstruent: *mzga* ‘bind’ (third person present) *mègsti* ‘to bind’. Its occurrence in loan words (nouns) would seem to indicate its purely phonological nature (Russian *Moskva* → Lithuanian *Maksvà*, see Kenstowicz 1971: 22).

A number of generalisations about metathesis can be made all of which have consequences for its manifestation.

(204)

a. Metathesis never obscures the grammatical structure of a word
b. Metathesis does not occur in inflections
c. Metathesis is only found in lexical stems

These three conditions are closely interrelated. The first is an observation made from cross-linguistic examinations but there is no way of claiming that it is a universal. The second and the third are mirror-images of each other. All three generalisations on metathesis derive from its grammatically afunctional nature.

2.9.4. Possible motivation for metathesis

Although metathesis has no grammatical function (Janda 1984) scholars often take the view that it has at least an incidental function on a phonological level. Thus when referring to metathesis of /r/ and a sibilant in Armenian, Winter (1962: 260-261) notes that ‘metathesis occurs when the regular developments would lead to a deviation from the established pattern’. When discussing historical French *r*-vowel metathesis Ultan (1978: 389) postulates that it was motivated by a tendency to develop open syllables in French (but Ultan also notes exceptions to this), e.g. *troubler* ‘to disturb’ ← *torbler,* but: *pour* ‘for’ ← *pro.*

In Classical Greek intervocalic /h/ (from Indo-European */s/) was lost but retained prevocalically. The preterite augment /e/ when added to a stem beginning with /h/ then metathesised to place /h/ in a prevocalic position even after prefixation: *heipomɛn* ‘we followed’ ← **ehepoman* (Ultan 1978: 385). Given the lack of intervocalic /h/ in Greek the metathesis restored /h/ after prefixation of /e/ to its preferred position before an initial vowel. Thus metathesis in this instance led to a structure more in keeping with the phonotactics of Greek.

In terms of syllable structure a voiceless fricative is preferred in initial position (in accordance with the greater degree of consonantality of syllable onsets) than in intervocalic medial position. An implication of this is that languages should show a tendency to lose intervocalic /h/; this is the case for example in Old English where /h/ was lost medially but retained initially, pre-Old English *sexan* Old English *seon* /se:on/ ‘to see’, *xus* → *hus* /hu:s/ ‘house’. In the Irish of Cois Fharraige /h/ has also been lost medially but retained initially (de Bhaldraithe 1945: 120), e.g. /boːhər/ → /bo:ər/ ‘road’ (*bóthar*), /hatə/ → /hatə/ ‘hat’ (*hata*). In Spanish /h/ which derives from the lenition of /s/ (Malmberg 1962: 64-65) is lost entirely in many dialects (Southern Spain and large parts of Spanish-speaking America); this /h/ only occurs in syllable final position: *pasta* [paʰta] → [pata] ‘pastry’, *casas* [kasaʰ] → [kasa] ‘houses’.

2.9.5. Metathesis and syllable structure

Principles of syllable structure state that onsets are maximally consonantal (Vennemann 1982: 283; Blevins 1995: 210-212; Goldsmith 2014: 180-182): they start with a strong segment, i.e. one which displays little sonority. The increase in sonority towards the nucleus implies a steady decrease in consonantal strength within the onset towards the nucleus. Thus fricatives are not to be expected before stops in syllable onsets, subject to the qualification that this does not hold for the fricative at the most natural point of articulation, the alveolar ridge. This means that one would not expect initial clusters such as /xt/, /hp/, /fk/ but that a cluster such as /sk/ or /sp/ is natural due to the place of articulation qualification. In this view /sp/ conforms to preferred syllable structure but /hp/ does not.

In Old Armenian Indo-European intervocalic */s/ was lost. The loss of /s/ in a language often implies a sequence /s/ → /h/ → Ø. Indo-European */sp/, however, appears as /p’/ in Old Armenian: *p’oit’* ‘haste’ (cf. Greek *spondḗ*) The ejective /p’/ can be regarded as having arisen from the absorption of /h/ (Ultan 1978: 385). But this could only have followed the stop if there was metathesis of a sequence /hp/, the preceding /h/ deriving from the lenition of /s/. The metathesis can be viewed as having been triggered by the relative unnaturalness of the syllable onset /hp/ in comparison to the onset /sp/ from which it arose.

Other instances of metathesis can be accounted for in terms of syllable structure, e.g. Persian *surx* ‘red’, *ʒarf* ‘deep’, cf. Zend[141] *suxra*, *jafra.* Ultan (1978: 388) maintains that apocope of the final vowels of the Persian forms led to metathesis of the earlier fricative and liquid cluster. But the question here is why should this be the case? If the forms *surx*, *ʒarf* and *suxra*, *jafra* have the following syllabification: /.surx./ vs. /.su.xra/, /.ʒarf./ vs. /.ʒa.fra/ then there is an obvious explanation in terms of syllable structure. A syllable coda, if it contains two or more segments, will preferably show an increase in consonantal strength away from the nucleus. The forms /.suxr./ and /.ʒafr./ would show a decrease in consonantal strength with the /r/. The attested metathesis, however, led to a syllable structure in which there is a gradual decrease in sonority from nucleus to right-most element in the coda, i.e. /-urx/ and /-arf/.

2.9.6. Metathesis across syllable boundaries

In the development of Semitic languages (and considerably later in Mandaic, Malone 1971: 394-395) sequences of /t/ and a sibilant were metathesised. These sequences were internal so that the metathesis can be interpreted as leading to preferred syllable structure with the stop providing the onset of the second syllable after metathesis: /- t.s -/ → /- s.t -/. This kind of metathesis is found in Anglo-Norman loanwords in Irish (see section II.2.9.10).

A certain phonological process may through its operation give rise to a heterosyllabic cluster which does not conform with the principles governing syllable contact. Such clusters arose in forms of Western Romance (Ultan 1978: 387-388) due to the loss of the second vowel in tri- and quadrosyllabic words. The resulting clusters sometimes show metathesis: *cumulu* > Spanish *colmo* ‘top, limit’.

Assuming that before metathesis the syncopated cluster was / - m.l - / then the coda was stronger than the onset of the following syllable. This imbalance was resolved by metathesis. But metathesis is not the only phonological means of redressing the balance. Clusters of a nasal and a liquid in French tend to develop an epenthetic stop before the liquid.

(205)

/-N.L-/		/-N.PL-/ (N = nasal; P = plosive; L = liquid)
Latin		French
camera	→	*chambre*

With this development one has both a stronger onset than the preceding nasal and a balanced onset will in general which have a stop before a liquid.

Because onsets are strong (Fallows 1981: 310) there is a documented tendency to (i) fortify onsets, (ii) reduce codas and (iii) partially reduce the onsets of non-initial syllables. These three situations can be evidenced by metathesis. In Grammont (1971 [1933]: 347) examples of metathesis are given from a one and a half to two year old infant who produced metatheses which fortified the onsets of word initial syllables (see Ultan 1978: 381 as well), e.g. French *ventre* /vãtr/ → *vrente* /vrãt/ 'belly', *porter* /pɔr.te/ → *proter* /prɔ.te/ 'carry'.

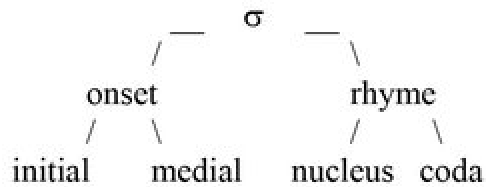
The liquid metathesis in Slavic languages (Sørensen 1952: 41; Leskien 1922: 33) had the effect of reducing the coda and strengthening the onsets of syllables: Proto-Slavic * *bergъ* 'height', cf. Polish *brzeg* 'edge; coast', Proto-Slavic **melka* → Old Church Slavonic *mleko* 'milk' and furthermore in some cases, such as /mle.ko/ just given, of creating an open syllable, a preferred phonotactic structure of Old Church Slavonic. However, it would be insufficient to say that Slavic liquid metathesis occurred due to the rise of an open syllable phonotactic type in Slavic (Ultan 1978: 388-389). This assertion does not point to the twofold benefit of this metathesis in terms of syllable structure: not only is the coda removed but the onset is strengthened by the liquid in the second position.

In Campidanian Sardinian[142] (Ultan 1978: 377) the metathesis of a stop and /r/ has the effect of providing the initial syllable with a coda and reducing the onset of the following syllable: *arbili* /ar.bi.li/ ← *aprile* /a.pri.le/, *sorgu* /sɔr.gu/ ← *socru* /sɔ.kru/. The simplification of codas is seen also in Middle English metathesis where words which had a short vowel and /r/ before a cluster of /xt/ underwent metathesis: *fyrʒt* → *fright*, *wyrhta* → *wright.* This metathesis resulted in an equilibrium between coda and onset. In those words which only have a single obstruent in the coda metathesis did not occur: *work* ← Old English *weorc.* Where both onset and coda contained one segment (excluding /r/) metathesis is found when the segment of the coda is homorganic with /r/ (see Ultan 1978: 393): Middle English *brid(de)* → *bird*, Old English *brinnen*[143] → Modern English *burn.*

2.9.7. More on syllable structure

A first division of syllable constituents would assume an onset, nucleus and coda (see section II.1.10 above). In Irish metathesis offers support for an extra slot, the medial, consisting of one or two consonants, yielding the following structure.

(206)



Irish can have mono-, bi- or trisegmental onset clusters and with the latter two the segments which constitute them must have sonority values which, if not adjacent to each other, are at least ordered consecutively on the following sonority scale going from right to left in an onset (and from left to right in a coda).

Table 30. Sonority cline for onset clusters in Irish

vowels	0
semi-vowels	1
/r/	2
/l/	3
nasals	4
voiced fricatives	5
voiceless fricatives	6
voiced stops	7
voiceless stops	8

This is true for all onset clusters with the one exception of those which have initial /s/ which can be followed by stops (*SP*- *ST*- *SK*-). Here the primacy of alveolar articulation among segments allows a consonantally weaker segment (/s/) to stand before stronger segments (stops) in syllable onsets.

Several instances of diachronic metathesis comply with the sonority cline towards the syllable nucleus,[144] e.g. /tar.snˠə/ → [tra.snˠə] ‘across’ (*trasna*) (Mhac an Fhailigh 1968: 168), /ʌrbəlˠ/ → [djrjʌbəlɣ] ‘tail’ (*urball*) (de Bhaldraithe 1945: 115). In the latter example two stages must be recognised. The first is *r*-vowel metathesis which reverses the order of these segments so that /r/ is now word-initial. But liquids are preferentially the medial. This leads to stop epenthesis which creates the preferential syllable structure.

(207)

initial	medial	nucleus	.	initial	nucleus	coda
dʲ	rʲ	ʌ	.	b	ə	lˠ

There are also cases of this where the syllable boundary is internal. Here there is no epenthesis involved but a redrawing of syllable boundaries such that the coda of the preceding syllable is made part of the onset of the next: /mʲɪlʲsʲɑːnˠ/ → [mji:ʃljɑ:nɣ] ‘sweet’ (*milseán*) (Ó Cuív 1944: 127). The /lj/ of the non-metathesised form represents the coda of the first syllable. On metathesis the syllable boundary is now before the /sj/ [ʃ] and the first vowel is lengthened: /mʲɪlʲ.sʲɑːnɣ/ → [mji:.ʃljɑ:nɣ].

In these forms two other changes accompany metathesis. In [djrjʌbəl] ‘tail’ (*urball*) the initial [djrj-] cluster is palatal. This is of no significance in this particular instance. In [m ji:ʃljɑ:nɣ] ‘sweet’ (*milseán*) the stressed vowel of the first syllable is lengthened on metathesis. Although it has no effect on the metathesis, this change is a consequence of the resyllabification caused by metathesis. Before the sonorant and sibilant metathesis in *milseán* ‘sweet’ the rhyme of the stressed first syllable consisted of a short vowel and a single consonant. As metathesis leads to the sonorant becoming the medial of the second syllable the short / ɪ/ of the first syllable is lengthened thus maintaining the quantity of the stressed syllable despite metathesis.

Some further cases of metathesis can now be considered, to see whether the syllable optimisation view can be substantiated. Take these three instances: /galˠ.rə/ → [gar.lɣə] 'disease' (*galra*), /lˠas.rəxiː/ → [lɣar.səxi:] 'matches' (*lasrachaí*), /mɪlʲ.nʲoːrʲ/ → [mɪnʲ.lʲo:rʲ] 'miller' (*muil-neóir*). The second two examples seem to confirm the view proposed because /r/ is more sonorous than both /l/ and /s/, that is the coda of the first syllable has been reduced and the onset of the second fortified in both forms.[145]

In addition to the above types, there are also numerous cases of metathesis which leave the syllable structure unaltered: /fʲɛ.lʲuː.nˠəx/ → [fjɛ.nju:.lɣəx] 'suitable' (*feiliúnach*). Furthermore, metathesis can affect syllables which are not in contact with each other, cf. /pa.rə.fʲɪnʲ/ → [fa.rə.pjɪnj] 'parafin' (Mhac an Fhailigh 1968: 169). Here /p/ (onset$_1$) and /fj/ (onset$_3$) are subject to metathesis. Their labiality and voicelessness is what they have in common. Thus sounds which metathesise must share phonetic features and the difference between the sounds can involve one feature maximally. Thus /lʲ/ and /nʲ/ in /mɪlʲ.nʲoːrʲ/ → [mɪnj.ljo:rj] differ in nasality; /fj/ and /p/ in continuance (assuming labial to be a cover feature including labial and labio-dental articulations).[146]

This condition whereby metathesising segments must share phonetic features applies to a further type of metathesis. Consider the Middle Irish example quoted by Grammont (1971 [1933]: 351) which involves the metathesis of an onset and a coda: /ɛs.kəp/ → /ɛs.pək/ 'bishop' (*easbog*, from Latin *episcopus* showing the original sequence of onset and coda, i.e. /k-p/). A Modern Irish example is /a.kəb/ → /a.pək/ ‘with-them’ (*acub*) (de Bhaldraithe 1945: 116).

If the segments in such cases of metathesis can only differ in one feature then one can conjecture that the devoicing of /b/ on metathesis was to conform with this restriction on metathesis. An example of such devoicing is also provided by Grammont in a case of onset-onset metathesis which he observed with a young child /bo.ku/ → /ko.pu/ ‘a lot, much’ (*beaucoup*) (Grammont 1971 [1933]: 348-349, see his second division of metathesis involving *ordre articulatoire* ‘articulatory order’).

There is a strict parallel here between metathesis and assimilation and dissimilation. With the latter two processes codas or onsets of different syllables or the same syllable either become more alike or more dissimilar. A case of assimilation is that offered again by Grammont (1971 [1933]: 342) for the verb ‘to remember’ in the French of Bagnères-de-Luchon (Haute-Garonne, Midi-Pyrénées).

(208)

membras	*bembras*	*brembas* ‘to remember’	
/mem.bras/	/bem.bras/	1)	onset$_1$-onset$_2$ assimilation
bembras	*brembas*		
/bem.bras/	/brem.bas/	2)	medial$_2$-medial$_1$ metathesis

Blevins and Garrett (1998: 526-527) cite this and similar forms and mention that their segments ‘occupy long durational windows which allow for their reinterpretation in nonhistorical positions’, something which they see as parallel to instances of perceptual CV metathesis such as Vr > rV or vice versa (see also Blevins 2006: 149-150). Their analysis does not appeal to syllable position as a possible trigger for metathesis as suggested here.

An example of intersyllabic coda$_1$-coda$_2$ dissimilation is found in Old English for example, with coda dissimilation:

Latin *purpura* Old English (Northhumbrian) *purple* /purpəl/ (Campbell 1959: 212-213).

2.9.8. Double, split-level and distant metathesis

So far metathesis has been treated as if it were a one step process. But forms such as [gar.l^{ɣ}ə] ‘disease’ show that metathesis can also involve two steps. For this form the steps can be given as follows (starting from the diachronically original form *galar*).

(209)

1) [gal^{ɣ}.ər] → [gal^{ɣ}.rə]
2) [gal^{ɣ}.rə] → [gar.l^{ɣ}ə] (de Bhaldraithe 1945: 116)

From the point of view of syllable structure these steps can be analysed as

(210)

1) coda$_2$ → onset$_2$
2) onset$_2$ → coda$_1$ + resyllabification

The analysis of [gar.lɣə] in two stages is justified by the attestation of the output of the first step *galra*. Another instance of this is the following: /ʌ.l^{ɣ}ər/ → [ʌl^{ɣ}.rə] → [ʌr.lɣə] ‘plural, multitude’ (*iolar*) (Mhac an Fhailigh 1968: 168; de Búrca 1958: 136).

In view of the independent existence of the intermediate stage it is appropriate to refer to such cases as double metathesis, the result of the first step being single metathesis.[147] The term ‘double metathesis’ is justified not only because there is, in some instances, an attested intermediate stage but because the two steps involve the same elements on both occasions.[148]

There are some cases also of what can be called split-level metathesis. This involves two shifts of elements which do not interact, e.g. /lˠʌ.xər.pɑːnˠ/ → [ljɛ.pjrjə.xɑːnɣ] ‘fledgling’ (*lucharpán*), (de Bhaldratihe 1945: 116). Apart from the palatalisation of the metathesised consonants and the attendant vowel change in the first syllable (nucleus assimilation to following consonant), two levels of metathesis can be identified.

(211)

1)	/lˠʌ.x--.pɑːnˠ/	→	/l$_j$ɛ.pʲ--.xɑːnˠ/
	onset$_2$+onset$_3$		onset$_3$+onset$_2$
2)	/--.-ər.---/	→	/--.-rʲə.---/
	nucleus+coda		medial+nucleus

The changes just listed cannot be ordered and are independent of each other, hence the description as ‘split-level metathesis’, though the output of (2) results in a preferential syllable onset /pjrj-/ because (1) leads to /pj/ forming the onset of the second syllable.

While double metathesis and split-level metathesis involve segments in contiguous syllables, another type occurs occasionally which can be termed ‘distant metathesis’ because the metathesis involves syllables not in contact with each other because of an intervening middle syllable. An example of this can be found in Grammont once again: /ta.pio.ka/ → /ka.pio.ta/ ‘tapioca’, which he says is a *forme fréquente dans le peuple* ‘form frequent among the people’ (Grammont 1971 [1933]: 351). Characteristic for distant metathesis is the skipping of a syllable (two syllable skipping is not attested, not least because of the rarity or nonexistence of non-compounded four syllable stems). In Irish there are no attestations of this due to the relative rarity of trisyllabic lexical stems. But there are cases which look rather like distant metathesis to start with:

/sʲa.rə.mo:nʲ/ → [ʃa.nɣə.mo:rj] ‘sermon’ (*searmóin*) (de Bhaldratihe 1945: 116). In this instance epenthesis occurs before the nasal /m/ as it is in both instances preceded by a further consonant (thus meeting the structural description for epenthesis). Epenthesis has as its result (and indeed its function) the resyllabification of phonotactically impermissible clusters so that the first underlying coda in the non-metathesised and metathesised forms occurs as an onset on the surface. The elements in the derivation can be given as follows (these are not to be understood as stages in the production of the phonetic form).

(212)

searmóin[149]	‘sermon’ /sjarmo:nj/
/sanˠmo:rʲ/	$coda_1 + coda_2$ metathesis
/sanˠəmo:rʲ/	epenthesis
/sa.nˠə.mo:rʲ/	syllabification

2.9.9. Metathesis and grammatical information

In the examples given so far some segments which metathesise do not change their value for [palatal], cf. the example just given: /s ʲar.moːnʲ/ → [ʃanɣə.mo:rj] ‘sermon’. After the metathesis, the final coda is still palatal. Somehow the feature [palatal] has been separated from the segment which it characterised. Because a change in the value for [palatal] in word-final position has grammatical significance Irish speakers associate such a change with an altered grammatical category. If metathesis with the word for ‘sermon’ involved the change: /r–nʲ/ → [nj–r] then this might suggest that the forms not only underwent metathesis but also a change from, say, nominative to genitive.

2.9.10. Metathesis with loanwords

The phonology of loanwords offers insights into how restrictions in the phonotactics of the borrowing language operate. These restrictions can be seen mostly clearly with a language which occupies a strong position vis à vis the donor language, i.e. where there is a robust community of speakers. This was the case with later medieval Irish (twelfth, thirteenth and fourteenth centuries) which was the main language during and after the introduction of Anglo-Norman and English into Ireland. The stronger a language, the more dominant is its phonotactics. In present-day-Irish many phonotactically non-occurring segments or sequences are found in loanwords (Hickey 1982), which reflects the weak position of Irish in comparison to today's dominant superstrate, English.

This is evident in an area which is of interest here, that of affricates. The English word *jacket*, for example, exists in two forms (Hickey 1982: 150) in present-day Irish, as an established and as a recent loan: *seaicéad* /sjakje:d/ and *jacket* /dʒakət/ respectively. The older form shows simplification of the initial affricate as Irish has no affricates phonemically; it also has devoicing as Irish has no voiced fricatives in the alveolar or palatal region (i.e. */z, ʒ/). Present-day Irish devoices /z, ʒ/ to /s, ʃ/ (Hickey 1982: 151) but allows the affricate /dʒ/.

With older loans which have a word-final affricate this is automatically devoiced and shows metathesis as well. The following is a selection of early loan words: *page* → /pɑ:sjtjə/ (*páiste*), *college* → /kʌlɣɑ:sjtjə/ (*coláiste*), *baggage* → /bʌgɑ:sjtjə/ (*bagáiste*), *orange* → /ʌrɑ:sjtjə/ (*oráiste*).[150]

Furthermore, in Early Modern Irish and Late Middle Irish (the earliest period of English, or Anglo-Norman, loanwords) there was a general tendency to disyllabify nouns by adding a word final schwa as in the following examples: *game* /gɑ:mə/ (*gáma*), *rope* /ro:pə/ (*rópa*), *arrest* /rjastə/ (*reasta*)

(MacEoin 1974: 63). This applied to a variety of Middle English loans including such ones where there is no question of a word final schwa still having been pronounced in the donor dialect of English in Ireland (e.g. *box*). There is also metathesis of stop and fricative clusters in such words: *box* → /bʌskə/ (*bosca*). All these examples can be accounted for by the principle of the optimalisation of syllable structure. If ideally the onset of a syllable is to be consonantally stronger than the coda of a preceding one then this may be achieved by metathesis. Consider some of the above forms in this light (again the steps here merely serve to indicate what happened between the English/Anglo-Norman input and the Irish output).

(213)

a.	*box*	/bɔks/		
		/bʌksə/	1)	final vowel epenthesis
		/bʌk.sə/	2)	syllabification
		/bʌs.kə/	3)	metathesis
b.	*page*	/pa:dʒə/		
		/pɑ:tʲsʲə/	1)	affricate devoicing
		/pɑ:tʲ.sʲə/	2)	syllabification
		/pɑ:sʲ.tʲə/	3)	metathesis

This argumentation is basically the same as that used for language-internal developments, e.g. /tɪgʲ.sʲənʲtʲ/ → [tɪsj.gjənjtj] ‘understanding’ (*tuisgint*) (de Búrca 1958: 137). This and other interpretations rely on considerations above the strictly phonetic level, something which is in keeping with the phonological nature of metathesis. Consider in this context the segment /h/ which in Irish is phonologically a voiceless fricative. This accounts for the coda-onset switches in the following cases: /lˠɑːh.rʲəx/ → /lɣɑ:rj.həx/ ‘immediately’ (*láithreach*), /feːh.lʲo:g/ → /fe:lj.ho:g/ ‘climbing plant’ (*féith-leog*), /ah.rʲiː/ → /ar.hi:/ ‘repentance’ (*aithrighe*, present-day spelling: *aithrí*), /ah.ruː/ → /ar.hu:/ 'was changed' (*athrughadh*, present-day spelling: *athrú*) (de Búrca 1958: 137; Ó Cuív 1944: 127).

These cases of metathesis are in agreement with the contact law (Murray and Vennemann 1983: 520) which specifies that an onset be consonantally stronger than the preceding coda. The interpretation of /h/ as a voiceless fricative is confirmed by viewing its phonotactic distribution: /h/ occurs before sonorants word initially /hlɣɑ:nɣ sje:/ 'he greeted' (*shlán sé*) as do all other voiceless fricatives: /f, s, x/.

2.9.11. Exceptions to metathesis

There are a few exceptions to the restriction on coda-onset sequences. These involve the reversal of a sonorant and /x/: /g^{j}ar.xal^{j}ə/ → [gjax.raljə] 'young girl' (*gearr-chaile*), (Mhac an Fhailigh 1968: 169). Such metatheses may have been rendered more likely by the previous existence of intrasyllabic coda-onset switching with sonorants and /x/ which are, however, in accordance with the principle of syllable structure optimisation: /i.l^{ɣ}əx/ → [i.xəlɣ] 'compulsion' (*iallach*).

Despite the apparent accountability of the various forms of metathesis considered so far there remain a considerable number of instances which do not seem to optimise syllable structure, rather the contrary: /pra.s^{j}əx/ →[par.ʃəx] 'mess' (*praiseach*), /bra.dɑːn^{ɣ}/ →[bar.dɑ:nɣ] 'salmon' (*bradán*). If the transition from the initial to the nucleus ideally has a medial to smooth it and if a coda is ideally minimal then these instances of metathesis do not conform to this patterning. However, these cases do conform to the syllable optimisation view: in the cases just quoted, metathesis has the effect of reducing the gap in sonority between a vowel-final first syllable and a consonantal initial in the second syllable, i.e. /rV.s,d/ →[Vr.s,d], albeit at the expense of removing the medial in the onset of the first syllable of each form.

The phonological restructuring of words which metathesis causes may be subject not only to the internal composition of syllables but also to the weight of syllables. Certainly where a language or variety does not have many instances of metathesis it would seem that *r*-vowel metathesis is the most common and that furthermore the vowel involved must always be short. This restriction has been noted by scholars before, e.g. by Campbell (1959: 184) when discussing Old English. In addition, unstressed syllables are more likely to be susceptible to metathesis than stressed ones, this is seen in supraregional Irish English where *r*-vowel metathesis in unstressed syllables is virtually the only type which is attested, cf. *modern* [mɒdrən]. The form *purty* [pə:rti] for *pretty*, with metathesis in the stressed syllable, is attested well into the nineteenth century (Hickey 2008: 237), but not in present-day Irish English (although it is found in Appalachian English due to dialect transportation in the eighteenth century, Hickey 2010).

2.9.12. Base form and metathesis

When trying to recognise metathesis the orthographic form of Irish words can be a help but it should not be assumed that this is the base form which is then subject to metathesis. Historically, the orthography usually reflects the original input and etymologically related words may well help in establishing this. For instance, the word *bolgam* 'mouthful' is found in variously metathesised forms as shown below. It is known from the word *bolg* 'belly' and various other words referring to swelling, e.g. *bolgán* 'bubble', that *bolgam* is etymologically original, but this does not necessarily mean that this form is that which speakers are exposed to and which is then subject to metathesis, [blɣʌ.gəm] or [bʌlɣ.məg], the first of these is recorded for Western Irish dialects (de Bhaldraithe 1945: 115; Mhac an Fhailigh 1968: 168; de Búrca 1958: 136) while the second is only recorded for Southern Irish (dialect of West Muskerry: Ó Cuív 1944: 128).

In Cois Fharraige Irish, for instance, one finds /kʌsʲ.kʲe:mʲ/ → [kʌmʲ .ʃeːkʲ] ‘footstep’ (*coiscéim*) which also has the intermediate stage [kʌʃ.mje:kj] (de Bhaldraithe 1945: 116). The occurrence of such forms poses the question whether speakers simply acquire a metathesised form without realising it or whether both metathesised and non-metathesised forms are features of their speech, i.e. part of language production.

2.9.13. Phonological processes and metathesis

In conclusion some of the processes which accompany metathesis or which are consequences of it should be mentioned as these frequently obscure surface forms. For instance, partial lenition is found when a voiceless fricative follows /r/ on metathesis.

(214)[151]

coda$_1$-onset$_2$ switch; /f/ → /h/

a.	/fiəf.riː/	→	[fiər.hiː]	‘ask’	(*fiafraighe*)[42]
b.	/koːf.rə/	→	[koːr.hə]	‘coffer’	(*cófra*)

Syncope and metathesis may co-occur within a single word as in the following example (Mhac an Fhailigh 1968: 168).

(215)

dílis	/dʲi:lʲəsʲ/	‘genuine, loyal’
	/dʲi:sʲəlʲ/	metathesis

	/dʲiːsʲəlʲə/	suffixation for comparative
	/dʲiːsʲlʲə/	syncope
níos dísle	/dʲiːsʲlʲə/	‘more genuine, loyal’

In Irish a final coda may not contain a cluster of sonorant and voiceless fricative; if it does so underlyingly an epenthetic vowel occurs to resyllabify the impermissible cluster, this has already occurred for the positive form /dji:ljəsj/ above.

Metathesis may lead to an alternation of allophony if the environment demands this, consider /n/ → [ŋ]/__/k/ in the following instance (Mhac an Fhailigh 1968: 168). Note that the first form is the historically original one (E. G. Quin ed. 1990: 204, col. 38)

(216)

deargnait	/dʲarg.nˠədʲ/	‘flea’
	/dʲrʲag.nˠədʲ/	*r*-vowel metathesis
	/dʲrʲanˠ.gədʲ/	$coda_1$ – $onset_2$ metathesis
dreancaid[152]	/dʲrʲanˠ.kədʲ/	post-sonorant fortition
	[dʲrʲæŋ.gədʲ]	allophonic adjustment of nasal post-palatal vowel fronting

Metathesis can also suspend a process: the non-metathesised input has the phonological realisation /dʲarg.nˠədʲ/ → [dja.rəg.nɣədj] with epenthesis and consequent resyllabification of the final cluster in the first syllable which underlyingly has an impermissible sonorant and voiced stop cluster. This does not apply to the metathesised form /dʲrʲaŋ.kədʲ/ as the sonorant + stop cluster is homorganic and contains a voiceless stop and straddles two syllables anyway.

Epenthesis may itself be triggered by metathesis as in the following instance (Mhac an Fhailigh 1968: 169).

(217)

an chroch chéasta	/(ə) xrʌxʲe:stə/ 'holy cross, crucifix'
/(ə) xʌrxʲe:stə/	media$_1$ -nucleus$_1$ metathesis
[(ə) xʌrəxʲe:stə]	epenthesis

The devoicing of velars can be triggered in the dialect investigated by Ó Cuív by a velar changing from internal to final position through metathesis (see Ó Cuív 1944: 127 where he uses the spelling *bolamac* for the final form).

(218)

bolgam	/bʌlˠ.gəm/		'mouthful'
	/bʌlˠ.məg/	1)	onset$_2$- coda$_2$ metathesis
	[bʌlˠ.mək]	2)	final velar devoicing
	[bʌlˠ.ə.mək]	3)	epenthesis

Lastly, because of position change relative to a stressed vowel diphthongisation may be triggered by metathesis (de Bhaldraithe 1945: 116).

(219)

admháil	/ad.vɑ:l^{j}/	(= [adwɑ:lj]) ‘receipt’
amhdáil	/av.dɑ:l^{j}/	metathesis
	[aw.dɑ:l^{j}]	glide realisation of /v/
	[au.dɑ:l^{j}]	glide and stressed low vowel fusion

2.9.14. Conclusion

The attested cases of metathesis examined in this section seem to support an interpretation based on syllable structure optimisation, e.g. metathesis never gives rise to clusters or sequences not elsewhere attested in the language/variety in question. However, the essential difficulty is that metathesis remains unpredictable: it occurs in some languages or varieties and not in others. Where it occurs it is attested in some words and not in others which are structurally similar. For instance, in many traditional dialects of English in Britain, Ireland, and by transportation overseas (Hickey 2010, 2004b), in varieties such as African American English, ASK-metathesis is common, i.e. *ask* is [æks]. But this seems to be the only case where it is found, other words like *task, mask, dusk, desk*, etc. do not show this metathesis.

Restrictions can be formulated on what segments metathesise. Apart from sequences of /s/ + stop, all other clusters must show considerable phonetic similarity, e.g. liquids and short vowels or obstruents which only differ in one feature, such as [+/-continuant]. In Irish there may be a case for positing a highly restricted unordered linearity for sonorants in syllables with short nuclei. Consider the word *feiliúnach* ‘suitable’ /fjɛlju:nɣəx/ which has a metathesised form /fjɛnju:lɣəx/ in which the two internal sonorants exchange positions. Unordered linearity would posit that either the lateral or the nasal can occur in the first internal sonorant position with the opposite sonorant then occupying the second position.

(220)

[+pal]	ε	[+pal]	u:	[-pal]	ə	[-pal]
$/f^j/$		$/l^j \sim n^j/$		$/n^\gamma \sim l^\gamma/$		/x/

The specification of [palatal] for the sonorant slots must be present because, irrespective of what element occurs in a given slot, the value for [palatal] remains constant. The order of the vowels is also fixed. There are no cases in Irish of shifting of vowels from one syllable to another. A vowel may metathesise with another segment, usually a liquid, within a syllable but it remains fixed with regard to the ordering of vowels in the word as a whole. Thus an important restriction on metathesis is that in the lexically stored form of a word the syllable position of vowels is fixed. These conditions can help to account for the attested cases of metathesis but no more. All the examples discussed in this section are retrospective analyses of metathesis which has already occurred.

3. Stress in Irish

Irish is a stress-timed language. Accented syllables are somewhat louder and longer than unaccented ones. The strong stress on accented syllables has meant that unaccented ones are phonetically reduced and throughout the history of the language this fact has led to considerable alteration in the phonological structure of words.

Word stress varies across the dialects of modern Irish (Ó Sé 1989). In the West and North, stress is almost invariably on the first syllable of a word, even if this is a prefix, while in Southern Irish stress is generally on the first long vowel in a word (if there is one), otherwise on the first syllable (Ó Sé 2008). In addition the syllable coda /-x(t)/ attracts stress.

Table 31. Long vowels in unstressed syllables

All dialects of Irish
athbhean (< *ath*+Lenition 'previous' + *bean* 'wife')
/ˈavʲanˠ/ 'former wife'
fíorchinnte (< *fíor*+Lenition 'true' + *cinnte* 'certain')
/ˈfʲiːrxʲɪnʲtʲə/ 'absolutely certain'
Southern Irish
cailín /kaˈlʲiːnʲ/ 'girl', *leanaí* /lʲaˈn$_ɣ$iː/ 'children'
ag ceannach /kʲaˈn$_ɣ$əx/ 'buying'
beannacht /bʲaˈn$_ɣ$əxt/ 'blessing'

In disyllabic words in which both syllables have long vowels, the first is generally accented in Southern Irish. However, there are exceptions, see the third example below.

(221)

a.	*úrscéal*	/ˈuːrʲsʲkʲeːl/	'novel'
b.	*nádúr*	/ˈnɑːduːr/	'nature'
c.	*prátaí*	/prɑːˈtiː/	'potatoes'

Historically, there was a change of stress which affected Southern Irish[153] and which becomes evident after the Norman conquest in the late twelfth century. In Anglo-

Norman stress was on the last syllable, or the penult if the final vowel was a weak *-e*. This stressed vowel was then rendered as long in Irish: *baggage* →*bagáiste* /baˈga:sjtjə/ 'baggage', *(a)vauntage* → *buntáiste* /bʌnɣˈta:sjtjə/ 'advantage' and received stress, thus breaking the absolute rule of initial stress inherited by all dialects from Old Irish (see Hickey 1997 for a detailed discussion).

There are two opposing views on the later development of word stress in Irish. O'Rahilly (1932: 86-92) believed that the shift of stress in Irish to the final heavy syllable of a word was due to Anglo-Norman influence, Risk (1971: 589) disagreed with this. The strongest argument for the contact view is that the accent shift only occurs in the dialects of Irish spoken in the regions with the greatest Norman influence in the late Middle Ages, i.e. in the province of Munster in the South and presumably in extinct forms of Irish spoken in Leinster in the East of the country. Rockel (1989: 59) goes even further and assumes that this situation is due to Normans who switched to Irish retaining the stress patterns of their native French, thus constituting a type of language change labelled by Guy (1990) as 'imposition'. Now while there may have been a section of the Norman population for whom this was the case this group would not have been numerically sufficient to effect an accent shift. The large section of native Irish must have adopted the stress patterns of Anglo-Norman, probably from those Normans who had begun to speak Irish.

As O'Rahilly (*loc. cit.*) rightly points out a language, such as Middle Irish, with a strong initial stress (not pitch) accent and long vowels in later syllables would show an inherent instability. This can be resolved in one of two ways. Either the long vowels in post-initial syllables are shortened (this is the solution in Ulster Irish, O'Rahilly, *loc. cit.*) or they remain long and attract the stress as in Munster Irish. Western Irish lies in between these two poles and would be, on O'Rahilly's analysis, an unstable system as it allows long vowels in unstressed syllables. However, the stress pattern of Western Irish has been retained to the present-day.

Before the onset of borrowing from Anglo-Norman long vowels had arisen due to compensatory lengthening (Hock 1986, Kiparsky 2011) with the loss of Old Irish intervocalic voiced fricatives during the Middle Irish period.

Table 32. Development of stress in early Irish

Old Irish (600 - 900)	Lexical root stress
Middle Irish (900 - 1200)	Long vowels develop through vocalisation of voiced fricatives in non-initial position. This leads to tension with initial short vowels and long vowels in later syllables.

Apart from the mechanisms of stress shift or vowel shortening in non-initial syllables, a third (partial) solution is possible in Southern Irish (and is also found in Western Irish): the short stressed initial syllable in words with a long, post-initial vowel is lost by syncope bringing the long vowel into initial position, e.g. *coláiste* → [klɣɑːʃtjə] 'college', *carráiste* → [krɑːʃtjə] (Ua Súilleabháin 1994: 481). The only condition here is that the cluster resulting from syncope be phonotactically permissible, hence the lack of syncope in a word like *scadán* [skəˈdɑːnɣ] 'herring' as an initial sequence [skd-] is not possible. In some cases procope may occur as in *taispeáin* 'show' [(?)spjɑːnj] (West and North; South: [təˈspjɑːnj]).

The Southern Irish stress shift to later syllables is opaque to grammatical information. Hence it also occurs with suffixes, moving stress away from the lexical root as in *deilbh* /djɛljvj/ 'shape' and *deilbhíocht* /djɛljˈvjiːxt/

‘morphology’. The stress shift is not to be found with long initial vowels, cf. *imnigh* /ˈi:mjnji:/ ‘worry’ a fact which would suggest that there is a correlation in Southern Irish between syllable weight (M. Gordon 2004) and stress placement (stress shift is blocked by heavy initial syllables).

In present-day Irish there is a dialectal distribution which shows various attempts at solving the inherent lack of equilibrium of those words with post-initial long vowels from Middle Irish. If the shift in Munster were purely a transfer phenomenon from Anglo-Norman then one would not expect any changes in the remaining dialects.

Table 33. Long vowels in unstressed syllables

North (Ulster)		Post-initial vowels are shortened	
West (Connaught)		Syncope moves long vowels to initial syllable	
South (Munster)		Stress is shifted to post-initial long vowels	
scadán ‘herring’			
	North	ˈV.VV → ˈV . V	/ˈskadanɣ/
	West	ˈV.VV → ˈV . V	/ˈskʊdɑ:nɣ/
	South	ˈV.VV → V . ˈV	/skəˈdɑ:nɣ/
sceireog ‘fib, lie’			
	North	ˈV.VV → ˈV . V	/ˈsjkjɛrjəg/
	West	ˈV.VV → ˈø . VV	/ˈsjkjɛrjo:g/
	South	ˈV.VV → V . ˈVV	/ˈsjkjrjo:g/

There is a certain reciprocality about the Southern Irish shift. Long vowels in non-initial syllables attracted stress and this exerted tension in Irish causing Anglo-Norman stressed vowels to be interpreted as long because speakers equated stress with vowel length. With words where the deletion of the first vowel results in an acceptable cluster this vowel may well be deleted, e.g. *sparán* → *sprán* [sprɑːnɣ] ‘purse’ (Ó Sé 1984: 173).

In Western and Northern Irish stress is always initial with very few exceptions, usually grammatical words which arose through syncope or fairly recent loans from English.

(222)

Non-initial stress in Western Irish

a.	*aríst*	/əˈrʲiːsʲtʲ/	‘again’
b.	*anois*	/əˈnˠɪsʲ/	‘now’
c.	*amháin*	/əˈvɑːnʲ/	‘once, only’
d.	*anocht*	/əˈnˠʌxt/	‘tonight’
e.	*tabac*	/təˈbak/	‘tobacco’

3.1. Word stress in Southern Irish

It is assumed here that there is agreement among native speakers on what syllables are stressed and what are not. The actual phonetics of word stress (Dogil 1999) is not an issue for the current discussion. In Irish stressed syllables involve greater phonation, i.e. they are longer and slightly louder, and it is this acoustic prominence which native speakers perceive as word stress. To achieve the contrast between stressed and non-stressed syllables the latter are backgrounded by being shorter and less loudly articulated.

For Southern (Munster) Irish stress generally falls on the first long vowel in a word (possibly in a non-initial syllable), otherwise on the first syllable; in addition the syllable rhyme /-ax(t)/ can attract stress. Where a lexical word consists of two syllables with long vowels, the second is generally stressed (first example below).

Table 34. Stress patterns in Southern Irish

disyllables with final stress on long vowels		
múinteoir	/mu:nʲˈtʲo:rʲ/	‘teacher’
tiomáint	/tʲʌˈmɑ:nʲtʲ/	‘driving’
beagán	/bʲʌˈgɑ:nɣ/	‘small amount’
cailín	/kaˈlʲi:nʲ/	‘girl’
disyllables with final stress on /-ax(t)/ coda		
portach	/pʌrˈtax/	‘bog’
mullach	/mʌˈl$_{ɣ}$ax/	‘summit’
fanacht	/faˈn$_{ɣ}$axt/	‘waiting’
tacaíocht	/taˈki:xt/	‘support’
trisyllables with final stress on long vowels		
árasán	/ɑ:rəˈsɑ:n$_{ɣ}$/	‘flat’
amadán	/aməˈdɑ:$_{ɣ}$n/	‘fool’
disyllables with penultimate stress		
ag chríochnú	/əg ˈxʲrʲi:xn$_{ɣ}$u:/	‘finishing’
críochnaigh	/ˈkʲrʲi:xn$_{ɣ}$i:/	‘finish.IMPERATIVE-SG’
trisyllables with penultimate stress		
(cluiche) iomána	/ɪˈmɑ:n$_{ɣ}$ə/	‘hurling (match)’
paróiste	/pəˈro:sʲtʲə/	‘parish’
oideachas	/əˈdʲaxəs/	‘education’

Stress in the final word above, *oideachas*, would seem to imply that the sequence /ax/ attracts stress irrespective of whether it can be unambiguously interpreted as forming a syllable rhyme (this is the case with words like *portach*).
There is synchronic evidence for the productivity of the stress rule in Munster Irish: when a long vowel arises due to suffixation of a genitive ending then this attracts stress.

(223)

a. *Gaillimh* [ˈgaljɪv] ‘Galway’
b. *Cuan na Gaillimhe* [kuənɣ nɣə gaˈlji:] ‘Galway Bay’

Stress on final /-əx/ can cause syncope of a short first vowel if the resulting cluster is phonotactically acceptable: *salach* /səˈl$_{\text{ɣ}}$ax/ → [słax] ‘dirty’; *fuireach* /fɪˈræx/ → [fræx] ‘wait, delay’ (An Rinn/Ring, Co Waterford). This syncope is general in Southern Irish, e.g. *turas* ‘journey’ /tʌrəs/ → [trʌs]. If syncope cannot occur for phonotactic reasons then the pre-stress vowel is just reduced, e.g. *bacán* [bəˈkɑ:nɣ] ‘crook, peg’ (Ua Súilleabháin 1994: 481).

Stress placement in Southern Irish is furthermore sensitive to metrical feet. There is a condition on foot structure that no two stressed syllables can follow on one another in a closely bound syntactic phrase such as a noun and an adjective. If this would happen, e.g. by applying the stress pattern of lexical words, then stress is shifted backwards to avoid the metrical clash of two stress syllables in succession within a syntactic phrase.[154]

(224)

a. *caiˈlín* + ˈ*óg* → ˈ*cailín* ˈ*óg* ‘a young girl’
b. *múinˈteoir* + ˈ*mór* → ˈ*múinteoir* ˈ*mór* ‘a big teacher’
c. *Conˈtae* + *Corˈcaí* → ˈ*Contae Chorˈcaí* ‘County Cork’

The stress system of Southern Irish has been the object of particular attention from phonologists in the last 20 years or so, see Blankenhorn (1981), Cyran and Rowicka (1996), Doherty (1991), Dubach Green (1996, 1997), Gussmann (1997). These various studies all try to offer specifiable phonological reasons for why stress in Southern Irish manifests itself just the way it does. Some of the studies, e.g. Cyran and Rowicka (1996) and Gussmann (1997) use, a particular model of phonology to account for the patterns which are found. The dialect studies from the 1940s which deal with Southern dialects have remarks on stress, see Ó Cuív (1944) and R. B. Breatnach (1947). Both authors refer to stressing long vowels or /-ax(t)/ in a second syllable. They almost always mention some common exceptions.

Table 35. Exeception to Southern Irish stress rules

(1) Ordinals
tríú 'third', *ceathrú* 'fourth; quarter', *cúigiú* 'fifth', etc. with initial stress Breatnach 1947: 77-78)

(2) Short-vowel verb stems where a suffix has a long vowel

bheidís	/ˈvʲɛdʲi:sʲ/	'they would be'
bheifí	/ˈvʲɛfʲi:/	'would be' (autonomous form)

(Ó Cuív 1944: 65)

The reasons for these apparent exceptions would seem to lie in the grammar of Irish. The ordinals are compounds of a cardinal number and the ending *-ú*, i.e. *tríú* 'third' consists morphologically of *trí* + *ú*, etc. For forms of the verb 'to be' the stem is *beidh* while /-i:sj/ and /fi:/ are inflectional endings.[155] However, the appeal to morphology is not sufficient in all cases, cf. *fan* 'wait', *fanacht* /faˈnɣaxt/ 'waiting' where final stress applies to the inflection /-ax(t)/.

Sjoestedt (1931: 132-137) in her study of Irish in Dunquin (Dingle Peninsula, Co. Kerry) also deals with stress (see

Chapter V, *L'accent*). She mentions the stressing of long final vowels and of syllables with an /-ax(t)/ rhyme and notes the many exceptions to stress placement on the latter type, e.g. *fathach* 'giant', *ocrach* 'hungry', *lasrach* 'flames', all with initial stress. In these cases the failure of stress attraction for /-ax(t)/ can be accounted for on purely phonological grounds.

Table 36. Stress attraction conditions for /-ax(t)/ rhymes

	Stressed	**Not stressed**
/.r-/		*ocrach* /ˈʌkrəx/ 'hungry'
/.h-/		*fathach* /ˈfahəx/ 'giant'
/.n-/	*misneach* /mɪsʲˈnʲax/ 'courage'	
/.d-/	*oideachas* /ʌˈdʲaxəs/ 'education'	

The generalisation here is that stress attraction to /-ax(t)/ rhymes only applies if the onset of the syllable is a nasal or a stronger segment (in terms of lack of sonority, or conversely, with regard to the degree of consonantality). The 'exceptions' which Sjoestedt mentions – *fathach*, *ocrach*, *lasrach* – are now seen not to be exceptions but to have syllable onsets which are too sonorous to trigger stress attraction.

There are still one or two words which appear to behave slightly differently, e.g. *seanachas* /ˈsjanɣaxəs/ 'lore; local knowledge' with initial stress. This may be a lexicalised exception, but the more probable reason is that the consonantal strength of the transition from the first to the second syllable, in this case just the nasal /-n-/ which closes the first syllable anyway, is not sufficient for the second to receive stress.

In those cases where there is tension between stress placement on a long vowel and on an /-ax(t)/ rhyme one can see that the former wins out. This justifies the following generalisation (see the two sample words where stress attraction to /-ax/ fails and the latter is then realised as [-əx]).

Table 37. Stress attraction strength for Southern Irish

Degree 1:	Long vowels		
Degree 2:	Syllable with /-ax/ rhyme		
a.	*iontach*	/ˈuːnˠtəx/	‘very good, great’
b.	*láithreach*	/ˈlɑːhrʲəx/	‘immediately’

III The morphonology of Irish

1. Origins and development of initial mutation

Initial mutation refers to series of changes made in the initial sounds of words for grammatical purposes. Typical for such mutations would be a switch from stop to fricative, from voiceless stop to voiced stop or from voiced stop to nasal under certain conditions, e.g. after a possessive pronoun, with the past tense of verbs, after many prepositions, etc. In such instances the mutation is the realisation of a particular grammatical category (Pilch 1986; Zimmer 2005). Originally, such changes were triggered by a preceding segment which stood in a sandhi relationship with it (Lewis and Pedersen 1937: 113-118). This put the initial consonant in intervocalic or post-nasal position at an earlier stage in the development of the Celtic languages (Hickey 1996a).

It is not possible to predict the functionalisation of sandhi changes once these have arisen in a language.[156] But there are certain processes which might trigger this functionalisation (Donegan 1993). Foremost among such processes would be the decay of inflectional endings which indicate grammatical categories. This may in time lead to a re-interpretation of sandhi changes as grammatically significant. This happened in the Celtic languages with first language learners re-analysing the grammatical system as one where initial mutations signal primary grammatical categories such as case with nouns and tense with verbs (Hickey 2003a).

The synchronic status of mutation in Irish and the other Celtic languages is that of a denaturalised phonetic process which has survived due to its functionalisation on a grammatical level. Its manifestation is essentially phonological, it involves the interchange of single segments, and evinces a large degree of regularity.

1.1. Divisions within Celtic

Before discussing historical developments in the Celtic languages it is necessary to introduce some nomenclature required for such a discussion. To begin with Proto-Celtic is the stage at which Celtic separated from the remaining Indo-European dialects. Unlike Germanic, there is no single major linguistic change which marks the initial stage of Celtic as a separate branch, i.e. there is no equivalent in Celtic to the Germanic Sound Shift (Harbert 2007: 42). However, there are a number of features which are common to all Celtic languages which are assumed to be inherited from the earliest stage of the branch. The most prominent of these features is the loss of IE **p* in all positions except adjacent to a homorganic obstruent (* *t*, **n*, **s* Hamp 1951: 230). This involved the weakening and ultimate deletion of the labial plosive. Most authors assume that this is an articulatorily weak position (Sommerfelt 1962b: 347). Inasmuch as labial articulation does not involve the tongue there may be some truth to this claim. Furthermore, one could refer to loss of **p* elsewhere in Indo-European, in Armenian, where it disappeared as part of a series of shifts from stop to fricative roughly along the lines of the Germanic sound shift.[157] However, there was probably a general leniting quality to early Celtic anyway (the ultimate source of later grammatical lenition) which would have encompassed the labial plosive. Recall furthermore that IE **b*, which would have formed a plosive pair with **p* and thus added stability to the latter, is a sound which probably did not exist anyway. Adhering to the glottalic theory (Hopper 1973, Gamkrelidze and Ivanov 1983), i.e. that **b* would have been an ejective, does not result in a labial plosive pair, but may account typologically for the lack of **b* which is interpreted as /p'/, i.e. not /b/ (Greenberg 1970: 127; Gamkrelidze and Ivanov 1973: 154-155).

Continental Celtic (Eska and D. E. Evans 1993: 26) is a term which is given to Celtic as spoken chiefly in Gaul and on the Iberian Peninsula (Tovar 1961: 76-90). From the few inscriptions one can say that this form of Celtic, which was probably introduced to the Iberian peninsula from 850 BCE onwards (Tovar 1961: 78), was more archaic than later forms of the Celtic languages. For instance, Celtiberian retained labio-velars and did not lose **p*. Both Tovar (1961: 80) and Lejeune (1955) are of the opinion that lenition existed in Celtiberian although the evidence is scanty.

The term 'Gaulish' refers to that variety of Continental Celtic which is attested from inscriptions up to the first few centuries CE. For comparative Celtic studies Gaulish has a certain referential value as it illustrates a stage of the language which is prior to both forms of Celtic which developed on the British Isles. At this stage the common language was still inflectionally complex as the Gaulish inscriptions attest (Schmidt 1957, Gray 1944).

By Insular Celtic (Isaac 1993, 2007) one means those forms of Celtic spoken in the British Isles. They fall into two groups, P-Celtic or Brythonic and Q-Celtic or Goedelic. The latter branch is confined to Ireland, the Isle of Man (now extinct) and Scotland, where Irish immigrants moved in the early centuries CE bringing their language with them. Breton is a form of Brythonic which arose due to emigration to Brittany by speakers of Celtic in the South-West of Britain as a consequence of the Germanic invasions which began in earnest as of the mid-fifth century CE, possibly linking up with remnants of Gaulish in the area.

1.1.1. P-Celtic and Q-Celtic

These designations derive from the treatment of Indo-European /k^w/ in Brythonic, the branch of Insular Celtic which led to Welsh, Cornish and Breton by transportation across the English Channel, and Goedelic, the branch which led to Irish, later Scottish Gaelic and Manx (Schmidt 1993). In Goedelic, Indo-European /k^w/ is retained whereas in Brythonic and Gaulish /p/, a consonant originally lost in all Celtic languages, was regained by the shift of /k^w/ to /p/, compare Old Welsh *map* (→ Modern Welsh *mab*) and Irish *mac* /mak/ 'son'; Modern Welsh *penn* /pɛn/ and Modern Irish *ceann* /kjɑːnɣ/ 'head'. Note that this closing of a labial element to a stop had a precedent in the shifting of Indo-European /g^w/ to /b/ in Celtic, cf. Old Irish *ben* 'woman' ← IE **gwena*; compare Old Irish *béo* 'alive' and Archaic Latin *uiuos* (Thurneysen 1946: 117).

Following Sarauw (1900), McManus (1984: 179) maintains that in Archaic Irish (before 500 CE) Latin *p* was replaced by k^w as long as a labialised version of the velar stop existed in the language. McManus (1983, 1984: 186-187) sees the main body of early Latin loans[158], the so-called *Cothrige* loanwords after an early form of the name *Patricius* 'Patrick', as showing the shift from *p* to *k*, cf. *puteus* > *cuithe* 'well', *planta* > *cland* 'children', the latter probably via another variety of Celtic as it shares the meaning 'children' with Brythonic, O'Rahilly (1971 [1957]).

After the mid sixth century, the *Pátraic* loanwords enter the language without the shift of labial to velar place of articulation, i.e. Irish reintroduced the bilabial stop /p/ via Latin loans like *pian* 'pain' or *póg* 'kiss' (← *osculum pacis* ‘kiss of peace’) and still later via Anglo-Norman loanwords like *píosa* ‘piece’ or *pláta* ‘plate’. [159]

Two other views of the reestablishment of /p/ in Q-Celtic are found (1) in O’Rahilly (1971 [1957]: 80-81) who maintains that the naturalisation of * *p* was facilitated by the fact that in the fifth century CE a Hiberno-Brythonic type of dialect was still spoken in Ireland which, being Brythonic, would have had **p* and (2) in Hamp (1958: 211) who assumes that *p* and *k* were allophones of each other at an early stage. For this assumption one would at least require some present-day case or a clearly attested historic instance in which *p* and *k* are allophonic, i.e. without systemic distinctiveness. Neither O’Rahilly’s view nor that of Hamp have gained wide support; rather the chronologically differentiated stance of McManus represents a consensus opinion.

1.1.2. Gaulish

The arrival of Celts in Britain can be assumed to have taken place in the fifth to fourth century BCE. By the end of the second century BCE the Romans had begun to expand into Gaul with the suppression of Celtic. The scant linguistic remains in Gaulish (Fowkes 1940; Gray 1944) are largely onomastic: inscriptions and references in classical sources.

Gaulish still maintained many inherited inflectional endings, particularly in the nominal area, e.g. for the nominative singular of nouns as evidenced in the following forms.

(1)

	Gaulish		Irish	Welsh	
a.	/-os/	*cattos*	*cat*	*cath*	‘cat’
b.	/-on/	*dúnon*	*dún*	*din*	‘castle’

Relative chronology of syncope and apocope It would appear that the syncope and apocope which occurred in

Celtic in the pre-written period must have followed lenition as the intervocalic environment for the latter must have been available. Consider the following forms.

(2)

Gaulish	Old Irish	Modern Irish	
nertomaros	*nertmar*	*neartmhar*	'strong'

The adjectival ending *-mhar* shows lenition which historically can only occur if the /m/, from which the /v/ *mh* derives, was originally in intervocalic position. This means that the lenition must have occurred very early before the preceding *-o-* was lost by syncope.

Cluster simplification This can be seen as part of the general tendency to reduce word forms phonetically, a tendency which can be taken to be connected with the strong initial stress accent which led to a natural weakening of unstressed syllables.

(3)

	Gaulish	Old Irish	Welsh
a.	*uxellos* (*x* = /ks/)	*úasal*	*uchel* (*ch* = /x/) 'noble'
b.	*vindos*	*find*	
		later N+stop → NN, Mod. Irish: *fionn* 'fair'	

1.2. Primitive Irish: Ogam

The Old Irish period is taken to have lasted from 700 to 900. This is a period of written remains. Primarily the Old Irish documents consist of glosses, those of the Epistles of Paul in the Codex Paulinus at Würzburg which date from the mid eighth century and those of Milan from the early ninth century (Thurneysen 1946: 4-10). Of the two collections the later is not as reliable as the former. For lexicographical purposes the mid-ninth century glosses from St. Gallen in Switzerland are of great value.

However, before these documents there are attestations of what is termed Primitive Irish[160] in a non-Latin alphabet called Ogam which was chiefly used for inscriptions. These are to be found in the South of Ireland and to a much lesser extent in the rest of the country and in Wales. There are a few hundred of these inscriptions which date mainly from the fifth and sixth centuries CE. Between these and the Old Irish glosses there is a gap of almost three centuries in which the language would appear to have changed quite radically, at least going on the representations of it in Ogam and in the earliest written documents, granting that the Ogam inscriptions were in all probability in a codified literary norm used by a very small literate section of the community and removed from the spoken language of the time. The primitive Irish of the Ogam inscriptions still shows full vowels without syncope.

(4)

a.	*senobena*	→	*senben*	→	*seanbhean*	‘old woman’
b.	*inigena*	→	*ingen*	→	*iníon*	‘daughter’
c.	*maqqos*	→	*maqq*	→	*mac*	/kk/ →/k/ ‘son’

An inscription like *Avii Corbbi* ‘grandson of Corbos’ contains *avios* ‘grandson’ (still preserved in the /o:/ in Irish surnames, cf. *Ó Murchú* ‘Murphy’) and also shows the initial *C* of the modified noun in intervocalic position. This lenites to /x/, the lenition here being characteristic to this day of the masculine genitive of nouns, e.g. *mac Choilm* ‘Colm’s son’.

1.2.1. Glides and labial obstruents

In Proto-Celtic, Indo-European *p was lost. This is one of the main defining characteristics of Celtic vis à vis the other major groupings of the Indo-European family. This loss left a system without a labial obstruent as IE * *b* in all probability did not exist either as a voiced stop or an ejective according to glottalic theory. In fact the only segments with a labial character were the glide *w* and the labial-velar stop k^w. The former was present as the initial sound in *wiros* ‘man’ for example, the latter in the word *maqqos* ‘son’. Thus at this early stage then there would appear to have been glides but no labial obstruents. This situation was later to change because of a number of developments.

1) *The closing of the labial element with glides and labial-velar stops* The closing can be seen quite clearly in a word like Old Irish *ben* ‘woman’ ultimately from IE **gwena*. In those instances where there was no preceding stop, i.e. with glides, the closing led to a voiceless fricative but not to a stop. *wiros* → *ver* → Old Irish *fer* ‘man’.

2) *Loss of labial element with voiceless velar stops* The labial off-glide of /k^w/ is lost through a development like *maqqos* /$makk^w$-/ → /makk/ →/mak/ ‘son’ in pre-Old Irish (MacEoin 1993: 102).

These developments eliminated glides from this stage of Celtic. The development of labial obstruents took its course with the borrowing of words with voiceless labial stops from Latin into Irish once the so-called *Pátraic* phase had been reached (Jackson 1953; McManus 1983).

1.3. Mutations in Old Irish

For the earliest stage of Old Irish one has a system which was regular in its operation and well on the way to being the main means for realising grammatical distinctions in the nominal area. For verbs the complex system of prefixes and infixes was still productive and mutation played a secondary role being a low-level phonetic phenomenon determined by the type of sandhi which arose with the verbal clitics.

(5)
Lenition
p → f
t → θ
k → x
s → h[161]
f → Ø
m → ṽ
b → v
d → ð
g → ɣ
rr → r
ll → l
nn → n

By this stage of Irish, palatalisation and de-palatalisation had become established as a means of nominal inflection on the right-hand margins of words. Thus the sound segment inventory had been divided in two with a whole series of non-palatal and palatal consonants (Thurneysen 1946: 96-99). The palatal # non-palatal dichotomy[162] applied to all consonants except /h/.

(6)
Nasalisation

(i)
p → b
t → d
k → g
f → v
V → n + V
(ii)
b → m

d → n
g → ŋ

The ontogenesis of the nasalisation mutation involves a number of issues. As can be seen from (6), the nasalisation of voiceless stops in Old Irish produced voiced stops[163] while that of voiced stops resulted in nasals. This would imply that nasalisation consisted first of all of the transfer of the feature of voice to the initial segment of the host lexical stem and only if this was voiced could nasalisation itself be transferred. The situation is different for Welsh (see the detailed discussion in IV.3 below) which has voiceless nasals, or sequences of nasal + /h/, as the result of nasalising voiceless stops. Irish thus contrasts with those languages which show prenasalisation as an initial mutation. This is found in a sub-group of Atlantic-Congo [164] languages (Heine and Fehn 2015), chiefly Fula and Serer (J. D. Sapir 1971: 67; Ladefoged 1968: 46); S. Anderson 1976; 1992: 348-349), see section IV.10 below.

In these cases nasalisation has not been broken up into voicing and nasalisation proper, i.e. voiceless stops prenasalise to clusters of nasal and voiced stop. For Scottish Gaelic Sommerfelt (1962d: 366) and Oftedal (1956: 166-167) have noted some cases of prenasalisation of voiceless and voiced consonants. This could be a timing phenomenon on a phonetic level. Both authors admit that the stop portion of clusters like /m^{b}/ is very short. Given the fact that Scottish Gaelic is historically a form of Northern Irish and given the tentative status of these post-nasal stops it would seem justified to regard them as later developments and not the original stops present before the rise of the nasalisation mutation.

1.4. Evidence for lenition in Celtic

In Insular Celtic the contrast lenition : non-lenition develops from an earlier stage of the language which had two consonant grades strong : weak. With lenition the weak series fricativises leaving no contrast in length among stops (see Harvey 1984: 90-91):

(7)

a.	pp, tt, kk	:	p, t, k
b.	p, t, k	:	f, θ, x

The dating of lenition has been a matter of controversy for as long as it has been a subject of investigation. Pokorny (1969 [1925]: 8), for instance, is non-committal about the dating of lenition and simply states that it arose 'vorhistorisch' (= in the prehistoric period). Pedersen (1909-1913: 436) states that 400 CE is the latest possible date for lenition. However, Pedersen sees evidence for lenition in Gaulish and indeed views the shift of i.e. *p to /f/ (and then to zero) as part of the original lenition process. This would place it in a period around 800 BCE. These dates suggest a time span of over 1000 years in which this phenomenon is taken to have arisen. Furthermore, there are narrow interpretations of lenition, which refer to the differential weakening of consonants in certain morphonological environments, and wider views, which treat any weakening attested in Celtic as lenition. The broader interpretation has the advantage of connecting to the loss of IE *p in Celtic (a common Celtic phenomenon, D. E. Evans 1977: 77) and possibly to the present-day lenition of /f/ as zero.

(8)

Indo-European	p	→	f	→	Ø
Irish			f (+ lenition)	→	Ø

Martinet advocates the narrower view of lenition. He excludes the loss of **p* (Martinet 1952: 196) on the grounds that it occurred unconditionally. Positing lenition at an early period helps to account for the appearance of mutations in all the Celtic languages as a common innovation and not instances of independent parallel developments. The latter view is difficult to sustain seeing how typologically unusual grammatical mutation is and that it is present in all attested Celtic languages. The earliest remains, Gaulish for Continental Celtic and Ogam for Insular Celtic, would then be interpreted as having low-level phonetic lenition but not showing it in writing. The lack of syncope and apocope in these early attestations is in keeping with the view that lenition must have preceded the loss of unstressed word-internal vowels and inflectional endings.

In his analysis of the linguistic evidence Gray (1944: 229) assumes that there was lenition in Gaulish and stresses that it probably arose much earlier than it was recorded. He notes cautiously that the instances of lenition in Gaulish are word-internal (Gray 1944: 224) and the four possible traces are probably spelling slips on the part of those recording the language.

The possible reasons for lenition occurring in Celtic are threefold according to Martinet (1952: 1) substratum influence on early Celtic; 2) appearance in one dialect and spreading to others and 3) its existence as a low-level phenomenon among all early varieties with later functionalisation in all dialects. In fact of these three options none is an explanation, they are rather descriptions of possible situations in which lenition arose. The ultimate reason why speakers weaken consonants in voiced environments (mainly intervocalically, word-finally or in sandhi contexts) would seem to lie in the preference for weak pronunciation variants of consonants in such contexts. This preference must of course come to the fore within a speech community (Yu 2013) and then spread throughout for it to become established as did lenition in the Celtic languages.

Martinet (1952: 216-217) furthermore refers to Celtic as a possible source of lenition in Western Romance (Brandão de Carvalho 2008), basically Romance dialects west of Italy. Ternes (1977), however, in his comprehensive study of both Celtic and Western Romance sandhi phenomena refuses to be drawn on possible ontogenetic connections between the two (1977: 51). The substratum view rests on significant parallels between forms Western Romance and Celtic but remains a matter of scholarly opinion.

1.4.1. Lenition in loanwords

For the status of lenition in the sound system of Irish the treatment of loanwords is of importance. One can see with Latin borrowings in Old Irish that lenition always takes place at the favoured phonetic sites, i.e. intervocalically or in the environment of voiced consonants.[165] This lenition in non-initial positions is carried to its natural conclusion in the further development of Irish loans as can be recognised from the following forms.

(9)

	Latin		Old Irish		Modern Irish	
a.	*liber*	→	*leber*	→	*leabhar*	'book'
b.	*opera*	→	*opair*	→	*obair*	'work'
c.	*sacerdos*	→	*sacart*	→	*sagart*	'priest'
d.	*regula*	→	*riagol*	→	*rial*	'rule'

The Latin loans have an intervocalic single consonant lenited from stop to fricative if the former is voiced. With geminates there is usually just a simplification without any further change in manner of articulation: *peccatum* → *peccad* → *peaca* 'sin'. This weakening is to be seen with later Scandinavian loans in intervocalic position: *markaðr* →

margadh ‘market’ (voicing of single consonant), *akkeri* → *acaire* ‘anchor’ (simplification of geminate).

With the Anglo-Norman loans, which turn up in the language from the twelfth century onwards, lenition is also to be found, but not in all cases as it was by then losing its force as an automatic rule on borrowing.

(10)

a.		*botel*	→	*buidéal*	‘bottle’
b.		*bacun*	→	*bagún*	‘bacon’
c.	but:	*super*	→	*suipéar*	‘supper’

The loanwords which are attested with lenition in the history of Irish are in keeping with the view that lenition involves a scale on which elements move one step towards the final stage of vocalisation. One can see with the Latin loans that geminates simplify to stops while voiceless stops weaken to voiced ones and the latter fricativise and are vocalised later in Irish.

(11)

1)	geminate stop	→	simple stop		
2)	voiceless stop	→	voiced stop		
3)	voiced stop	→	voiced fricative	(→	vowel)

The lenition attested for the above loanwords is different from the lenited reflexes of voiceless stops from early Celtic: these stops become voiced in Latin and later loanwords while the simple voiceless stops of pre-Insular Celtic become voiceless fricatives in Irish, e.g. /t/ becomes /θ/ and not /d/ as in later stages of the language. In Welsh the position is different as these stops remain plosives but become voiced (like the later Irish loanwords). Indeed if one looks at lenition in a broader perspective then one finds repeatedly that the lenition cline forks at voiceless stops and takes one of two possible pathways: either these segments fricativise directly and retain their voicelessness or they retain their occlusion and acquire voice.

1.5. The origin of mutation in Celtic

Any consideration of the origin of mutation (Hannahs 2013b) must start with phenomena which are purely realisational.[166] For the earliest phase of Celtic the assumption is that certain low-level phenomena were present in the language (Martinet 1952: 196). The argument has been put forward that morphonological alternations can arise ex nihilo without any source on a phonetic level, see discussion of Ford and Singh (1983) in Comrie (2002: 74).[167] But as Comrie points out, there is no typological support for alternations as complex as the Celtic initial mutations arising without there having been phonetic sandhi as an initial trigger (Comrie *loc. cit.*).

For early Insular Celtic there was a traditional assumption (i) that early lenition was a single process and (ii) that there is no evidence for this in the inscriptions handed down from the first few centuries CE. These assumptions were lent the authority of Kenneth Jackson in his monumental *Language and History in Early Britain* (Jackson 1953: 543 + 560-561). But as Sims-Williams (2003: 48) points out, it was the lack of orthographic evidence for lenition which led to Jackson dating the rise of lenition to the late fifth century. Concerning the two basic types of lenition – (1) fricativisation of voiced stops and (2) voicing of voiceless stops – Sims Williams (2003: 48) supports the more recent view, as proposed, for example, by Ziegler (1994: 163) in her examination of Ogam inscriptions, that lenition consisted of two separate stages with (1) preceding (2) chronologically. [168] Because of the nature of the inscriptions examined by Sims-Williams, the phonotactic positions offering evidence for lenition were word-internal and so the chronology of phrase-level phonetic sandhi, which is the ultimate source of the initial mutations, cannot be reconstructed on the basis of inscriptional data.

For the discussion in this book a basic distinction between two phonotactically motivated types of lenition is necessary (without at this point further distinguishing between fricativisation and voicing).

(12)

a. Type (1): Intervocalic/word-final lenition
   Effect: compensatory lengthening, diphthongisation, reduction of unstressed syllables to schwa
b. Type (2): Word-initial lenition due to sandhi
   Effect: later functionalisation as initial mutation with preceding particles (articles, pronouns, etc.)

Type (1) continued throughout the entire history of Irish and is amply attested in Modern Irish, e.g. *saol* /si:lɣ/ 'life' < *saoghal* /si:ɣəlɣ/. Type (2) led to the weakening of the initial segments of lexical stems and is the ultimate source of alternations such as *ceist* /kjɛsjtj/ 'question' ~ *an cheist* /xjɛsjtj/ 'the question' in Modern Irish.

Low-level phonetic lenition is a necessary precondition for the development of morphonological lenition at a later point. Given the stress-accent in Celtic the inherited inflectional endings became indistinct, due to the same process of lenition word-initially. At some later point language learners of Q-Celtic then reanalysed the word-initial lenition as systemic and the word-final inflections continued to become blurred, disappearing entirely in many instances.[169] It is important to note here that a single low-level phenomenon – phonetic blurring – is ultimately responsible for the typological reorientation (Andersen 1980) from a largely inflectional language to one with initial mutation as a premier device for the indication of central grammatical categories .

Together with the decline of inflections one has a phonetic levelling of grammatical modifiers in pre-nominal position, e.g. the definite article, which came to lose its distinctiveness. This situation may well have been the trigger for language learners at this time to reanalyse the external sandhi (lenition and nasalisation) as systemic (Hickey 2003b), thus maintaining grammatical distinctions inherited in the language albeit by very different means (mutation rather than inflection).

1.5.1. Relative chronology of mutation

The relative chronology of mutation in the Celtic languages poses a problem for the historical linguist much as does umlaut in Germanic. Recall that umlaut is a phenomenon which is present in West and North Germanic to varying extents (it is one of the major changes which sets these branches of Germanic off from Gothic). Yet umlaut, in Old English for example, must be placed in the pre-written period of Old English at the very earliest after the palatalisation of velar stops (Lass 1987: 124) otherwise the pronunciation of, say, the word for ‘king’ would be [tʃɪŋ], that is, the process of palatalisation would have appeared to have become inactive before *i*-umlaut set in so that those words which experienced *i*-umlaut did not go through palatalisation.

(13)
Palatalisation and *i*-umlaut in Old English

1)	palatalisation	*cinn*	→	*chin* [tʃɪn]
2)	*i*-umlaut	*kuning*	→	*kyning* [kynɪŋ]

For the Celtic languages the mutations are shared innovations on a grammatical level as various early attestations of these languages show pre-mutation forms, cf. the many early Celtic inscriptions which do not show lenition (fricativisation or voicing). What is probably the case is that the weakening of consonants and the presence of external sandhi was part of the phonetic makeup of the Celtic languages in their common stage, like incipient umlaut in Germanic, but that this was only phonologised after they split and diverged.

Initial mutation, if it is to be utilised grammatically, must affect the onsets of stressed syllables, i.e. in Celtic the beginnings of lexical stems. This is possible if external, phrase-level sandhi phenomena develop in which the segmental composition of syllable onsets varies according to context. Such variation can involve changes which are different in type. In sonority terms these may move in one of two directions, representing weakening or strengthening of a segment respectively.

(14)

1) Lenition

a.	Voicing	Irish, Welsh
b.	Fricativisation	Irish, Welsh
c.	Degemination	Old Irish
d.	Deletion	Irish /f/

2) Fortition

a.	Devoicing	Breton

b.	Plosivisation	Breton
c.	Gemination	Old Irish
d.	Nasalisation	Irish, Welsh

Lenition can be interpreted as a general directive to weaken a segment (Bauer 1988: 381). Its actual manifestation in a particular language will depend on the constellation of segments which that language shows. A comparison of Irish and Welsh illustrates this point clearly.

(15)

Common Celtic	Irish	Welsh
/p, t, k/	/f, θ, x/	/b, d, g/

The shift of the voiceless plosives must be seen in the context of the other obstruents in both languages. Martinet (1952: 200) sees the shifts of Common Celtic obstruents as a matrix in which lenition is sensitive to the arrangement of these segments in the sound system of each language.

(16)

Common Celtic		Irish	Welsh	
/tt/	→	/t/	/t/	
/t/	→	/θ/	/d/	
/dd/	→	/d/	–	(rare in Brythonic)
/d/	→	/ð/	/ð/	

As word-internal voiced geminates[170] were common in Irish (Kuryłowicz 1957) due to the assimilation of sequences of stop plus nasal and as these simplified to single voiced stops (degemination as the lenition of geminates, Feuth

1983) the voiceless stops did not become voiced but fricativised.

Once lenition is functionalised it does not normally continue phonetically, i.e. its further development is arrested. Thus /ɣ/ does not disappear word-initially as it does word-internally in Irish (though it has in Welsh). /s/ does not lenite beyond /h/ as opposed to Andalusian Spanish which has /s/ → /h/ → Ø, e.g. *las casas*, [lah kasah], [la kasa]. /k/ does not assibilate to /ʃ/ via /tʃ/ as it has done in French, for instance.[171]

1.5.2. A case in point: the third person possessive pronouns

On the assumption that the mutations arose as sandhi phenomena one can determine what final consonants, if any, were to be found at the end of particles which later triggered mutation in the Celtic languages. The case considered here is that of third person possessive pronouns which have their origin in the genitive of personal pronouns. For Indo-European the reconstructed forms are as follows (Szemerényi 1989: 219).

(17)

SG MASC.	SG FEM.	PLURAL
esjo	*esjās*	*eisōm*

Reflexes of these can be seen for the singular in Sanskrit *asya*, *asyās* and in Greek in the non-reflexive personal pronouns. However, by the beginning of the Old Irish period (Thurneysen 1946: 278) these forms had all been reduced to a single sound *a* /ə/. After the loss of distinctiveness with the original pronouns there was obviously systemic pressure to find alternative means of indicating gender and number leading to reanalysis with language learners.

In the inherited framework of Indo-European morphology, the third person showed three distinctions, two in the singular between masculine and feminine and a plural form different from both of these. Given the pre-Old Irish reduction of *esjo*, *esjās*, *eisōm* to *a* whatever alternative means of category distinction in this area of the language's morphology was to be employed, it would have to involve a tripartite distinction to maintain the inherited distinctions.

The sandhi phenomena provided the means to retain distinctions by transferring them from the form of the possessive pronoun itself to the effect it had on the following word. Consider the types of sandhi found with the third person possessive pronouns in Irish.

(18)

a.	Masculine:	Lenition
b.	Feminine:	Gemination
c.	Plural:	Nasalisation

Historically, the gemination (Jackson 1956) is due to the final /-s/ of a preceding clitic. With a following vowel, the /-s/ lenited to /-h/ in a regular development and this /h/ then became attached to the initial vowel of the following lexical stem.

The situation in Welsh is somewhat more complicated. In Middle Welsh one has a group of stressed possessive pronouns with fuller forms, masc. *eidaw*, fem. *eidi* and pl. *eidunt* (D. S. Evans 1964: 54) which result in Modern Welsh *eiddo*, *eiddi*, *eiddynt* respectively. However, the corresponding forms to the Irish ones are the so-called dependent pronominal forms (S. Williams 1980: 48) which show two forms: masc. and fem. *ei* and pl. *eu* in the modern language.[172] When used as a masculine form *ei* causes the voicing ('soft') mutation and when it occurs as a feminine form it evokes the fricativising mutation ('aspiration'). The plural form *eu* causes no mutation with a consonant-initial word but prefixes *h* to a vowel at the beginning of a word.

(19)
Welsh

tad	/ta:d/	'father'	
ei dad	/i da:d/	'his father'	Soft Mutation
ei thad	/i θa:d/	'her father'	Spirant Mutation
eu tad	/i ta:d/	'their father'	Zero Mutation

The differences in mutations between the various Celtic languages is largely due to the manner in which the original sandhi was manifested. Thus in Irish the final /-s/ of the feminine third person possessive pronoun, 'her', led to gemination but in Welsh and in Breton (Jackson 1967a: 319) it led to spirantisation, i.e. in Irish the length of the /-s/ was transferred to the initial segment of the following word, yielding a geminate, while in Brythonic (Welsh and Breton) the fricative character of the /-s/ was carried over to the next segment in sandhi with it, fricativising it.

With Brythonic *eson* 'their' (from IE *eisōm*) the continuant character of the final nasal caused fricativisation of the next segment in Breton, hence the spirant mutation after *o* 'their'. In Welsh no mutation followed except for prefix-*h* with vowel-initial forms (Jackson 1967a: 320). In this respect Irish represents the language which in its sandhi effect is closest to the original form of the plural possessive pronoun, i.e. it triggered a nasal mutation.

(20)
Breton

ti	/ti:/	‘house’	
e di	/e di:/	‘his house’	Lenition
he di	/he zi:/	‘her house’	Spirantisation
o zi	/o zi:/	‘their house’	Spirantisation

Breton has *e* ‘his’ with soft-mutation; *he* ‘her’ has spirant-mutation (I. Press 1986: 38-55). Recall that in Irish lenition is always spirantisation (or glottalisation or deletion if the input to the mutation is an independent fricative, i.e. /s/ or /f/ respectively). The soft-mutation of Welsh and Breton, as a phonetic process, is part of the morphonological mutation nasalisation in Irish, i.e. voiceless stops ‘nasalise’ to voiced stops: /p, t, k/ → /b, d, g/, one step on the way to nasals, so to speak. This step in Welsh is realised by voiceless nasals: /p, t, k/ → /m̥, n̥, ŋ̊/.

In Breton there is a further mutation called ‘provection’ which is a kind of phonological strengthening, e.g. it converts voiced stops into voiceless ones.[173] It is not realised with voiceless segments as these are voiceless anyway. This restricted applicability of provection may be a reason for its relative rarity in Celtic.

It is important to stress that the type of mutation found at a paradigm point is not dependent on the grammatical category but simply on the type of phonetic ending in the preceding particle in any word group involved in sandhi, compare the three types of sandhi attested for Old Irish.

(21)

a.	-V#___	Lenition	(V = vowel)
b.	-O#___	Gemination	(O = obstruent)
c.	-N#___	Nasalisation	(N = nasal)

Now consider the reanalysis necessary if the inflectional endings of a language become increasingly blurred. It can change to a more synthetic type and use relational particles like prepositions and render the word-order less flexible (this has happened in English, for instance). For the Celtic languages the solution has been to retain a largely inflectional type but to change this to a kind of stem inflection in which the beginnings of lexical stems carry grammatical information by allowing systemically relevant variation in phonetic form. And this has been done while flouting the principle of root constancy which is cross-linguistically preferred.

Sandhi and cliticisation The external sandhi which resulted in the Celtic mutations implies a removal of lexical status from the mutating particle and the docking of this to the noun which serves as host. The particle loses stress and phonetic distinctiveness, a movement which is similar to the morphologisation cline *lexical item* → *clitic* → *affix* assumed by Hopper and Traugott (1993: 132). But while the mutating particles in Celtic share many of the characteristics of cliticisation (low phonetic profile, for instance) they do not undergo affixation. If they were absorbed by the host this would lead to loss of grammatical function unless some other method was available in the language to take over this function. The mutations of Celtic were not a response to lexical stems moving on the morphologisation cline but to a loss of phonetic distinctiveness.

1.6. A minimal system of mutation

Those languages which have promoted low-level phonetic phenomena, similar to those in early Celtic, to system status often have a tripartite arrangement of distinctions. Given, say, initial voicing with lexical stems due to external sandhi one then has a bipartite distinction between voiceless and voiced segments in a word-initial position. A third distinction can be arrived at in a number of ways. A common means is for the initial segment of a lexical stem to be geminated under specific circumstances. This is the situation, which one has in Tuscan Italian with the *gorgia toscana*, (Catellani 1960, Contini 1960) which is the fricativisation of stops (Gianelli 1976: 19-25), and the more general *raddoppiamento sintattico* which leads to the gemination of stops in initial position under certain syntactic conditions (R. Hall 1964). Historically, these occur with such preceding particles as ended in a consonant in Latin and whose length finds a reflex in the initial gemination of the first segment of the following word, e.g. *a* ‘to’ ← Latin *ad*.

(22)

a. *di casa* [di hasa] ‘from home’
b. *per casa* [per kasa] ‘for home’
c. *a casa* [a kkasa] ‘to home’

(Lepschy and Lepschy 1986: 77)

In Tuscan Italian these initial changes have not been functionalised. They are allophonic[174] just as Germanic umlaut was before the loss of the endings which triggered it. As long as the prepositions which cause the *gorgia toscana* are phonetically distinguishable the systemic load, from a grammatical point of view, cannot be said to be carried by these sandhi phenomena.

In pre-Old Irish, that is in the formative period for the grammatical mutations, the three distinctions now present in Tuscan Italian were also available (though gemination is no longer to be found in Modern Irish). A further distinction was available in Insular Celtic, namely that of nasalising a word-initial segment (if voiced) or voicing it (if voiceless). Other phonetic changes, particularly in secondary articulation, are conceivable, such as palatalisation/velarisation, labialisation, aspiration, etc. In the case of Nivkh (Jakobson 1971) aspiration is used as a variant in initial mutation (see section IV.9 *Nivkh* for a detailed discussion).

A further option consists of strengthening a weak segment. This can mean plosivising a fricative or unvoicing a voiced consonant. Fortition, which is traditionally termed ‘provection’ (Hemon 1975: 11-12) and which occurs in Breton, must be distinguished from the lack of a change, i.e. zero mutation.

Fortition and zero mutation Final /-s/ was a common occurrence in the inflections of Indo-European. This fricative was lost in all instances through an intermediary stage /h/ (Holmer 1947: 125) just as in those dialects of Spanish which lose final /-s/ (see sections under IV.6 *Spanish* below). As long as final /-s/ was still present in the Celtic languages it had the effect of geminating the initial consonant of a following word. A geminated segment when subject to lenition[175] was reduced to a single segment but was not further weakened. When phonological length was lost in Irish in the Middle Irish period, the inflection with a former final /-s/ had the effect of triggering no mutation. What started out as a fortition mutation (gemination)[176] resulted in a zero mutation for later stages of Irish down to the present-day, e.g. *a bronntanas* ‘her present’ with no mutation.

1.6.1. Regularity and scope of mutation

With the promotion of mutation to the grammatical level of a language an analogical spread to phonetic contexts, in which it would not occur phonetically, is common. Indeed it may well be that this regularisation is a pre-condition as otherwise a typological shift is unlikely to establish itself in a language.[177] In Old Irish lenition is just a change in manner of articulation but not of place. Furthermore, a mutation must in principle be applicable to several segments in the language. Again in Old Irish the contrast of geminates and non-geminates offered an axis on which contrast could be achieved with most consonants in the language.

Once mutation has been promoted in a language it obtains a certain degree of stability due to its functionalisation. In Old Irish the phonetic process of lenition was more or less frozen in initial position although it continued in word-medial and word-final position to its natural conclusion, i.e. it led to the vocalisation of voiced fricatives. This arresting of phonetic developments on functionalisation also applied to palatalisation in Irish: none of the palatalised velars proceeded to become affricates or sibilants as they did in many of the Romance languages.

1.6.2. Disruption of the system

It is not true to maintain that change never takes place once phonetic phenomena are promoted to system status. There is often a degree of tension between phonetic shifts and the nature of the language's grammar. Thus in Irish the system has survived a considerable degree of disruption which resulted from two major sound changes: 1) the simplification of geminates and 2) the loss of the ambi-dental fricatives /θ/ and /ð/ in the Middle Irish period. The fricatives were compensated for by a shift in place of articulation on lenition so that an earlier change of /t/ → /θ/ was rearranged as /t/ → /h/ while /d/ → /ð/ was rerouted as /d/ → /ɣ/. This led to some homophony but contextualisation was and is sufficient for disambiguation. The loss of geminates was in part compensated for by shifting the distinctions in length to ones in secondary articulation, i.e. by realigning the axis for contrast in the language's phonology (see section III.1.7.3 *Realignment of oppositions* below).

1.6.3. Overlap and zero in mutation

Two further aspects of mutation might seem to militate against its functionalisation in a language. The first of these is overlap. Consider the following examples from Modern Irish.

(23)

a.	*a phian*	/ə fiən$^{\text{ɣ}}$/	‘his pain’
	a fion	/ə fiən$^{\text{ɣ}}$/	‘her wine’
b.	*a dhuais*	/ə ɣʊəs^{j}/	‘his prize’
	an ghuais	/ə ɣʊəs^{j}/	‘the hazard’

With the first two words the result of leniting /p/ is the same as the independent /f/ which is unchanged after the feminine form of the third person possessive pronoun. With the second word pair one has the same phonetic shape for two words which have different unmutated forms. This results from the fact that both /d/ and /g/ yield /ɣ/ on lenition. These homophonous forms occur within the same word class but as speakers always know the unmutated form of the lexical stem and given sufficient contextualisation the homophony does not disrupt the functioning of the language.

Equally a language with mutation can accommodate an alternation between a segment and zero. In Irish this is, and always has been, the case with /f/ which lenites to zero: *a fhíon* /ə iənɣ/ ‘his wine’. This alternation is consistent with both the principle of lenition and the phonological structure of Irish. Given that stops lenite to fricatives then the latter if they lenite must either change their place of articulation and realise lenition as a different fricative or they must lenite to zero. Both options can be observed in Irish: /s/ lenites to /h/, but as the language has no further independent fricative, /f/ lenites to zero. Note that an independent, non-mutated segment does not result in a segment which is itself the result of a mutation operating.

This blocks a shift like /f/ → /x/ in principle as the velar fricative does not occur initially in lexical stems, i.e. it only appears in this position as the result of leniting /k/.

1.7. Effectiveness of mutation systems

If one views the Celtic languages as instances where phonetic mutation has been raised to a grammatical level then one can note certain characteristics of the systems they embody and tentatively conclude that these have general validity for languages which might functionalise such phonetic phenomena. Considering the Celtic mutations from the point of view of their effectiveness one notes that they involve minimal phonological alternations within the individual languages.

Feature change
By ‘phonologically minimal’ is meant here that only one feature of a segment is altered for a mutation. For instance, the change from voiceless stop to voiced one or of voiced stop to voiced fricative are single-step changes, in feature terms from [-voice] to [+voice] or [-continuant] to [+continuant]. The shift from voiceless stop to voiced fricative as in the Northern Sardinian dialects of Logudoro (Lüdtke 1953: 412) involves two steps, i.e. [-voice, -continuant] changes to [+voice, +continuant]. For a language which functionalises mutation, single-step alterations are generally preferred as this leads to a system which is economic in its feature changes. The more economy there is, the more distinctions can be encoded using mutations, the more effective the resulting system is for the morphology of the language concerned. This can be seen quite clearly by comparing Welsh with the Sardinian dialects just mentioned. The latter collapse voicing and fricativisation into a two-step change whereas the former

has these as separate processes which have become historically the *soft mutation* (i.e. voicing) and the *spirant mutation* (i.e. fricativisation) allowing a greater number of grammatical categories to be realised than a system with two-step changes like Sardinian.

Phonetic exponence
For a language which has functionalised mutation the manifestations of a mutation may subdivide into different kinds of phonetic realisation. For instance, in Old Irish the mutation lenition had at least two exponents (see III.1.7.3 *Realignment of oppositions* below for a further option which arose after Old Irish); nasalisation in Modern Irish also has two exponents.

(24)

a.	Old Irish lenition	→	(a)	Degemination
			(b)	Fricativisation
b.	Irish nasalisation	→	(a)	Voicing
			(b)	Nasalisation

1) *The notion of zero mutation.* For any language with a system of mutation there is an implicit contrast between a mutated form and a non-mutated form. The lack of a mutation may not only be characteristic of citation forms of lexical stems but also of an inflectional category. Thus in Modern Irish the genitive singular of nouns normally shows lenition, but with feminine nouns there is no mutation, e.g. *an chéir* ‘the wax’, *dath na céarach* ‘the colour of the wax’. In such instances one can speak of a zero mutation which is typical of this category.

2) *Absence of double mutation.* In the Celtic languages there is a prohibition on applying more than one mutation to a single segment. Synchronically, the explanation for this is

that the mutations are associated with mutually exclusive grammatical categories. From a diachronic standpoint double mutation could well have developed but would have obscured the steps of the individual mutations.

3) *Functional load of mutation system.* The main function of the mutational system in Irish lies in the nominal area. With verbs the three tenses and moods are indicated by inflectional means though in some cases syncretism has arisen due to the phonetic indistinctiveness of these endings. In such instances mutation, in this case lenition, can become the sole distinguishing factor between two verbal categories (see section III.1.8.2 *Analytical trends in Irish* below).

The core distinctions in the nominal area are those of case and number. The latter is indicated by palatalisation or by an inflection. The set of inflectional endings is becoming increasingly regularised reducing, for instance, the allomorphy of the plural (Hickey 1985a).

There are three cases in Modern Irish which show mutation. The main distinction is between the nominative and genitive. The vocative is functionally a pragmatic category, used when addressing a speaker or when attracting a speaker's attention. It is formed with the particle *a* /ə/ which takes lenition: *a Sheáin* /ə hɑ:nj/ 'John. VOCATIVE'. The separate accusative and dative forms of the older language have long lost their phonetic distinctiveness and the categories have been abandoned as can be seen from the syncretism with the following forms.

(25)

NOMINATIVE	*céle*	'fellow'
GENITIVE	*céli*	

DATIVE	*céliu*

All of these became *céle/céli* (Lewis and Pedersen 1937: 73) where the final *e/i* is taken to indicate /ə/ after a palatal consonant.

The loss of the accusative and dative left just the nominative and genitive (the ablative, locative and instrumental of Indo-European are not attested in Celtic). The mutations associated with them depend on the gender of the noun in question. In Modern Irish there is a mirror-image relationship of the mutation required with each of these cases for the two genders.

(26)

a.	Zero mutation	MASC NOM	FEM GEN
b.	Lenition	FEM NOM	MASC GEN

For the plural there are no distinctions of gender so that the nasalisation which is characteristic of the genitive plural applies to both genders and in the nominative both have zero mutation.

1.7.1. Analogical spread and regularity

It is a commonplace that a phonetic phenomenon which is functionalised tends to spread to instances within a paradigm or lexical category to which it did not apply before. For instance, umlaut in German originally affected nouns with *r*-plurals so that pairs like *Lamm* : *Lämmer* ‘lamb’ : ‘lambs’ are genuine umlaut plurals. Nouns which formed their plurals in *-a* did not have umlaut to begin with (there was no phonetic motivation as the inflection contained a low vowel) so that instances like *Baum* : *Bäume* ‘tree’ : ‘trees’ are cases of analogical spread of umlaut as a plural formation device (Lockwood 1965: 95).

The mutational system in the Celtic languages has experienced the same type of analogical spread to achieve paradigm regularity (Thurneysen 1946: 149). An example is provided by Latin loanwords. Recall that IE **p* was lost in Celtic so that Irish has no inherited lexical stems with an initial /p-/. However, with the Christianisation of the country many Latin loans entered the language, some of which had an initial /p-/. For these the shift of /p/ to /f/ on lenition was introduced on the analogy of the fricativisation of other voiceless stops on lenition, *peccad* 'sin' (← Latin *peccatum*), *do pheccad* /do φekað/ 'your sin' (*do* 'your' takes lenition).

The second kind of analogical levelling concerned not the introduction of a new lenition type (/p/ → /f/) but the regularisation of the output with a given input. Normally /s/ was lenited to /h/. There existed, however, instances of /s/ which went back to one of two clusters, either * *sw* or **sp* (Thurneysen 1946: 117). Both the /p/ and the /w/ after /s/ disappear (compare Old Irish *siur* 'sister', lenited as *fiur*, with Old English *sweoster* or German *Schwester*). The remaining /s/ lenited, however, to /f/ in accordance with the lost labial segment and not to /h/ (Thurneysen 1946: 84). Now there are two possible situations which may have obtained in Old Irish. The first is that there was a phonetic difference between a normal /s/ and one derived from a cluster **sw* or **sp* (perhaps a difference in rounding for the latter). The second situation is that both kinds of /s/ were phonetically identical but that speakers stored the lenited form of lexical stems as exceptions in words which had an initial * *sw* or **sp*. If this were the case then the later levelling of lenition for all types of /s/ to /h/ had the advantage that there was no need to store lenited forms of stems as exceptions.

Regularisation in the nominal area The mutations which occur with nouns are historically determined by the nature of the clitic preceding them. As part of the process of regularising the nominal area adjectives were made to agree in mutation with the nouns they accompanied.

(27)

a.	*an fear bídeach*	/b/ *b*	'the tiny man'
b.	*an bhean chainteach*	/k/ *c* → /x/ *ch*	'the talkative woman'

This situation is quite different from that which applied when lenition arose. Originally mutation was triggered directly by a clitic preceding a lexical host. The connection with a certain grammatical category was indirect as a mutating clitic was always associated with a given category. With adjective agreement the mutation is determined by the gender and case of the associated noun and not by any clitic preceding the adjective.

The mutation of adjectives on the grounds of the gender of the governing noun would seem to have applied to Old Irish (Strachan 1976 [1949]: 21). However, this point does not appear to have attracted the attention of linguists (though Thurneysen 1946: 228 does not mention it).

There are instances in which nasalisation affects adjectives in a similar way to that of lenition just outlined. According to the standard this does not occur but many dialects attest the nasalisation of adjectives where nouns are nasalised due to a preceding nasalising segment: *seolta na mbád mbána* 'the sails of the white boats' (Ó Siadhail 1989: 131).

In keeping with the VSO syntax of Irish, adjectives followed and still follow the nouns they govern. However, there are instances of preposing in which the adjective is so closely bound to the noun stem that the resulting form can be considered a compound. Here lenition always occurs, i.e. the preposed adjective lenites the nominal host. With some frequent adjectives there exist (from Old Irish, Thurneysen

1946: 230) special preposed forms. The lenition of the noun bears no relation to its gender and the preposed adjective is not lenited unless a further clitic, which would require lenition, precedes the entire compound. The gender of the compound is governed by the host noun.

(28)
Preposed adjective in compound forms: *droch*- ‘bad’

drochbhlas	(*droch* + *blas*)	‘bad taste’	(MASC)
an drochbhlas		‘the bad taste’	(MASC-NOM)
drochbheatha	(*droch* + *beatha*)	‘bad, immoral life’	(FEM)
an dhrochbheatha		‘the immoral life’	(FEM-NOM)

Postposed adjective:	*olc*	‘bad’
	leabhar olc	‘bad book’

1.7.2. Lenition of sonorants

Coronal sonorants are produced in the dental-alveolar-palatal area and exclude the labial nasal *M* which can lenite normally in Irish to *V* and the velar nasal *ŋ* which only occurs as the outcome of nasalising *G*. As the result of a mutation can never be the input to a further mutation (synchronically speaking), the velar nasal need not be considered here.

Analogical lenition In Southern Irish there is just a two-way distinction with sonorants as opposed to the three-way one found in the West and North. This has meant that the reduction of polarised articulation [178] which has been used in the remaining dialects for lenition with sonorants is not possible in Munster.

(29)

		West, North	South	
a.	*náire*	/nˠɑːrʲə/	/n$_\gamma$ɑːrʲə/	‘shame’
b.	*a náire*	/ə n$_\gamma$ɑːrʲə/	/ə n$_\gamma$ɑːrʲə/	‘his shame’

In the South, in South-West Cork, a type of analogical lenition arose for /r/ which involved palatalising the sound as opposed to depolarising it: *beir* /bjɛrj/ ‘bear’, *rug* /rʌg/ → [rʲʌg] ‘bore’ (after Ó Cuív 1944: 49).

1.7.3. Realignment of oppositions

The coronal sonorants *n*, *l*, *r* have no obvious fricative equivalents unlike /p, t, k/ which have the continuant forms /f, θ, x/ so it is worth considering whether lenition can apply to coronal sonorants and if so what phonetic exponents were and possibly are used. Consider the following options.

(30)

a. devoicing of sonorants (Welsh)
b. geminate simplification (up to and including Old Irish)
c. depolarisation (as of Middle Irish)

The situation in Welsh is the most regular because the opposition voiceless # voiced applies to all obstruents and sonorants, bar /s/ (as opposed to Irish which only has the opposition for obstruents). Lenition of sonorants is phonologically devoicing so that Welsh has two sonorant series, i.e. /m, n, ŋ, l, r/ and /m̥, n̥, ŋ̊, l̥, r̥/, see section IV.3.

The second option of degemination is one which held for Old Irish, i.e. as long as the language had a general distinction of length among consonants. However, phonological length was lost in the Middle Irish period. This, like the loss of inter-dental fricatives, led to a disruption of the mutational system as lenition with sonorants was threatened with the loss of phonetic exponence. But by the Old Irish period all geminate sonorants in the language were either palatal or non-palatal, with the latter phonetically velarised as their reflexes are today and because, viewed typologically, a language with a palatal # non-palatal opposition tends to velarise the non-palatals to increase their articulatory distinctiveness, e.g. Russian (Jones and Ward 1969: 133; Padgett 2001). A similar development took place in Classical Arabic (Versteegh 2002: 87-90) which developed a contrast of non-emphatic ~ emphatic for four consonants /t, s, d, ð/. On these Versteegh (2002: 20) notes: 'In Arabic, these [the emphatic consonants - RH] are articulated by a process of velarisation: the tip of the tongue is lowered, the root of the tongue is raised towards the soft palate, and in the process the timbre of the neighbouring vowels is shifted towards a posterior realisation'.[179]

The reverse is found in Persian (Bhat 1978: 77) where the velars /k, g/ were palatalised after the rise of pharyngealisation. Thus the reactive polarisation (velarisation) of non-palatal segments in Irish can be linked to the establishment of palatalisation as a grammatical device in the language, i.e. one had originally [n] (before palatalisation), then [n] : [n^{j}], later /n/ : /nj/, phonetically [nɣ] : [nj]. The velarisation of sonorants is particularly audible (given their sonority) but is equally present with obstruents, e.g. [$t^{ɣ}$] : [tj], [sɣ] : [sj].

With the loss of phonological length the articulatory polarisation on a palatal-velar axis acquired a new status in the language. From this point onwards lenition became a new manifestation of the reduction of an articulation on this palatal-velar axis.[180] If one uses the term ‘polarisation’ to refer to either articulatory extreme on the axis in question then ‘depolarisation’ can be taken to refer to a movement away from either extreme to a midway neutral articulation. To give a concrete example, both [nj] and [nɣ] are polarised articulations in a palatal and velar direction respectively; a depolarised articulation is represented by a neutral, alveolar [n].

(31)

	palatal		alveolar		velar
a.	l^j	→	l	←	l^{γ}
b.	n^j	→	n	←	n^{γ}
c.	r^j	→	r	←	r^{γ}

The tripartite distinction of sonorants can be formulated in terms of articulatory gesture: the tongue can either be arched upwards for palatal or downwards for velarised sonorants. If there is no salient arching of the tongue the configuration is neutral. This results in the following distribution.

(32) *Relative tongue configuration*

a.	convex	–	neutral	–	concave
b.	(palatal	–	alveolar	–	velar)

Relative tongue position is a classification of sonorants according to secondary articulation. The primary articulation is always apical (or possibly laminal) with lateral release for *l*-sounds and nasal release for *n*-sounds. For convex tongue (palatal) configuration the primary closure is between the blade and immediately behind the lower teeth and for concave tongue (velar) configuration the primary closure is between the apex and immediately behind the upper teeth.

For Old Irish the decline of phonological length in effect meant that the opposition long # short was realigned as polarised # non-polarised as of Middle Irish. This was possible as the former geminate sonorants had a secondary articulation which tended to one or other extreme on a palatal-velar axis. The non-geminate sonorants were not affected, these retaining to this day their neutral, alveolar articulation.

(33)[181]

a.	*baile* [-l_j-][26] 'town'	:	*buille* [-l^j-] 'mad'
b.	*duine* [-n_j-] 'people'	:	*bainne* [-n^j-] 'milk'

There are parallels to this situation in other languages which have undergone changes with former geminates. An example which Martinet has cited (1952: 204) is the development of Latin geminate sonorants to palatals in the development of West Romance languages, Spanish *caña* ← Latin *canna* 'reed' with single consonants having identical reflexes: *luna* ← *lūna* 'moon'. According to Martinet it is the longer duration and 'superior energy' which is 'transmuted' into a 'wider application of the front of the tongue' (*loc. cit.*), i.e. which results in palatalisation.

This view would seem to harbour the clue to the switch from length to polarisation with sonorants in Irish. Consider that sonorants with extreme tongue configuration (concave or convex) require more time for such a position to be adopted than neutral alveolar consonants do. As long as a language has phonological length distinctions elsewhere, for obstruents for instance, this length can be interpreted systemically. By these means one can account for the double spellings of polarised sonorants found in Old Irish

(Thurneysen 1946: 85-86). With the general demise of distinctive length in Irish, polarisation is promoted within the phonological system as length no longer distinguishes certain types of sonorants.

Polarisation is a general phenomenon and applied not just to former geminates but also to clusters of sonorant and obstruent and to those sonorants which stood in the onset of stressed syllables. This last environment is revealing as it is one of phonological strength. The polarised segments are traditionally referred to as ‘tense’, a feature which is without a direct phonetic interpretation but which covers the polarisation being discussed here and correlates on a phonological level with the notion of ‘strength’. Recall that lenition is a general directive to weaken a segment. In view of the phonological strength of polarised articulations, depolarisation can be seen as obvious weakening and thus be a phonetic exponent of morphonological lenition.

Polarisation with r With *r* there is least tendency for a tripartite realisation [rɣ, r, rj]. In the dialects of Irish a two-segment arrangement [rɣ, rj] is usually found. In addition, depolarisation as an exponent of lenition has long been on the wane and scholars are uncertain as to whether it exists robustly in any of the dialects today (Ó Cuív 1986: 398).

Traditional treatments of sonorants In Irish linguistics velarisation for obstruents is not generally indicated but in the case of sonorants capital letters for polarised forms are used, e.g. L (= $l^{ɣ}$), L′ (= lj); N (= nɣ), N′ (= nj). In this transcription system lowercase letters denote depolarised or non-polarised articulations.[182] In fact what results is a putative fourway distinction for the sonorants *l* and *n*. Consider the examples given by D. Ó Baoill (1979: 90) for

Donegal Irish (I have adapted his transcription and not indicated consonant length).

(34)

a.	*mionna* /m^{j}ʌn$^{\gamma}$ə/	/-N-/	‘oath’
b.	*miona* /m^{j}ʌn$_{\gamma}$ə/	/-n-/	‘minor.PL’
c.	*i mbinn* /ɪ m^{j}in^{j}/	/-N´/	‘in, on a cliff’
d.	*i min* /ɪ m^{j}in$_{j}$/	/-n´/	‘in meal’

The transcription is intended to illustrate the lack of palatalisation in the form *i min*; the word *miona* does not show velarisation either. But there is no possibility of contrast between the two types of nasals in these two words in any context. So the transcription system implies a systemic contrast which is not backed up by the phonetics. Wagner (1979 [1959]: 20) actually said this without touching on the distribution, contrast and functional load of these segments in the variety he was investigating (Irish in Teileann, South Donegal).[183]

1.7.4. Tenseness, gemination and lenition

The origin of geminates in the older stages of the Celtic languages raises certain issues of interpretation. Consider Common Celtic at the time before the mutations had developed into the types of alternations known from later history. It is assumed that the initial consonants of lexical stems were in some phonetically unspecified sense ‘tense’ and so distinct from medial consonants which were then later lenited. Assuming that tenseness was phonetically gemination then geminates would have occurred stem-initially in the pre-lenition phase. These geminates could in fact be found, according to Lewis and Pedersen (1937: 48-49), in three different positions.

(35) ‘

Tense’ consonants and syllable position

(i) In absolute initial position, i.e. in the onset of stressed syllables without lenition: *lá* /lɣɑ:/ ‘day’, also after /s/: *slua* /slɣuə/ ‘crowd’ (Modern Irish).

(ii) In word-final position where, because of cluster simplification, a geminate arose in final position: *cland* → *clann* /klɣɑ:nɣ/ ‘children’.

(iii) In medial clusters with dental obstruents e.g. /dl/ in *codladh* ‘sleep’ (later simplified: Modern Irish /kʌlˠə/).

These segments would appear to be somewhat different in nature. The types found in (ii) and (iii) were geminates in the true sense as they arose from the assimilation of a stop by a sonorant and the absorption of its length yielding a phonologically long segment. The first type is treated as a geminate not because of any assimilatory phenomena but because it does not lenite like medial non-geminate segments.

The medial geminates known to have existed in Celtic arose in many cases because of assimilation of one element in a cluster to another.[184] The geminates in forms such as **kattos* ‘cat’ in intervocalic position remain in the pre-lenition period and then simplify with the development of fricatives for simple stops in this position (Greene 1966: 116).

(36)

	Lenition CC → C; C → F		
	Proto-Celtic	Irish	
a.	**kattos*	*cat*	'cat'
b.	**katus*	*cath*	'battle'
c.	**tōtū*	*túath* /tu:əθ/	'people'
d.	**abbero*	Welsh *aber*	'confluence'

This leads to an equivalence of geminates with later unlenited consonants and single consonants with lenited ones (Greene 1973: 129). This can be seen quite clearly in the treatment of Latin loanwords, which also demonstrate this equivalence (Harvey 1984: 94).

If one allows for a stronger articulation in initial vis à vis medial position in the pre-lenition stage then one is talking of a realisational difference – a view supported by Greene[185] – which, if present, was a correlate of position in relation to a stressed syllable: the strength is a result of the initial position of the stop and the medial position is what leads to a further reduction of the stops yielding fricatives which later attain systemic status.

Greene following Sommerfelt (1954: 116-117) denies that there could have been a threefold systemic distinction among stops of the kind geminate : 'tense' stop : simple stop (in their notation KK : K : k where capitalisation is intended to imply 'tenseness').[186] Threefold distinctions did in fact later arise for sonorants (see III.1.7.2 *Lenition of sonorants* above) but they were defined not on a length axis but on one of relative palatality, i.e. by exploiting the distinction between palatal - neutral - velarised.

The status of loanwords is relevant here. Sommerfelt (1962e: 326) has pointed out that Norse loans in Irish tend to have voiceless stops replaced by voiced ones and quotes uncontroversial loans like Old Norse *brók* > Irish *bróg* 'shoe'. Sommerfelt's argumentation is that the Norse voiceless stops were perceived as weak like the Irish voiced stops and uses this to strengthen his claim that the Irish voiceless stops were tense and fortis, more so than in other languages. However, it may simply have been that the Norse stops were not perceived as voiced but that they were subject to the productive process of post-vocalic voicing on or after borrowing. This account is in keeping with the application of other mandatory operations on loans such as the obligatory metathesis which affected all words with voiced affricates which were borrowed from Anglo-Norman into late Middle Irish (Risk 1971, 1974, Hickey 1997): *page* → *páiste* ‘child’, *orange* → *oráiste* ‘orange’.

1.8. Extensions after functionalisation

With the establishment of mutation as the main grammatical device in the nominal area certain extensions arose which helped supplement paradigms at those points where mutation was not to be found. The lack of mutation had two basic reasons. Either (i) no mutation applied because there was no historical justification for it and no analogical spread had occurred into a particular point in a paradigm or (ii) mutation would have applied but the particular noun contained an initial segment which was either immutable, e.g. *an lacha* /ənɣ lɣ-/ ‘the duck’ (fem.), or mutation was blocked due to the homorganic nature of the final consonant of the mutating element and the initial consonant of the following noun, e.g. *an dúil* /ənɣ d-/ ‘the desire, hope’, cf. *an cháin* /ənɣ x-/ ‘the punishment, fine’.

These two situations provoked two different responses. The first was the morphologisation of the lack of mutation. Thus zero mutation has become in the course of the history of Irish a signal for the genitive singular of feminine nouns and for the nominative of masculine nouns. The second response was the introduction of new devices which played the same role as mutation. Examples of these are the prefixed segments *T*, /h/ and *N*. The instances discussed below all refer to Modern Irish which is used for purposes of illustration. Similar phenomena are found in present-day Welsh and Breton.

Prefix-T

By this term is meant the prefixation of *T* to vowel-initial stems in certain contexts. Because lenition cannot be shown before vowels and some consonantal clusters any grammatical category which relies on lenition to be indicated will not be seen with vowel-initial stems.

Prefix-*T* occurs with masculine nouns in the nominative case. This is not a leniting environment but the *T* helps to distinguish gender as it is not present with feminine nouns in the same case. With consonant-initial stems the distinction between the two genders is realised by zero mutation in the masculine and lenition in the feminine.[187]

(37)

a.	MASC	*an ceann*	'the head'
		an t-arán	'the bread'
b.	FEM	*an bháisteach*	'the rain'
		an áit	'the place'

Use of pre-vocalic /h/

There is no phonotactic prohibition on the juncture of two vowels between word forms in Irish. Nonetheless, /h/ occurs in the nominative plural and genitive singular of vowel-initial nouns for reasons which are now morphonological (determined by grammatical category).[188]

(38)

a.	NOM PL	*na héin*	'the birds.MASC'
b.		*na háiteanna*	'the places.FEM'
c.		*muintir na hÉireann*	'the people of Ireland'

The reason for assuming that there is no such automatic phonotactic rule for vowel juncture is that the presence or absence of an intervocalic /h/ is exploited in those cases where different mutations may occur in the same environment. The feminine possessive pronoun of the singular takes zero mutation, i.e. it triggers neither lenition nor nasalisation as opposed to the masculine singular and the plural forms of the same pronoun.

(39)

a.	*a* 'his'	*a athair*	/a ahər^{j}/	'his father'
b.	*a* 'her'	*a hathair*	/a hahər^{j}/	'her father'
c.	*a* 'their'	*a n-athair*	/a nahər^{j}/	'their father'
d.	*a* 'his'	*a chara*	/a xarə/	'his friend'
e.	*a* 'her'	*a cara*	/a karə/	'her friend'
f.	*a* 'their'	*a gcara*	/a garə/	'their friend'

This pronominal paradigm shows that hiatus /h/ is the exponent of zero mutation (i.e. neither lenition nor nasalisation) with vowel-initial forms, while the lack of this /h/ corresponds to lenition with consonant-initial words. Nasalisation has the regular manifestation of an *N*-prefix before a vowel-initial form.

Prefix-*T* occurs furthermore with the nominative singular of feminine nouns beginning with *S* followed by a sonorant (i.e. *S + L, N, R*) or a vowel. It is also to be found with masculine nouns in the genitive case. Note that these cases are both points in the nominal paradigms where lenition is required for both genders.

(40)

a. FEM NOM

an tseachtain 'the week'

	an bhróg	'the shoe'
b.	MASC GEN	
	i lár an tsamhraidh	'in the middle of the summer'
	hata an mharcaigh	'the hat of the rider'

Nasalisation and vowel-initial nouns
In present-day Irish a prefixed *N* is found before nouns at those points in a paradigm where nasalisation occurs. By these means nasalisation has a realisation before nearly all segments, including vowels, but not before sonorants or clusters of *S* and a further segment.

(41)

GEN PL	
ceol na n-éan	'the music of the birds'
dath na mbád	'the colour of the boats'
fad na scéalta	'the length of the stories'

1.8.1. Redundancy with mutation

There are cases where the mutation is redundant because the phonetic shape of the mutating element is sufficiently delimited from other such elements. A case in point is offered by simple prepositions as shown in the following.

(42)

a.	*ar an mbord*	~	*ar an bhord*	'on the table'
b.	*ag an bhfear*	~	*ag an fhear*	'with the man'

These prepositions can take either lenition or nasalisation according to the standard of Modern Irish. There would seem to be a dialectal preference with lenition occurring more in the South and nasalisation in the West and North (MacEoin 1993: 113).

Note that with simple prepositions the article intervenes between these and the lexical stem which is mutated. This is a further exception to the situation in Irish that a mutating particle immediately precedes the word it affects: *ar an mbaile seo againne* [on the town this to-us] 'in our town'. The variation in mutation only applies to those prepositions which take the nominative case. Apart from these there is a large number which govern the genitive in which case the mutation is that which is required by the gender of the noun in question.

(43)

a. *trasna na farraige* 'across the sea' (FEM + Ø)
b. *timpeall an chnoic* 'around the mountain' (MASC + L)

The situation with prepositions is quite different from that of particles which take different mutations according to function. Here there is one and the same phonetic form but a difference in the effect on the following lexical stem. The simplest case is *a* /ə/ which apart from its function as a possessive pronoun of the third person also plays a role as a vocative particle with lenition and as an indirect relative particle with nasalisation.

(44)

a. *a Sheáin* 'John. VOCATIVE'

b.	*an fear a bhfuil amuigh*	‘the man who is outside’

Loss of clitics and the deconditioning of mutations
In some instances, mutation has come full circle in Irish as clitics which trigger it are often deleted in spoken Irish. A case in point is the interrogative particle *an*. This is placed before a verb and nasalises it. It is so weakly stressed that it quite often disappears in colloquial language (Ní Chiosáin, Welby and Espesser 2012: 82). The net result is that for interrogatives the nasalisation of the initial segment of the verb is the only segmental cue, i.e. it has been deconditioned with the loss of *an* and hence morphologised.

(45)

a.	*Tuigeann sé an cheist*.	‘He understands the question.’
	[tɪgʲənˠ ʃe: ə çɛʃtʲ]	
b.	*An dtuigeann sé an cheist*.	‘Does he understand the question?’
	[ən dɪgjənɣ ʃe: ə çɛʃtʲ] →	[n̩dɪgʲənˠ] → [dɪgʲənˠ]

1.8.2. Analytical trends in Irish

Simplification in the verbal system In the course of the history of Irish the blurring of phonetic forms has meant that with verbs, formerly contracted forms were resolved in Early Modern Irish into two separate forms.

(46)

a.	*atá*	→	*atá mé*	'I am'
b.	*ataoi*	→	*atá tú*	'you are', etc.

The many infixed pronouns became separated from verb forms and postposed in much the same fashion.

(47)

a.	*am mac*	→	*is mac mé*	'I am a son'
b.	*ro-m-chráid*	→	*ro-chráid mé*	'he has harassed me'

The complex system of verbal affixes (prefixes and infixes) was simplified. For instance, the prefix *ro* which rendered a preterite form perfect (Strachan 1976 [1949]: 154) was absorbed into the stem of the verb yielding a grammatically opaque form as with the dependent past of 'to be': Old Irish *ro-bá* → Modern Irish *raibh* (Strachan 1976 [1949]: 71). During the Middle Irish period many affixes were reduced to a single prefix *do*. This was originally a verbal proclitic with the meaning 'to, for' which became attached to the beginning of a verb. The stress remained on the verb stem, hence the term 'deuterotonic' for such compound forms which had stress on the second syllable (E. G. Quin 1975: 40). *Do-* always induced lenition as in Old Irish *do-beir* /doˈvjerj/ 'he gives' and later became in some cases absorbed into the verbal stem leading to a group of irregular verbs in the modern language. A more widespread development of *do* was its generalisation as the signal of the past tense. It retained its leniting quality and was thus responsible for the lenition which is seen at the beginning of verbal stems in past and conditional forms in present-day Irish. As a separate prefix, *do* was retained well into the modern period and has only recently been lost before non-vocalic verb stems. In this position it is not included in the standard of Modern Irish (Government of Ireland 1958), nor is it found in Western and Northern Irish but it is general in

Southern Irish. Inasmuch as *do* is lost, lenition is not a conditioned change in the past of verbs but an integral part of the forms which indicate this category. This interpretation is strengthened by the fact that with vowel-initial verb forms and those which begin in *F*, a contracted form[189] of *do* is still mandatory.

(48)

a.	*ithim*	'I eat'	:	*d'ith mé*	'I ate'
	/ɪhɪmʲ/			/dʲɪ mʲe:/	
b.	*fágaim*	'I leave'	:	*d'fhág mé*	'I left'
	/fɑ:gimʲ/			/dɑ:g mʲe:/	

Both vowel-initial forms and those with a lenited initial *F* have no consonantal onset in the past, because in the latter category the *F* is reduced to zero as the regular manifestation of lenition. The signal of the past at the beginning of the word stem is thus /d/, the reduced form of *do.* It is true that with initial *F* the contrast *F* : Ø for the present and past respectively would be sufficient to separate the tenses but the past forms of *F*-initial verbs would seem to have been associated with vowel-initial stems and obtained the *D*-prefix like these. These are really cases of double marking: *F* plus lenition and then *D*-prefixation where in fact the lenition would have been sufficient.

In the history of Irish there would appear to have been some confusion between vowel-initial forms and those with initial *F* as the latter lenites to Ø. When language learners were exposed to a vowel-initial word they would not have been sure (without a context) whether this was vowel-initial in the citation form, i.e. without mutation, or whether it began with *F*. Thus one has many instances of *F* being added to the beginning of a word although there is no historical justification for it as speakers assumed the form, as they were exposed to it, was lenited.[190]

(49)

a.	*uacht*	→	*fuacht*	'cold.NOUN'
b.	*uar*	→	*fuar*	'cold.ADJ'
				(Lewis and Pedersen 1937: 130)
c.	*adhb*	→	*fadhb*	'problem'
		(de Bhaldraithe 1945: 114, Thurneysen 1946: 134)		

2. Initial mutation in Modern Irish

The consonants of Irish phonology divide into independent and dependent groups. The former can occur in word-initially without initial mutation, but the latter cannot, for example there are no lexical words whose citation forms which have an initial *V*, *ɣ* or *X*. These consonants are dependent on a syntactic environment which triggers an initial mutation, hence the designation 'dependent'.[191] Notationally, this can be represented as follows.

(50)
x ^ *V* x ^ *ɣ* x ^ *X*

where 'x' stands for the particular mutation - lenition or nasalisation - which gives rise to the segment it is linked to by the caret. There are a number of generalisations which can be made concerning initial mutations in Modern Irish (Ellis 1965; Grijzenhout 1995; Duffield 1997; Dubach Green 2006).

1) In general, consonants cannot have both dependent and independent status when the mutation is lenition. The two exceptions to this are (i) *F* which occurs in citation forms, i.e. independently, and also as the result of leniting *P*, i.e.

dependently: *a fíon* /ə fi:ənɣ/ ‘her wine’, *a phian* /ə fi:ənɣ/ ‘his pain’ and (ii) /h/ which occurs independently (see section II.1), as in *híleáil* ‘heel’ (verb), and dependently as in *a shaol* /ə hi:lɣ/ ‘his life’.

When the mutation is nasalisation, there are far fewer restrictions. All voiced stops can occur independently and dependently, e.g. *deoch* ‘drink’, *a dteachtaireacht* /ə djaxtərjəxt/ ‘their message’. The same is true for all nasals except the velar nasal, e.g. *neart* /ə njart/ ‘strength’, *a ndath* /ə nɣa/ ‘their colour’; but: *a ngairm* /ə ŋarmj/ ‘their profession’.

2) A dependent segment is the result of the application of one and only one mutation. Lenition and nasalisation cannot have the same segment as output, with the exception of *V* which is both the result of leniting *B* and of nasalizing *F*: *a bhia* /ə vjiə/ ‘his food’ (← /bj/ + L), *a bhfia* /ə vjiə/ ‘their deer.SG’ (← /fʲ/ + N).

3) A mutation can never be applied twice to the same segment; there is no such thing in Irish as a segment which can be lenited and nasalised in the same syntactic environment. The mutually exclusive nature of mutation is of importance for phonological analysis.

2.2. Prefixation of consonants

2.2.1. Prefix H

Initial mutation and prefixation are related phenomena in Irish morphonology. The former implies changes to a phonological segment and the latter the addition of phonetic substance to the beginning of a word, including the prefixing of /h/ to a word which begins with a vowel under certain grammatical conditions, for instance after the feminine possessive pronoun: *a harm* /ə harm/ ‘her arm’. Prefixing /h/ is structurally equivalent to zero mutation with words which do not begin with a vowel, e.g. *a gúna* /ə gu:nɣə/ ‘her dress’ (*/ə ɣu:nɣə/).

2.2.2. Prefix T

There are two further cases of prefixation which involve the prefixation of *T* and *D* respectively under certain grammatical conditions (see section II.1.8 above). *T* is added initially to words beginning with a vowel or *S* when followed by a homorganic sonorant *L*, *N* or *R*. The requirement for the sonorant following *S* to be homorganic can be confirmed by viewing words where this is not the case, e.g. *an scian* ‘the knife’ (FEM), **an tscian*; *an smailc* ‘the mouthful, bite’ (FEM), **an tsmailc*, but *an tsladaíocht* ‘the plundering’ (FEM). This means that prefix *T* applies to S_1 (see section II.1.7.2.2 *Sibilants, Subdivision of S*), i.e. *S + L, N, R*, but not to S_2 = *S + M, P, T, K, F*.

Prefix-*T* serves the function of distinguishing masculine and feminine nouns: it is found before masculine nouns beginning with a vowel, though not before feminine nouns, contrast *an t-arán* /ə tarɑ:n/ ‘the bread’ (MASC) and *an ascaill* /ənɣ askɪlj/ ‘the armpit’ (FEM). It is furthermore found with feminine nouns beginning with *S + L, N*, or *R* in the nominative singular and with masculine nouns also beginning with *S + L, N* or *R* in the genitive singular as seen in *an tslí* /ənɣ tjlji:/ ‘the way’ (FEM NOM SG) and *barr an tsléibhe* /bɑ:r ənɣ tjlje:vjə/ ‘top of the mountain’ (MASC GEN SG).

In synchronic terms prefix-*T* is grammatically equivalent to lenition as the environments which call for it are those which normally require lenition while their converses – feminine genitive singular and masculine nominative singular respectively – neither have lenition nor prefix-*T* (before initial *S + L, N*, or *R*).

(51) *Matrix for prefix T*

	masculine	**feminine**
NOM	*T* + V-	V-
	masculine	feminine
NOM	–– ––	$T + S_1$
GEN	$T + S_1$	–– ––

The exclusion of sequences of *S* + non-homorganic sonorants/stops from *T*-prefixation is in keeping with the non-application of lenition to words with such initial sequences, e.g. *an smig* ‘the chin’ (FEM), **an shmig*; *an scoth* ‘the blossom; tip’ (FEM), **an shcoth*.

Table 1. Occurrence of prefix-*T* and prefix-*h*

Prefix *h*	(i)	before vowels	
Prefix *t*	(ii)	*S* + vowel / sonorant	

Non-leniting environment

1a)	*an t-arán*	‘the bread’	(MASC, NOM)
2a)	*an siopa*	‘the shop’	(MASC, NOM)
	(*an fear*	‘the man’)	
3a)	*ar fud na háite*	‘all over the place’	(FEM, GEN)
4a)	*i rith na seachtaine*	‘during the week’	(FEM, GEN)
	(*fad na cuairte*	‘the length of the visit’)	
5a)	*a haois*	‘her age’	(POSS PRO, FEM)

Leniting environment

1b)	*an áit*	‘the place’	(FEM, NOM)
2b)	*an tseachtain*	‘week’	(FEM, NOM)
	(*an chuairt*	‘the visit’)	
3b)	*blas an aráin*	‘the taste of the bread’	(MASC, GEN)
4b)	*in aice an tsiopa*	‘beside the shop’	(MASC, GEN)

	(*culaith an fhir*	'the man's suit')	
5b)	*a aois*	'his age'	(POSS PRO, MASC)

2.2.3. Prefix D

The prefixation of *D* also replaces lenition in a certain environment, this time word-initially in the past tense of verbs. Normally verbs have lenition here but if they begin with a vowel a *D* is prefixed to them which is then the signal of the past tense, cf. *chaith sé* /xa sje:/ 'he threw' (← *caith* /ka/), *d'imigh sé* /djɪmjə sje:/ 'he left' (← *imigh* /ɪmjə/).

D can occur with verbs which have been lenited if the result of lenition is zero. In effect, this means that the non-lenited consonant was *F*. It is the only case where prefixation and mutation can apply to the same form.

(52)

fág	/fɑ:g/		'leave'
fhág	/ɑ:g/	1)	lenition
d'fhág	/dɑ:g/	2)	*D*-prefixation
d'fhág sé	/dɑ:g s^je:/		'he left'

2.3. Analysis of the mutations

When looking at the status of mutations, the absence of initial change, zero mutation, has grammatical significance (see section III.1.6 above). However, when analysing the sound changes engendered by the mutations, zero mutation is not relevant. This leaves the two other mutations, lenition and nasalisation. The first implies that the segment which undergoes it is phonologically weaker as a result, in effect that the segment is fricativised. But lenition is not simply the fricativisation of a stop. Firstly some sounds when lenited also undergo a shift in place of articulation and secondly the fricatives *S* and *F* can also be lenited.

The second term ‘nasalisation’ is not entirely satisfactory, but for want of a better term it will be retained for the present discussion. [192] It implies that its output is a nasal. Now while many sounds which undergo nasalisation are converted into nasals others result in stops. It is most appropriate to view the two mutations as a series of steps which are taken with a given input. The steps are in all cases a change in value for a single feature. There are restrictions in the ordering of steps and the number taken. Before attempting to formulate these, the inputs and outputs to the two mutations of Irish can be given.

(53)

Summary of the Irish mutations[193]

Lenition			Nasalisation		
K	→	*X*	*P*	→	*B*
P	→	*F*	*T*	→	*D*
F	→	Ø[3]	*K*	→	*G*
B, M	→	*V*	*B*	→	*M*
D, G	→	ɣ	*D*	→	*N*
T, S	→	/h/	*G*	→	ŋ
			F	→	*V*

2.3.1. Nasalisation

Certain generalisations about nasalisation are possible. Voiceless stops change to voiced ones and voiced stops change to homorganic nasals.[194] For convenience sake, the third person plural possessive pronoun – a nasalising particle – can be used for illustration: *peaca* /pjakə/ ‘sin’, *a bpeacaí* /ə bjaki:/ ‘their sins’, *béal* /bje:lɣ/ ‘mouth’, *a mbéil* /ə mje:lj/ ‘their mouths’. A direct shift from voiceless stop to nasal is not possible as this would involve a change of two features: [-voice] > [+voice] > [+nasal] and only one step in this sequence is permissible in Irish nasalisation, hence once has either (i) [-voice] > [+voice] or (ii) [-nasal] > [+nasal], depending on the segment to which nasalisation is applied, see the examples just given.

As opposed to lenition, nasalisation does not involve a shift in place of articulation. The change of *B* to *V* is not regarded as a change in place of articulation, labio-dental articulation is more common for fricatives and bilabial for stops. Furthermore, a true bilabial [ß] is sometimes found for non-palatal /v/ when followed by a vowel or a pause in Connemara Irish. The two steps of nasalisation in Irish are then as follows.

(54)

(i)	[-voice, -nasal]	+	N	→	[+voice, -nasal]
(ii)	[+voice, -nasal]	+	N	→	[+voice, +nasal]

2.3.2. Lenition

Irish lenition is basically the fricativisation of a stop. In feature terms it is [-continuant] →[+continuant]. But this change of feature only applies in this simple form to *K* and *G* and to *P* and *B* (if the bilabial to labio-dental shift in articulation for the latter is neglected). On the lenition of *D* there is a shift in place of articulation to ɣ, something which has its origin in Middle Irish (O'Rahilly 1932: 56 + 211). This may have had a drag effect on some cases of initial *D* in grammatical forms leading to them being realised as voiced velar stops, e.g. *do* /gə/ 'for', *de* /gə/ 'from' and the prepositional pronouns deriving from them (de Bhaldraithe 1953a: 142).

It is probable that lenited *D* shifted from the assumed dental position to a velar one with the general decline of ambi-dental fricatives in Irish. For the present-day language one need only note that *T* and *D* both have outputs from lenition although neither /θ/ nor /ð/ have existed in Irish since the Middle Irish period. The fact that *G* regularly fricativises to *ɣ* means that a dental-velar merger takes place among voiced segments under lenition: *a dhuais* /ə ɣu:sj/ 'his prize', *a ghuais* /ə ɣu:sj/ 'his fear, dismay'. There is no merger under nasalisation as *D* nasalises to *N* and *G* to *ŋ : a nduais* /ə nɣuəsj/ 'their prize', *a nguais* /ə ŋuəsj/ 'their fear, dismay'.

A further merger under lenition is found with *S* and *T* which both shift to /h/ in coda position. The shift of a voiceless alveolar fricative to a glottal one is commonly attested for a wide variety of languages: Indo-European */s/ becomes /h/ prevocalically in Greek (and Armenian), later disappearing in the latter (Buck 1933: 132), /s/ is reduced to /h/ in Spanish frequently in Southern dialects and in Latin America (Malmberg 1962: 64-65). If lenition is seen as a progressive weakening in articulation then *S* →/h/ can be accounted for as the removal of the oral gesture, i.e. the reduction of the fricative led to the loss of primary articulation (dental frication) with glottal friction remaining as the remnant of the voicelessness of the original fricative.

There is also reduction in the manifestation of the distinction between palatal and non-palatal sounds with *S* and *T*. Both of these become /h/ on lenition. The normal realisation of the glottal fricative is [h] as in the following cases: *Séamus* [ʃe:məs] 'James', *a Shéamuis* [ə he:məʃ] 'James.VOC',[195] *tonn* [tu:nɣ] 'a wave', *faoi thoinn* [fwi: hu:nj] 'under water', *saol* [si:lɣ] 'life', *anonn ina shaol* [hi:lɣ] 'later in his life'. These examples are of /h/ deriving from (1) a non-palatal /s/ or /t/ before all vowels and (2) /h/ deriving from palatal /sj/ (or /tj/) before front vowels (first example). However, a palatal fricative [ç] is not found where the lenited /sj/ or /tj/ occurs before a front vowel. It also appears when the following vowel is low or back as in these forms: *Seán* [ʃɑ:nɣ] 'John' *a Sheáin* [ə çɑ:nj] 'John.VOC', *tiocfaidh* *mé* [tʲʌkə mje:] 'I will come', *thiocfainn* [çʌkɪnj] 'I would come', *teocht* [tjo:xt] 'heat', *dá theocht* [ço:xt] *an aimsir* 'however hot the weather', *siúil* [ʃu:lj] 'to walk', *shiúil mé* [çu:lj mje:] 'I walked'.

When the *T* (or *S*) is lenited to /h/ the realisation of the immediately following vowel is distinctive (when low) so that /t/ + /s/ and /tʲ/ + /sj/ are kept apart after lenition: *a theach* [ə hæ:x] 'his house', *a thalamh* [ə ha:lɣə] 'his land'.

2.3.2.1. Mergers on lenition

Apart from the merger of *S* and *T* on lenition there is also that of *B* and *M* to *V*. The bilabial nasal loses its nasality on fricativisation. This is the case in present-day Connemara Irish though in some isolated cases speakers may retain slight nasality when *V* derives from *M* by lenition (de Bhaldraithe 1945: 31): *a mhála* [ə wɑ:lɣə] ~ [ə ŵɑːlˠə] 'his bag'. For all intents and purposes *B* and *M* can be taken as merging on lenition. It is significant here that *M* is treated in Irish as a stop from the point of view of lenition. But not all nasals are lenitable although they all share oral closure: *ŋ* occurs as the output of nasalizing *G*. It is only found before a velar stop so that it can be interpreted as the realisation of *N* before *K* and *G*.

The lenition of *F* involves the shift to zero: *fírinne* /fji:rjɪnjə/ 'truth', *an fhírinne* /ənɣ i:rjɪnjə/ [ənji:rjɪnjə] 'the truth'. There is also evidence for an /f/ → /h/ shift in Irish with the reflexive pronoun *féin* which is pronounced /he:nj/ (O'Rahilly 1932: 81). The reduction of /h/ to Ø simply involves loss of glottal friction and is common, for example in informal varieties of urban British English (Wells 1982: 253) and in many traditional dialects as well. But *F* differs from *S* and *T* in Irish in that the latter lenite to /h/. This fact led Pedersen (1897: 19) to believe that lenited *F* was reduced to zero before *F* arose from the fortition of *V* in pre-Old Irish. Pedersen imagines the correspondence *V* ~ Ø to be more likely than *F* ~ Ø so that only non-lenited independent *V* was shifted to *F* originally.

2.3.2.2. Permanent lenition

There are a few words which begin with lenited segments, even in their citation forms. These are all cases of unstressed grammatical words which have been fossilised with permanent lenition.

(55)

cheana	'already'	*bhur*	'your'
choíche	'ever'	*dhá*	'two'
chomh	'as'	*dhuit*	'for you'
		de, di [jɛ, jɪ]	'for him; for her'
chun	'for'	*chuig*	'to'

Lenition is found with the object form of second person singular pronouns as opposed to subject forms with zero mutation: *tú* /tu:/ (subject), *thú* /hu:/ (object). Phonetic lenition of weakly stressed forms is found elsewhere in spoken Irish and would appear to affect /d/ more than any other segment. Instances are seen in the paradigm of the preposition *do* [gʌ] 'to' and all its compound forms, e.g. *dhom* [ɣʌm] 'to me', *dhuit* [ɣɪtj] 'to you', etc. One or two lexical items are affected by this low-level lenition. The word *dháiríre* [ɣɑ:ˈrji:rjə] 'seriously' is an example (Fife 1993: 12), this word also has stress on the second syllable (it is normally on the first in Western Irish) and this fact may be connected to the lenition the word shows.

3. Organisation and principles

Grammatical categories in Irish can be indicated by one of two basic means, either by the addition of grammatical suffixes or by changes to the segments at the margins of lexical words. The latter type of change encompasses mutation of the initial segment and/or a shift in the value for [palatal], by reversing polarity, at the end of a word. These latter types of change can be classified under the heading 'morphonology' because they involve alternations of single phonological segments, rather than the addition of grannatical endings.[196]

It is not easy to classify Irish with conventional labels of linguistic typology. It is closest to the inflectional type, but this designation might imply that grammatical categories in Irish are chiefly indicated by means of grammatical suffixation as in the previous stages of many Indo-European languages. But this is not the case. On the contrary, inflections play a minor role, instead grammatical categories are indicated by altering the bases of words, by initial mutation or by reversing the polarity of [palatal] at the end. This principle of altering the margins of bases had already established itself by the Old Irish period through a re-analysis of low-level phonetic phenomena at the beginnings and ends of words (Hickey 2003b). This reanalysis was doubtlessly triggered by the phonetic blurring of inherited inflections and grammatical formatives in the development of the Celtic languages.

The term *base margin alteration* can be used for the new typological principle which arose in the pre-written period of Irish. This involved the functionalisation of phonetic phenomena at the beginning and end of words, hence the qualifier 'margin'. The alterations affected the base form of words in virtually all grammatical paradigms.

3.1. Base and root

Base Taking the first major category, nouns, the base is the unaltered form of the noun: nominative case, singular number, i.e. the citation form for a dictionary. It may consist of one or two syllables (in rare cases three). If there are two or more then the second (and third) is a *root extension*. The latter does not usually carry any lexical information and only has a few manifestations, typically /əx/.

Root This is a base minus its extension, if present. The lexical part is the root and is identical with the base in monosyllabic forms.

Transparent root extensions /əx/ →/iː/ + palatalisation (historically /ɪç/ → /ɪj/ → /iː/)

(1)

a.	*marcach*	:	*marcaigh*	‘rider.NOM’ : ‘rider.GEN’
b.	*báisteach*	:	*báistí*	‘rain.NOM’ : ‘rain.GEN’

Other examples of root extensions are the following, all of which vary in their degree of semantic transparency: *-as*, e.g. *eolas* ‘knowledge’ (cf. *eol* ‘known’); *-adh*, e.g. *samhradh* ‘summer’; *-án*, e.g. *árasán* ‘apartment, flat’ (cf. *áras* ‘residence’), *ardán* ‘stage, platform’ (cf. *ard* ‘high’), *amadán* ‘fool’.

Some root extensions may be old Latin loans as in *peaca* ‘sin’ (from *peccatum*), *airgead* ‘silver’ (from *argentus*), *anam* ‘soul’ (from *anima*). Others are Anglo-Norman loans which entered Irish after the twelfth century *séipéal* ‘church’ (from *chapele*), *seomra* ‘room’ (from *chambre*).

Depalatalisation can include the adding of a root extension as with fifth declension nouns: *riail* : *briseadh na rialach* ‘rule’ : ‘breaking of the rule’:

3.2. Type of alteration

The base form of any paradigm can undergo various alterations when different grammatical categories are indicated. At the beginning of a base a *mutation* can occur. At the end a change in *consonantal quality* may appear. This is a change from palatal to non-palatal or vice versa with any consonant, i.e. a reverse in polarity for the feature [palatal]. Phonetically, non-palatal sounds are velarised to improve phonetic contrast. Palatalisation in Irish only involves a change in palatality. As Irish has palatalisation as a grammatical phenomenon, e.g. when forming the genitive or plural, non-palatal consonant quality is systemically significant. In the following simple examples the final consonant in the singular is non-palatal, /k/ and /x/ respectively. In the plural this changes to /k^{j}/ [c] for the first word but to /i:/ for the second so that with many words in present-day Irish, grammatical palatalisation is no longer phonetic palatalisation due to various historical changes (here: vocalisation of a final palatal fricative).

(2)

	Singular (non-palatal)		Plural (palatal)	
a.	*cnoc*	‘hill’	*cnoic*	‘hills’
b.	*marcach* /-əx/	‘rider’	*marcaigh* /-i:/	‘riders’

3.3. Base margin alteration

Left margin alteration	(1) Lenition	(2) Nasalisation
Right margin alteration	(1) Palatalisation	(2) De-palatalisation

Base onset	*Base nucleus*	*Base coda*
C-	-V-	-C
voiceless stop	palatal	(affects any type
voiced stop	non-palatal	of consonant)
fricative		
nasal		
initial mutation with manner of articulation affected (rarely place)	final mutation where consonant quality alters; pairs exist for all consonants (except /h/)	

The changes in the quality of the base coda do not affect the quality of preceding long vowels but do with short vowels. These shift from a front to a back articulation or vice versa depending on the nature of the quality change of the final consonant or cluster, e.g. *olc* [ʌljk] 'evil.NOM' versus *oilc* [ɛljkj] 'evil.GEN'. The front vowel can furthermore be realised as a diphthong, e.g. *oibre* 'work.GEN' [aɪbjrjə] (in Western Irish).

(3)

Base: *bás* 'death'		*Onset*	*Coda*
a bhás	'his death'	Lenited	Non-palatal
a bás	'her death'	Neutral	Non-palatal
a mbás	'their death'	Nasalised	Non-palatal
am a bháis	'time of his death'	Lenited	Palatal
am a báis	'time of her death'	Neutral	Palatal
am a mbáis	'time of their death'	Nasalised	Palatal

The above table represents an ideal case. But in spoken Irish there is much fluctuation and increasingly lenition after *a* [ə] (third person possessive pronoun) is found even when 'her' is meant, a location where zero mutation is found historically.

Older loans in the language always undergo base margin alteration as if they were native words.

(4)

a.	*páipéar : dath an pháipeír*	'paper' : 'colour of the paper'
b.	*peann : barr an phinn*	'pen' : 'tip of the pen'

3.4. Root extension and remnants of older patterns

If root extension for an inflected form of a word coincides with base margin alteration (compared with the uninflected form), then this always involves a transparent extension, in effect /əx/ in modern Irish as in the following examples.

(5)

a.	*cáin : méid na cánach*	'tax' : 'the amount of tax'
b.	*beoir : blas na beorach*	'beer' : 'the taste of the beer'
c.	*traein : uimhir na traenach*	'train' : 'the number of the train'

Base extension may trigger syncope if the phonotactic conditions are right, e.g. when an obstruent and sonorant come together.

(6)

eochair : cuma na heochrach	'key' : 'the shape of the key'

There are historical remnants of a consonantal inflection (*n* or *d*) for a few nouns which are added to a base which already has schwa extension (-*a* /-ə/) as in:

(7)

a.	*comharsa :*	*iníon na comharsan*
	'neighbour' :	'the neighbour's daughter'
b.	*cara :*	*ainm an charad*
	'friend' :	'name of the friend'

3.5. Scope of base margin alteration

Base margin alteration applies above all to the area of nouns and adjectives. With verbs there still are inflections, but these are often uniform for a tense. All verbs which begin with a lenitable consonant show lenition in the past.

The principle of base margin alteration applies to those categories which involve a binary distinction such as nominative versus genitive. An inflectional short vowel, typically schwa or /iː/ (above all in Northern Irish) is often retained for a third category, usually the plural.

With verbs base margin alteration is used to distinguish between present and past. The latter has lenition as its distinctive inflection. The future is irregular, very often with suppletive forms or an ending /(h) ə/ which is added to the root. In a system where lenition is a marker of a certain tense, here the past, the absence of lenition becomes significant. Thus the lack of lenition in the present and future in Irish contrasts with the situation for past forms.

Initial mutation, or the lack of it, applies to all persons and both numbers in verbal paradigms. It also applies to the two main conjugational types of modern Irish. These differ in the inflections they show, but agree in the use of initial mutation. Lenition is the only mutation used for indicating tense or mood with verbs. However, nasalisation is frequently triggered by pre-verbal particles such as clause relativisers or interrogative particles.

(8)

a.	*Lenition*	Past, Imperfect, Conditional
b.	*No lenition*	Present, Future, Subjunctive

A consequence of base margin alteration is that consonants have been foregrounded in Irish phonology and vowels backgrounded accordingly. Vowel length is still systemically relevant in Irish but among short vowels there is only a binary difference between a front vowel [ɪ] or [ɛ], governed by a following palatal consonant, and a low front/central to back vowel [æ,a] or [ʌ], governed by a following non-palatal (velarised) consonant.

(9)

a.	*fliuch* [f^{j}l$_{j}$ʌx] ‘wet’	*níos fliche* [f^{j}l$_{j}$ɪçə] ‘wetter’
b.	*moch* [mʌx] ‘early’	*níos moiche* [mɪçə] ‘earlier’
c.	*glas* [gl^{ɣ}as] ‘lock.NOM’	*glais* [gl^{ɣ}ɛʃ] ‘lock.GEN’

d. *fear* [f^{j}ær] ‘man.NOM’ *fir* [f^{j}ɪr^{j}] ‘man.GEN’

3.6. Anomalies in the mutation system

Sonorants These segments undergo neither lenition nor nasalisation and so grammatical categories must be recognised from the context.

(10)

a.	*a rún*	‘his, her, their secret’
b.	*a lámha*	‘his, her, their hands’

While it is true that lenition does not affect words (i) beginning in a vowel or (ii) beginning in a cluster with an initial *S*, the pressure of morphonological distinctiveness would seem to have led to the use of prefix-*T* before certain noun forms, see discussion in III.2.2 above.

(11)

a.	*an t-arán* /ən^{ɣ} tara:n^{ɣ}/	‘the bread’	(MASC, NOM)
b.	but: *an áit* /ən^{ɣ} a:t^{j}/	‘the place’	(FEM, NOM)

Certain prepositions cause this rule to be waived.

(12)

a.	*an t-am*	‘time’,	but: *ag an am*	‘at the time’
b.	*an t-im*	‘butter’,	but: *leis an im*	‘with the butter’

Prefixation to a noun with an *S*-initial cluster only applies where there would be lenition, i.e. with feminine nouns in the nominative. Prefix-*T* overrides lenition in cases of *S* + V.

(13)

a.	*an tsláinte*	‘health’	(FEM, NOM)
b.	*an tslí*	‘the way’	(FEM, NOM)
c.	*in aice an tsiopa*	‘beside the shop’	(MASC, GEN)
d.	*an tseachtain*	‘the week’	(FEM, NOM)
	not: *an sheachtain*		

Nasalisation is realised before vowels as prefix-*N*. Here it contrasts with lenition, which has no manifestation before vowels. However, the lack of lenition before a vowel-initial word is realised as *h* in the genitive of nouns, in the plural and with the possessive pronoun ‘her’ as seen in the following examples.

(14)

a. *a aois* ‘his age’ : *a haois* ‘her age’ : *a n-aois* ‘their age’
b. *trasna na habhann* ‘across the river’ (FEM, GEN)
c. *na heaglaisí* ‘the churches’

3.7. Canonical word form

Lexical roots in Irish are generally monosyllabic. Words of more than one syllable can arise in a number of ways. (1) A root can take a prefix, e.g. *ceart* ‘correct’, *mícheart* ‘incorrect’ (< *mí*-Lenition ‘not’ + *ceart*), *an-bheag* ‘very small’ (< *an*-Lenition ‘very’ + *beag*), *ró-bheag* ‘too small’ (< *ró*-Lenition ‘too’ + *beag*). (2) A word can consist of a root and a grammatical suffix, e.g. *brisim* (< *bris* + *im*) ‘I break’. (3) A word may consist of more than one syllable (usually two), deriving historically from a compound which has long since become opaque, e.g. *freagair* ‘answer’. (4) A word may be a compound, consisting of two or more parts, e.g.

teangeolaíocht 'linguistics' from *teanga* 'tongue, language' and the ending -*eolaíocht* 'study, discipline', itself deriving from *eolas* 'knowledge'. (5) A word may be a borrowing from English and hence consist of more than one syllable. This can be a complex matter: older loans have been integrated into the language and may have received native grammatical material in the form of a suffix, e.g. English *rogue* appears in Irish as *rógaire* with the animate ending -*aire.* (6) A root may have a root extension (Hickey 2003b). By this is meant an extra syllable (usually a suffix) which is added to a monosyllabic root. These extensions may have arisen from independent grammatical, and ultimately lexical, elements through univerbation, but for the present-day language they are opaque. Perhaps the most common root extension is *-(e)ach* /-əx/, e.g. *báisteach* 'rain'. This suffix is semantically empty with nouns, though it is grammatically functional in that with certain feminine nouns it occurs in the genitive case, e.g. *cáin* 'tax', *méid na cánach* 'the amount of tax'. It is the sole carrier of grammatical meaning with many adjectives where it is suffixed to a nominal stem, e.g. *clúmh* 'down, fur', *clúmhach* 'feathery, hairy'.

Root extension is intricate in the history of Irish. In previous stages of the language, notably during the early modern period, there is an observable tendency to use /-ə/ as a root extension and disyllabify nouns. This can be seen with many borrowings which entered the language from Anglo-Norman in the period after the initial settlement of the late twelfth century, e.g. *gúna* /gu:nɣə/ 'dress' ← *gune*, *póca* /po:kə/ 'pocket' ← *poke*, *mála* /mɑ:lɣə/ 'bag' ← *male.* In these examples the second vowel could have come from the final schwa in the original words, if this was still pronounced in the variety of Anglo-Norman brought to Ireland. However, many Anglo-Norman and English loans show an additional vowel where this cannot have originated in the donor language. Words like English *box* and Anglo-Norman *curs,* *terme* have phonotactically unacceptable codas for Irish /-ks, -rs, -rm/ and appear in disyllabic form in Irish: *box* →*bosca* (with /ks/ > /sk/ metathesis), *curs* → *cúrsa* 'course', *terme* → *téarma* 'term'. Other examples of this syllabic extension are: *hat* → *hata*, *wall* → *balla*, *coat* → *cóta.* Adding schwa caused resyllabification, yielding acceptable structures: /bɔs.kə/, /cu:r.sə/, /te:r.mə/, etc. The argument from phonotactics would seem to be strengthened from those few instances where an extra vowel is not added: Anglo-Norman *sorte* → *sórt* 'sort' as the coda /-rt/ is common in native Irish words, cf. *gort* 'field'.

4. Origins and development of palatalisation

The functionalisation of palatalisation had already taken place before the first attestations of Old Irish. This palatalisation is a specifically Q-Celtic phenomenon and is not found in P-Celtic languages such as Welsh. It occurs in Scottish Gaelic and Manx because these languages are derived from Irish through movements of speakers away from Ireland to Scotland and the Isle of Man in the first millennium CE.

The origin of palatalisation[197] lies in the spread of 'highness' and 'frontness' from vowels to adjacent consonants in early Q-Celtic.[198] The consonants anticipated the quality of the vowels by adopting the relevant tongue position for them in advance. This type of secondary articulation was the precondition for later functionalisation. By the Old Irish period there are two established 'qualities' with a third which is regarded as peripheral.

(15)

a. palatal before *i* and *e*

b. neutral (non-palatal) before *a* and *o*
c. *u*-quality before *u*[199]

Irish palatalisation never involved further changes in articulation. Specifically, consonants which were palatalised have not developed into sibilants/affricates as in Slavic languages. This fact is important when considering that in Irish there have been cases of depalatalisation, e.g. /gʲ/ changing to /g/ as in *uiliug* [ɪˡljʌg] 'every' (Western Irish, de Bhaldraithe 1953a: 154).

Note that for Old Irish, and presumably for forms of Celtic which preceded it, vowels were affected by the quality of consonants which preceded or followed them. In present-day Irish the quality of short vowels is only affected, to any substantial extent, by following consonants, compare *clog* [klɣʌg] 'clock.NOM' and *cloig* [klɣɛgj] 'clock.GEN'. A preceding palatal does not cause vowel retraction, e.g. *pioc* [pʲʌk] 'pick' with a back vowel before the non-palatal /k/. The opposite case, a non-palatal consonant preceding a short front vowel, something like *[lˠɛk], does not occur in Irish.

4.1. Spread of palatalisation

In the earlier stages of palatalisation the segments affected were coronals (dentals/alveolars) and velars. Of the two groups the one most subject to palatalisation was the coronal group.[200] This is a view supported by Greene (1973: 128) and by Kuryłowicz, who maintains that labials and velars were not palatalised before a front vowel if a back vowel preceded (Kury łowicz 1971: 67-68). The view of the process starting from a favoured phonetic site, the dental area, and then encroaching on the orally peripheral areas helps to explain why certain instances of non-palatalisation are to be found in Old Irish. Labials are specifically excluded from the initial effect of palatalisation on consonants. Tense and spread lips are phonetic correlates of phonemic palatalisation which later developed to distinguish palatal labials from non-palatal ones (Jackson 1967b: 181).

In addition once palatalisation had become an exponent of grammatical categories (case, number, adjective comparison, etc.) it attained systemic prominence and tended to spread at the expense of non-palatal consonants. This is what Greene takes Kuryłowicz to mean by the utilisation of marked phonemes (Greene 1973: 135) and quotes such instances as the alternative *luinge* ‘ship.GEN’ in Middle Irish for Old Irish *lungae* (from the base *long* ‘ship’) where the palatalisation has spread by analogy with the accusative-dative *luing* to the genitive thus increasing the manifestations of the genitive in a final -C^jə.

This spread of special consonant quality is connected to its origin. Consider the fact that initially, as long as the inflectional syllables which could have caused a change in consonant quality were present, this change tended not to take place. Greene (1973: 29) quotes the instance of *u*-infection (velar quality for consonants adjacent to high back vowels) which spread to consonants when the ending with the high back vowel was lost but not when this was still present: *fer* ‘man’ has the dative singular *fiur* with velar quality from an earlier * *firuh* but the accusative plural *firu* in Old Irish does not have velar consonant quality as the high back vowel is still present. One can thus postulate three principal steps for the spread of consonant quality.

(16)

1) neutral consonant + vowel in ending
2) vowel quality transferred to consonant on loss of ending
3) spread of consonant quality through paradigm to achieve regularity

The origin of palatalisation in the assimilation of consonants to the vowels of inflectional endings means that in terms of relative chronology morphonological palatalisation arose first, with palatality (and non-palatality) as a property of the phonological structure of lexical words appearing later.

(17)

(i) Rise of morphonological palatalisation
Assimilation to high vowels in suffixal inflections Systemic interpretation of palatal consonants on blurring and later loss of inflections Reactive velarisation of non-palatals
(ii) Incorporation of the palatal # non-palatal distinction into the lexical structure of words.

The extension of polarised articulations (here: palatal and non-palatal /verlarised) to the lexical structure of words is characteristic of those languages which have complete or near-complete series of polarised articulations. This is true of Russian, for instance, where obstruents participate in the palatal # non-palatal distinction in the lexicon.

4.2. Contrast in lexical words

As all consonants in Irish, bar /h/, are marked as being either palatal or non-palatal then all words in the lexicon will carry specifications for [palatal]. The simplest case of contrast is with initial consonants: *dán* /dɑ:nɣ/ ‘poem’, *deán* /djɑ:nɣ/ ‘channel in strand’. Here the value for [palatal] is part of the phonological structure of the word as is the value for [voiced] in English word pairs like *pat* versus *bat*. Palatality contrast in the lexicon can also occur word-finally, e.g. *úr* /u:r/ ‘fresh’, *úir* /u:rj/ ‘soil’, but is much less common.

Originally, palatal consonants could only occur in the environment of front vowels. Equally non-palatals were only to be found before low and back vowels. A number of developments led to a contrast of consonant quality and vowel type in lexical stems. With the functionalisation of the process phonetic palatalisation was arrested: Few new cases arose and, importantly, the further phonetic development of palatals to affricates and sibilants (as in the Slavic and the Romance languages) did not occur.

But there remains an essential question: if palatalisation was originally phonetic, with consonants agreeing in [palatal] with the vowels which surrounded them, how did contrast arise in lexical stems with palatal consonants occurring before back vowels and vice versa? The answer to this question lies in further language-internal developments which led to this contrast.[201]

1) *Shift of Old Irish diphthongs* In particular the diphthong written *ao* was shifted to a vowel value /i:/, or /e:/ in Southern dialects, in the modern period, leaving non-palatal consonants flanking the high vowel: *caol* /ki:lɣ/ 'slender' (Greene 1976: 39-40). New lexical contrasts arose because of this as can be seen in Modern Irish *síl* /sji:lj/ 'think', *saol* /sɣi:lɣ/ 'life'.
2) *Shift of stress on diphthongs* The diphthongs which contained a first element *é* in Old Irish such as *béo* (from **bew* 'live', Greene 1976: 31) experienced a shift of the stressed vowel to a back vowel of equal height (changing a falling diphthong to rising one; it is not, however, possible to date this shift Greene 1976: 42), cf. Modern Irish *beo* /bjo:/ 'alive'. The reflex of the /e:/ is seen in the continued palatal quality of the preceding consonant. If the vowel is word-initial then only the orthography can offer a hint of the former

pronunciation, e.g. *eochair* /ʌxɪrj/ (with present-day /ʌ/ deriving from an earlier *o* /o/). The former /e/ at the beginning of such words has the effect of still palatalising the consonant of a preceding word (see II.1.3 above).

3) *Shift of nucleus to glide* Any syllable which has a glide adjacent to its nucleus can experience a shift of stress from the latter to the former. The result of this is that the glide achieves acoustic prominence and becomes the nucleus of the syllable. This process can be recognised when comparing Scottish Gaelic (Gillies 2009) with Irish, the latter often having the shift where the former does not, e.g. *uisce* [ɯˡʃkʲə] → [ɪʃkʲə].[202]

There is a parallel in the early shift of Common Slavonic * *e* to Russian /o/ in initial position, cf. *odin* ‘one’ and *ozero* ‘lake’ and the much later the tendency in Russian for stressed /e/ before a non-palatal consonant to be retracted to /o/.[203] It is shifts like these which add to the possibility of contrast between palatal and non-palatal consonants in the onset of stressed syllables in Russian, cf. *nes* [njos] ‘carried’ versus *nos* [nɣos] ‘nose’ (Comrie 1990: 70).

The situation with short vowels in Irish is somewhat different. These vowels are affected in their quality by following consonants, mostly in Western and Northern dialects of the present-day language. Indeed this assimilation of a nucleus vowel to the value for [palatal] of the coda can be used as a defining characteristic of short vowels in Irish (see section II.1.9.5 above). The general rule is that following palatals associate with /ɪ/ and /ɛ/ and non-palatals pair with /a/ and /ʌ/, e.g. *cur* /kʌr/ ‘put.NON-FINITE’, *cuir* /kɪrj/ ‘put.IMPERATIVE’, *troid* /trɛdj/ ‘fight.NOM’, *troda* /trʌdə/ ‘fight. GEN’.

However, if the consonant which precedes such a short vowel is different in quality to that which follows the vowel then a contrast in the value for [palatal] arises for the syllable onset. This can be seen quite clearly with the lowering of /e/ before non-palatal sonorants as Old Irish *fer* ‘man’ → /fjar/ (Modern Irish *fear*); *ben* ‘woman’ → /bjanɣ/ (Modern Irish *bean*).

The transcriptions given here are systemic. But there is an undoubted phonetic influence of palatals or non-palatals on vowel realisation in Modern Irish: the vowel in a word like *caol* is quite retracted [kɨ:lɣ] and that of *fear* is quite front [fjær]. This realisation of vowels can play a crucial role as a phonetic cue to consonant quality, particularly with secondary palatals such as labials.

The assimilation of short vowels to adjacent consonant quality is often a feature of languages with series of palatal consonants. Russian has a similar phenomenon known as *ikan’e* which neutralises the difference between unstressed *e*, *i, o* and sometimes *a* after palatal consonants, e.g. *devica* /djɪˈvjitsə/ ‘maid’, *časy* /tʃɪˈsɨ/ ‘watch’.

4.3. Palatalisation and morphonology

The high vowels of inherited Indo-European inflectional endings palatalised coda consonants in pre-Old Irish and then fell away. What started as a non-systemic process accompanying inflections came to be functionalised when the inflections were lost. Because many of the inflections with high front vowels were often used for the genitive singular or nominative plural in Common Celtic (Lewis and Pedersen 1937: 167) the palatalisation of stem-final consonants, which resulted from these, came to signal these categories on the loss of the inflections. The nominative singular was normally not affected by palatalisation (because it showed no inflection), non-palatal stem codas then providing a contrast to those of the genitive singular or nominative plural.

But not all nominative singulars were unaffected by palatalisation. Because of this a contrast arose in citation forms with those instances where palatalisation had no effect, cf. the Modern Irish words *duais* /du:sj/ ‘prize’, *cluas* /klɣu:s/ ‘ear’, *bán* /bɑ:nɣ/ ‘white’, *cáin* /kɑ:nj/ ‘tax’. But in those cases where palatalisation was found in the nominative it was absent in the genitive. This led to the division of nouns into declensional types and one of these, the fifth declension (Christian Brothers 1977: 54-56), is characterised by the reversal of palatalisation in the genitive, e.g. *beoir* /bjo:rj/ ‘beer’, *blas na beorach* /blas nə bjo:rəx/ ‘the taste of the beer’.

Because of the grammatical function of palatalisation in Modern Irish, contrast in final consonant quality between citation forms of nouns is practically unknown as such contrast is reserved for case distinctions in the nominal area or adjective comparison, cf. *bán* ‘white’ : *níos báine* ‘whiter’. Nonetheless, examples can be found, e.g. *toll* ‘hollow’ (adjective), *toill* ‘to fit, contain’ (verb), but here there is still a difference in word class.

4.4. Palatalisation in syllable rhymes

The syllable rhyme of lexical words in Irish has quite a different status from the onset as the latter cannot alter its value for palatality in any grammatical process.[204] Furthermore, the nucleus is affected if this is short.

(18)
Possible alteration of lexical words in Irish morphology

Unalterable *Alterable* *Alterable*

Onset Coda Nucleus of short vowels

Nucleus assimilation rule
The nucleus of the syllable adopts the value for palatality of the coda in the same syllable if the nucleus consists of a short vowel.

The nucleus assimilation rule will account for cases such as *blas* /blɣas/ 'taste.NOM' ~ *blais* /blɣɛsj/ 'taste.GEN' and not affect cases like *bád* /bɑ:d/ 'boat.NOM' ~ *báid* /bɑ:dj/ 'boat.GEN' where the stem vowel is long. There are also diphthongs which derive from mid short vowels and which are subject to the nucleus assimilation rule as in the following forms.

(19)

a.	*poll*	'hole.NOM'
	/pʌlˠ/	(underlying)
	/paulˠ/	mid-vowel diphthong before non-palatal coda
b.	*poill*	'hole.GEN'
	/pɛlʲ/	(underlying)
	/pailʲ/	mid-vowel diphthong before palatal coda

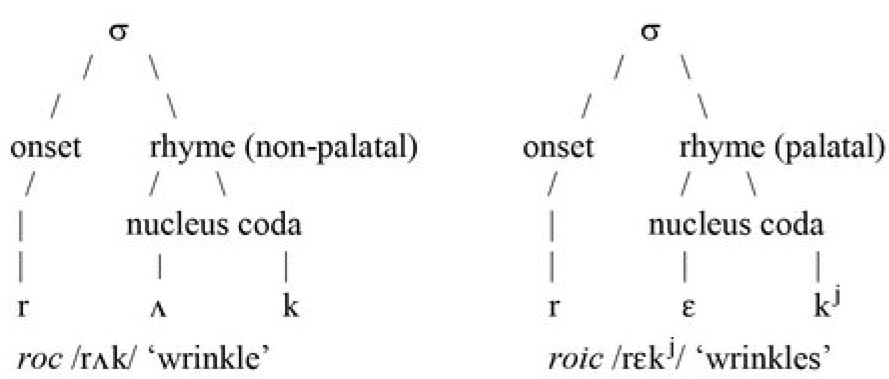
The validity of these derivations can be seen in those cases where there is variation between a mid vowel and a diphthong across dialects, e.g. /ɛbj(ə)rjə/ (Southern Irish) ~ /aibjrjə/ (Western Irish) 'work.GEN'. Alternations within grammatical paradigms, e.g. *solas* /sʌlɣəs/ 'light' ~ *soilse*

/sailjsjə/ ‘lights’, offer further evidence of the clear association of diphthongs with short vowels.

The assimilation triggered by palatalisation in Irish is not bidirectional: a palatal syllable coda only effects the realisation of a nucleus vowel before it. Consider the forms in *saibhreas* /sɛvjrjəs/ (*/sɛvjrjɪsj/) ‘wealth’, *caidreamh* /kadjrjəv/ (*/kadjrjɪvj/) ‘intimacy’. A reason for this might be that if there were forward assimilation then there could be a collapse of grammatical distinctions, e.g. if the cluster /vjrj/ in /sɛvjrjəs/ triggered forward assimilation and produced /sɛvjrjɪsj/ then this would be homophonous with the existing genitive form *saibhris* /sɛvjrjɪsj/ which has vowel assimilation in the second syllable due to the following palatal in the final coda.

Thus palatalisation can only effect segments to the left; but it does not extend to the initial segment of a word. This means that changing the value for [palatal] of a syllable coda will effect the entire rhyme of the syllable but not the onset of this syllable, consider the following example: *roc* /rʌk/ ‘wrinkle’, *roic* /rɛkj/ ‘wrinkles’.

(20)



5. Palatalisation in Russian and Irish

Russian is the Slavic language with a system of palatalisation closest to Irish, both on a lexical and on a phonological level. In Russian the functionalisation of palatalisation has been much discussed in the literature and the present analysis takes cognisance of the studies in this field. [205] Throughout this section[206] explicit comparisons are made with Irish to hopefully throw light on the typology of palatalisation (see Stadnik-Holzer 2002, especially pp. 81-93 on palatalisation in Slavic languages).

The history of the Slavic languages is characterised by a series of palatalisations which shaped the sound system of each of them (Cubberley 2002: 18-21; Sussex and Cubberley 2006: 137-152). There are two major palatalisations which phonetically involved the fronting and assibilation of velars to palatal sibilants in a number of stages (Comrie 1990: 58).

(21)

a.	*First palatalisation* before original front vowels	g, k, x	→	dʒ, tʃ, ʃ
b.	*Second palatalisation* before front vowels arising from monophthongisations	g, k, x	→	dz, ts, s (ʃ)

In addition to this, Common Slavonic developed palatal sonorants from original sequences of sonorant plus /j/. i.e. /nj, lj, rj/ →/nj, lj, rj/. There is a third palatalisation in Slavonic which in its effect is the same as the second but its environment is different: it occurs after front vowels. In addition to full vowels, Slavonic had a special development of inherited **i* and **u* which gave rise to reduced vowels known as *yers*. The later development of the *yers* depended on their position. If strong, e.g. as the nucleus of a stressed syllable, then they resulted in a later full vowel, Russian *o* : *son*, Polish *e*: *sen* ‘sleep’, from a strong back *yer* (Entwistle and Morison 1964 [1949]: 171, Cubberley 2002: 87). Both a weak and a strong *yer* can be seen in Russian *den'* (from *dьnь*) 'day' where the strong *yer* in the first syllable resulted in /e/ and the weak *yer* at the end disappeared, after palatalising the preceding /n/ to /nj/. If weak, e.g. word-finally, *yers* were lost. The front yer has a reflex as palatalisation of the consonant which preceded it, Russian *brat'* 'take'. In this respect the weak front *yers* are like the high front vowels of inflectional endings in Irish before apocope set in during the pre-Old Irish period.

As back or front vowels could occur in many morphemes palatalisation came to be a feature of morphonological alternations in Russian. Palatalisation was not the only feature in such cases as the inflection with the front vowel was usually retained.

Russian, like Irish, has a virtually complete and complementary series of palatal and non-palatal consonants (Avanesov 1972: 100). Labial, dental and velar points of articulation can be recognised, all of which are of relevance to the palatal ~ non-palatal distinction.[207] Labio-dental articulations are subsumed under labials.

In Irish the glottal fricative /h/ exists both as an independent phoneme in some English loans, like *hata* 'hat', and as the output of leniting /s/ or /t/, e.g. *a sheoladh* 'his address', *a theach* 'his house'. The voiceless palatal fricative [ç] is a possible realisation of /h/, when it represents the lenited form of /s/ and when found before a low or back vowel, see second example below. When /h/ is an independent phoneme, i.e. not the result of lenition, it always has the realisation [h]. [ç] may also occur as the palatal counterpart to /x/, i.e. as the phonetic realisation of /x^j/. In this case it belongs to the set of palatal/non-palatal consonant pairs.

(22)

a.	/h/ (independent):		
	a h-iníon	[ə hɪnʲiːənˠ]	‘his daughter’
b.	/h/ (< /s/ + fricativisation):		
	a Sheáin	[ə çɑːnʲ]	‘John.VOCATIVE’
c.	/kʲ/ (independent):		
	ceol	[kʲoːlˠ]	‘music’
d.	/xʲ/ (< /kʲ/ + fricativisation):		
	gan cheol	[gənˠ çoːlˠ]	‘without music’

In Russian the situation is simpler. It has no /h/; in those loanwords which contain /h/ it has been replaced by /x/, e.g. *xoll* /x-/ ‘hall’, and occasionally by /g/, e.g. *gavan’* ‘harbour’ < Dutch or Middle Low German *Haven*.

For the present analysis of palatalisation in Irish and Russian four points of articulation can be recognised in the oral cavity.

(23)

1) labial
2) dental
3) palatal
4) velar

These can be organised as either of two groups. The first rests on the classification of the sounds as systemically palatal or non-palatal.

(24)

Phonologically palatal	**Phonologically non-palatal**
1) Palatal	1) Labial
	2) Dental

3) Velar

The second grouping is determined by whether the tip or the blade of the tongue is the active articulator for a given sound. All sounds for which this is the case are classified as central; those where it is not the case are non-central or peripheral:

(25)

Central	**Non-central (peripheral)**
1) Dental	1) Labial
2) Palatal	2) Velar

5.1. Palatalisation: origin, extent and function

In languages which have a series of palatal consonants these normally stem from an original phonetic assimilation whereby a high vowel following a consonant caused the point of articulation to be shifted to the area of the palate immediately behind the alveolar ridge. At the initial stage in the development of palatalisation there is a convergence of point of articulation on the hard palate, usually from behind with a shift from velar to palatal as with [k] to [c], but also with the retraction of [t̪, t] to [tj] (Padgett 2003a, 2003b). The convergence points from both directions do not necessarily meet, indeed retraction of dentals and alveolars is usually accompanied by assibilation so that [tj] can frequently be [tʃ]. This affricate can arise through velar fronting but only as a second step after the rise of [c]. Assibilation is not a necessary correlate of palatalisation, cf. Russian which shows it and Irish which does not.

In a language with restricted inventory of palatal sounds, the palatal segments are frequently lateral or nasal. For instance, in (Tuscan) Italian both /lj/ and /nj/ (as geminates) exist (Lepschy and Lepschy 1977: 87-88); in French /nj/ is found while /lj/ existed also before its vocalisation to /j/ (Rothe 1978: 78, 140-141); in Spanish both /nj/ and /lj/ exist for those dialects which do not have *yeísmo* (MacPherson 1975: 76-77).

As sonorants have a clearly discernible formant structure during their articulation palatalisation can be recognised acoustically. By this is meant that an /i/-quality can be heard during the articulation of the sonorant. This /i/-quality is perceptible in the relatively high second formant (around 2,000 Hz), which stands in contrast to a /u/-quality which can be found with velarisation as secondary articulation in which the second formant lies in the region of 800 to 1,000 Hz. The claim that a language with restricted palatalisation will have this for sonorants is not absolute and there may be mixed situations, e.g. Hungarian has /tj/, /d^j/[208] and /nj/ but not /lj/ (Benkő and Imre 1972: 74). Palatalisation with the sonorant /r/ is rarer than with /l/ and /n/ for articulatory reasons, there is no direct contact with the palate apart from possible trilling.

After sonorants the consonants most likely to be affected by palatalisation are /k, g/ and /t, d/ as they can have a shift in their primary articulation towards the hard palate. With /s/ and /z/ palatalisation may be accompanied by a grooved tongue configuration with possible lip-rounding, yielding /ʃ/ and /ʒ/ respectively. This has been the situation in Irish which has, as its palatal counterpart to /s/, the phonetic realisation [ʃ], a sound to be interpreted phonologically as /sj/. Because obstruents do not have the clear formant structure of sonorants the cue for their point of articulation comes from the bending of the second formant on release of the stop or on the transition from fricative to a following vowel (Fry 1979: 138).

The palatalisation of labials, where it occurs, is usually a matter of analogy (M. McKenna 2001) to dental and velar palatals as when labial palatalise before front vowels, though contrast can also arise between palatals and following back vowels, for instance if front vowels are retracted or if a suffix with a back vowel is attached to a stem with a final palatal consonant.

In both Russian and Irish the palatal ~ non-palatal contrast plays a significant role in morphology, both inflectional and derivational and in the phonological structure of lexemes (Hickey 1985b; Unbegaun 1969: 29). Again in both languages palatalisation has a common origin as co-articulatory assimilation to a following high vowel (Kury łowicz 1971; Lunt 1956). The phonetic motivation for palatalisation is no longer necessary so that palatal consonants can occur without any following high vowel triggering them, e.g. *cleas* /kjljas/ 'trick.NOM', *chlis* /xjljɪsj/ 'trick.GEN', *corp* /kʌrp/ 'body.NOM', *choirp* /xɪr^{j}p^j/ 'body.GEN'.

Each of these word pairs has palatalisation as the marker of the genitive just as Russian has palatalisation as a morphonological alternation between, for example, the singular and plural with certain nouns (Lomtev 1972: 66-67) and within items of a verbal paradigm as in *stlat'* /stlatj/ 'to spread', *stelju* /sjtjɪˈlju/ 'I spread', *gnat'* /gnatj/ 'to drive', *gonju* /gʌˈnju/ 'I drive'. There need not, however, always be a contrast in palatality with such alternations, cf. *lovit'* /lʌˈvjitj/ 'to catch', *lovlju* /lʌvˈlju/ 'I catch', *kolotit'* /kəlʌˈtjitj/ 'to beat', *koloču* /kəlʌˈtjʃju/ 'I beat'.

5.2. Palatalisation as a cover feature

When palatalisation applies to the entire or nearly the entire consonant inventory, its phonetic realisation will vary depending on the affected segment. For the grammar of the language in question [palatal] is a *cover feature*. What is important is that the phonological segments form pairs which are distinguished by phonetic features. For palatalisation this feature is point of articulation for dentals and velars; for labials is can be tenseness and spreading of the lips, as in Irish.

This situation is parallel to that in other languages which have a cover feature which can distinguish most consonants by its presence or absence. An example of this is the feature ‘intensity’ for many Caucasian languages (Catford 1977b: 289), where the elements which carry this feature have different but related realisations, for example tense and aspirated initial and final consonants, geminate intervocalic segments (in the Lezgian language) and affrication of stops with prolonging of the fricative portion of affricates in Avar and most Andi languages. In all languages lenis consonants are short and non-aspirated. A similar distinction is found for certain Mesoamerican Indian languages such as Zapotec, which have fortis and lenis consonants. The phonetic correlates of this phonological distinction are voicelessness with fortis stops, increased intensity with fricatives and length with fortis sonorants (Nellis and Hollenbach 1980: 92; Jaeger and van Valin 1982: 126).

5.3. The position of /j/

With regard to [j] Russian and Irish differ. In Irish it is always the realisation of the phoneme /ɣj/ as in: *giobach* /gjʌbəx/ ‘untidy’, *an-ghiobach* /anɣ ɣjʌbəx/ [anjjʌbəx] ‘very untidy’. A peculiarity of [j] is not just that it is a dependent phoneme, i.e. that it only occurs as the result of lenition but that it can be the result of leniting both /gj/ and /dj/ as in: *díol* /dji:əlɣ/

'sale', *a dhíol* /ə ɣjiːəlɣ/ [əjiːəlɣ] 'his sale'. There is no independent segment /j/ in Irish. In Russian /j/ exists phonologically, cf. the initial sound in *jazyk* /jɪˈzɨk/ 'language' and *jug* /jug/ [juk] '(the) South'. In addition, /j/ can occur immediately after a palatal consonant as in: *p'janyj* /pjjanij/ 'drunk, intoxicated', confirming its status as an independent segment in Russian (DeArmond 1975).

5.4. Segment inventories

The following tables show the entire sets of palatal and non-palatal segments for Russian and Irish. In Russian there are a few asymmetries such as the long fricative[209] and the affricate, /ʃjʃj/ and /tjʃj/ respectively, which do not have non-palatal counterparts, the fricatives /ʃ, ʒ/ are always non-palatal and is it normally assumed that the affricate /ts/ does not have a palatal counterpart. In Irish there is no palatal counterpart to /h/.

(26)

a. Non-palatal/palatal consonant pairs in Russian

/p/	:	/p^j/	/b/	:	/b^j/
/f/	:	/f^j/	/v/	:	/v^j/
/m/	:	/m^j/			
/t/	:	/t^j/	/d/	:	/d^j/
/ts/	:	Ø			
/s/	:	/s^j/	/z/	:	/z^j/
/ʃ/	:	Ø	/ʒ/	:	Ø

/l/	:	/l^{j}/	/n/	:	/n^{j}/
/r/	:	/r^{j}/			
Ø	:	/ʃ^{j}ʃ^{j}/	Ø	:	/t^{j}ʃ^{j}/
/k/	:	/k^{j}/	/g/	:	/g^{j}/
/x/	:	/x^{j}/			

b. Non-palatal/palatal consonant pairs in Irish

/p/	:	/p^{j}/	/b/	:	/b^{j}/
/f/	:	/f^{j}/	/v/	:	/v^{j}/
/m/	:	/m^{j}/			
/t/	:	/t^{j}/	/d/	:	/d^{j}/
/s/	:	/s^{j}/			
/l/	:	/l^{j}/	/n/	:	/n^{j}/
/r/	:	/r^{j}/			
/k/	:	/k^{j}/	/g/	:	/g^{j}/
/x/	:	/x^{j}/	/ɣ/	:	/ɣ^{j}/
/h/	:	Ø			

The above sets of sounds can be grouped as pairs and with the use of consonant pair notation (see II.1.5 *Pairwise notation* above) can be represented in rows and columns. Here the most obvious difference between the two languages becomes apparent: Russian has voiced sibilants while Irish does not and Irish has voiced velar fricatives while Russian does not.

(27)

a. Consonant pairs in Russian

P	:	*B*	;	*F*	:	*V*
M						
T	:	*D*	;	*S*	:	*Z*
L	:	*N*				
R						
K	:	*G*	;	*X*		

b. Consonant pairs in Irish

P	:	*B*	;	*F*	:	*V*
M						
T	:	*D*	;	*S*		
L	:	*N*				
R						
K	:	*G*	:	*X*	:	ɣ

Various sounds are excluded from these tables. In Russian neither of the affricates, /tjʃj/ and /ts/, forms a consonant pair. The grooved fricatives /ʃ/ and /ʒ/ do not form pairs either. It is true that a palatal voiceless fricative, /ʃjʃj/, exists in Russian but, as it is a geminate and as it does not alternate morphologically with /ʃ/, it cannot be interpreted as the palatal counterpart of the latter.

5. 4.1. The palatalisation of labials

As mentioned at the outset above, the consonant inventories of Russian and Irish can be split into a labial and non-labial group. The former has a palatal ~ non-palatal contrast. It is known from Irish and the developments in various dialects (including Scottish Gaelic) that labials were palatalised by analogy with the already existing palatals with the tongue as active articulator (Jackson 1967b: 180). Furthermore, there does not seem to be any language which has a palatal ~ non-palatal contrast only for labials (Bhat 1978: 68-70).

The secondary nature of palatalised labials is due to the fact that the tongue obviously cannot be involved in their articulation (at least directly) and so they can only arise if there is pressure from the grammatical system for some feature of secondary articulation to be adopted as the realisation of morphonological palatalisation. This may be later lost as in Ukrainian (Beloded and Zhovtobrjux 1966:

130) with the retention of the reflexes of primary palatalisation in the affricates which exist in this language.

In both Irish and Russian the phonetic correlates of palatalisation of labials is similar, a high front vowel position for the tongue during the articulation of the labial and a glide from this tongue position on the release of the labial (Jones and Ward 1969: 93). In both languages palatalised labials are characterised by tensing and spreading of the lips and a scarcely perceptible glide on their release.

(28)
/pj/ = [+tense, +spread lips]
Russian *pjat’* /pjatj/ ‘five’, Irish *peann* /pjɑːnɣ/ ‘pen’

In Russian the timing of the release of the stop and the glide from the high front vowel position has been exploited to give a contrast between a palatal consonant (Cj) and a palatal consonant followed by a /j/-glide (C^j+/j/). The labial consonant in question may be either a stop or a fricative as in *pes* /pjos/ ‘dog’, *p’eš’* /pjjoʃ/ ‘you.SG drink’ or *vedro* /ˈvjedrʌ/ ‘bucket’, *v’eš’* /vjjoʃ/ ‘you.SG plait’.

Vowel quality provides the acoustic cue for palatalisation with labials in medial and final position.[210] Again in both Irish (de Bhaldraithe 1953a: 43) and Russian (Jones and Ward 1969: 93) a brief on-glide [ʲ] to a palatal is heard with low and back vowels.

(29)

a. Irish *cóimheas* /koːvʲəs/ → [ko:jvjəs] ‘comparison’
b. Irish *cóip* /koːpʲ/ → [ko:jpj] ‘copy’
c. Russian *opera* /ˈopʲirə/ → [ˈojpjirə] ‘opera’
d. Russian *top’* /topʲ/ → [tojpj] ‘marsh, bog’

The examples just given concern palatalised labials after stressed vowels. Due to abundant stress variation in Russian, as opposed to (Western) Irish, which has fixed initial stress in all but a handful of words, there are many instances of medial palatalised labials where the stressed vowel follows. In these cases there is no on-glide, instead a degree of tenseness and spreading of the lips renders their phonological identification unambiguous: *obida* /ʌˈbjidə/ ‘insult’.

5.4.2. The palatalisation of velars

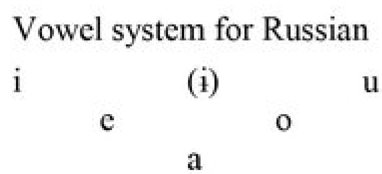
In Russian palatalised velars occur frequently and can in some instances also stand in front of low or back vowels. They are, however, far more restricted in their occurrence than in Irish. Two sources of velars before low and back vowels can be identified. The first is that of loanwords; here velars before a back vowel have been used, for instance, as the rendering of French /ky-/ as in: *k’juvet* /kjuvjet/ ‘ditch’ or of a fronted velar before a low vowel onset for a diphthong as in *gjaur* /gjaur/ ‘non-Muslim’ (from Turkish). The second source is formed by a small class of verbs with monosyllabic bases, for instance *tkat’* ‘to weave’ and its derivatives (Zaliznjak 1980: 669) which, on suffixation of the second person singular morpheme /joʃ/, do not palatalise: *tkeš’* /tkjoʃ/ ‘you.SG weave’ (Daum and Schenk 1971: 634). All other verbs with stem-final /k/ show palatalisation of the stem-final consonant, e.g. *tečeš’* /tetjʃjjoʃ/ ‘you.SG flow’, (cf. *teku* /tjɪˈku/ ‘I flow’).

The Irish palatalised velars are those which have existed in the language since the earliest attestations; they are true palatal stops (phonetically /kj/ = [c]; /gj/ = [ɟ]), which have not, and do not, show any signs of developing into affricates as has happened at various stages and places in the development of the Romance languages (see Price 1971: 49 on palatalisation in French). Nor did any changes, such as those which led to affricates in Russian, even occur in the history of Irish.

5.4. 3. The interpretation of [i] and [ɨ]

A long-standing issue for Russian scholars divided the Leningrad (St Petersburg) and Moscow phonological schools (Halle 1968: 8-9; Cubberley 2002: 63). This is whether the vowels [i] and [ɨ] (Leed 1963; Plapp 1996) should be treated as two phonological units or two realisations of the same unit with the Moscow school favouring the latter analysis yielding a five, not six, vowel system. Their argument is based on the complementary distribution of the vowels [i] and [ɨ], the latter only occurring after non-palatal consonants and the former only after palatal consonants.

(30)



The sound written <AO> in Irish corresponds to the [ɨ] of Russian written ы. It shows precisely the same distribution as its Russian counterpart: it only occurs after a non-palatal consonant and so also shows a complementary distribution with the high front vowel. The only difference is that the Irish vowels are both phonologically long, i.e. [ɨ:] (*naoi* [nɣɨ:] ‘nine’) and [i:] (*ní* [nji:] ‘(small) thing, something’) with the latter contrasting with a short [ɪ] as in *bith* ‘any, ever’ # *bí* ‘be.IMPERATIVE’.

The vowels [i:] and [ɨ:] are not treated as separate phonological units in Irish. The parallel with Russian is that the [ɨ:] of Irish is, bar its length, pronounced exactly as the vowel indicated in Russian by ы. The discussion of the status of the high central or front vowels appears to be more motivated in Irish than in Russian. In the latter [ɨ] can be seen as a post-velar realisation of a general phoneme /i/. Consider the example *s Igorem* [sɨgərjimj] ‘with Igor’ (Jones and Ward 1969: 36) where initial [i] changes to [ɨ] after a velarised /s/, i.e. the non-palatal equivalent of /sʲ/. This is a case of assimilation, here realised as depalatalisation. The few cases of word initial [ɨ] in (non-Russian) place names and its occurrence in the name of the letter ы (*y* in transliteration) is hardly enough to ascribe it phonological status.

In Irish, however, there are several instances of initial [ɨː] in native words, e.g. *aon* [ɨ:nɣ] 'one', *aos* [ɨ:s] 'people, group', and it occurs on its own in *aoi* [ɨː] 'guest'. Furthermore, it triggers non-palatal assimilation in a preceding consonant, e.g. *in aon a ráite* [ɪnɣɨ:nɣ ...] 'on the point of saying' (*in* = /ɪnj/), *an t-aon* [ənɣ te:nɣ] 'the first'[211]. The second form shows a velar consonant /t/ (= [tɣ]) which remains velarised in the position before [ɨ]. In Irish progressive assimilation is normal so that if the vowel following [t] was palatal then this consonant would be realised as [tj], compare: *an t-imeacht* [ənj tjɪmjəxt] 'the parting'. Thus *an t-aon* [ənɣ tɣɨ:nɣ] 'the first' is an example of non-palatal assimilation due to the non-palatal vowel [ɨː].

5.4.4. Coronal consonants

The next group of segments to be considered in Russian is the large set of central consonants. By these are meant non-labials and non-velars. They include dentals, alveolars and palatals and can be grouped together as 'coronals'. The sounds have in common that they involve the tip or blade of the tongue as the active articulator. In a particular language these sounds may group in a different manner on a phonological plane: thus dentals and alveolars do not contrast phonologically in Irish[212] or Russian, although they do, in Irish English (Hickey 1984b), for example. or in certain Australian languages, such as Guugu Yimidhirr, North Queensland (Dixon 1980: 132-136), Dravidian languages (Krishnamurti 2006: 53-54, on Modern Tamil, see Anamalai and Steever 1998: 102) and in the Pomoan languages of Northern California (Mithun 1999: 15-16 + 473). However, non-palatalised dentals and palatalised dentals contrast phonologically in both Irish and Russian. There is also acoustic evidence that the set of sounds which contrast with coronals, i.e. labials and velars, group together, but rarely with central consonants (Hickey 1984b). Consider, for instance, the genitive ending *-ogo* /-əvə/ of Russian and occasional words like *segodnja* /sjɪˈvodnjə/ ‘today’ which show a labial fricative from a much earlier velar stop (Cubberley 2002: 131).

The inventory of coronals for Irish and Russian are as follows.

(31)

Coronal consonants in Irish

/t/	/t^j/	/l/	/l^j/
/d/	/d^j/	/n/	/n^j/
/s/	/s^j/	/r/	/r^j/

Coronal consonants in Russian

/t/	/t^j/		/l/	/l^j/
/d/	/d^j/		/n/	/n^j/
/s/	/s^j/		/r/	/r^j/
/z/	/z^j/			
/ts/	/t^{j}ʃ^j/	/ʃ/	/ʃ^{j}ʃ^j/	/ʒ/

Irish has no voiced alveolar or palatal fricatives. Phonologically, there are no affricates in Irish either in contrast to Russian. [tʃ] does, however, occur due to assimilation: *tiocfadh sé* /tjʌkəx s^{j}eː/ → [tjʌkətʃe:] ‘he would come’. This can be seen, as the result of two processes, one is assimilation of point of articulation, /x/ to /s/, and the other is fortition of /s/ to /t/ as there is an absolute rule in

Irish phonotactics that no two fricatives can occur in succession (fricative dissimilation). [tʃ] remains a sandhi phenomenon in Irish and is not connected with non-palatal ~ palatal alternations. In Irish palatal /sj/ is realised as [ʃ] (the same sound as in English *shoe*), that is Irish uses the distinction between the narrow-grooved [s] and the broad-grooved [ʃ] to carry the phonological distinction non-palatal # palatal. In Russian the two fricatives in *rosa* /rʌˈsa/ 'dew', *osel* /ʌˈsjol/ 'ass' are both narrow-grooved, the second being pronounced with phonetic palatalisation. Equally, there is a distinction between the phonetic [ʃ] of Irish and the phonemic /ʃ/ and /ʒ/ of Russian as the latter are velarised (Jones and Ward 1969: 133): *nozh* /noʒ/ →[noʃ] 'knife', *shit'* /ʃitʲ/ → [ʃitj] 'to sew (imperfective) '. The [ʃ] from /sj/ in Irish may in its turn be palatalised to [ʃʲ] if it occurs contiguously with a further palatal consonant whose articulation is phonetically palatal: *sneachta* /sʲnʲaxtə/ → [ʃjnjæ:xtə] 'snow'.

The two coronal stops of Irish, /t/ and /d/, are similar to those of Russian in articulation as both are dental. However, all voiceless stops in Irish are aspirated, much as in English. The coronal palatal stops, /tʲ/ and /dj/, are again similar in both languages. In Russian there is no neutralisation of the contrast between /tj/ and the voiceless palatal affricate /tjʃj/, cf. *tina* /tjinə/ 'ooze, mud', *čina* /tjʃjinə/ 'rank.GEN'. In Irish the affrication of palatal stops varies. The Northern dialects show increasing affrication: in Erris, which is intermediate between the North and West, palatal stops have the apex behind the lower teeth, a slight sibilant release and no lip-rounding (Mhac an Fhailigh 1968: 36) in Donegal, which represents the North proper, /tj/ and /dj/ are widely realised as affricates [tʃ] and [dʒ] (Wagner 1979 [1959]: 9).

5.4.5. Coronal sonorants

With the sonorants /l/ and /n/ palatality is most easily perceptible as there is a recognisable formant structure during the articulation of the sonorant. Not only is palatality clear with /lj/ and /nj/, but the reactive velarisation is also obvious with /l/ and /n/ (= [lɣ] and [nɣ]), e.g. *lán* [lɣɑ:nɣ] 'full', *nó* [nɣu:] 'or'. This velarisation is recognised traditionally in Irish phonetics and these sounds are transcribed as **L** and **N** respectively (see de Bhaldraithe 1945: 37-41 for these in the Cois Fharraige dialect of Western Irish). But it is equally present with all other non-palatal consonants, as can be seen from the following narrow transcriptions: *tuí* [tɣɨ:] 'straw', *daor* [dɣɨ:ɹ] 'dear'. With obstruents there is no audible formant structure during their articulation and so this velarisation is perceived as formant bending on their release; the tongue configuration for [tɣ] and [nɣ], however, is exactly the same, the only difference being the nasal release and voice. In Russian a similar situation obtains. The laterals in *el* [jelɣ] 'ate' and *dolg* [dolɣk] 'duty' are clearly velarised (Jones and Ward 1969: 109-110) as are the nasals, cf. *nužnyi* [nɣuʒnɣɨj] 'necessary' where both nasals are velarised. In both Russian and Irish the primary articulation is still dental; velarisation is achieved by lowering the body of the tongue while simultaneously raising the back.

In Irish, as in Russian, both /r/ and /rj/ exist but the realisation of them is different. Both of them are apical but in Irish /r/ is either a frictionless continuant, like English [ɹ], but velarised, when it occurs initially, medially (intervocalically) or word-finally after a vowel, e.g. *rua* [ɹuə] 'red-haired', *árasán* [ɑ:ɹəsɑ:nɣ] 'flat, apartment', *cur* [kʌɹ] 'putting', or it is a flap when it comes after a stop and before a stressed vowel *brón* [bɾo:nɣ] 'sorrow'. /r/ is not rolled in Western Irish (unless for emphasis) although in the North a rolled [r] occurs as the articulation of /r/ with some older speakers; this has generally been replaced by a non-trill realisation.

The only similarity between Irish and Russian here is the tap articulation, which in Russian may occur when /r/ is found intervocalically and does not immediately precede a stressed vowel, e.g. *gorod* [ˈgoɾət] ‘town’, *xorošo* [xəɾʌˈʃo] ‘good’. Otherwise a slight trill is the normal realisation of non-palatal /r/.

The palatal /rj/ of Russian maintains its trill character with simultaneous raising of the tongue body towards the palate. This raising of the tongue is accompanied by closing of the jaws and slight lip spreading (Bolla 1981: 99-100, plate 77). On the whole Russian shows greater consistency in the realisation of /r/ and /rj/. In Irish palatal /rj/ is realised with considerable tongue body raising and is never a flap or trill. It is found at its clearest in intervocalic or pre-stress, post-stop position: *amáireach* [əˈmɑːɹʲəx][213] ‘tomorrow’, *tríd* [tjɹji:dj] ‘between’.

The functional load of the /r/ # /rj/ contrast in pre-vocalic initial position is slight in Irish. The few instances of this which exist tend to neutralise the distinction in favour of /r/: *reangach* [ɹæŋgəx] ‘wiry, sinewy’, *reatha* [ɹæə] ‘run.GEN’. The palatal allophone [æ] of the low vowel /a/ remains as an indicator of the phonological palatality of the preceding [ɹ]. A similarity to this situation is found in Belarusian which, like Russian itself, has a palatal # non-palatal series of consonants but has lost /rj/ (Birillo et al. 1966: 163).

Furthermore there is no evidence in Irish for the alternative of sequentialising the apico-alveolar and palatal articulations so that the sequence /r/+/j/ does not occur nor are there any cases of palatal /rʲ/ plus /j/ as in Russian *r’janyj* /rjjanij/ ‘zealous’.

5. 4. 6. Labialisation of vowels

In Russian an additional factor plays a role in the realisation of non-palatal consonants. Velarised consonants, when they occur before high or mid back vowels, i.e. /u/, /o/ or /i/ (= [ɨ]) are also labialised in this position (Jones and Ward 1969: 79). In Irish this labialisation is slight as consonants and back vowels do not have the labial to non-labial transition during their articulation which is characteristic of Russian and most evident with /o/, e.g. *noskij* [n'oskʲij] 'durable (of clothes)'. Velarisation of nasals in Russian is clearest before /i/ ([ɨ]), /u/, /o/ and to a certain extent before /a/, e.g. *nam* [nɣam] 'us. DATIVE', *ona* [ʌˈnɣa] 'she' (when stressed). As [nɣ] does not occur before [i] (this being the allophone of /i/ after a palatal consonant) the issue does not arise nor does it with /e/ as non-palatal nasals (or other consonants, Jones and Ward 1969: 198) cannot precede the mid front vowel. This statement must be qualified somewhat to include the position of loanwords and loan-morphemes. Thus in Russian examples of non-palatals before the mid-vowel /e/ are found with loanwords such as *tennis* [ˈtenjis] 'tennis'.

In Irish non-palatals are found before front vowels, irrespective of the origin of the words in question. They are correspondingly velarised to optimise the non-palatal # palatal distinction: *Gaeilge* [gɣe:ljgjə] 'Irish', *naoi* [nɣi:] 'nine', *néal* [nje:lɣ] 'cloud'.

5. 4, 7. Inflection in Irish and Russian

The phonological parallels between Irish and Russian are not matched by similarities in the morphology of both languages. In the case of Russian one has variable stress, frequently falling on inflections and thus retaining them with their full phonetic value. Irish has strong initial stress and inflections have become increasingly indistinct in the course of the language's history. The weight of exponence has been shifted to consonants with the opposition palatal # non-palatal. This system is supported by the completeness and regularity of the opposition in Irish phonology and by the fact that it does not involve further changes such as the assibilation characteristic of the Slavic palatalisations (Carlton 1990: 97). As inflections are never stressed there is no possibility of an alternation between stem vowel and zero which one has quite frequently in Russian as with so-called unstable vowels in oblique cases of nouns (Pulkina 1981: 25-26), e.g. *rot* ~ *rta* ‘mouth’, *lev* ~ *l’va* ‘lion’.

V Appendixes

1. The history of Irish

The divisions given here are those which are generally accepted by Celtic scholars, beginning with Thurneysen (1946: 1) and continued in the major work on the history of Irish, *Stair na Gaeilge* (McKone et al., eds, 1994).

Time	***Period***	***Language***
4c	Primitive Old Irish	Written remains of the language are not yet available. This is the period of Christianisation in Ireland (in 432 by St. Patrick according to tradition). The early Celtic Christian church is particularly strong in Ireland and Scotland.
6c	Early Old Irish	Attested in Ogam inscriptions (standing stones with personal names etched on the edge in a particular script).
7c	Old Irish	Documented in glosses to religious works found in monasteries in continental countries, especially Germany, Switzerland and Italy.
800-	Old Irish	Linguistic influence is seen in borrowings from Old Norse.

900-	Middle Irish	Available in legal texts, sagas as well as in works of literature contained in famous manuscript collections. Viking settlements are established in Ireland.
1169	Middle Irish	Coming of the Normans (military conquest). Introduction of Norman French to Ireland; English speakers came in the retinue of the Norman lords.
1200-	Early Modern Irish	An increasingly fossilised form of language is found in praise poetry and emulations of older literary styles.
1600	Modern Irish	Dialectal divisions become obvious (North-West, West, South-West). Appearance of regional differences in writing. Separation of Scottish Gaelic and Manx is complete.
1601	Modern Irish	Irish and Spanish forces are defeated at Kinsale, Co. Cork. During the seventeenth century a vigorous policy of plantation is pursued, chiefly by Cromwell in the late 1640s and early 1650s. This led to a concentration of Irish speakers in the poorer regions of the west of the country. Speakers from Ulster are also settled in North Connacht.
Early 19c	Modern Irish	Rapid decline of the Irish language sets in despite Catholic Emancipation in 1829.

1845-8	Modern Irish	Potato famine occurs, affecting the poorer, mostly Irish-speaking areas; about one million people die.
late 19c	Modern Irish	Decline is furthered by mass emigration in the ensuing exodus from the countryside. More than a million people emigrate during and immediately after the Great Famine.
1850-	Modern Irish	Irish-speaking areas no longer geographically contiguous.
20/21c	Modern Irish	Present-day Irish is spoken natively in areas now greatly reduced in size. There
		are now three main regions (Donegal, Connemara, Kerry with remnants in South-West Cork and West Waterford) with not very much more than 20,000 native speakers left in the Gaeltacht. However, there is a much greater number of non-native speakers, with varying degrees of competence in the language.

1.1. Studies of Irish

The first treatise in Irish on questions of language is *Auraicept na n-Éces*, literally ‘the poet’s primer’ (Calder 1917, Ahlqvist 1983), an uneven work, containing many unfounded speculations on the origin of Irish and of alphabets alongside reasonable comments on the structure of the language. It was composed in sections, the earliest of which reach back to the seventh century (although the manuscripts date from the fourteenth century and afterwards) and in which many of the terms later found in the language were introduced. The text itself is quite short, less than 200 lines, but the manuscript contains much extraneous comment, resulting in a size of some 1600 lines for the entire work. It is not known who the original author was, although there is no lack of speculation, such as that of O'Donovan (1845: 55) who sees the work as having been composed by one Forchern who is supposed to have flourished in Ulster in the first century AD.

More recent authors such as Ó Cuív (1965: 158) see *Auraicept na nÉces* as arising under the influence of Isidore of Seville's (c 560-636) *Etymologiae* (something also noted by Thurneysen 1928: 303) and ventures that the latter accounts for the liking for etymologies and explanations which one finds in many of the later glossed manuscripts. It is not until very much later that one has grammars on Irish.

1.2. The bardic tracts

The Irish *Bardic Tracts* is a collective term (L. McKenna 1979 [1944]) given to a series of treatises for instructing professional writers in the grammar of Irish. They belong to the period from 1200 to 1600 (Classical Modern Irish, Ó Cuív 1965:141) during which a uniform type of language was used in professional praise-poetry for Irish local rulers. This written register was far removed from spoken speech and one of the chief purposes of the bardic tracts was to instruct potential writers in a form of the language which for them would have been quite archaic. Most of the material in the tracts stems from the sixteenth and seventeenth centuries (Adams 1970: 158) but some of it survives in manuscripts which were written in the eighteenth and nineteenth centuries. The earliest of the tracts may, in the opinion of Ó Cuív and Bergin go back to 1500 or possibly earlier.

Linguistically, the bardic tracts are far superior to the *Auraicept na nÉces* . They contain terms which are both derived from Latin and devised to deal with the special features of Irish, for instance the well-known three parts of speech: *focal* 'noun', *pearsa* 'verb' (later replaced by the indigenous term *briathar*) and *iairmbéarla*, literally 'hindspeech' a term used to refer to unstressed proclitics (Adams 1970: 158).

1.3. Early grammars of Irish

In 1571 there appeared the *Alphabeticum et Ratio legendi Hibernicum*, *et Catechismus in eadem Lingua* by John Kearney. At the beginning of the seventeenth century Giolla Brighde Ó hEodhasa [O'Hussey] (c 1575-1614), a Franciscan monk working in Louvain, produced a grammar entitled *Rudimenta Grammaticae Hibernicae* (de Clercq and Swiggers 1992: 87-91). Later in the seventeenth century, in 1677, the *Grammatica Latino-Hibernica* , *nunc compendiata* by Francis O'Molloy appeared and somewhat earlier, in 1643, Mícheál Ó Cléirigh had produced an elementary Irish dictionary again in Louvain.

At the beginning of the eighteenth century one finds *The Elements of the Irish Language*, *grammatically explained in English*, *in fourteen chapters* by Hugh MacCurtin which was printed in Louvain in 1728. By the same author there exists an *English-Irish Dictionary* (Paris, 1732). In keeping with the profession practised by many of these authors, one often has grammatical comment as an interspersion or an appendix in a religious work. Thus Andrew Donlevy appended a chapter entitled 'The elements of the Irish language' to his *Irish-English Catechism* of 1742. Towards the end of the eighteenth century one finds an Irish grammar (*Grammar of the Iberno-Celtic*, *or Irish language*) by Charles Vallancey in 1773 which was printed in an enlarged edition in 1782. By the beginning of the nineteenth century more grammars begin to appear, the most comprehensive being *A Grammar of the Irish Language* by John O'Donovan in 1845. By this time the interest of Indo-European scholars had been directed towards Celtic languages, consider Johann Kaspar Zeuss's *Grammatica Celtica* of 1853 (revised by H. Ebel in 1871) and academic articles by scholars like Heinrich Zimmer. This period also saw Alfred Holder's *Alt-celtischer Sprachschatz* (3 vols. 1896-1907) and Franz Nikolaus Finck's *Die araner mundart*. *Ein beitrag zur erforschung des westirischen* (1899). The beginning of the twentieth century saw the monumental *Vergleichende Grammatik der keltischen Sprachen* by Holger Pedersen (1909-1913 and Rudolf Thurneysen's standard work *Handbuch des Altirischen* (1909, translated into English and published in 1946) along with the *Grammaire du Vieil-Irlandais* by Joseph Vendryes (1908), the *Manuel d'irlandais moyen* by Georges Dottin (1913, 1980) and Julius Pokorny's *A Concise Old Irish Grammar and Reader* (1914).

1.4. Bibliographical information on Irish

An early bibliographical compilation of available publications and manuscripts was made by Richard Best which covers the period 1913-1941 (Best 1942). A later publication is Baumgarten (1986) which brought Best's compilation up to 1971. On the website of the Dublin Institute for Advanced Studies there is a section dealing with the electronic *Bibliography of Irish Linguistics and Literature* (URL: http://bill.celt.dias.ie/). A book with an emphasis on the external development of Irish and its status in Irish society since the late eighteenth century is Edwards (1983).

The standard print collection of early Irish manuscript material is Stokes and Strachan 1975 [1901].

2. The sound system of Modern Irish

The following tables gather together summary information, about the consonants and vowels of Irish, which is contained in various sections of the present book. The purpose of doing that here is to provide a complete overview in one section for easy reference by readers.

2.1. The consonant inventory

The consonants of Irish come in pairs of palatal and non-palatal segments, the phonetic realisations of the latter being noticeably velarised in Western and Northern Irish.

Table 1. Consonants of Modern Irish

	labial	alveolar	velar	glottal
1: stops	p, p^j; b, b^j	t, t^j; d, d^j	k, k^j; g, g^j	
2: fricatives	f, f^j; v, v^j	s, s^j	x, x^j; ɣ, $ɣ^j$	h
3: nasals	m, m^j	n, n^j		
4: liquids		l, l^j; r, r^j		

Because all consonants, bar /h/, occur in palatal/non-palatal pairs one can achieve a more parsimonious description by using a pairwise notation – an italic capital – which acts as a cover symbol for both segments of a pair. This notation has the additional advantage of capturing an important generalisation about the sound structure of Modern Irish:

initial mutation applies equally to both palatal and non-palatal consonants, i.e. the distinction in polarity between segments of a consonant pair is invisible to mutation.

non-palatal member			**palatal member**		
P	p	*post*, ‘job’	*P*	p^j	*pioc*, ‘pick’ (v.)
B	b	*bó*, ‘cow’	*B*	b^j	*beo*, ‘alive’
M	m	*mála*, ‘bag’	*M*	m^j	*meabhair*, ‘mind’
F	f	*fág*, ‘leave’	*F*	f^j	*fionn*, ‘bright, ‘fair’
V	v	*bhog*, ‘moved’, *an-mhór*, ‘very big’	*V*	v^j	*bhí*, ‘was’ *an mhí*, ‘the month’
T	t	*tóg*, ‘take’	*T*	t^j	*teach*, ‘house’
D	d	*dubh*, ‘black’	*D*	d^j	*deoch*, ‘drink’
N	$n^{ɣ}$	*naoi*, ‘nine’	*N*	n^j	*neart*, ‘strength’
L	$l^{ɣ}$	*luí*, ‘lying’	*L*	l^j	*léamh*, ‘read’
l	l	*abhaile* ‘home’	*n*	n	*tháinig* ‘came’
R	r	*roinnt*, ‘somewhat’	*R*	r^j	*trí*, ‘three’
S	s	*súil*,‘eye’	*S*	s^j	*siúl*, ‘walking’
K	k	*cá*, ‘where’	*K*	k^j	*ceart*, ‘correct’
G	g	*gach*, ‘every’	*G*	g^j	*gearr*, ‘short’
X	x	*chun*, ‘in order to’	*X*	x^j	*a cheann*, ‘his head’ (lenited)
ɣ	ɣ	*ghlac*, ‘took’	*ɣ*	$ɣ^j$	*a ghiall*, ‘his jaw’

ŋ	ŋ	*a nglór*, 'their voice'	*ŋ*	ŋʲ	*a ngeall*, 'their promise'
(h		*a hainm*, 'her name'	h		*a hiníon*, 'her daughter')

2.2. The vowel inventory

All dialects of Irish distinguish between long and short vowels. In addition, there are three height levels – high, mid and low. Short vowels may change when the polarity of consonants of the syllable coda is reversed due to a change in grammatical category, e.g. from nominative to genitive or singular to plural, e.g. *fear* /fjar/ 'man.NOM' ~ *fir* /fjɪrj/ 'man.GEN'. This phenomenon is termed 'vowel gradation' in the present book (see section II.1.9.4).

Table 2. Vowels of Modern Irish

	Front		Mid	Back	
Level 1	iː	ɪ			uː
Level 2	eː	ɛ	ə	ʌ	oː
Level 3			a		ɑː

Long vowels	/iː/, /eː/, /ɑː/, /oː/, /uː/
	bí /bʲiː/ 'be', *cé* /kʲeː/ 'although', *tá* /tʲɑː/ 'is', *bó* /boː/ 'cow', *tú* /tuː/ 'you.SG'
Short vowels	/ɪ/, /ɛ/, /a/, /ʌ/
(i) stressed	*ar bith* /bʲɪ/ 'at all', *te* /tʲɛ/ 'hot', *cath* /ka/ 'battle', *scoth* /skʌ/ 'choice (adj.)', *luch* /lˠʌx/ 'mouse'
(ii) unstressed (schwa)	post-stress position: *domhanda* /daunˠdə/ 'global' pre-stress position: *arís* /əˈrʲiːsʲ/ 'again'

3. Lexical sets for Irish

In studies of varieties of English it is common practice to use lexical sets when discussing realisations of vowels across different forms of the language. A lexical set consists of all words which show a particular sound, e.g. a long high front vowel. Following the practice established by Wells (1982) a keyword is written in capital letters to refer to a lexical set. In the case just mentioned the keyword is FLEECE, the keyword for the long mid front vowel is FACE, for the short low front vowel is TRAP, etc.

This system has been applied to the sounds of Irish (see Hickey 2011a) and is useful when describing sounds which have varying realisations across different dialects, e.g. TÁ which refers to the long low vowel, realised as [ɑː] in Southern and Western Irish but as [æː], [ɛː] or even [eː] in forms of Northern Irish.

Table 1. Lexical sets: Consonants

	Non-palatal	Palatal
Labials		
Stops		
P	/p/ POST 'job'	/p^{j}/ PIOC 'pick'
B	/b/ BUIDÉAL 'bottle'	/b^{j}/ BEO 'alive'
Fricatives		
F	/f/ FADA 'far'	/f^{j}/ FIÚ 'even'
V	/v/ BHOG 'moved' AN-MHÓR 'very big'	/v^{j}/ BHÍ 'was' AN MHÍ 'the month'
Coronals		
Stops		
T	/t/ TÓG 'take'	/t^{j}/ TEACH 'house'
D	/d/ DUBH 'black'	/d^{j}/ DEOCH 'drink'
Fricative		
S	/s/ SÚIL 'anticipation, eye'	/s^{j}/ SIÚL 'walking'
Velars		
Stops		
K	/k/ CÁ 'where'	/k^{j}/ CEART 'correct'
G	/g/ GACH 'every'	/g^{j}/ GEARR 'short'
Fricatives		
X	/x/ A CHARR 'his car'	/x^{j}/ A CHEANN 'his one'
ɣ	/ɣ/ DHÁ 'two' A GHUTH 'his voice'	/ɣ^{j}/ DHÍOL 'sold' A GHIALL 'his jaw'
Sonorants		
Nasals		
M	/m/ MÁLA 'bag'	/m^{j}/ MEALL 'pile'
N	/n$^{\gamma}$/ NAOI 'nine'	/n^{j}/ NEART 'strength'
n	/n/ THÁINIG 'came'	
ŋ	/ŋ/ A NGLÓR 'their voice'	/ŋ^{j}/ A NGEALL 'their promise'
Liquids		
L	/l$^{\gamma}$/ LUÍ 'lying'	/l^{j}/ LÉAMH 'read'
l	/l/ ABHAILE 'at/to home'	
R	/r/ ROINNT 'part'	/r^{j}/ TRÍ 'three'

In some cases two keywords are given because a sound may have two sources. This is the case with *V* which can represent the outcome of leniting

B or *M* and with ɣ which results from leniting either *D* or *G*.

Table 2. Lexical sets: Vowels

	same as preceding consonant for [palatal]	**different from preceding consonant for [palatal]**
Short vowels		
/ɪ/	FIOS 'knowledge'	BUIDÉAL 'bottle'
/ɛ/	TE 'hot'	—
/a/	SLACHT 'neatness'	TEACHT 'coming'
/ʌ/	COR 'turn'	SIOC 'frost'
	TURAS 'journey'	FLIUCH 'wet'
Short unstressed vowel		
/ə/	BALLA 'wall'	
Long vowels		
/iː/	LÍON 'fill (v.)'	BAOL 'danger'
/eː/	MÉ 'me'	GAELACH 'Irish'
/ɑː/	TÁ 'is'	BREÁ 'good'
/oː/	BÓ 'cow'	BEO 'alive'
/uː/	CÚ 'hound'	FIÚ 'even, still'
Diphthongs		
/ai/	AGHAIDH 'face'	BEIDH 'will be'
/au/	CABHAIR 'help'	LEABHAR 'book'
/iə/	BIA 'food'	—

/uə/	CRUA ‘hard’	—

Table 3. Vowels before heavy sonorant codas

NN			
	FONN ‘desire’	GLEANN ‘valley’	BINN ‘summit’
NG			
	LONG ‘ship’	MOING ‘mane’	
M			
	TROM ‘heavy’	AM ‘time’	IM ‘butter’
RR, RD			
	CORR- ‘odd’	BORD ‘table’	AIRDE ‘height’
LL			
	MALL ‘slow’	POLL ‘hole’	MOILL ‘delay’

4. The transcription of Irish

The present book, following on the practice found in Hickey (2011a), uses the transcription system corresponding to the International Phonetic Alphabet (revised to 2005, see chart below). There are differences between this and the transcriptions found in other published works on Irish and these are outlined here.

IPA and Irish phonetic transcription

Irish phonetic transcriptions are given in bold typeface without bracketing, e.g. **ba:Lə** *balla* ‘wall’. This means that a distinction between phonological segments, enclosed in obliques //, and phonetic realisations, enclosed in square brackets [], is not made in Irish phonetic transcription.

Secondly, palatality is indicated differently in both transcription systems. The IPA uses a superscript yod [^{j}]

whereas in Irish phonetics the prime symbol [ʼ] is placed after the segment it qualifies.

Thirdly, there is a difference in the use of capital letters in the IPA and Irish phonetics. The latter uses uppercase N and L to indicate a nasal or liquid which is pronounced with maximum secondary articulation along a palatal-velar cline. Where an uppercase N or L is used with the prime symbol this indicates a strongly palatal *n*- or *l*-sound; without the prime it indicates a strongly velarised *n*- or *l*-sound.

Table 1. Traditional Irish and IPA transcription in comparison

	Traditional sign/indication	Irish sample		IPA sign	IPA sample	
Palatality	prime	*bhí*	**v′i:**	^{j}	*bhí* /v^{j}i:/	‘was’
Velarity	uppercase	*naoi*	**Ni:**	^{ɣ}	*naoi* /n^{ɣ}i:/	‘nine’

The options (i) uppercase or (ii) lowercase and (iii) prime or (iv) no prime yield a fourway distinction for *n*- or *l*-sounds in Irish.

(1)

Possible fourway distinction for sonorants in Irish

palatal	**N′**	**n′**	**n**	**N**	velarised
	L′	**l′**	**l**	**L**	

In this system the two lowercase transcriptions represent a weakly velarised sonorant (without a prime) and a weakly palatalised sonorant (with a prime).

The IPA transcription used in Hickey (2011a) allowed for a similar fourway distinction among sonorants. This possibility did not mean that a fourway phonological distinction was assumed for any dialect. Because the dialects have sonorants which show varying degrees of palatalisation and velarisation, the fourway system is necessary to refer to individual sounds. The high polarity sonorants are indicated by a *superscript* diacritic and the low polarity are shown by a *subscript* diacritic as can be seen in the following table.

Table 2. Polarity cline for *N*- and *L*-sounds (after Hickey 2011a)

maximal palatality			maximal velarity
n^j, l^j	n_j, l_j	n_γ, l_γ	n^γ, l^γ
high polarity —	low polarity	—	high polarity

The greatest degree of phonetic contrast in this system is between the two extremes of palatality and velarity. The sounds in the centre of the continuum cannot show high phonetic contrast as they are more similar phonetically than those at either end of this continuum. In articulatory terms the four distinctions on the polarity cline can be described as follows.

(2)

[n^j]	The body of the tongue is arched upwards towards the palate with dorsal contact; the tip of the tongue is behind the lower teeth (convex tongue configuration).
[$n^ɣ$]	The body of the tongue is arched downwards away from the palate; the tip of the tongue is behind the upper teeth (concave tongue configuration).
[n_j]	There is apico-alveolar contact with slight raising of the body of the tongue towards the palate.
[$n_ɣ$]	There is apico-alveolar contact with slight lowering of the body of the tongue away

from the palate.

Because the analysis in this book is based largely on the Irish of Connemara in the West of Ireland two groups of low polarity sonorants, i.e. /n_j, l_j/ and /nɣ, lɣ/ are not necessary. For Connemara Irish there are three types of sonorants as follows.

Table 3. N- and *L*-sounds in Western Irish (Connemara)

maximal palatality n^j, l^j		n_j, l_j		maximal velarity n^{γ}, l^{γ}
high polarity	—	low polarity	—	high polarity

This means that one has transcriptions like *neart* /njart/ ‘strength’, *duine* /dɪnjə/ ‘person’ and *naoi* /nɣi:/ ‘nine’ and *leaba* /ljabə/ ‘bed’, *baile* /baljə/ ‘town’ and *luach* /lɣuəx/ ‘price’ .

For *R* sounds only two distinctions are necessary because there is no phonetic difference between a low polarity *r* and a high-polarity velarised *r* (this contrasts with *N*- and *L*-sounds). The binary opposition /r^j/ – /r/ is thus sufficient. There is no need to transcribe velarised /r/ as /r $^{\text{ɤ}}$/ just as there is no need to transcribe velarised /t/ as /t $^{\text{ɤ}}$/, unless phonetic detail is being discussed.

Table 4. R-sounds in Western Irish (Connemara)

maximal palatality r^j [r^j]		**maximal velarity** r [$r^{\text{ɤ}}$]
high polarity	—	high polarity

Vowel quantity and quality

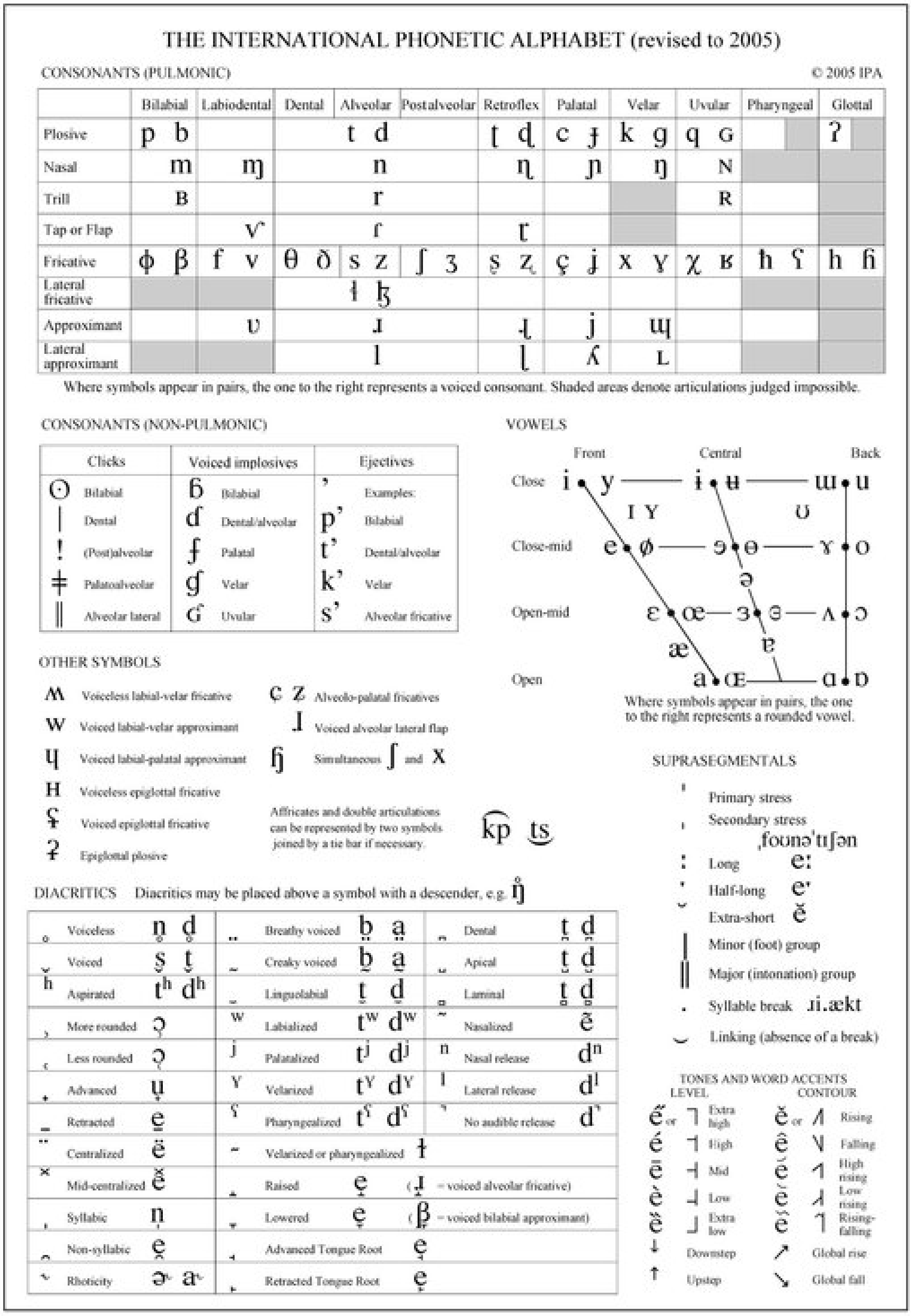
In Irish phonetic studies, a typographical distinction is not normally made between long and short vowels, apart from the length mark with the former. However, in Irish short vowels are clearly centralised, hence the indication of this quality in this book.

(4)

		Irish	IPA	
a.	*te*	**t´e**	/t^{j}ɛ/	'hot'
b.	*ar bith*	**b´i**	/b^{j}ɪ/	'at all'

Another difference in the transcription of vowels concerns orthographic <*u*> and <*o*>, as in *fliuch* 'wet' and *moch* 'early' respectively. In Irish dialect studies these are represented as /u/ and /o/ respectively. However, on a phonological level it is doubtful whether there are two separate units, at least for Western and Southern Irish. The majority realisation of both <*u*> and <*o*> is [ʌ] (mid back unrounded short vowel) as in *fliuch* [fjljʌx] and *moch* [mʌx] and for that reason /ʌ/ is the transcription used in this book. Conditional variants are found, e.g. [ʊ] in Western Irish, e.g. *rug mé* [rʊg] 'I was born' (before velar stops), or [ɔ] in Northern Irish, e.g. *cor* [kɔr] 'turn, movement' (before liquids). Note that in word-final position, where conditioning by a following segment can be ruled out, one has [ʌ] always, e.g. *scoth* [skʌ] 'choice, select' (adj.); *guth* [gʌ] 'voice'. See the discussion in section II.1.9.5 above.

Low vowels are transcribed phonologically as /a/ and /ɑ:/ (long and short respectively). Especially the short vowel can be allophonically fronted after a palatal consonant, e.g. *teas* [tjæs] 'heat' and can often be lengthened, especially in Cois Fharraige Irish, i.e. /t jas/ = [tjæ:s].



5. Scottish Gaelic

Gaelic was introduced to Scotland from Ireland in approximately 500 CE. The settlers who brought the language came to dominate the Picts who had until then enjoyed a wide distribution across Scotland. Under linguistic pressure from the newly arrived Irish, Pictish receded to finally die out perhaps as late as the ninth century. The strong position of Irish lasted up to the twelfth century after which it came to be replaced by Scots in the Lowlands, nonetheless lasting in Galloway up to the seventeenth century (Gillies 2009: 230-231). By the late middle ages Scotland was distinctly bilingual with a Gaelic-speaking area covering the Highlands and Islands and a Scots-speaking Southern area, largely consisting of the Lowlands. These areas are known as the *Gàidhealtachd* and the *Galldachd* respectively from the terms *Gàidheal* for Gael and *Gall* for non-Gael or foreigner.

The severing of ties with Ireland and the contact with the Viking settlers as of the ninth century led to the development of independent traits in Scottish Gaelic. Evidence of these is scanty in the early period of Gaelic in Scotland and it is not until after the early modern period (twelfth to seventeenth century) that these are clearly recognisable. By the end of the Middle Irish period (900-1200), Scottish Gaelic, as attested in entries in the twelfth-century *Book of Deer* (associated with the abbey of Deer in Buchan), is taken to have diverged significantly from Irish (Jackson 1972), something confirmed later by the early sixteenth century *Book of the Dean of Lismore*.

The main reason for the invisibility of Scottish Gaelic features is that the Classical Irish of the early modern period in Ireland (twelfth to seventeenth century) was used as a type of literary koiné throughout the Gaelic-speaking world (Ireland and Scotland) much in the style of West Saxon in the late Old English period.

Separation from Irish

The term ‘Common Gaelic’ (Jackson 1952) is used to refer to a stage of Q-Celtic in which Scottish Gaelic had not separated from Irish through the development of features of its own. According to Jackson this stage lasted until approximately 1200 (Jackson 1952: 82). Certainly the phonology of Scottish Gaelic (and Manx) shows the same velarisation and glottalisation which is attested in Irish after the loss of interdental fricatives as the lenited forms of *T* and *D*.

(1)

Common Gaelic				as of thirteenth century
/t/	→	/θ/	→	/h/
/d/	→	/ð/	→	/ɣ/

A further feature of Scottish Gaelic which links it up with the Irish of the Middle Ages is the shift of /n/ to /r/ in clusters of stop plus nasal as in *cnoc* with /kr-/ for an earlier /kn-/. There is, however, great variation in the realisation of /rj/ in Scottish Gaelic as it has been vocalised to /j/ or fricativised to /z/ or /ð/.

Dialect areas

As part of a general recession of Gaelic-speaking areas, the focus of Gaelic has been pushed to the North West of Scotland with the demise of the language in the Central and Eastern Highlands. Today speakers of Gaelic are concentrated on the islands and a few pockets on the Western coast and a small but well-researched pocket on the North-East in Sutherland (Dorian 1978a, 1978b). Up to the middle of the twentieth century the Gaelic-speaking continuum spread as far South as Arran and Kintyre, the dialects of which were examined by Holmer (1957, 1962). For the purpose at hand there would seem to be some justification in discussing the Gaelic of the Hebrides as a representative form of the language. It is quite close to the literary norm and is that which is preferred in print (Gillies 2009: 231). Where necessary deviating details will be mentioned.

Dialect studies

The examination of Scottish Gaelic dialects was chiefly a provenance of Scandinavian scholars who at the middle of the twentieth century carried out a number of investigations of the main dialect areas. The main works in this area are: Borgstrøm (1937, 1940, 1941), Holmer (1938, 1957, 1962), Oftedal (1956). Further studies are MacGillFhinnein (1966), M. Ó Múrchú (1989), Watson (1983, 1994), Ó Maolalaigh (1995-6, 1997, 2003), Stewart (2004a, 2004b) and Ternes (2006 [1973]) as well as handbook chapters such as McInnes (1992) and studies of specific issues such as Rogers (1980). A similarly comprehensive work to Wagner's atlas of Irish dialects has been compiled by Cathair Ó Dochartaigh (ed., 1994-97). The demise of Scottish Gaelic (Mackinnon 1993), especially in the area of East Sutherland, has been studied intensively by Nancy Dorian, cf. Dorian (1977, 1978a, 1978b), for remarks on sociolinguistic variation within a Scottish Gaelic community, see Dorian (1994).

Dictionaries of Scottish Gaelic

There are many dictionaries for Scottish Gaelic, some of which are reprints of older works with much etymological information, e.g. Dieckhoff (1998 [1932]), MacBain (1982 [1911]) and Maclennan (1979 [1925]). There is also an initiative by a number of Scottish universities to compile an historical dictionary of Scottish Gaelic, see the relevant website at http://www.faclair.ac.uk/.

5.1. Linguistic features

Word-internal h

This is sometimes found as a reflex of historic /θ/ as in *fitheach* /fi(h)əx/ which can contrast with words like *fiach* /fiəx/ 'debt' which contains the diphthong /iə/ which arose through breaking from Old Irish /eː/.

Voiceless # voiced opposition

Scottish Gaelic has for all intents and purposes lost the distinction in voice for its series of stops. Instead the distinction is realised as aspirated # non-aspirated much as in present-day Icelandic. The aspiration has two realisations (1) as post-aspiration after stops in the onsets of stressed syllables and (2) as pre-aspiration in the codas of such syllables (Nance and Stuart-Smith 2013). Aspiration occurs between the margin and the nucleus of a stressed syllable which means after the onset or before the coda stop.

Development of geminates

Scottish Gaelic has lost the geminate sonorants of Old Irish which occurred in medial and final positions but shows a long vowel or diphthong reflex, e.g. *meall* /mjaul/ 'ball, lump' (de Búrca 1977/8: 403), a feature it shares with Southern Irish at the opposite end of the Gaelic continuum. In the case of *r* Scottish Gaelic has an additional distinction not known in Irish with the possible exception of parts of Donegal: both the non-palatal and the palatal *r* can be realised as trills.

Vowel system of Scottish Gaelic

There are more vowel contrasts in Scottish Gaelic than in Irish for two principal reasons. The first is that there is a distinction in the mid area between closed and open vowels, i.e. /e:/ # /ɛ:/ and /o:/ # /ɔ:/. This would apply to long vowels but it is not certain whether the short vowels /e, ɛ/ and /o, ɔ/ are not simply contextually determined variants dependent on the quality of flanking consonants.

(2)
Maximal vowel system for Scottish Gaelic after Gillies (2009: 236)

i(ː)	ɯ(ː)	u(:)
e(:)		o(:)
ɛ(:)	ə(:)	ɔ(:)
	a(:)	

High unrounded vowels

The second reason for the size of the Scottish Gaelic vowel inventory is that there exists a high unrounded back vowel /ɯː/. This can be seen with the reflexes of the Old Irish diphthong <AO> as in *caob* /kɯːb/ 'scoop, dollop'. However, the status of this vowel is debatable in such words. Consider the cognate word in Irish *scaob* /ski:b/. This is transcribed with /i:/ but in fact the vowel is quite retracted, phonetically the form is [sk ^{ɯ}ɨ:b]. The retraction is determined by the flanking non-palatal consonants. In a parsimonious phonological analysis the vowel can be assigned to /i ː/ and the retraction attributed to the velarity of the preceding and following consonants. This interpretation would work for Scottish Gaelic as well. With the short form of the vowel the same situation applies; *duine* /dɯn^{j}ə/ 'person', *tuig* /tɯg^{j}/ 'understand' contains the same retracted vowel as does the identical Irish word which is taken to contain a variant of the / ɪ/- vowel.

Vowel quality and flanking consonants

Attributing vowel quality to adjacent consonants would lead to the collapse of a systemic distinction between closed and open short vowels. Forms such as *deich* /djexj/ 'ten' and *each* /ɛx/ 'horse' would be seen to contain conditioned variants of a short /ɛ/ vowel. This becomes all the clearer if one considers the morphophonemic alternations which occur in nominal paradigms : *each* /ɛx/ 'horse', *eich* /eç/ 'horses'.

The distinction for the long versions of mid vowels is better established as it is not dependent on the quality of flanking consonants. Of the two vowels /e:/ and /ɛ:/ the former is lexically more frequent while with the back vowel pair /ɔ:/ is more common than /o:/. The difference between the two vowels in each set is indicated orthographically, a fact which would point to a difference of some age: *feum* /fe:m/ 'need', *sèimh* /sjɛ:v/ 'mild', *mòr* /mo:r/ 'big', *òr* /ɔ:r/ 'gold'.

Short back vowels

The status of back vowels must be decided with reference to the quality of the surrounding consonants, particularly of the segment following the vowel in question. For Scottish Gaelic a stressed schwa is assumed as it occurs in a word like *goid* /gədj/ 'theft' (Gillies 1993: 149). The vowels in the cognate Irish forms are: *goid* /gɛdj/ ~ /gʌdj/ (de Bhaldraithe 1945: 12). There is variation in the vowel used for the short syllable in this and other words with a similar structure. Such words could be viewed as containing a short stressed mid-vowel which has a realisation on a horizontal scale / ɛ - ə - ʌ/. The Scottish vowel is roughly in the middle of this line with pronunciations from either extreme possible in (Western) Irish. Such an interpretation would of course shift systemic status from the short vowel to the right-flanking consonant.

The above view gains support from observations within Scottish Gaelic where vowel alternations co-occur with changes in the value for [palatal] for final consonants in short monosyllables, e.g. *cat* /kat/ ‘cat’ : *cait* /kɛtj/ ‘cats’. Such cases are parallel to Irish ones like *troid* /trɛdj/ ‘fight.NOM’ : *troda* /trʌdə/ ‘fight. GEN’.

The net effect of these considerations is that the different short back vowels can be grouped according to environmental criteria. Even the rounded short /ɔ/, which is obvious in a form like *loch* ‘lake’, can be linked to the presence of velars or velarised segments like [1ˠ] which here, like in Irish (de Bhaldraithe 1945: 14), tend to favour rounding. Compare in this respect the lack of shift of Early Modern English /ʊ/ to /ʌ/ observable in words like *bull, pull* in Modern English.

(3)
Horizontal levels for short vowels

1)	high:	/ɪ - ɨ - ʊ/
2)	mid:	/ɛ - ə - ʌ/
3)	low:	/æ - a - ɑ/

Vocalisation of fricatives

Internal lenition has been carried to its conclusion in Scottish Gaelic and voiced fricatives have merged with the vowel which precedes them to form new diphthongs. The type of diphthong which results depends on whether the former fricative was palatal or non-palatal.

(4)

/v, ɣ/	→	/u/		
/uɣ/	→	/uː/	*ùghdar*	‘author’
/av/	→	/au/	*samhradh*	‘summer’
/vʲ, ɣʲ/	→	/i/		
/uvʲ/	→	/ui/	*cuimhne*	‘memory’
/oɣʲ/	→	/əi/	*oidhche*	‘night’

The cluster /xt/ In this case the stop has assimilated in place to the preceding velar fricative yielding /xk/ as in *uachdar* ‘top’. Onomastic evidence shows that this is a late development, cf. the placename element *Auchter*- with /xt/ (Gillies 1993: 163).

/m/ in Scottish Gaelic The stop-like behaviour of /m/, found in Irish, is also to be seen in Gaelic. The clusters /sp, st, sk/ and /sm/ are not subject to lenition (Gillies 1993: 167).

For information on initial mutation in Scottish Gaelic, see section IV.2 above. For comparative information on Ulster Irish and Scottish Gaelic, see O’Rahilly (1932: 122-191) and C. Ó Baoill (1978; 2000). A detailed discussion of the tone-accent character of Scottish Gaelic is to be found in Ternes (2006: 129-145).